

Polytopes at Surreal Scales

Lifting compression, truncation histories, affine residues,
lattice counts and invisible extension complexity

AI-assisted research manuscripts of 30 September 2026,
merged in the batch-58 write phase

Manuscripts 02–07 of batch 58 of the ProveIt collection, pinned to 6996fee (02, 03, 04, 06, 07) and 5e0f4e0 (05)

Abstract

A finite polytope with surreal coordinates has an ordinary real realizable face lattice: real-closed-field transfer rules out new finite combinatorial types. What the surreal numbers add is the organization of that finite geometry across infinitesimal and transfinite scales, and this report collects six independent answers to the question of how much of it can be classified. Part I fixes a real configuration of n points of affine dimension d and varies only the heights: arbitrary surreal heights compress to at most $n - d - 1$ real layers that keep the leading term of every real affine-dependence evaluation and every common initial normal-form history; the complete sign type is an oriented partial flag whose minimum polynomial germ degree is exactly $m - 1$ (or $m - r_0$ with a prescribed standard part), while finite lifted data need degree at most one. Part II shows that truncating the *coordinates* instead is uncontrolled: four bounded planar points realize every prescribed triangle/quadrilateral history, and $\omega^{-\omega}$ selects the final hull independently. Part III, over any set-sized real closed field $K \supseteq \mathbb{R}$ with its natural valuation, introduces affine residues and scale lattices and proves that simultaneous inner segments on a finite rational grid preserve every pure mixed-volume valuation. Part IV counts lattice points: a quadrilateral with the vertex shadows of the unit square can have counting function $(n + 1)^2 - \lceil \alpha n \rceil$ for any irrational $0 < \alpha < 1$, whose generating function has the unit circle as a natural boundary. Part V constructs polygons whose coordinates agree to all orders of an infinitesimal with a polygon of logarithmic extension complexity, while their own complexity grows like a power of n ; such gaps exist exactly when the infinitesimal is not an order unit of the value group.

Status and scope. Six AI-assisted, unrefereed research manuscripts, merged without selection: every result, proof, example, question and limitation of each is printed, a result proved by two sources once with both named. The underlying tools (Conway normal forms, real-closed-field transfer, secondary polytopes, curve selection, mixed volumes, Ehrhart theory, extension complexity) are classical; each Part states what it claims. [write] No statement of this report is formalized in Lean or Rocq, and the report makes no claim to literature-wide priority. Labels carry the prefix `poly:`; conventions, the relation between the manuscripts and the provenance are in [section 1](#) and [Appendix A](#); text added in the merge is marked [write].

Contents

1	Guide to the report	5
I	Liftings of a fixed real configuration	9
2	The problem, its scope, and its relation to earlier work	11
3	Questions, answers, and provenance	12
4	Foundations, normal forms, and the affine quotient	15
5	Ordered fields, finite configurations, and normal forms	16
6	A finite coefficient-pivot theorem	18
7	Support-adapted finite-rank compression	19
8	Oriented flags classify the complete dependence sign type	21
9	Polynomial realization and exact minimum degree	22
10	Finite combinatorics: determinants, slacks, and a linear path	23
11	Fixed real bases: circuits, liftings, and certificates	26
12	Exactly which lifting histories occur?	30
13	Shadows and canonical refinement hierarchies	33
14	Standard parts: exact identities and lost faces	34
15	Higher-rank tropical minimizers: an elementary bridge	35
16	Examples that locate the exact boundary	37
17	A toric application: preserving the whole initial ideal	38
18	General surreal polytopes: what transfer gives and what it does not	40
19	A finite Puiseux model with the same residue	41
20	Corrections to the earlier convexity draft	42
21	Computation, certificates, and a formalization route	44
22	Further research questions and concrete routes	46
23	Conclusion	47
24	Dependency and verification ledger	47
25	Source and reproduction notes	48
II	Coordinate-truncation histories	49
26	Four moving points can encode arbitrary histories	49
27	Algorithms, verification, and proof boundaries	53

28 Further research questions	55
29 Conclusion	57
30 An exact recursion for the universal construction	58
31 Reproducibility and source inventory	58
III Affine residues and mixed-volume scales	59
32 Research target, principal results, and status	61
33 Set-sized foundations and finite geometry	63
34 Algebraic volume and reduction to the residue field	65
35 The scale lattice and canonical affine residue	67
36 Parallelotope models and finite determinant certificates	69
37 Simultaneous inner segments and the valued Grassmannian	71
38 Full profiles, cloned segments, and higher-rank exchange	73
39 Recovering a polytope's scale class from probes	76
40 Examples, counterexamples, and what the invariants forget	78
41 Finite higher-rank certificates and real specialization	80
42 Exact verification and a formalization route	83
43 Further research questions and topics	84
44 Conclusion	86
45 Proof dependencies and audit boundaries (Appendix A of manuscript 06)	86
IV Real traces and lattice-point counting	87
46 The research questions and their answers	89
47 Fields, finite polytopes, shadows, and traces	90
48 The classical limit: no new finite face lattices	92
49 A finite lexicographic normal form for real traces	93
50 One-parameter algebraic descent preserving the entire trace	95
51 Face implantation and boundary universality	97
52 An integral-shadow quadrilateral with a natural boundary	98
53 The universal surface coefficient	101
54 The full attainable interval, with fixed combinatorics	103
55 A rationality criterion that restores Ehrhart behavior	105

56	Repository integration and the exact proof boundary	107
57	Exact computational checks and reproducibility	108
58	Further research questions	110
59	Conclusion	112
60	Additional scope checks and potential failure modes (Appendix A of manuscript 03)	113
61	Reproducibility and source ledger (Appendix B of manuscript 03)	113
V	Extension complexity beyond all orders	114
62	The questions, the answer, and the research boundary	117
63	Fields, polytopes, lifts, and power-jet information	120
64	The explicit separation theorem	125
65	Geometry of the cap and stability beyond all powers	126
66	An explicit logarithmic-size reflection lift	128
67	Generic lower bounds over real closed fields	129
68	Explicit independence and the hard lower bounds	132
69	Identical initial slack matrices and identical jets	134
70	A second construction: regular and cyclic polygons on one circle (source 04)	135
71	What the asymptotic data cannot recover (source 04)	140
72	A common field and a sharp valuation-rank boundary	142
73	Residue descent for positive ranks and polytope lifts	146
74	Prescribed precision and finite conservativity	149
75	The same separation in every fixed dimension	154
76	A rational-chart generalization (source 04)	155
77	Exact computation versus discarded information (source 04)	156
78	Reproducibility and final conclusions	158
79	Source and dependency audit (Appendix A of manuscript 07)	160
80	Repository, literature, and verification audit (Appendix A of manuscript 04)	160
81	Notation and possible misreadings (Appendix B of manuscript 04)	162
82	Further research questions and concrete next steps	162
	Formalization roadmaps, provenance and references	168
83	Formalization roadmaps (proposals only)	168

A Provenance of the report	173
References	177

1 Guide to the report

[write] This section, the formalization chapter’s framing, the provenance appendix and every paragraph marked [write] were written when the six manuscripts were merged. Everything else is the manuscripts’ text.

1.1 Six answers to one question

Manuscripts 02–07 of batch 58 were written independently, on 30 September 2026, in answer to one prompt about finite polytopes over the surreal numbers. They share a setting: a set-sized real closed field $\mathbb{R} \subseteq K \subseteq \mathbf{No}$ (or, in manuscript 05, an ordered subfield), its ring of finite elements, and the standard part st . They share three basic facts, each proved by several of them and printed here once: $\text{st conv} = \text{conv st}$ ([proposition 13.1](#)), real-closed-field transfer of a finite configuration ([theorem 18.1](#)), and an explicit real threshold for finitely many sign conditions ([lemma 10.3](#)). Beyond that they answer three different questions, the three spines of this report.

Part	Subject	Sources (base first)	Labels
I	Liftings of a fixed real configuration	05, 02	<code>poly:</code> , <code>poly:ts:</code>
II	Coordinate-truncation histories	02	<code>poly:ts:</code>
III	Affine residues and mixed-volume scales	06	<code>poly:mv:</code>
IV	Real traces and lattice-point counting	03	<code>poly:ehr:</code>
V	Extension complexity beyond all orders	07, 04	<code>poly:ic:</code> , <code>poly:bao:</code>

- *The lifting spine* ([Parts I](#) and [II](#)). Manuscripts 05 and 02 prove the same theorem: surreal heights on a fixed real base compress to at most $q = n - d - 1$ real layers. Manuscript 05 is the base (weaker hypotheses, stronger conclusions, complete classification of the invariant); 02’s compatibility with every common initial cut, its history classification, planar sharpness, polar and band identities, Puiseux model and query-model bound are printed beside it. 02’s coordinate-truncation theory is [Part II](#).
- *The residue and volume spine* ([Parts III](#) and [IV](#)). Manuscripts 06 and 03 are complementary: they share no theorem beyond the three basic facts. Manuscript 06 moves the coordinates and keeps volume information; manuscript 03 counts lattice points.
- *The complexity spine* ([Part V](#)). Manuscripts 07 and 04 prove the same phenomenon: explicit polygons that agree to all orders of an infinitesimal with a polygon of small extension complexity but have much larger complexity themselves. Manuscript 07 is the base (every $n \geq 4$, linear and semidefinite lifts, descent for every convex valuation ring); 04’s field- and scale-general criterion, its same-circle regular and cyclic family with a two-sided $\sqrt{n}$ order, and its rank-one and factorization results are printed beside it.

1.2 How the manuscripts fit together

No manuscript contradicts another. Three places look like conflicts and are not.

Remark 1.1 (Three minimum degrees). [write] Part I prints three “minimum degree” or “number of layers” statements that can look incompatible: exactly $m - 1$ ([theorem 9.1](#)), at most $r \leq p$ layers ([theorem 12.2](#)), and one real vector or degree at most one ([theorem 10.4](#) and the paragraph after [figure 1](#)). They are three different invariants of one lifting:

- (a) the *complete real dependence sign type* χ_h (definition 8.1): its rank is m (05's letter; 02's p), and its minimum polynomial germ degree is $m - 1$, or $m - r_0$ when the standard part is prescribed (theorems 9.1 and 9.2);
- (b) the *strict geometric history* of lower subdivisions along the common initial cuts: it has $r \leq m$ strict steps and needs exactly r layers (theorems 11.2 and 12.2);
- (c) the *final finite data* (the lifted face lattice, the marked subdivision, finitely many signs): one real height vector reproduces it, and with a prescribed standard part an affine-linear path of degree at most one does (theorem 10.4; the paragraph after figure 1).

The tempting false reading is that 05's "degree $m - 1$ " and 02's " r layers" or "one real vector" measure the same thing with different answers. They do not: (a) asks for *every* real dependence sign, including the uncountably many that no finite face test sees; (b) asks only for the finitely many subdivisions met along the cuts; (c) asks for one of them. Theorem 10.6 shows that (a) and (c) can be arbitrarily far apart. In 02's hexagon, (a) and (b) both need three layers, while (c) needs one real vector.

Remark 1.2 (Valuation rank three and rank two). [write] Manuscript 07 builds its polygons in one field of valuation rank three, while manuscript 04 uses fields K_n of rank two, and 07 also proves that rank two is sharp for gaps invisible to all-order jets. These are consistent. An invisible gap needs a value group of rank at least two: the jet scale, and elements beyond all its powers. Manuscript 07's third archimedean layer comes only from keeping $\varepsilon = \omega^{-1}$ as a separate layer above its jet scale $\omega^{-\omega}$, for its extra assertion that each polygon is flat-close to a fixed triangle at the larger scale ε ; 07's own rank-two variant (corollary 72.4) drops that assertion, and 04's construction never makes it. 04's fields K_n all lie in one rank-two field, the real closure of $\mathbb{R}(\omega^{-1}, \omega^{-\omega\sqrt{q}} : q \text{ prime})$ (remarks 70.8 and 72.5). The criterion common to both (Part V) says that invisible gaps exist exactly when the valuation of the jet scale is not an order unit of the value group, which already happens in rank two.

Remark 1.3 (Directions of the transfer statements). [write] Several Parts pass between K and $\mathbb{R}$, in different directions. Transfer (theorem 18.1) keeps a finite combinatorial type exactly but forgets the scales. Standard part and residue maps keep the scale data of Part I (proposition 13.3) but can only coarsen, and in Part V can only lower extension complexity; manuscript 04's real-algebraic transfer keeps the exact extension complexity. Manuscript 03's descent theorem (Part IV) lands in the real closure of $\mathbb{R}(t)$, a field of valuation rank one where the all-order jet kernel of Part V vanishes; 03 makes no valuation-theoretic claim, so it does not conflict with Part V.

Part III's theory holds for every set-sized real closed field $K \supseteq \mathbb{R}$ with its natural convex valuation; its surreal specifics are examples, and the word "surreal" in manuscript 06's title is not a hypothesis.

1.3 Conventions and notation

Each Part keeps its source's definitions and hypotheses and opens with a paragraph "Notation in this Part". The following conventions hold throughout; the table lists the letters and words that mean different things in different Parts, with the reading to avoid.

- **No** is the class of surreal numbers, ω^γ Conway's monomials. Every construction takes place in a set-sized subfield K with $\mathbb{R} \subseteq K \subseteq \mathbf{No}$ (real closed unless a Part says otherwise).
- $\varepsilon = \omega^{-1}$ and $\tau = \omega^{-\omega}$ wherever a fixed surreal is meant; sources also use ε for an arbitrary positive infinitesimal in statements valid for all of them, and say so. The one exception

is [Part III](#), which keeps manuscript 06's η for $\omega^{-\omega}$ because there τ names mixed-volume profiles. The renamed source letters are listed in the Parts' notation paragraphs and in [Appendix A](#).

- st is the standard part on the ring of finite elements ($\mathcal{O}$; manuscript 02 writes $\mathcal{O}_K$, 03 $\mathcal{O}_K$). As in the collection's notation file, it is the scalar map $\text{st} : \mathcal{O} \rightarrow \mathbb{R}$, applied coordinatewise.

Symbol or word	Meanings, by Part	Reading to avoid
ε	ω^{-1} wherever a fixed surreal is meant (I–V); an arbitrary positive infinitesimal in statements valid for all of them (I; in IV, 03's ε, δ are always arbitrary)	that 03's ε is ω^{-1}
τ	$\omega^{-\omega}$ in I, II and V (in V it replaces 07's ρ of its main construction; in III, 06's η is kept); in III, $\tau_I, \tau(\alpha)$ are 06's mixed-volume profiles	that a profile τ_I is a power of $\omega^{-\omega}$
η	I: explicit thresholds $\eta(f)$ (05), η (02), and the band width of proposition 14.2 ; III: $\omega^{-\omega}$ (06's letter, kept); IV: a real cutoff; V: 07's perturbation monomials η_ℓ , 04's flat bound or rational tolerance	that III's η is a threshold
t	I: the real parameter of polynomial models, $t \rightarrow 0^+$ (05); II and 02's introduction: the power-series variable, evaluated at $t = \varepsilon$; III: formal Hahn monomials t^γ ; IV: the real parameter or infinitesimal of $\mathbb{R}(t)$	$t = \omega^{-1}$ in Part I
m	I: the comparison rank $\dim_{\mathbb{R}} L_h(V)$ (05) $= p$ (02), but in 02's blocks the stage index of H_m ; III: the number of bodies; V: an ordinary integer exponent (07) and $n = 2^m$ (04)	that 02's m is the rank
p	I, II: 02's comparison rank ($=$ 05's m)	
q	I: $n - d - 1 = \dim V$; III: $q(B) = v \det W_B$; V: 07's easy vertices q_i , 04's primes q_i	
r	I: 02's strict history length $r \leq p$; 05's $r_0 \in \{0, 1\}$; 02's gauge residuals $r_i(h)$; III: affine dimensions r, r_i	that $r_i(h)$ are history steps
L	I: 05's map L_h , 02's space L_A ; III: the scale lattice $L(P)$, an $\mathcal{O}$ -module; IV: the lattice-point count $L_P(n)$	$L_A = L_h$; $L(P) = L_P$
V	I: the dependence space $\ker \tilde{A}$ (05), a configuration (02's transfer and Puiseux theorems); III: mixed volume V ; V: 07's coarse valuation ring V_τ	
S, T	I: a sign cell, the tail sum $T(f)$; III: segments S_i, T_i ; IV: the real trace set $S = \text{Tr}_{\mathbb{R}}(P)$, an implanted polytope T ; V: the slack matrix S , 07's triangle T , a nonnegative factor T in $S = TU$	
ρ	II: a sign $\rho \in \{-1, 1\}$; III: the threshold ρ_g ; IV: separated real scales $\rho_1 \gg \dots \gg \rho_r$; V: a jet scale that 07 re-chooses (corollary 72.4 , theorem 74.2)	that III's ρ_g is a jet scale
residue	I, II: 02's word for the standard part; III: the affine residue $\text{Res}(P)$, renormalised and dimension-preserving; V: residue maps of convex valuation rings and their residue fields k (07)	$\text{Res}(P) = \text{st}(P)$ (III's thin triangle has $\text{st } P$ a segment and $\text{Res } P$ a triangle)

Symbol or word	Meanings, by Part	Reading to avoid
shadow	I: the standard part of heights or of a configuration (05); IV, V: $\text{st}(P)$; V: 04's "algebraic finite-accuracy shadows" are real-algebraic approximants and are printed as approximants	
trace	IV: the real trace $\text{Tr}_{\mathbb{R}}(P) = P \cap \mathbb{R}^d$; V: the matrix trace tr	that $\text{Tr}_{\mathbb{R}}(P)$ is $\text{st}(P)$ or tr
valuation	I: $\text{le}(x)$, the leading exponent (larger means larger); III: v with larger value meaning smaller element, value group Γ not necessarily inside $\mathbb{R}$; IV: none used; V: v , with larger value meaning smaller element (07; 04's v renamed)	a common sign convention across Parts
$\mathcal{O}$	the ring of finite elements, one ring in every Part ($\mathcal{O}_K$ in 02, $\mathcal{O}_K$ in 03, $\mathcal{O}$ in 04, 06, 07), with maximal ideal $\mathfrak{m}$	
flat ideal	V: $\mathfrak{j}_x = \bigcap_{m \geq 1} x^m \mathcal{O}$ for a positive infinitesimal x (07's $\mathfrak{j}_s$; 04's $\mathcal{I}_\varepsilon$ is printed $\mathfrak{j}_\varepsilon$)	
minimum degree	I: three different invariants (remark 1.1)	that $m - 1$, r and "one real vector" conflict
rank	I: the comparison rank $m = p$; V: the valuation rank (the number of archimedean layers of the value group), matrix rank, nonnegative rank rank_+ and psd rank rank_{psd}	valuation rank three versus two (remark 1.2)

1.4 What is claimed, and what is not

Every Part states its own claims and non-claims, and the merge keeps all of them. In summary: every theorem has a written conventional proof in its source; the verification programs check finite examples only; no manuscript claims literature-wide priority or the solution of a named published problem; the classical ingredients are attributed in each Part. [write] None of the report's results is formalized. The only ProveIt declaration that any Part uses as an input is `SurrealFoundations.SignSequence.omnific_bounded_iff` (Algebra/SurrealNumbers/Surreal/Foundations/OmnificUnits.lean; bounded omnific integers are ordinary integers), in [Part IV](#). Manuscripts 05 and 02 inspected the coordinate-geometry file `Algebra/SurrealNumbers/Surreal/Algebra/Geometry.lean` as background; no result depends on it. Placement of this report beside the collection's Lean development confers no formal status on it.

Manuscripts 05 and 07 correct a privately supplied draft, *Finite Convexity and Exact Lexicographic Optimization over the Surreals* (22 September 2026), which is not part of ProveIt and could not be checked ([section 20](#) and [Part V](#)).

1.5 Labels, numbering and marks

Every label carries the prefix `poly::`; material from manuscripts 02, 03, 04, 06 and 07 carries the sub-prefixes `poly:ts:`, `poly:ehr:`, `poly:bao:`, `poly:mv:` and `poly:ic:`, followed by the delivered label unchanged. Sections are numbered consecutively through the report, and theorems, equations and questions within sections, so a source's numbers are not those printed here. A result printed once for two sources carries both labels. Citation keys carry the prefixes `fs:` (05), `ts:` (02), `ehr:` (03), `bao:` (04), `mv:` (06) and `ic:` (07); the references are listed at the end, grouped by Part.

Part I

Liftings of a fixed real configuration

Source and scope. [write] This Part prints manuscript 05 of batch 58, *Finite-Scale Geometry of Surreal Liftings: Sharp dependence compression, exact polynomial degree bounds, and linear realization of finite combinatorics* (research manuscript prepared for Vladimir Reshetnikov, an AI-assisted contribution to the ProveIt research program, 30 September 2026; a 25-page A4 PDF; pinned to 5e0f4e0), whole and in its order. It is the base of this Part and of the report. Merged into it is the lifting material of manuscript 02, *Surreal Polytopes at Finite and Transfinite Scales: Sharp lifting compression and universal coordinate-truncation histories* (research article prepared for Vladimir Reshetnikov in connection with the ProveIt research repository, 30 September 2026; a 28-page Letter PDF; pinned to 6996fee): its introduction, its working-field conventions, its cut-compatible compression theorem, its fixed-base lifting and history sections, its standard-part and Puiseux sections, its certificate workflow and its query-model bound. Manuscript 02’s Section 6, on coordinate truncation, is [Part II](#).

The two manuscripts prove one theorem from two sides: surreal heights on a fixed real base compress to at most $q = n - d - 1$ real layers that preserve every real affine-dependence sign. Manuscript 05 is the base because it assumes no more (a set-sized ordered subfield $\mathbb{R} \subseteq K \subseteq \mathbf{No}$, with real closure only where a statement needs it, against 02’s real closed K) and concludes more (exact leading terms rather than signs, the complete flag classification, and exact germ degrees). A result proved by both is printed once, with both sources named; a genuinely different proof from 02 is kept as a second proof; the statements of 02 that go beyond 05 (compatibility with every common initial cut, the classification of histories, planar sharpness, the polar and band identities, the Puiseux model and the query-model bound) are printed whole. Each block from 02 opens with a [write] note that names it and relates it to 05.

Notation in this Part. [write] (The report-wide table is [section 1.3](#).) ε is ω^{-1} wherever a fixed surreal is meant; both sources also write ε for an arbitrary positive infinitesimal in statements that hold for every one, and say so. Manuscript 02 writes $t = \omega^{-1}$; in the 02 material of this Part that letter is replaced by ε , because here t is manuscript 05’s *real* parameter of polynomial models ($H(t)$, $t \rightarrow 0^+$). Only 02’s introduction ([section 3](#)), which previews [Part II](#), keeps t : there, as in [Part II](#), t is the variable of power series evaluated at $t = \varepsilon$. $\tau = \omega^{-\omega}$ (02’s letter; 05’s ε_2 in [section 16.5](#), renamed). The comparison rank is written m by 05 and p by 02; it is one number, $m = \dim_{\mathbb{R}} L_h(V) = \text{rank}_{\mathbb{R}}(c \mapsto c \cdot h) = p$ ([corollary 6.3](#) and [theorem 11.2\(d\)](#)). The letter p is kept in 02’s material because 02 uses m for the stage index $0 \leq m \leq p$ of its polynomials H_m : in 02’s blocks m is *not* the comparison rank. 05’s dependence space $V = \ker_{\mathbb{R}} \tilde{A}$ is 02’s L_A (02’s L_A is a space, not 05’s map L_h); in 02’s transfer and Puiseux theorems V is a configuration of points. 05’s slack $s_{B,i}(h)$ is 02’s $\delta_{B,i}(h)$, 05’s equality set $I_B(h)$ is 02’s cell $D_B(h)$, and 05’s gauge residuals g_i are 02’s $r_i(h)$ (not the history length r). 02’s ring $\mathcal{O}_K$ of finite elements is the ring written $\mathcal{O}$ elsewhere in the report. In 02, “residue” means the standard part st ; it is not the affine residue Res of [Part III](#). Three different minimum degrees occur in this Part; they are compared in [remark 1.1](#).

Abstract of manuscript 05. A finite surreal polytope has an ordinary realizable face lattice, but its standard part can erase faces and their incidences. We isolate a setting in which the scale information can be classified exactly: surreal heights on a fixed real configuration of n labeled points of affine dimension d . Let V be the real affine-dependence space and let $L_h(c) = \sum_i c_i h_i$. If $m = \dim_{\mathbb{R}} L_h(V)$, then $m \leq n - d - 1$. After an affine height is removed,

at most m monomial coefficient slices preserve the leading term of *every* real dependence evaluation, including its exact vanishing. This remains true for arbitrary set-sized Conway supports, not just finite series. The complete dependence sign type is classified by an oriented partial flag of length m .

A real polynomial height vector represents that entire sign type as a germ at 0^+ with minimum degree $m - 1$ when $m > 0$. For bounded heights, requiring the correct standard part changes the exact minimum to $m - r_0$, where $r_0 \in \{0, 1\}$ is the rank of $\text{st} \circ L_h$. By contrast, preserving only the finite lifted combinatorics and the standard part always requires degree at most one. For integral bases, the same linear deformation can also preserve the entire toric initial ideal. We prove both bounds sharply and explain why they do not conflict. We give determinantal and barycentric certificates, an explicit rational specialization threshold, counterexamples to naive truncation and shadow arguments, and corrections to two statements in an earlier convexity draft. Exact verification code and a formalization plan accompany the text. The underlying normal-form, ordered-field, and symbolic-perturbation tools are classical; the proposed contribution is this sharp, unified lifting package, not a claim to have invented higher-rank secondary geometry.

Status (manuscript 05). The statements below have mathematical proofs in this manuscript. The program checks finite examples only. No new Lean proof or independent refereeing is claimed. Literature-wide priority of the package has not been established.

Abstract of manuscript 02. (It covers this Part and [Part II](#).) Finite polytopes over the surreal numbers have no new face-lattice types: real-closed-field transfer already gives real algebraic realizations. The interesting question is therefore not which single finite combinatorial types exist, but how much information an infinitesimal or transfinite hierarchy can carry.

We establish a precise contrast. For a fixed real configuration of n points spanning dimension d , an arbitrary surreal lifting vector has a canonical support-adapted compression with at most $n - d - 1$ real coefficient vectors. This compression preserves the signs of *all* real affine-dependence functionals and the complete history of regular subdivisions obtained by common initial normal-form truncation. The number of effective comparison layers is an intrinsic real rank; the number of strict subdivision changes is at most this rank. Every marked coherent refinement chain occurs, and parabolic polygons attain the bound already in dimension two. Explicit finite sign certificates turn the compressed history into real polynomial liftings. These positive statements are constructive consequences and refinements of classical lexicographic-order and secondary-polytope ideas, not new versions of those foundational theories.

By contrast, four bounded planar points suffice to encode an arbitrary prescribed triangle/quadrilateral history under common coordinate truncation, from degree two onward. The construction uses reciprocal convergent power series with rational coefficients. A monomial beyond all ordinary powers makes the full configuration uniform and independently selects its final hull type without changing any finite truncation. An elementary rational example already alternates indefinitely. We also prove standard-part identities, explain the failure of a naive map on faces, and give a semialgebraic Puiseux model for any finite collection of incidence data. Full conventional proofs, an exact-arithmetic verification program, a formalization roadmap, and further research questions are supplied.

Status (manuscript 02). AI-assisted, unrefereed research manuscript. The new arguments have not been independently certified or checked in Lean. Finite computations are regression

tests, not substitutes for the proofs. No claim of exhaustive literature priority or resolution of a named published major conjecture is made.

2 The problem, its scope, and its relation to earlier work

2.1 What can genuinely change over the surreals?

A finite list of points with surreal coordinates may exhibit arbitrarily separated scales. Nevertheless, its face lattice is controlled by finitely many polynomial signs. Real-closed-field transfer therefore prevents new *finite realizable combinatorial types* from appearing solely because the ambient field is **No**. This observation does not make the geometry empty. It redirects attention to specialization, leading terms, and which comparisons must be retained simultaneously.

Three different demands must not be conflated:

- (i) retain the finite face structure, or the finite regular subdivision;
- (ii) retain the signs of all real affine-dependence evaluations;
- (iii) retain their actual leading monomials and leading coefficients.

The second demand involves an uncountable family of tests, even when the point configuration is finite. The third is stronger still. This manuscript shows that these distinctions lead to different, exactly computable notions of scale complexity.

The real-base hypothesis is essential. Fix

$$A = (a_1, \dots, a_n), \quad a_i \in \mathbb{R}^d, \quad \text{aff}\{a_1, \dots, a_n\} = \mathbb{R}^d.$$

We may take the points distinct; every label is retained, including labels that are not vertices of a resulting polytope. Only the heights $h_i \in \mathbf{No}$ vary. The lifted configuration consists of $p_i = (a_i, h_i)$. Determinants involving its height row are *linear* in h . This simple fact is the source of the sharp results. General moving surreal coordinates lead to polynomial, not merely linear, sign conditions; Section 18 treats the corresponding weaker conclusion.

2.2 Connection with the repository and the previous convexity draft

The inspected ProveIt source `Algebra/SurrealNumbers/Surreal/Algebra/Geometry.lean` develops coordinate geometry over an ordered base field, including dot and cross products, Gram identities, and Ptolemy-related identities [8]. The surrounding project documents a constructed surreal field, normal forms, and its real closedness; its guide explicitly separates unrefereed research manuscripts from statements mapped to Lean declarations [9]. The present paper adds a higher-dimensional *affine-dependence and lifting* layer. It does not depend on the more speculative results elsewhere in that collection, and it does not infer machine verification from a repository README. [write] The inspected source is unchanged at the report's placement: the blob of `Geometry.lean` is 450fd4b at 5e0f4e0, at 6996fee and at b30441a. Its declarations in the namespace `Surreal.Complexify` include `dot`, `cross`, `cross_def`, `cross_swap`, `dot_sq_add_cross_sq` (the Gram identity), `ptolemy_identity`, `ptolemy`, `triangleArea`, `heron_factorization` and `heron`. No result of this report uses them, and none of the report's results is formalized (section 1).

An earlier companion draft, *Finite Convexity and Exact Lexicographic Optimization over the Surreals*, dated 22 September 2026, already treats finite objective hierarchies and a sufficiently-small-parameter compiler [10]. We do not present that compiler as new. Our questions instead

concern compression of arbitrary height normal forms, preservation of *all* real dependence tests, and optimal polynomial degree. Section 20 also corrects its global validity-interval statement and clarifies the field of convex combinations in its shadow statement. [write] That draft is not part of ProveIt; see the note at the head of section 20.

2.3 Literature boundary

Conway normal forms and the ordered-field foundations are classical [1]. Regular liftings, subdivisions, and secondary polytopes are standard subjects [2]. Joswig and Smith develop higher-rank tropical geometry and explicitly ask how regular-subdivision duality, multiplicities, and combinatorial structure should extend; their rank-two decomposition already exhibits successive minimizer data [3].

The literature check must include a very recent overlap: Chirivì, Costa Cesari, Fang, and Littellmann’s *Higher rank Gelfand–Kapranov–Zelevinsky fans*, arXiv:2604.20250v2, revised 13 September 2026, develops higher-rank GKZ cones and fans and associated quasi-valuations and degenerations [4]. Thus this article does *not* claim the first higher-rank secondary theory or a complete solution of the broader questions in [3]. The explicit scale and degree classification below has a narrower scope. Its linear-algebraic mechanism is elementary once stated correctly; its priority as a package requires further literature and referee scrutiny.

2.4 Main answers at a glance

Put $q = n - d - 1$. The quantities m and r_0 will be defined precisely below. For $m > 0$, the conclusions are:

Information to preserve	Exact bound or conclusion
All real dependence leading terms	$m \leq q$ selected monomial slices
All real dependence signs, as a polynomial germ	Minimum degree $m - 1$
Same signs and the labeled real shadow	Minimum degree $m - r_0 \leq q$
Finite lifted sign data and the shadow	Minimum degree 0 or 1
A fixed positive real specialization for every real dependence	Impossible when $m \geq 2$

The first count is taken *after removing an affine height*. The polynomial models preserve signs, not numerical surreal exponents. The finite-data degree is zero exactly when the shadow already satisfies all the required signs. These qualifications are part of the theorems, not exceptions left implicit.

3 Questions, answers, and provenance

[write] This section is the introduction of manuscript 02, printed whole. It previews both the fixed-base lifting results, printed in this Part beside manuscript 05’s, and the coordinate-truncation results of Part II; in its second subsection $t = \omega^{-1}$ is the variable of the power series of Part II. Its three operations (i)–(iii) are the settings of theorem 18.1 (for (i)), of the lifting results of this Part (for (ii)), and of Part II (for (iii)).

Replacing real coordinates by surreal coordinates introduces exact infinite and infinitesimal scales. It does not automatically introduce new finite combinatorial objects. A polygon with one infinitesimal edge is still a polygon; its limiting picture may forget an edge, but the edge is present before taking a residue. To obtain a meaningful research problem one must specify which information is retained and which operation changes the scale.

This article separates three operations:

- (i) replacing one finite configuration by another with the same signed incidences;
- (ii) keeping a real base configuration fixed and truncating its lifting heights at common initial normal-form cuts;
- (iii) truncating the coordinates of the base configuration itself.

The first operation has a classical real-algebraic transfer theorem. The second admits a sharp finite-dimensional compression. The third can have arbitrary infinite combinatorial histories even with four planar generators. Conflating these operations would erase the main phenomenon.

3.1 The main positive statement

Let $A = (a_1, \dots, a_n) \in (\mathbb{R}^d)^n$ affinely span $\mathbb{R}^d$, and let $h \in \mathbf{No}^n$ be a vector of heights. Its lower lifted hull defines a marked regular subdivision of A . Set

$$L_A = \left\{ c \in \mathbb{R}^n : \sum_i c_i = 0, \quad \sum_i c_i a_i = 0 \right\}, \quad q = \dim L_A = n - d - 1.$$

Theorems 7.1 and 11.2 construct vectors $w^1, \dots, w^p \in \mathbb{R}^n$, with $p \leq q$, such that

$$\text{sgn}(c \cdot h) = \text{lexsgn}(c \cdot w^1, \dots, c \cdot w^p) \quad (c \in L_A). \quad (3.1)$$

The same list, read through its initial segments, preserves *every* initial normal-form lifting subdivision, including those reached only after transfinitely many redundant coefficient positions. The least possible number of vectors preserving the complete profile (3.1) is

$$p = \text{rank}_{\mathbb{R}}(L_A \longrightarrow \mathbf{No}, c \longmapsto c \cdot h).$$

There are at most p strict subdivision changes, and q can occur in the plane. Theorem 12.2 characterizes the subdivision histories, with repetitions suppressed, as the finite marked regular refinement chains starting at the trivial subdivision.

3.2 The main negative statement

For a positive infinitesimal $t = \omega^{-1}$, consider four points

$$A_0 = (0, 0), \quad B = (1, f(t)), \quad C = (g(t), 1), \quad D = (0, 2).$$

Theorem 26.3 shows that, given any binary sequence from degree two onward, one can choose reciprocal convergent power series f, g with rational coefficients so that the hull of the degree- N coordinate truncations is a triangle or a quadrilateral according to the prescribed N th bit. Theorem 26.6 changes the full endpoint to either uniform configuration type by inserting $\omega^{-\omega}$, without changing any finite truncation.

This disproves a natural extension of the positive theorem: there is no bound depending only on n and d for the number of combinatorial changes under arbitrary common coordinate truncation. It also shows that requiring the full surreal configuration to be uniform is insufficient. By contrast, uniform ordinary power-series configurations do stabilize; their relevant nonzero determinants have finite orders (Proposition 26.7).

3.3 What is classical, and what is claimed here

The surreal normal form and its leading-coefficient order are classical; we use Gonshor’s account [1]. Real-closed-field transfer is Tarski’s theory [11]. Linear programming and finite convex-hull arguments over ordered fields, including finite Puiseux specialization, are established tools; see Joswig–Loho–Lorenz–Schröter [12]. Finite-dimensional total orders on real vector spaces have lexicographic models; Wortel’s Proposition 4.4 is a convenient precise reference [13]. The secondary polytope has dimension $n - d - 1$ and encodes regular subdivisions by refinement; Lee–Santos, Theorem 16.4.1, summarizes this classical structure [14].

Consequently, neither the existence of lexicographic representations nor the dimension bound on coherent refinement chains is claimed as a new foundational discovery. The contribution developed here is a single explicit framework that starts with arbitrary surreal normal-form coefficients, preserves all support-initial histories, gives a sharp intrinsic comparison rank and finite certificates, and contrasts this with a universal four-point coordinate-truncation construction. The arbitrary prescribed history, its convergent reciprocal realization, and the independent uniform transfinite endpoint are the principal specific constructions advanced by this manuscript. They come with proofs; their historical priority has not been established by an exhaustive search.

The questions answered here are formulated in this article. We do not attribute them to a published conjecture unless explicitly stated, and we do not reclassify classical transfer as the solution of a new open problem.

3.4 Connection with the repository

The inspected repository snapshot is

6996fee43cc97b6c16351def7507d59a95bf62f0.

Its commit timestamp is 30 September 2026, 16:07:39 UTC. The Algebra/SurrealNumbers subproject connects surreal normal forms, ordered-field algebra, finite geometry, and research reports. In particular, we inspected the actual source

Surreal/Algebra/Geometry.lean

within that subproject. It defines dot and cross products over a general coefficient ring and supplies Gram, Ptolemy, and Heron results under explicitly stated ordered-field and square-root hypotheses [16]. Its `cross_def` and `cross_swap` identities are exactly the kind of finite determinant algebra used in our planar examples.

The present article extends this direction toward finite configurations, lifting circuits, normal-form comparison ranks, and scale-sensitive convex hulls. It does not rely on unreviewed repository research claims as axioms. The repository’s claim ledger and individual source declarations must be consulted separately from its narrative reports. We did not run its Lean build, and no theorem of this article is presented as already formalized. The focused source inspection is not a proof that no related idea appears elsewhere in the large repository. Detailed source identities and the verification boundary are recorded in the accompanying notes. [write] Those notes are shipped with this report as `02-transfinite-scales-sources.md` and `02-transfinite-scales-proof_audit.md` (delivered as `notes/sources.md` and `notes/proof_audit.md`).

4 Foundations, normal forms, and the affine quotient

4.1 Finite mathematics inside a class field

Although **No** is a proper class, each normal form has set-sized support, and this paper uses only finitely many numbers at a time. Statements about convex hulls can be interpreted in a set-sized ordered subfield K containing $\mathbb{R}$ and the finitely many coordinates in question. When needed, enlarge K to its real closure inside **No**. Selected monomials can also be added to this field. We do not apply first-order model theory directly to a proper-class structure, or appeal to compactness of a surreal interval.

For points $v_1, \dots, v_N \in K^D$, our convention is

$$\text{conv}_K(v_1, \dots, v_N) = \left\{ \sum_{i=1}^N \lambda_i v_i : \lambda_i \in K_{\geq 0}, \sum_i \lambda_i = 1 \right\}.$$

A nonempty exposed face is the set maximizing a K -linear functional. For a finite polytope these give the usual faces. One elementary justification is finite linear elimination: feasible nonnegative combinations and supporting inequalities admit finite certificates, and redundant inequalities can be removed until facet inequalities remain. Alternatively all fixed-size assertions can be transferred from real closed fields. The later lifting certificates will be proved directly.

4.2 Normal forms and standard part

Every nonzero surreal has a unique Conway normal form

$$x = \sum_{\gamma \in S} x_\gamma \omega^\gamma, \quad x_\gamma \in \mathbb{R} \setminus \{0\}, \quad (4.1)$$

where S is reverse well-ordered: every nonempty subset has a greatest member [1]. We write

$$\text{le}(x) = \max S, \quad \text{LT}(x) = x_{\text{le}(x)} \omega^{\text{le}(x)}, \quad \text{LT}(0) = 0.$$

Throughout, ω^γ denotes the Conway omega-map monomial in normal form; no identity with $\exp(\gamma \log \omega)$ is assumed for general surreal γ . The sign of x is the sign of its leading coefficient. A finite union of reverse well-ordered supports is reverse well-ordered, so real linear combinations of a finite list are computed coefficientwise. These are the only infinite-series operations needed in the central proof.

A surreal x is *finite* when $|x| < M$ for some real $M > 0$. Its normal form has no positive exponent, and its standard part is its coefficient at exponent zero. The map st on finite surreals is an order-preserving ring homomorphism to $\mathbb{R}$. In particular $x \geq 0$ implies $\text{st}(x) \geq 0$, not necessarily $\text{st}(x) > 0$. The loss of strictness is responsible for geometric collapse.

4.3 Removing affine heights

Let

$$\tilde{A} = \begin{pmatrix} 1 & \cdots & 1 \\ a_1 & \cdots & a_n \end{pmatrix}, \quad V = \ker_{\mathbb{R}} \tilde{A}, \quad \dim_{\mathbb{R}} V = q.$$

An element $c = (c_i) \in V$ is a real affine dependence: $\sum_i c_i = 0$ and $\sum_i c_i a_i = 0$. Define

$$L_h : V \longrightarrow \mathbf{No}, \quad L_h(c) = \sum_i c_i h_i, \quad m = \dim_{\mathbb{R}} \text{im } L_h. \quad (4.2)$$

Thus $0 \leq m \leq q$. Adding the values of any surreal-valued affine function on $\mathbb{R}^d$ to the heights leaves L_h unchanged.

Choose an affine basis $B \subseteq \{1, \dots, n\}$ with $|B| = d + 1$. Write the real barycentric coordinates of a_i relative to B as

$$a_i = \sum_{b \in B} \lambda_{B,i,b} a_b, \quad \sum_{b \in B} \lambda_{B,i,b} = 1.$$

They are uniquely defined, without a positivity requirement. Set

$$\ell(a_i) = \sum_{b \in B} \lambda_{B,i,b} h_b, \quad g_i = h_i - \ell(a_i). \quad (4.3)$$

The function ℓ is affine and $g_b = 0$ for $b \in B$.

Lemma 4.1 (The affine gauge). *Projection $c \mapsto (c_i)_{i \notin B}$ is a real-linear isomorphism $V \simeq \mathbb{R}^q$. Under it,*

$$L_h(c) = \sum_{i \notin B} c_i g_i, \quad m = \dim_{\mathbb{R}} \text{span}_{\mathbb{R}} \{g_i : i \notin B\}.$$

The shear $(a, z) \mapsto (a, z - \ell(a))$ preserves the labeled lifted combinatorics and the projected lower subdivision.

Proof. The equations $\widetilde{A}c = 0$ determine c_B uniquely from c_{B^c} because the columns indexed by B form an invertible square matrix. Affine functions annihilate affine dependencies, yielding the displayed formula. The shear is an invertible affine transformation with positive determinant one. An inequality $z \geq u \cdot a + v$ becomes another inequality with coefficient $+1$ on z , so lower supporting hyperplanes remain lower supporting hyperplanes. General faces are preserved by affine invertibility. $\square$

5 Ordered fields, finite configurations, and normal forms

[write] Manuscript 02's conventions. Its working field is a set-sized *real closed* K , whereas manuscript 05 (section 4) needs only a set-sized ordered subfield and passes to the real closure where a statement requires it; every 02 result printed in this Part is stated over 02's K . The common initial normal-form cut (5.2) is the operation on which 02's cut-compatibility clauses (theorem 7.1(c),(d) and theorem 11.2(b)) and all of Part II rest.

5.1 A set-sized working field

Although **No** is a proper class, the data used by any one construction here form a set. Choose a set-sized real-closed subfield

$$\mathbb{R} \subseteq K \subseteq \mathbf{No}$$

containing those data. When all initial normal-form truncations of finitely many coordinates are used, include that set of truncation values as well. One obtains such a field by taking the real closure inside **No** of the field generated by $\mathbb{R}$ and the specified set. Whenever needed, include $\varepsilon = \omega^{-1}$ and $\tau = \omega^{-\omega}$.

For $v_1, \dots, v_n \in K^d$, write

$$P_K = \text{conv}_K(v_1, \dots, v_n) = \left\{ \sum_{i=1}^n \lambda_i v_i : \lambda_i \geq 0, \sum_i \lambda_i = 1 \right\}.$$

All linear maps displayed with target **No** below have image in such a working field; their ranks are ordinary set-sized real vector-space ranks. All faces in this article are exposed polytope faces, including the empty and full faces. A proper nonempty face is the maximizer set of a linear functional. The usual equivalence with the algebraic notion of a face holds for finite polytopes over an ordered field. Supporting-hyperplane existence, lower-hull descriptions, and linear feasibility can be proved by finite elimination or the ordered-field Farkas lemma [12]. Nothing here uses compactness in the order topology of K .

The finite face labels are independent of enlarging the real-closed working field: whether a specified finite system of linear equalities and inequalities has a solution is unchanged by a real-closed extension. This justifies speaking informally of a finite “surreal polytope” while carrying out every set-theoretic construction in K .

5.2 Common initial normal-form truncation

Every surreal number has a Conway normal form with real coefficients and reverse-well-ordered monomial support [1]. For a finite vector $h \in \mathbf{No}^n$, take the union of the coordinate supports and write

$$h = \sum_{\alpha < \lambda} v_\alpha \omega^{\gamma_\alpha}, \quad v_\alpha \in \mathbb{R}^n, \quad \gamma_\alpha > \gamma_{\alpha'} \quad (\alpha < \alpha'). \quad (5.1)$$

Here λ is a set ordinal, and zero entries in a coefficient vector are allowed. The finite union of the supports is again reverse well ordered. For $0 \leq \beta \leq \lambda$, define

$$h^{<\beta} = \sum_{\alpha < \beta} v_\alpha \omega^{\gamma_\alpha}. \quad (5.2)$$

The cut is common to all coordinates and retains the large monomials first. It is not a truncation of Conway sign sequences. Nor is it an operation that deletes leading terms and retains a tail.

A real linear functional commutes with this normal-form sum. After any cancellations, the sign of its value is the sign of its first nonzero coefficient. This elementary fact, rather than real-closedness, supplies the positive compression theorem. In a power-series example we write $\varepsilon = \omega^{-1}$ [write] (in Part II: t , evaluated at $t = \varepsilon$) and use the increasing exponents $0, 1, 2, \dots$

5.3 Lexicographic sign

For a finite real tuple $(b_1, \dots, b_p)$, let $\text{lexsgn}(b_1, \dots, b_p)$ be the sign of its first nonzero entry, or zero when all entries are zero. The empty tuple has sign zero. If ε is a positive infinitesimal, then

$$\text{sgn}\left(\sum_{j=1}^p b_j \varepsilon^{j-1}\right) = \text{lexsgn}(b_1, \dots, b_p).$$

This identity holds for every real tuple. Replacing ε by a positive *real* number preserves it for a prescribed finite collection of nonzero tuples when that number is sufficiently small, but not uniformly for all real tuples. This quantifier distinction will matter.

6 A finite coefficient-pivot theorem

Theorem 6.1 (Simultaneous leading-term compression). *Let $g = (g_1, \dots, g_q) \in \mathbf{No}^q$. There exist distinct exponents $\gamma_1 > \dots > \gamma_m$ and linearly independent real vectors $b_1, \dots, b_m \in \mathbb{R}^q$ such that*

$$g^\# = \sum_{j=1}^m b_j \omega^{\gamma_j} \quad (6.1)$$

satisfies

$$\text{LT}(\lambda \cdot g^\#) = \text{LT}(\lambda \cdot g) \quad \text{for every } \lambda \in \mathbb{R}^q. \quad (6.2)$$

Here $m = \dim_{\mathbb{R}} \text{span}_{\mathbb{R}}\{g_1, \dots, g_q\}$. The exponents and vectors may be chosen among the actual coefficient slices of g . Moreover, the nonzero real combinations of the coordinates of g have exactly m different leading exponents.

Proof. Take the finite union S of the supports of the coordinates and write

$$g = \sum_{\gamma \in S} b_\gamma \omega^\gamma, \quad b_\gamma \in \mathbb{R}^q \setminus \{0\}.$$

If S is empty, the conclusion holds with $m = 0$. Otherwise choose $\gamma_1 = \max S$. Having chosen $\gamma_1, \dots, \gamma_j$, put $U_j = \text{span}_{\mathbb{R}}\{b_{\gamma_1}, \dots, b_{\gamma_j}\}$. If every coefficient vector belongs to U_j , stop. If not, the set

$$\{\gamma \in S : b_\gamma \notin U_j\}$$

is nonempty and has a greatest element; choose that as γ_{j+1} . Each choice strictly increases the real dimension of U_j , so at most q choices are made. They have strictly decreasing exponents. Write $b_j = b_{\gamma_j}$.

Fix $\lambda \in \mathbb{R}^q$. If $\lambda \cdot b_j = 0$ for every selected vector, then λ annihilates every coefficient vector, by the stopping condition. Hence $\lambda \cdot g = \lambda \cdot g^\# = 0$. Otherwise let j be the first selected index with $\lambda \cdot b_j \neq 0$. Every coefficient vector at an exponent greater than γ_j belongs to U_{j-1} : this is precisely the maximality in the choice of γ_j . Those coefficients vanish after pairing with λ . The coefficient at γ_j is $\lambda \cdot b_j \neq 0$. Consequently both combinations in (6.2) have leading term $(\lambda \cdot b_j) \omega^{\gamma_j}$.

The maps $\lambda \mapsto \lambda \cdot g$ and $\lambda \mapsto (\lambda \cdot b_1, \dots, \lambda \cdot b_m)$ have the same kernel. The latter has rank m since the selected vectors are independent. Rank-nullity identifies m with the dimension of the span of the coordinates of g . Finally, for every j there exists a real λ annihilating U_{j-1} but not b_j . Thus γ_j actually occurs as a leading exponent. No other leading exponent can occur by the preceding argument. $\square$

Remark 6.2 (Finite rank is not finite support scanning). The proof makes finitely many *rank-increasing choices*. It need not inspect finitely many consecutive terms of each input normal form. There may be an infinite block of coefficient slices in U_j before the next slice outside U_j . Without an effective representation and a certificate that no further independent slice exists, the theorem is not an algorithm for arbitrary surreal names. Section 21 gives the exact finite-support algorithm that is implemented here.

Corollary 6.3 (Sharp affine-height compression). *For a lifting h of A , remove an affine height as in (4.3). There is a height vector*

$$h_i^\# = \ell(a_i) + \sum_{j=1}^m b_{j,i} \omega^{\gamma_j}, \quad b_{j,b} = 0 \quad (b \in B),$$

with $m \leq n - d - 1$, such that

$$\text{LT}(L_{h^\#}(c)) = \text{LT}(L_h(c)) \quad (c \in V).$$

Exactly m distinct leading exponents occur among the nonzero $L_h(c)$. In particular m is the minimum number of distinct monomial scales in a real-coefficient representation preserving all these leading terms modulo affine heights.

Proof. Apply Theorem 6.1 to the q residual coordinates, and use Lemma 4.1. A representation with s monomial scales can have at most s different leading exponents under real linear combinations; the last assertion of the theorem proves the lower bound $s \geq m$. $\square$

Example 6.4 (Sharpness in every codimension). For any integer $q \geq 1$, take $d = 1$, $n = q + 2$, and $a_i = i - 1$ for $1 \leq i \leq q + 2$. Use $B = \{1, 2\}$ and set

$$h_1 = h_2 = 0, \quad h_{j+2} = \omega^{-j} \quad (1 \leq j \leq q).$$

The residual heights are real-linearly independent, since their leading exponents are distinct. Hence $m = q$. The fundamental dependence for the j th nonbasis point evaluates to ω^{-j} . This is a sharpness example for a full dependence invariant; it does *not* assert that a face lattice requires q independent infinitesimals.

7 Support-adapted finite-rank compression

[write] This section is manuscript 02's form of [theorem 6.1](#). Both select, among the coefficient vectors of a normal form and in decreasing order of exponents, each one that increases the real span. Clauses (a) and (b) of [theorem 7.1](#) and (7.1) restate [theorem 6.1](#) for signs (manuscript 05 proves the stronger equality of leading terms and the exact count of leading exponents), and the minimality assertion at its end is the lower bound of [corollary 6.3](#). Clauses (c) and (d), agreement with every common initial cut (5.2) and exactly $p + 1$ distinct profiles along the cuts, are new against 05 and are proved by the same argument as (b); so the theorem is printed whole. Here E is any finite-dimensional real vector space; [theorem 11.2](#) applies the theorem to the quotient Q_A .

7.1 A linear theorem independent of convexity

Let E be a real vector space of finite dimension q . A normal-form E -vector can be interpreted by choosing a basis of E and using a finite vector of surreal coordinates. This interpretation is independent of the basis.

Theorem 7.1 (Greedy support compression). *Suppose*

$$x = \sum_{\alpha < \lambda} v_\alpha \omega^{\gamma_\alpha}, \quad v_\alpha \in E,$$

with the exponents ordered as in (5.1). Select the least nonzero coefficient, and thereafter the least coefficient outside the real span of the previously selected coefficients. Let the selected indices be $\alpha_1 < \dots < \alpha_p$, and put $w^j = v_{\alpha_j}$. Then:

(a) $p \leq q$ and $\text{span}_{\mathbb{R}}\{w^1, \dots, w^p\} = \text{span}_{\mathbb{R}}\{v_\alpha : \alpha < \lambda\}$;

(b) for every $\ell \in E^*$,

$$\text{sgn } \ell(x) = \text{lexsgn}(\ell(w^1), \dots, \ell(w^p));$$

- (c) for every initial cut β , the sign profile of $x^{<\beta}$ is that of the initial selected list consisting of the w^j with $\alpha_j < \beta$;
- (d) the complete real-linear sign profiles along these cuts have exactly $p + 1$ distinct values, including the zero profile.

If $T_x : E^* \rightarrow \mathbf{No}$ sends ℓ to $\ell(x)$, then

$$p = \text{rank}_{\mathbb{R}} T_x. \quad (7.1)$$

Moreover, any list of real vectors producing the same complete sign profile as x has length at least p .

Proof. At every selection the span strictly increases. Thus at most q coefficients are selected. If an unspanned coefficient remains, the well-order on coefficient positions provides a least such coefficient, so the procedure cannot stop before the selected vectors span all coefficients. This proves (a), including the possibility $p = 0$.

Fix ℓ . If it kills every selected vector, it kills their span and hence every coefficient, so $\ell(x) = 0$. Otherwise let j be the first selected index for which $\ell(w^j) \neq 0$. By the definition of the selection, every coefficient before α_j belongs to the span of the earlier selected vectors. It is therefore killed by ℓ . The first nonzero coefficient of $\ell(x)$ occurs at α_j and equals $\ell(w^j)$, proving (b). Exactly the same argument with the coefficient positions restricted to $\alpha < \beta$ proves (c).

Whenever w^j is added, there exists a real functional that vanishes on $w^1, \dots, w^{j-1}$ but not on w^j . Its sign changes from zero to nonzero. No other profile change is possible by (c). This proves (d).

Finally, $\ker T_x$ is precisely the annihilator of the coefficient span. Its codimension in E^* is p , proving (7.1). If a list of r vectors gives the same sign profile, its common annihilator is $\ker T_x$, since zero is one of the signs that must be preserved. The map from E^* to $\mathbb{R}^r$ obtained by evaluation on that list has rank at most r and kernel of codimension p . Hence $r \geq p$. $\square$

Remark 7.2 (Scope of the word canonical). The selected *positions and coefficient vectors* are uniquely specified once the normal form and quotient space are fixed. Representatives of a quotient vector, coordinate gauges, and other equivalent lexicographic lists need not be canonical. The rank p is intrinsic. We do not assert that this construction preserves valuations, monomial magnitudes, or multiplicative relations.

7.2 Transfinite delays are not extra rank

Example 7.3. Let w^1, w^2 be independent real vectors. With $\varepsilon = \omega^{-1}$ and $\tau = \omega^{-\omega}$, put

$$x = \frac{w^1}{1 - \varepsilon} + \tau w^2 = w^1 + \varepsilon w^1 + \varepsilon^2 w^1 + \dots + \tau w^2.$$

There are two selected coefficient positions. The second occurs only after all the ordinary power-series positions, but the compression is simply $w^1 + \varepsilon w^2$. Thus the theorem does not say “keep the first q terms.” It says “keep the first q possible rank increases,” and a rank increase can be delayed through a transfinite interval of redundant positions.

The underlying existence of a finite lexicographic description is consistent with classical ordered-vector-space theory [13]. What the present proof adds to that abstract description is explicit compatibility with the *given* normal-form initial cuts, together with an elementary optimality certificate through the kernel of T_x .

8 Oriented flags classify the complete dependence sign type

Definition 8.1. The *complete real dependence sign type* of a lifting is the function

$$\chi_h : V \longrightarrow \{-1, 0, 1\}, \quad \chi_h(c) = \operatorname{sgn} L_h(c).$$

It is not just the list of signs of the finitely many circuits of A .

By the compression theorem, there are independent functionals $f_1, \dots, f_m \in V^*$ such that

$$\chi_h(c) = \operatorname{sgn}_{\operatorname{lex}}(f_1(c), \dots, f_m(c)), \quad (8.1)$$

where the sign on the right is that of the first nonzero entry, or zero if all entries vanish. The selected coefficient rows give $f_j(c) = \sum_i c_i b_{j,i}$. Define

$$W_0 = V, \quad W_j = \bigcap_{k=1}^j \ker f_k \quad (1 \leq j \leq m).$$

Each inclusion has codimension one. On the one-dimensional quotient W_{j-1}/W_j , the restriction of f_j specifies a positive ray.

Theorem 8.2 (Flag classification). *The function χ_h determines, and is determined by, the oriented partial flag*

$$V = W_0 \supsetneq W_1 \supsetneq \dots \supsetneq W_m = \ker L_h.$$

Every such flag on V occurs for a surreal lifting of A . Two independent lists $f_1, \dots, f_m$ and $f'_1, \dots, f'_{m'}$ produce the same complete sign type if and only if $m = m'$ and

$$f'_j = \alpha_j f_j + \sum_{k < j} \beta_{jk} f_k, \quad \alpha_j > 0. \quad (8.2)$$

Proof. Given the oriented flag, find the first j for which $c \notin W_j$; the orientation of $c + W_j$ in W_{j-1}/W_j determines its sign. Vectors in W_m have sign zero, so the flag determines (8.1).

Conversely, suppose the sign type is not identically zero and let $P = \{c : \chi_h(c) = 1\}$. In the ordinary finite-dimensional real topology,

$$\overline{P} = \{c : f_1(c) \geq 0\}.$$

Indeed positive points lie in this closed halfspace. Every point with $f_1 > 0$ is positive, and each point with $f_1 = 0$ is a limit of points with $f_1 > 0$. Thus χ_h determines $\ker f_1$ and the positive orientation of its quotient. Restrict the sign function to this kernel and repeat. After exactly m steps it is identically zero. This reconstructs the oriented flag without choosing a coordinate system.

For two lists defining the same flag, the restrictions of f'_j and f_j to W_{j-1} have the same kernel and the same positive ray. They differ there by a positive factor α_j . Their difference annihilates W_{j-1} and is therefore in the span of $f_1, \dots, f_{j-1}$, proving (8.2). These transformations plainly preserve the first nonzero sign, proving the converse.

Finally extend functionals realizing a prescribed flag to coefficient rows in $\mathbb{R}^n$, or use the affine-gauge identification $V \simeq \mathbb{R}^q$. Choose heights with residual vector $\sum_{j=1}^m b_j \omega^{-j}$. Their dependence type is the required flag. $\square$

Remark 8.3 (What is canonical and what is not). The flag and its orientations are canonical for χ_h ; the normalized functionals are not. The actual leading exponents are additional data not determined by the sign type. Replacing ω^{γ_j} by any other strictly decreasing list of surreal monomials leaves (8.1) unchanged. Formula (8.2) is also the familiar triangular invariance of lexicographic orders, not a new group action discovered here.

9 Polynomial realization and exact minimum degree

For a real polynomial height vector $H(t) \in \mathbb{R}[t]^n$, define $\chi_{H,0^+}(c)$ to be the eventual sign of $L_{H(t)}(c)$ as $t \rightarrow 0^+$. A polynomial is eventually zero precisely when it is identically zero. We assign degree zero to constant paths, including the zero vector. The quantifiers are

$$\forall c \in V \exists \eta_c > 0 \forall t \in (0, \eta_c) : \text{sgn } L_{H(t)}(c) = \chi_h(c).$$

They do not assert a uniform η_c for all real c .

Theorem 9.1 (Exact unrestricted polynomial degree). *If $m > 0$, the minimum degree of a real polynomial height vector H with $\chi_{H,0^+} = \chi_h$ is exactly $m - 1$. Such a vector is*

$$H(t) = b_1 + b_2 t + \cdots + b_m t^{m-1}, \quad (9.1)$$

where the coefficient rows vanish on the chosen affine basis. If $m = 0$, a constant affine height vector suffices.

Proof. For every c , the first nonzero coefficient of $L_{H(t)}(c)$ is the first nonzero $f_j(c)$. This gives the upper bound.

For the lower bound write an arbitrary degree- D candidate as $H(t) = u_0 + u_1 t + \cdots + u_D t^D$. Its identically vanishing dependence space is

$$\bigcap_{j=0}^D \ker(c \mapsto c \cdot u_j).$$

Equality of complete sign types makes this space exactly $\ker L_h$, which has codimension m . An intersection of $D + 1$ linear kernels has codimension at most $D + 1$. Thus $m \leq D + 1$. $\square$

Theorem 9.2 (Exact degree with prescribed standard part). *Suppose all h_i are finite and put $h^0 = \text{st}(h)$ componentwise. Define*

$$r_0 = \text{rank}_{\mathbb{R}}(\text{st} \circ L_h : V \rightarrow \mathbb{R}) = \text{rank}_{\mathbb{R}} L_{h^0} \in \{0, 1\}.$$

The minimum degree of a real polynomial height vector satisfying

$$H(0) = h^0, \quad \chi_{H,0^+} = \chi_h$$

is exactly $m - r_0$. This includes $m = 0$, where the minimum is zero.

Proof. Use the affine gauge. All residual heights are finite, so the selected exponents satisfy $\gamma_j \leq 0$.

If $r_0 = 1$, the coefficient slice at exponent zero in the residual vector is nonzero. It is the first pivot, so $\gamma_1 = 0$ and $b_1 = \text{st}(g)$. Then

$$H_i(t) = \text{st}(\ell(a_i)) + b_{1,i} + \sum_{j=2}^m b_{j,i} t^{j-1}$$

has constant term h_i^0 and degree $m - 1$.

If $r_0 = 0$, then $\text{st}(g) = 0$. All pivots, if any, have strictly negative exponents. Set

$$H_i(t) = \text{st}(\ell(a_i)) + \sum_{j=1}^m b_{j,i} t^j.$$

Its constant term is h_i^0 and its degree is m . In both cases the first nonzero dependence coefficient has the required sign.

For a degree- D candidate, the coefficient at t^0 induces the fixed map L_{h^0} of rank r_0 . Each of the remaining D coefficient rows increases the rank of the joint coefficient map by at most one. Since that map must have the same kernel as L_h , its rank is m , yielding $m \leq r_0 + D$. This proves optimality. $\square$

Corollary 9.3 (Sharp uniform bounds). *For real-base liftings, the complete dependence type has a polynomial model of degree at most $q-1$ when it is nonzero. For bounded liftings with the correct shadow, degree at most q always suffices. Both bounds are attained by Example 6.4; in its shadow-preserving version $r_0 = 0$.*

Proposition 9.4 (A fixed real parameter cannot do the same job). *If $m \geq 2$, there is no real height vector $w \in \mathbb{R}^n$ such that $\text{sgn } L_w(c) = \chi_h(c)$ for every $c \in V$. In particular no single positive real specialization of a polynomial model can preserve all real dependence signs at once.*

Proof. The real-valued linear map L_w has rank at most one. Equality of the sign types would force $\ker L_w = \ker L_h$, which has codimension $m \geq 2$. This contradicts rank-nullity. $\square$

For a concrete picture, compare $x + \varepsilon y$ at a positive infinitesimal ε with $x + t_0 y$ at a fixed real $t_0 > 0$. The real pair $(x, y) = (-t_0, 1)$ makes the second expression zero and the first strictly negative. The exceptional test depends on t_0 . This is exactly why pointwise stabilization for every fixed test is compatible with failure of a uniform real specialization.

10 Finite combinatorics: determinants, slacks, and a linear path

10.1 Determinants of a lifted configuration

The augmented matrix of the lifted points is

$$M(h) = \begin{pmatrix} 1 & \cdots & 1 \\ a_1 & \cdots & a_n \\ h_1 & \cdots & h_n \end{pmatrix}.$$

For an ordered $(d+2)$ -subset J , expansion along the last row gives

$$\det M(h)_J = \sum_{i \in J} c_{J,i} h_i = L_h(c_J), \quad c_J \in V. \quad (10.1)$$

The coefficients are real signed cofactors. The relation $\tilde{A}c_J = 0$ follows by expanding a determinant with two equal rows. Some such vectors may be zero, and those minors vanish for every height vector.

Lemma 10.1 (Finite determinant signs determine the labeled faces). *Suppose $m > 0$. The signs of the finitely many determinants in (10.1) determine the labeled face lattice of the lifted polytope. When $m = 0$, the lifted polytope is an affine copy of the base polytope and its labeled face structure is already fixed by A .*

Proof. When $m > 0$, the height row is not in the span of the rows of $\tilde{A}$, so $M(h)$ has rank $d+2$ and the lifted affine dimension is $d+1$. An ordered list I of $d+1$ lifted points is affinely independent exactly when it extends to a nonzero augmented maximal minor. For such I , the signs of $\det M(h)_{I \cup \{j\}}$, with a consistently fixed column order, tell on which side of its affine

hyperplane each other label lies. The hyperplane supports a facet precisely when all nonzero signs agree. The zero labels are exactly the labels on that facet. Every facet contains an affine basis of this size, so this enumerates all facets with their labels. Every proper nonempty face is an intersection of facets, completing the reconstruction. If $m = 0$, Lemma 4.1 identifies the whole configuration with the graph of an affine function on A . $\square$

In particular equality of complete dependence types implies equality of the finite lifted sign data. The converse fails dramatically: finitely many hyperplanes cannot detect all the levels of an oriented flag.

10.2 Barycentric certificates for lower cells

For each affine basis B of A and each label i , define

$$s_{B,i}(h) = h_i - \sum_{b \in B} \lambda_{B,i,b} h_b. \quad (10.2)$$

This is another real dependence evaluation. Its coefficient vector is a fundamental circuit vector when zero entries are removed; the zero vector occurs for $i \in B$.

Proposition 10.2 (Exact lower-cell certificate). *A basis B supports a lower cell of full base dimension precisely when $s_{B,i}(h) \geq 0$ for all labels i . Its equality-label set is*

$$I_B(h) = \{i : s_{B,i}(h) = 0\}.$$

The maximal marked cells of the induced regular subdivision are exactly the distinct sets $I_B(h)$ arising this way, interpreted geometrically as $\text{conv}_{\mathbb{R}}\{a_i : i \in I_B(h)\}$ together with their labels. Their faces supply the remaining cells. This convention includes the full cell when all heights are affine.

Proof. The graph of the affine function interpolating h on B is a lower supporting hyperplane exactly when every lifted point lies on or above it. These are the inequalities in (10.2). The equalities identify its contact labels, and the projection has dimension d because it contains B . Conversely every full-dimensional projected lower face contains such a base affine basis, so every maximal cell is obtained. The resulting finite sign system has a real specialization by the polynomial construction and the finite threshold below; thus the asserted subdivision and its face incidences are also those of an ordinary real regular lifting. This avoids any appeal to topological compactness over **No**. $\square$

We use *marked* subdivisions deliberately. A point lying in the interior of a real cell may be on its lower supporting hyperplane, or may lie above that hyperplane. The two possibilities need not alter the underlying geometric region but are distinguished by I_B .

10.3 An explicit simultaneous real threshold

Lemma 10.3 (Finite sign stabilization). *For a nonzero real polynomial $f(t) = \sum_{j=0}^D a_j t^j$, let $k = \min\{j : a_j \neq 0\}$ and $T(f) = \sum_{j>k} |a_j|$. Set*

$$\eta(f) = \begin{cases} 1, & T(f) = 0, \\ \min\{1, |a_k|/(2T(f))\}, & T(f) > 0. \end{cases}$$

Then $\text{sgn } f(t) = \text{sgn } a_k$ whenever $0 < t < \eta(f)$. For a finite list of polynomials, the minimum of these bounds for its nonzero members preserves every sign, with identically zero members remaining zero.

Proof. For $0 < t < 1$,

$$\left| \sum_{j>k} a_j t^{j-k} \right| \leq t \sum_{j>k} |a_j| = tT(f).$$

When the tail is nonzero this is less than $|a_k|/2$ under the stated bound. The sign of a_k is therefore unchanged; multiplication by $t^k > 0$ finishes the proof. A finite minimum of positive real bounds is positive. $\square$

Apply the lemma to the determinant polynomials and all the slack polynomials of any model from Section 9. The resulting η_0 gives the same labeled lifted faces and the same marked regular subdivision for every $0 < t < \eta_0$. Rational coefficients give a rational η_0 and a rational specialization $t_0 = \eta_0/2$.

The threshold is sufficient, not asserted optimal. Section 20 explains why even an optimal *initial* interval must not be confused with the entire set of valid parameters.

10.4 The finite-data degree collapses to one

Theorem 10.4 (Linear realization with the exact shadow). *Let $h \in \mathbf{No}^n$ be finite and $h^0 = \text{st}(h)$. Let $\mathcal{F} \subseteq V$ be any finite collection of real dependence vectors. There exists $w \in \mathbb{R}^n$ such that the affine-linear path*

$$K(s) = (1 - s)h^0 + sw \tag{10.3}$$

satisfies

$$\text{sgn } L_{K(s)}(c) = \text{sgn } L_h(c) \quad (c \in \mathcal{F}, 0 < s \leq 1).$$

In particular one path preserves the entire labeled lifted face lattice, all its determinant signs, and the marked regular subdivision while starting at the exact labeled standard part.

Among polynomial paths starting at h^0 and realizing the chosen finite sign data for all sufficiently small positive parameters, the minimum degree is zero when h^0 already has those signs, and one otherwise.

Proof. Choose a shadow-preserving polynomial model H from Theorem 9.2. The finite sign lemma provides a real $t_0 > 0$ such that $w = H(t_0)$ has the required signs on $\mathcal{F}$. For any $c \in \mathcal{F}$ with $L_h(c) > 0$ we have

$$L_{h^0}(c) = \text{st}(L_h(c)) \geq 0, \quad L_w(c) > 0.$$

Their convex combination in (10.3) is strictly positive for $0 < s \leq 1$. The negative case follows by replacing c with $-c$. If $L_h(c) = 0$, both real endpoint values are zero. This proves the simultaneous claim. Use the finite list of determinants and slacks for the geometric consequence.

A constant path starting at h^0 can realize only the signs of h^0 . When those signs suffice, take that constant path. Otherwise degree zero is impossible and the exhibited path has degree one. $\square$

Remark 10.5 (Why the conclusion is stronger than a limit argument). The path has the correct signs on the *whole* punctured segment $0 < s \leq 1$, not merely on an unspecified small interval. After the real endpoint has been chosen, this is a convexity argument in a finite linear sign chamber. It does not claim to preserve χ_h on all of V . Proposition 9.4 rules that out when $m \geq 2$.

Theorem 10.6 (Arbitrarily large separation of the two degrees). *For every real configuration with $q = n - d - 1 \geq 1$, there is a bounded lifting with affine shadow for which the minimum degree for the complete dependence type with that shadow is q , whereas the minimum degree for all finite lifted determinant and slack signs with that shadow is one.*

Proof. There are finitely many nonzero determinant and slack vectors in V . Choose $f_1 \in V^*$ nonzero on all of them: a finite union of proper hyperplanes does not cover V^* . Extend f_1 to a basis $f_1, \dots, f_q$ of V^* , represented by real height rows b_j in the affine gauge. Put

$$h = \varepsilon b_1 + \varepsilon^2 b_2 + \dots + \varepsilon^q b_q, \quad \varepsilon = \omega^{-1}.$$

Then $m = q$ and $h^0 = 0$, so $r_0 = 0$. Theorem 9.2 gives minimum complete-type degree q . Every nonzero vector in the finite test list already has its sign decided by f_1 , so $K(s) = sb_1$ realizes that list for all $s > 0$. At least one lifted determinant is nonzero because $m > 0$; at the shadow every such determinant is zero. Hence a constant path cannot realize the finite list, and its exact minimum degree is one. $\square$

This separation is one reason not to use the phrase “the number of surreal scales of a polytope” without specifying the invariant. Even a single nontrivial combinatorial degeneration can carry an arbitrarily long complete dependence flag.

11 Fixed real bases: circuits, liftings, and certificates

[write] Manuscript 02’s fixed-base sections. The setting is 05’s (a fixed real configuration A of n distinct points spanning $\mathbb{R}^d$, heights $h \in \mathbf{No}^n$), over 02’s real closed working field; 02’s comparison rank p is 05’s m , and its space L_A is 05’s V . Its slack certificate is [proposition 10.2](#), proved by both manuscripts (02’s formulation is [remark 11.1](#) below); its threshold [theorem 11.4](#) is [lemma 10.3](#) with 02’s constants; the remark after that theorem is [proposition 9.4](#). New against 05 are clause (b) of [theorem 11.2](#), the positive-sum [lemma 12.1](#), the classification [theorem 12.2](#) of support-initial histories, and the sharp planar family [theorem 12.3](#) with its hexagon.

Throughout this section, $A = (a_1, \dots, a_n)$ is a fixed configuration of *distinct real points* spanning $\mathbb{R}^d$. We use labeled cells even when some points are interior to the convex hull. The statements extend to repeated labels with suitable conventions, but no such extension is needed here.

11.1 Marked lower cells

For heights $h \in K^n$, lift a_i to $(a_i, h_i) \in K^{d+1}$. A maximal lower cell is the label set

$$C = \{i : h_i = \ell(a_i)\}, \quad h_i \geq \ell(a_i) \text{ for every } i,$$

for an affine function $\ell : K^d \rightarrow K$, with $\dim \text{Aff}\{a_i : i \in C\} = d$. The resulting collection of maximal marked cells is denoted $\mathcal{S}_A(h)$, or $\mathcal{S}(h)$ when A is fixed. Their convex hulls cover $\text{conv}_K(A)$ and meet along faces. This is the marked regular subdivision induced by h .

A point of A may be omitted from all lower cells. Even if it lies geometrically in another cell, its omission is part of the marked data. The trivial subdivision has the single marked cell $\{1, \dots, n\}$. We say that $\mathcal{T}$ *refines* $\mathcal{S}$ when each maximal marked cell of $\mathcal{T}$ is contained as a label set in a maximal marked cell of $\mathcal{S}$. Strict refinement means that the collections differ. For configurations in convex position this is the usual geometric refinement relation. Keeping the marking avoids an ambiguity when interior points are inactive.

11.2 A finite test using affine bases

Let B be any affine basis of A , that is, a $(d + 1)$ -element index set with affinely independent points. For each $j \notin B$, let $\lambda_{B,j,b} \in \mathbb{R}$ be the unique barycentric coefficients satisfying

$$a_j = \sum_{b \in B} \lambda_{B,j,b} a_b, \quad \sum_{b \in B} \lambda_{B,j,b} = 1.$$

Define a slack functional

$$\delta_{B,j}(h) = h_j - \sum_{b \in B} \lambda_{B,j,b} h_b = c_{B,j} \cdot h. \quad (11.1)$$

Its coefficient vector $c_{B,j}$ lies in L_A . Its support is the fundamental affine circuit associated with B and j , with zero coefficients omitted. It is harmless to retain all these vectors even when several represent the same circuit.

Remark 11.1 (Manuscript 02's slack certificate). [write] Manuscript 02 states [proposition 10.2](#) (source 05) as its Lemma "Complete slack certificate", whose label `poly:ts:lem:slack` is attached to [proposition 10.2](#), in the notation (11.1): *An affine basis B lies on a lower facet exactly when $\delta_{B,j}(h) \geq 0$ for every $j \notin B$. In that case the associated maximal marked cell is*

$$D_B(h) = B \cup \{j \notin B : \delta_{B,j}(h) = 0\}. \quad (11.2)$$

Enumerating all lower bases and removing duplicate sets (11.2) recovers $\mathcal{S}_A(h)$. Consequently, the signs of the finite family of functionals (11.1) determine the whole subdivision. Thus $D_B(h) = I_B(h)$ and $\delta_{B,j} = s_{B,j}$. Manuscript 02's proof, which is the proof of [proposition 10.2](#), reads: There is a unique affine function ℓ_B taking the prescribed values h_b on B . At a_j its value is $\sum_b \lambda_{B,j,b} h_b$. The condition that its graph support the lifted points from below is exactly the nonnegativity of all the slacks. The equality labels are exactly (11.2). They span dimension d since they contain B , so they form a maximal lower cell. Conversely, every maximal lower cell contains an affine basis, and its supporting function agrees with the interpolating function of that basis.

This proof works over any ordered extension of the real coefficient field. It is the direct finite algebra behind the usual lifting and circuit tests; no topological limiting argument is hidden in it.

11.3 The sharp comparison-rank theorem

The vectors of affine height values form the subspace

$$\text{Aff}(A) = \{(b + u \cdot a_i)_{i=1}^n : b \in \mathbb{R}, u \in \mathbb{R}^d\} \subseteq \mathbb{R}^n.$$

Set $Q_A = \mathbb{R}^n / \text{Aff}(A)$. Its dimension is $q = n - d - 1$, and its dual is naturally L_A . Adding affine heights does not change the lower subdivision or any functional in L_A .

[write] [Theorem 11.2](#) is [corollary 6.3](#) (source 05) in manuscript 02's form. Its clauses (a) and (d) are the sign and minimality parts of [corollary 6.3](#), with $p = m$; 05 also keeps exact leading terms and exact leading exponents. Clause (b), agreement with every common initial cut, is new against 05. Clause (c) is 05's canonical refinement chain [proposition 13.4](#) read along the cuts: 05's chain keeps the first k pivot rows, 02's the first m selected positions, and both have at most $m = p$ strict steps.

Theorem 11.2 (All-scale lifting compression). *Let $h \in \mathbf{No}^n$ have common normal form (5.1). Apply Theorem 7.1 to the coefficient classes in Q_A , and choose representatives $w^1, \dots, w^p \in \mathbb{R}^n$ of the selected classes. Define*

$$H_m(\varepsilon) = \sum_{j=1}^m \varepsilon^{j-1} w^j \quad (0 \leq m \leq p), \quad H_0 = 0.$$

Then:

(a) $p \leq n - d - 1$, and for every $c \in L_A$,

$$\text{sgn}(c \cdot h) = \text{sgn}(c \cdot H_p(\varepsilon)) = \text{lexsgn}(c \cdot w^1, \dots, c \cdot w^p);$$

(b) every common initial normal-form cut $h^{<\beta}$ has the same complete L_A sign profile, and hence the same marked subdivision, as $H_m(\varepsilon)$, where m is the number of selected positions before β ;

(c) the sequence

$$\mathcal{S}_A(H_0), \mathcal{S}_A(H_1(\varepsilon)), \dots, \mathcal{S}_A(H_p(\varepsilon))$$

is a refinement chain, possibly with repetitions;

(d) p is the least number of real coefficient vectors that preserve all L_A signs. It equals the real rank of $c \mapsto c \cdot h$.

In particular, the whole initial-cut lifting history has at most $n - d - 1$ strict changes.

Proof. Assertions (a), (b), and (d), except for the subdivision conclusion, are Theorem 7.1 applied to Q_A . The subdivision conclusion follows from proposition 10.2, because each slack belongs to L_A .

To prove (c), take an affine basis B that is lower for $H_{m+1}(\varepsilon)$. If any slack for $H_m(\varepsilon)$ were negative, its first nonzero coefficient would stay negative when one smaller coefficient is appended. This would contradict the choice of B . Thus B was also lower for $H_m(\varepsilon)$. Furthermore a nonzero slack at stage m cannot become zero at stage $m + 1$: its first nonzero coefficient is unchanged. Therefore

$$D_B(H_{m+1}(\varepsilon)) \subseteq D_B(H_m(\varepsilon)).$$

Each maximal cell at the new stage has a lower basis, so this is exactly the marked refinement assertion. There are p transitions, proving the last bound. $\square$

Remark 11.3 (Zero rank and invisible comparison layers). If $q = 0$, all heights are affine and the only subdivision is trivial. More generally $p = 0$ means that h is affine over its working field. For $p > 0$, the complete comparison profile changes at every selected position, but the subdivision need not change there. A comparison not visible in the lower hull can acquire a new nonzero sign. Thus the full profile rank p and the strict geometric history length r are different invariants; only $r \leq p$ is asserted.

11.4 A concrete quotient gauge

Choose one affine basis B_0 . Subtract the affine interpolant through its height values. For every $i \notin B_0$, the residual is

$$r_i(h) = h_i - \sum_{b \in B_0} \lambda_{B_0, i, b} h_b. \quad (11.3)$$

The q residual coordinates identify Q_A with $\mathbb{R}^q$ at the coefficient level. The comparison rank is also

$$p = \dim_{\mathbb{R}} \text{span}_{\mathbb{R}} \{r_i(h) : i \notin B_0\} \subseteq \mathbf{No}. \quad (11.4)$$

Indeed the vectors $c_{B_0,i}$ form a basis of L_A , so the image of $c \mapsto c \cdot h$ is the span in (11.4). For finite rational coefficient arrays this yields an exact Gaussian elimination algorithm: scan the gauged rows in their supplied order, retaining precisely the rank-increasing rows.

11.5 An explicit real specialization bound

The polynomial with infinitesimal parameter preserves every real affine comparison. For the finite geometry, an actual real parameter suffices, and can be certified explicitly.

[write] [Theorem 11.4](#) and [lemma 10.3](#) (source 05) give the same kind of explicit threshold with different normalizations: 02 caps it at $\frac{1}{2}$ and allows $0 < s \leq \eta$, while 05 caps it at 1 and requires $0 < t < \eta(f)$. Both are valid, and they agree on the report's numerical examples. Recomputed exactly in the write phase over all basis slacks and all augmented determinants, both give $\eta = 1/34$ for 02's hexagon (12.4) and $\eta_0 = 1/10$ for 05's three-scale example in [section 16](#). 02's form is kept because it holds simultaneously for all stages H_m of a history.

Theorem 11.4 (Simultaneous finite sign specialization). *Let $C \subseteq L_A$ be a finite family, for example all the slack vectors $c_{B,j}$. Put $a_{c,j} = c \cdot w^j$. For each c with some nonzero entry, let k_c be its first such index and put*

$$B_c = \sum_{j > k_c} |a_{c,j}|.$$

With an empty inner minimum omitted, define

$$\eta = \min \left(\frac{1}{2}, \min_{c: B_c > 0} \frac{|a_{c,k_c}|}{2B_c} \right) > 0. \quad (11.5)$$

For every real s with $0 < s \leq \eta$, every $c \in C$, and every $0 \leq m \leq p$,

$$\text{sgn}(c \cdot H_m(s)) = \text{lexsgn}(a_{c,1}, \dots, a_{c,m}).$$

Taking all slack vectors therefore realizes the complete compressed subdivision history by the real polynomial liftings $H_m(s)$.

Proof. If all entries through m vanish, the assertion is exact equality to zero. Otherwise $m \geq k_c$, and

$$s^{1-k_c} c \cdot H_m(s) = a_{c,k_c} + \sum_{j=k_c+1}^m a_{c,j} s^{j-k_c}.$$

Since $s \leq 1/2 < 1$, the absolute value of the tail is at most sB_c . If $B_c = 0$ there is no tail. If $B_c > 0$, the choice of η makes the tail at most $|a_{c,k_c}|/2$, so it cannot change the leading sign. The argument is simultaneous in m and c . $\square$

When A and the selected height rows are rational, the bound is a positive rational number obtained by exact arithmetic. The list of basis slacks has at most

$$\binom{n}{d+1} (n-d-1)$$

entries, with fewer when some candidate bases are dependent. Enumeration is a finite certificate construction, not a claimed dimension-uniform polynomial-time algorithm.

Remark 11.5 (A real parameter cannot preserve all comparisons). [write] This is [proposition 9.4](#) (source 05) in manuscript 02's notation. The finite restriction on C is essential. If $p > 1$, no single real height vector can reproduce all L_A signs of h . Its evaluation map $L_A \rightarrow \mathbb{R}$ has rank at most one, whereas $c \mapsto c \cdot h$ has rank p . Equality of all signs would imply equality of the kernels, which would contradict their codimensions. Thus Theorem 11.4 preserves finite geometry, not an entire non-Archimedean ordered vector space inside $\mathbb{R}$.

12 Exactly which lifting histories occur?

12.1 Positive sums of compatible heights

Lemma 12.1 (Compatible positive sums). *Suppose real height vectors u, v induce marked subdivisions $\mathcal{S}, \mathcal{T}$, respectively, and $\mathcal{T}$ refines $\mathcal{S}$. For positive scalars a, b in any ordered extension of $\mathbb{R}$,*

$$\mathcal{S}_A(au + bv) = \mathcal{T}.$$

Proof. Let C be a maximal cell of $\mathcal{T}$, contained in a maximal cell D of $\mathcal{S}$, and choose an affine basis $B \subseteq C$. Every slack $\delta_{B,j}(u)$ and $\delta_{B,j}(v)$ is nonnegative. For $j \in C$ both vanish, since $C \subseteq D$. For $j \notin C$, the second is strictly positive, because the zero set for v is exactly C . Consequently the lower cell of $au + bv$ containing B is exactly C .

All maximal cells of $\mathcal{T}$ therefore occur as maximal lower cells of the sum. Their convex hulls cover the base polytope. There can be no additional different maximal lower cell: on a full-dimensional overlap, the two supporting affine functions would agree and hence define the same equality labels. Equivalently one may use the usual finite lower-envelope description. The finite covering by real cells remains a covering over an ordered real-closed extension, by linear elimination. The same elementary description proves the assertion over any ordered extension for these real cell data. $\square$

Theorem 12.2 (Classification of support-initial lifting histories). *Fix A . After repetitions are suppressed and the initial trivial subdivision is included, the histories of common initial normal-form surreal liftings are exactly the finite strict chains*

$$\mathcal{S}_0 > \mathcal{S}_1 > \cdots > \mathcal{S}_r, \quad \mathcal{S}_0 = \{\{1, \dots, n\}\}, \quad (12.1)$$

of marked regular subdivisions, ordered from coarse to fine. Every such chain has $r \leq n - d - 1$. It has a realization

$$h = u_1 + \varepsilon u_2 + \cdots + \varepsilon^{r-1} u_r \quad (12.2)$$

with real u_j inducing $\mathcal{S}_j$, and the u_j are independent modulo $\text{Aff}(A)$. If A is rational, the u_j can be chosen rational. The least number of nontrivial coefficient layers required to realize the strict chain is r .

Proof. Theorem 11.2 shows that every support-initial history is a finite refinement chain. Its subdivisions are realizable over $\mathbb{R}$ by Theorem 11.4, so they are regular in the ordinary sense.

Conversely choose real u_j realizing $\mathcal{S}_j$. The refinement relation and Lemma 12.1 show inductively that every positive linear combination of $u_1, \dots, u_j$ with all coefficients positive induces exactly $\mathcal{S}_j$. Apply this to the coefficients $1, \varepsilon, \dots, \varepsilon^{j-1}$. Each initial segment of (12.2) therefore realizes the prescribed stage. Notice that no smallness bound is needed for this compatibility argument: positivity is enough.

For independence, consider the strict transition from $\mathcal{S}_{j-1}$ to $\mathcal{S}_j$. There is a maximal marked cell D of $\mathcal{S}_{j-1}$ on which the entries of u_j are not the values of one affine function. Otherwise, take a full-dimensional new cell inside each old cell. Its supporting affine function

would agree with u_j on every label of the old cell, forcing that entire old cell to persist; the refinements would not be strict. Linear algebra now provides a relation $c \in L_A$, supported in D , with $c \cdot u_j \neq 0$. Every earlier u_i is affine on D , since D lies inside a cell of $\mathcal{S}_i$. Hence $c \cdot u_i = 0$ for $i < j$. Thus u_j cannot lie in the span of its predecessors modulo $\text{Aff}(A)$. Induction gives r independent classes and therefore $r \leq q$.

For rational A , a height vector realizing a specified marked regular subdivision can be described by a finite rational system of linear conditions. One way to see this without a general secondary-fan theorem is to choose a real witness and retain the signs of all its basis slacks. The zero conditions define a rational linear subspace, and the other conditions are strict rational inequalities. Rational points are dense in that subspace, so a sufficiently close rational point is a witness with the same subdivision. Choose such a point for each u_j .

Finally, each appended coefficient layer can cause at most one new stage. A strict chain of length r therefore needs at least r layers, and (12.2) achieves that number. $\square$

The dimension assertion is also visible from the secondary polytope: its face poset is the marked coherent refinement poset and its dimension is $n - d - 1$ [14, Theorem 16.4.1]. The proof above isolates the precise coefficient mechanism and handles arbitrary surreal normal-form starting data without treating that classical dimension fact as new.

12.2 Sharpness in dimension two

Theorem 12.3 (A sharp parabolic polygon family). *For every $n \geq 4$, let*

$$a_i = (i, i^2), \quad 0 \leq i \leq n - 1.$$

For $2 \leq k \leq n - 2$, define

$$w_i^{(k)} = \max\{0, i(i - k)\}. \quad (12.3)$$

These $n - 3$ height vectors are independent modulo affine heights. The successive prefixes of

$$h = w^{(2)} + \varepsilon w^{(3)} + \dots + \varepsilon^{n-4} w^{(n-2)}$$

have exactly $n - 3$ strict subdivision changes. At stage j , for $0 \leq j \leq n - 3$, the maximal cells are the triangles

$$\{0, \ell, \ell + 1\} \quad (1 \leq \ell \leq j)$$

together with the remaining polygon

$$\{0, j + 1, j + 2, \dots, n - 1\}.$$

Thus both the comparison-rank bound and the strict-history bound are sharp already for planar configurations in convex position.

Proof. All the a_i are vertices of their convex hull: the graph of x^2 is strictly convex, so each point has a supporting tangent line meeting no other listed point. The convex hinge

$$\psi_k(x, y) = \max\{0, y - kx\}$$

takes the values (12.3) on the vertices. Its crease cuts the polygon along the chord from a_0 to a_k . On each side it is affine, and its convexity puts its graph below every convex combination of lifted vertices. Since the two vertex polygons cover the base polygon, it is exactly the lower piecewise-affine roof for these heights.

The chords from a_0 to $a_2, \dots, a_{n-2}$ do not cross in their relative interiors. A positive sum of the corresponding hinges has the common refinement of their domains of affinity. No crease disappears: across the interior of one chord only its own hinge changes gradient, and its coefficient is positive. The first j chords give precisely the listed fan triangles and remaining polygon. This proves the subdivision statement and its strictness.

For independence, suppose a real linear combination of the height vectors is affine on A . Each height vector vanishes at $i = 0, 1, 2$. An affine function of (i, i^2) is a polynomial in i of degree at most two; those three zeros force it to vanish identically. Evaluate the purported zero combination at $i = 3$: only $w^{(2)}$ can be nonzero and its value is 3, so its coefficient vanishes. Evaluation at $i = 4$ then isolates $w^{(3)}$, with value 4. Continue up to $i = n - 1$. Every coefficient vanishes, giving independence modulo affine heights. $\square$

12.3 A fully explicit hexagon

For $n = 6$, the three rows are

$$\begin{aligned} w^1 &= (0, 0, 0, 3, 8, 15), \\ w^2 &= (0, 0, 0, 0, 4, 10), \\ w^3 &= (0, 0, 0, 0, 0, 5). \end{aligned} \tag{12.4}$$

The cells are shown in Table 2. The all-basis-slack certificate (11.5) gives $\eta = 1/34$. Thus the ordinary rational height vectors w^1 , $w^1 + w^2/34$, and $w^1 + w^2/34 + w^3/34^2$ provide a single explicitly certified real specialization of the three stages. This bound is not optimal. In this compatible hinge family every choice of positive layer weights works, as the geometric proof shows.

Table 2: The maximal marked cells of the compressed hexagon history. Braces contain vertex labels.

Stage	Maximal marked cells
0	$\{0, 1, 2, 3, 4, 5\}$
1	$\{0, 1, 2\}, \{0, 2, 3, 4, 5\}$
2	$\{0, 1, 2\}, \{0, 2, 3\}, \{0, 3, 4, 5\}$
3	$\{0, 1, 2\}, \{0, 2, 3\}, \{0, 3, 4\}, \{0, 4, 5\}$

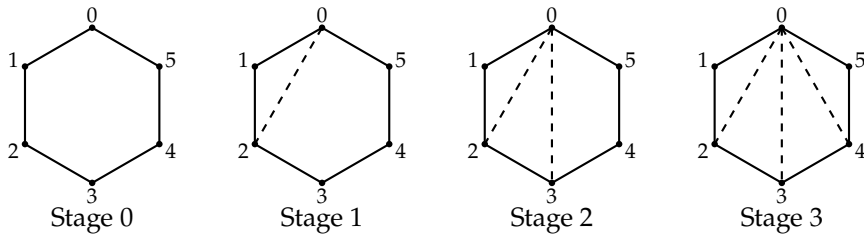


Figure 1: Successively introducing three fan diagonals. This regular-hexagon drawing is a combinatorial schematic; the actual rational coordinates are (i, i^2) , and the heights are (12.4).

The example also distinguishes two tasks. To preserve only the final triangulation, a single ordinary real height vector is enough. To preserve all three strict transitions requires three layers. To preserve every real affine comparison of the original non-Archimedean height vector requires its full comparison rank. Those numbers agree in this example, but need

not agree in general. [write] These are three of the invariants compared in [remark 1.1](#): the final triangulation (one real vector here; with a prescribed shadow, degree at most one by [theorem 10.4](#)), the strict geometric history (r layers, [theorem 12.2](#)), and the complete sign type (rank $p = m$, minimum germ degree $m - 1$ by [theorem 9.1](#)). [Theorem 10.6](#) shows how far apart they can be.

13 Shadows and canonical refinement hierarchies

Proposition 13.1 (Standard part of a finite convex hull). *For finite points $v_1, \dots, v_N \in K^D$,*

$$\text{st}(\text{conv}_K(v_1, \dots, v_N)) = \text{conv}_{\mathbb{R}}(\text{st}(v_1), \dots, \text{st}(v_N)). \quad (13.1)$$

Proof. Every convex coefficient lies between zero and one and is therefore finite. Standard part takes a convex combination to the convex combination with real coefficients $\text{st}(\lambda_i)$. Conversely every real convex combination of the shadows is the standard part of the combination of the original points with those same real coefficients. $\square$

[write] Manuscript 02 proves the same identity (its Proposition “Standard part of a finite hull”, for points of $\mathcal{O}_K^d$) by the same argument; its labels `poly:ts:prop:stconv` and `poly:ts:eq:stconv` are attached here. [Parts III to V](#) use the identity and cite it here.

The proposition is an equality of convex sets after applying `st`. It does not say that standard part sends each face to a face.

Example 13.2 (A face whose shadow is not a face). Lift the four corners of a unit square with heights $(0, 0, 0, \varepsilon)$, where the ordering is $(0, 0), (1, 0), (0, 1), (1, 1)$ and $\varepsilon > 0$ is infinitesimal. The first three lifted points form a lower triangular facet. The shadow of the entire polytope is a square in the plane $z = 0$, whereas the shadow of that facet is the filled triangle with vertices $(0, 0), (1, 0), (0, 1)$. That triangle is not a face of the square. Thus there is no unrestricted face-lattice map $F \mapsto \text{st}(F)$.

Proposition 13.3 (The shadow subdivision is a coarsening). *For a bounded lifting of a fixed real base, its marked regular subdivision refines the one induced by $h^0 = \text{st}(h)$: every maximal source cell is contained in a maximal shadow cell, including inclusion of its contact labels.*

Proof. Choose a supporting basis B of a maximal source cell. All $s_{B,i}(h) \geq 0$, so all $s_{B,i}(h^0) = \text{st}(s_{B,i}(h)) \geq 0$. Thus the same basis supports a shadow cell. Every source equality remains an equality after standard part, giving $I_B(h) \subseteq I_B(h^0)$. $\square$

More generally the complete flag yields a finite hierarchy, even when the heights are not bounded. Let $b_1, \dots, b_m$ be its coefficient rows, and for $0 \leq k \leq m$ use the lexicographic height with the first k rows. Equivalently take the polynomial $H_k(t) = \sum_{j=1}^k b_j t^{j-1}$ at a sufficiently small positive real t , with $H_0 = 0$. Denote its marked subdivision by $\mathcal{S}_k$.

Proposition 13.4 (Canonical finite refinement chain). *The subdivisions $\mathcal{S}_k$ do not depend on the choice of coefficient functionals realizing the same oriented flag, and*

$$\mathcal{S}_0 \geq \mathcal{S}_1 \geq \dots \geq \mathcal{S}_m = \mathcal{S}(h),$$

where $\geq$ means “is coarser than.” Every subdivision in the chain is an ordinary real regular subdivision of A . There are at most m strict refinement steps. The oriented flag need not be recoverable from this chain.

Proof. Positive triangular transformations (8.2) preserve the first-nonzero signs of every truncated list, so they preserve all truncated slack signs. Hence the chain depends only on the flag. If B supports a lower cell for the first $k + 1$ rows, every slack has nonnegative lexicographic sign through that level. Its first k entries are either all zero or have the same positive first nonzero entry. Thus B also supports a lower cell at level k , and its equality labels can only increase when the last row is removed. This proves refinement. Finite sign stabilization realizes every level over $\mathbb{R}$; a common real parameter can be chosen for the finitely many levels and slacks.

For the last assertion, choose f_1 nonzero on every nonzero slack vector as in Theorem 10.6. All later rows are invisible to the subdivision, so $S_1 = \cdots = S_m$ while the remaining flag on $\ker f_1$ can vary whenever its dimension permits. $\square$

This is a flag-refined version of the usual secondary data, not a claim that the secondary fan is the entire space of complete dependence types. Already in a generic real secondary chamber, further infinitesimal coefficients may encode nontrivial orders on its invisible real dependence directions.

14 Standard parts: exact identities and lost faces

[write] Manuscript 02's standard-part section. Its first proposition, $\text{st conv} = \text{conv st}$, is [proposition 13.1](#) (both sources) and is not repeated. The finite polar identity, the lost face and the optimality band are new against 05. Manuscript 05's [example 13.2](#) and 02's (14.2) are two instances of one failure, a face whose standard part is not a face; [Part III](#) adds a pentagon ([section 40](#)), and [proposition 14.2](#) bears on the questions of [Part III](#) about face specialization.

Let $K \supseteq \mathbb{R}$ be an ordered real-closed working field containing a positive infinitesimal. Write

$$O_K = \{x \in K : |x| < m \text{ for some } m \in \mathbb{N}\}$$

for its ring of finite elements. Completeness of $\mathbb{R}$ gives a unique standard part $\text{st}(x) \in \mathbb{R}$ for each finite x : the difference $x - \text{st}(x)$ is infinitesimal. Standard part is a weakly order-preserving ring homomorphism on O_K . Apply it coordinatewise.

14.1 Convex hull and polar commute with residue

[write] The hull identity, 02's first proposition here, is [proposition 13.1](#), printed once above for both sources.

Proposition 14.1 (Finite polar residue). *Let $P = \text{conv}_K(v_1, \dots, v_n)$ with all v_i finite, and define $P^\circ = \{y \in K^d : y \cdot x \leq 1 \text{ for all } x \in P\}$. No assumption that 0 is interior is required. Then*

$$\text{st}(P^\circ \cap O_K^d) = (\text{st } P)^\circ. \quad (14.1)$$

Proof. If y is a finite polar point, standard part of each inequality $y \cdot v_i \leq 1$ gives the inclusion from left to right. For the reverse inclusion, let $y \in (\text{st } P)^\circ \subseteq \mathbb{R}^d$ and put

$$\delta = \max(0, \max_i (y \cdot v_i - 1)).$$

All terms are finite and their standard parts are nonpositive, so δ is a nonnegative infinitesimal. The point $y' = y/(1 + \delta)$ satisfies $y' \cdot v_i \leq 1$ for all i , is finite, and has standard part y . Thus $y' \in P^\circ \cap O_K^d$. $\square$

The finite-part restriction in (14.1) matters because a polar can be unbounded and standard part is not defined on infinite coordinates. The proof does not need a Hausdorff convergence theorem or a metric on all of K^d .

14.2 A face need not have a face as its standard part

Consider

$$P = \text{conv}_K\{(-1, 0), (0, -\varepsilon), (1, 0), (1, 1), (-1, 1)\}. \quad (14.2)$$

Its unique point maximizing $(x, y) \mapsto -y$ is the vertex $(0, -\varepsilon)$. The standard-part hull is the rectangle $Q = [-1, 1] \times [0, 1]$. The standard part of that vertex is $(0, 0)$, which lies in the relative interior of the bottom edge of Q and is not a face by itself.

Thus the assignment $F \mapsto \text{st}(F)$ is not a map from the face lattice of P to the face lattice of Q . This is stronger than the observation that two vertices can merge. Even an exposed face can map to a proper nonface subset of a limiting face.

14.3 Near-optimal points recover the lost limiting face

Proposition 14.2 (An exact infinitesimal optimality band). *Let $P = \text{conv}_K(v_1, \dots, v_n)$ be finite, $Q = \text{st}(P)$, and $c \in \mathbb{R}^d$. Put $M = \max_i c \cdot v_i$ and $m = \text{st}(M)$. There is a positive infinitesimal η such that*

$$\text{st}\{x \in P : c \cdot x \geq M - \eta\} = \arg\max_{y \in Q} c \cdot y. \quad (14.3)$$

Proof. Standard part commutes with a finite maximum, so $m = \max_{y \in Q} c \cdot y$. Let

$$J = \{i : c \cdot \text{st}(v_i) = m\}, \quad D = \max_{i \in J} (M - c \cdot v_i).$$

The set J is nonempty, and D is a nonnegative infinitesimal. Choose any positive infinitesimal $e \in K$ and set $\eta = D + e$.

A point in the band on the left of (14.3) has standard-part objective value at least m , hence exactly m because its standard part lies in Q . Conversely, every v_i with $i \in J$ belongs to the band, and so does their K -convex hull. Its standard part is the real convex hull of the maximizing standard-part vertices. That is the entire maximizer face of Q , proving the other inclusion. $\square$

For (14.2), exact optimization remembers just the infinitesimally lower vertex, but an appropriate infinitesimal optimality band recovers the whole bottom edge. This suggests that residue geometry should retain near-optimal fibers or scale-decorated cells rather than attempt to force a false homomorphism between ordinary face lattices.

15 Higher-rank tropical minimizers: an elementary bridge

This section records a finite-rank interpretation of the hierarchy, not a new theory of higher-rank tropical multiplicities. It is useful for comparing the lifting package with [3, 4] without identifying different notions of topology or convexity.

Take real height rows $b_1, \dots, b_r \in \mathbb{R}^n$. For $x = (x_1, \dots, x_r) \in (\mathbb{R}^d)^r$, compare the vectors

$$\Phi_i(x) = (b_{1,i} + a_i \cdot x_1, \dots, b_{r,i} + a_i \cdot x_r) \in \mathbb{R}^r$$

in lexicographic order. Define $I_0 = \{1, \dots, n\}$ and successively

$$I_j(x) = \arg \min_{i \in I_{j-1}(x)} (b_{j,i} + a_i \cdot x_j). \quad (15.1)$$

The final minimizers of $\Phi_i(x)$ are exactly $I_r(x)$.

For nonempty $T \subseteq S \subseteq \{1, \dots, n\}$ set

$$\mathcal{D}_b^S(T) = \{y \in \mathbb{R}^d : \arg \min_{i \in S} (b_i + a_i \cdot y) = T\}.$$

This is a relatively open polyhedron, possibly empty. Indeed choose $i_0 \in T$, impose equalities

$$(a_i - a_{i_0}) \cdot y = b_{i_0} - b_i \quad (i \in T),$$

and strict inequalities saying that every $i \in S \setminus T$ has a larger value. If nonempty, it has dimension $d - \dim \text{aff}(A_T)$.

Proposition 15.1 (Product strata for a fixed minimizer history). *For a specified chain of nonempty sets $I_0 \supseteq I_1 \supseteq \dots \supseteq I_r$, the points x with this exact successive minimizer history form*

$$\mathcal{D}_{b_1}^{I_0}(I_1) \times \mathcal{D}_{b_2}^{I_1}(I_2) \times \dots \times \mathcal{D}_{b_r}^{I_{r-1}}(I_r). \quad (15.2)$$

A nonempty such stratum has dimension

$$rd - \sum_{j=1}^r \dim \text{aff}(A_{I_j}).$$

There are finitely many histories, and the lexicographic tie locus is the disjoint union of those strata with $|I_r| \geq 2$.

Proof. Lexicographic comparison first discards all labels not minimizing the first coordinate, then discards nonminimizers of the second among the survivors, and continues. For a prescribed chain, the j th condition involves only x_j , which proves the product formula. The equalities defining a nonempty factor have rank $\dim \text{aff}(A_{I_j})$. Its strict inequalities cut out an open subset of that affine solution space and do not decrease its dimension. Dimensions add under products. The finiteness and disjointness follow because every x has a unique chain of finite sets. $\square$

This derivation makes no claim that the Euclidean closures of these strata constitute a pure polyhedral complex. Higher-rank examples can fail the naive closure and purity expectations; such issues are treated explicitly in [3]. Its rank-two product decomposition is a close precedent for (15.2). The newer framework [4] uses higher-rank GKZ cones and its own face and coface structure. A scalar-field Farkas lemma over K and a separation statement for lexicographically ordered matrix halfspaces are different assertions; one cannot import one as the other without checking the ambient structure.

The present finite compression concerns the *height dependence space*. It does not preserve the full ordered multiplicative group of all surreal monomials. Nor does it supply tropical weights, balancing, or the algebraic degeneration machinery of [4]. Those are additional structures beyond the sign and leading-term invariants proved here.

16 Examples that locate the exact boundary

16.1 Identical leading heights, opposite square diagonals

For the square ordered as above, the affine-dependence space is spanned by $c = (1, -1, -1, 1)$. Consider

$$h^+ = (1, 1, 1, 1 + \varepsilon), \quad h^- = (1, 1, 1, 1 - \varepsilon), \quad 0 < \varepsilon \ll 1.$$

All four individual heights have leading term 1 in both liftings and both standard parts are $(1, 1, 1, 1)$. Yet

$$L_{h^+}(c) = \varepsilon, \quad L_{h^-}(c) = -\varepsilon.$$

The positive case uses diagonal a_2a_3 , with maximal cells $\{1, 2, 3\}$ and $\{2, 3, 4\}$. The negative case uses diagonal a_1a_4 , with maximal cells $\{1, 2, 4\}$ and $\{1, 3, 4\}$. These exact contact-label sets are also returned by the supplied checker.

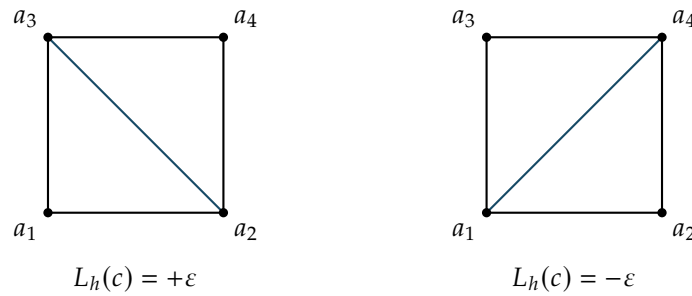


Figure 2: The coordinatewise leading heights and the real shadows agree; the first nonzero affine-dependence coefficient determines different regular triangulations. The figure shows the projected lower cells.

16.2 A nontrivial three-scale compression

Take $A = (0, 1, 2, 3, 4)$ in one dimension, and gauge the first two heights to zero. On the last three coordinates put

$$g = (1, 1, 0)\varepsilon + (2, 2, 0)\varepsilon^2 + (0, 1, 1)\varepsilon^5 \\ + (1, -2, -3)\varepsilon^6 + (1, 0, 1)\varepsilon^9, \quad \varepsilon = \omega^{-1}.$$

The coefficient at ε^2 is twice the first; the coefficient at ε^6 is $(1, 1, 0) - 3(0, 1, 1)$. The coefficient-pivot procedure therefore selects powers 1, 5, 9, and

$$g^\# = (1, 1, 0)\varepsilon + (0, 1, 1)\varepsilon^5 + (1, 0, 1)\varepsilon^9.$$

The three selected rows are independent. Every real linear combination of the three original coordinates has exactly the same leading term after compression, not merely the same sign on the finitely many circuits. A shadow-preserving polynomial model is

$$H(t) = (0, 0, t + t^3, t + t^2, t^2 + t^3).$$

Its minimum degree for the complete sign type with zero shadow is three. The exact checker gives the sufficient bound $\eta_0 = 1/10$, the sample $t_0 = 1/20$, and lower maximal contact sets $\{1, 2\}$ and $\{2, 5\}$. Thus a degree-one path from zero to $H(1/20)$ already retains the finite lifted geometry, while losing some of the full real dependence type.

The gaps between the powers could be replaced by much larger surreal exponent gaps. The proof depends on their order and coefficient spans, not on the existence of a smallest step in the exponent group.

16.3 Unsigned determinant valuations also lose combinatorics

Let $\delta > 0$ be infinitesimal and $u = (1, 1)$. For a real configuration A put $p_i = u + \delta a_i$. Every coordinate of every p_i has leading term 1. Every nonzero augmented 3×3 determinant is δ^2 times the corresponding real determinant. Compare

$$A = \{(0, 0), (2, 0), (0, 2), (2, 2)\}$$

with

$$A' = \{(0, 0), (2, 0), (0, 2), (1/2, 1/2)\}.$$

In both configurations every triple is affinely independent, so all these determinants have the same leading exponent. Nevertheless one hull has four vertices and the other has three, with a labeled interior point. Neither coordinate leading terms nor the unsigned determinant valuations recover the labeled face lattice. The signs of leading coefficients matter.

16.4 Finite truncation can destroy an exact equality

Let $y = \varepsilon$ and $x = \varepsilon + \varepsilon^2 + \cdots = \varepsilon/(1 - \varepsilon)$. The exact algebraic relation $(1 - y)x - y = 0$ holds. Replacing x by the first N terms gives

$$(1 - \varepsilon)(\varepsilon + \cdots + \varepsilon^N) - \varepsilon = -\varepsilon^{N+1} \neq 0.$$

The pivot theorem preserves real *linear* combinations. It does not preserve arbitrary polynomial relations between the coordinates, and it must not be advertised as a finite truncation algorithm for a surreal field.

16.5 One parameter does not preserve the entire scale field

The positive infinitesimals $\varepsilon = \omega^{-1}$ and $\tau = \omega^{-\omega}$ satisfy $\tau < \varepsilon^N$ for every positive ordinary integer N . Replacing them by t^a, t^b with fixed positive real a, b in a rank-one monomial group cannot preserve all these inequalities: choose $N > b/a$. Our finite-height models preserve a specified real-linear dependence type, not every multiplicative comparison in the field generated by the original scales. This is a boundary on what the theorem says, not a contradiction to its polynomial realization conclusion.

17 A toric application: preserving the whole initial ideal

Suppose now that the base points are integral, $a_i \in \mathbb{Z}^d$, and let k be any field. The homogeneous toric ideal is

$$I_A = \ker(k[x_1, \dots, x_n] \longrightarrow k[z_0, z_1^{\pm 1}, \dots, z_d^{\pm 1}], \quad x_i \longmapsto z_0 z^{a_i}).$$

For a polynomial, define its h -initial form to be the sum of the terms whose weights $\sum_i u_i h_i$ are *maximal*. These weights lie in the ordered additive group of $\mathbf{No}$; no infinite sum or coefficient valuation is involved. The ideal generated by the initial forms of all members of I_A is denoted $\text{in}_h I_A$.

Put $\Lambda = \ker_{\mathbb{Z}} \tilde{A}$. For $c \in \Lambda$ write $c = c^+ - c^-$ with disjoint nonnegative integer supports, and set $B_c = x^{c^+} - x^{c^-}$. Its initial form depends only on $\chi_h(c)$. A vector $g \in \Lambda \setminus \{0\}$ is *conformally indecomposable* when it is not a sum of two nonzero lattice vectors lying in its closed orthant, with their absolute values adding coordinatewise. The set G_A of these vectors, with either both

signs or one representative of each pair, is the Graver basis. Its relationship with universal Gröbner bases is classical; see, for example, [7]. We include the finite argument needed for arbitrary ordered weights.

Lemma 17.1 (A finite certificate for arbitrary ordered weights). *The set G_A is finite. For heights in any totally ordered abelian group,*

$$\mathrm{in}_h I_A = \langle \mathrm{in}_h B_g : g \in G_A \rangle.$$

Proof. In each orthant, conformal indecomposables are the minimal nonzero elements of the lattice under coordinatewise comparison of absolute values. Dickson's lemma makes this set finite. One proof of that lemma is induction on the number of coordinates: any infinite sequence in $\mathbb{N}^n$ contains an infinite subsequence whose first coordinate is nondecreasing; apply the induction hypothesis to the remaining coordinates to find a comparable pair. An infinite set of distinct minimal elements would give an infinite sequence with no such pair.

Repeatedly splitting a decomposable lattice vector strictly decreases the positive integer ℓ^1 -size of each resulting summand. Thus every $c \in \Lambda \setminus \{0\}$ has a conformal decomposition $c = g_1 + \cdots + g_s$ into Graver vectors. Let $\alpha_j = h \cdot g_j$. If $h \cdot c > 0$, then some $\alpha_j > 0$. The monomial $\mathrm{in}_h B_{g_j} = x^{g_j^+}$ divides $x^{c^+} = \mathrm{in}_h B_c$. The negative case is symmetric. If $h \cdot c = 0$ and not all α_j vanish, there are both a positive and a negative α_j . The two terms of B_c are then separately divisible by initial monomials of Graver binomials. If all $\alpha_j = 0$, use the identity, valid for conformal sums,

$$B_{g+g'} = x^{g'^+} B_g + x^{g^-} B_{g'}.$$

It expresses B_c in the ideal generated by the tied Graver binomials. Hence every initial form of a lattice binomial belongs to the stated ideal. A common monomial factor handles binomials whose exponents overlap.

It remains to justify that binomials suffice. Decompose any $f \in I_A$ into its $\tilde{A}$ -homogeneous parts. Within each part, the coefficients sum to zero, because all monomials have the same image under the toric map. If the terms of one part have different weights, every maximal-weight monomial is the initial monomial of its difference with a lower-weight monomial in that same part. If all its terms have equal weight, the whole part is a linear combination of tied differences with one fixed monomial. Thus its initial form belongs to the ideal of initial binomials. The initial form of the original f is a sum of the initial forms of those homogeneous parts whose maximal weights attain the global maximum, so it belongs as well. The reverse inclusion follows because every B_g belongs to I_A . $\square$

Theorem 17.2 (Toric compatibility of compression and real shadows). *For an integral base configuration, the compressed lifting satisfies*

$$\mathrm{in}_{h^\#} I_A = \mathrm{in}_h I_A.$$

A polynomial model of its complete dependence type has the same initial ideal for every sufficiently small positive real parameter. If h is bounded, a real affine-linear path starting at $\mathrm{st}(h)$ can preserve this initial ideal, all the lifted determinant signs, and the marked lower subdivision simultaneously on its whole punctured segment.

Proof. Compression preserves the sign of $L_h(g)$ for every $g \in G_A \subseteq V$, so Lemma 17.1 gives the first equality. The finite sign lemma applied to these Graver comparisons proves stabilization along a polynomial model. For the last assertion, add G_A to the finite determinant and slack test list in Theorem 10.4. $\square$

This preserves the initial ideal itself, not only its radical or the underlying subdivision. The initial algebra and its scheme-theoretic invariants therefore agree. The statement is an application of classical finite Graver control together with our lifting construction; it is not a new construction of the higher-rank degenerations of [4]. For the square, $I_A = (x_1x_4 - x_2x_3)$: the two infinitesimal choices above give initial ideals (x_1x_4) and (x_2x_3) , respectively.

18 General surreal polytopes: what transfer gives and what it does not

Theorem 18.1 (Classical finite realization and shadow arcs). *Every finite surreal point configuration has the same labeled oriented matroid, and hence the same labeled polytope face structure, as a real configuration. Without prescribed real coordinates or other real parameters, a realization can be chosen with real algebraic coordinates. If the original coordinates are finite, there is a real semialgebraic arc of configurations with the same finite determinant signs that converges to their labeled standard parts. Additional finitely many polynomial sign conditions can be preserved along the same arc. [write] In manuscript 02's form (its Theorem "Classical real-algebraic transfer"): for any finite labeled configuration $V = (v_1, \dots, v_n)$ in a real-closed field, there is a labeled configuration of real algebraic points with the same signed affine-hyperplane patterns and the same affine dimension. In particular, the two convex hulls have isomorphic labeled face lattices.*

Proof. List the signs of all minors of the homogeneous coordinate matrix, keeping each zero as an equality. This is a finite system of polynomial equalities and strict inequalities with integer coefficients. It has a solution in a set-sized real closed subfield of **No** containing the coordinates. Completeness of the theory of real closed fields transfers the existential sentence to $\mathbb{R}$, and also to the real algebraic numbers when there are no prescribed real parameters. The minor signs determine the rank and oriented matroid; the facet reconstruction in Lemma 10.1, in its appropriate affine rank, gives the face structure. These are standard real-algebraic facts [5].

For the bounded assertion, let v^0 be the vector of all standard-part coordinates and let S be the corresponding real semialgebraic sign cell, including the extra predicates when present. For every positive real r , the surreal tuple lies in the ball of radius r about v^0 and satisfies the sign conditions. Transfer over the named real parameters shows that S contains a real tuple within that ball. Thus $v^0 \in \bar{S}$. If a constant configuration already suffices, take a constant arc. Otherwise semialgebraic curve selection yields an arc entering S and tending to v^0 [5, 6]. Its one-variable semialgebraic coordinates are algebraic-function branches; one may use a finite power reparameterization to obtain analytic branches at the endpoint. The determinant signs on the punctured arc are the desired ones. $\square$

Second proof of the realization assertion, in manuscript 02's covector form (source 02). For each $\sigma \in \{-1, 0, 1\}^n$, let $\Psi_\sigma(V)$ be the formula

$$\exists u_1, \dots, u_d, b \bigwedge_{i=1}^n \text{sgn}(u \cdot v_i + b) = \sigma_i.$$

Replace each sign assertion by the corresponding polynomial equality or strict inequality. There are only 3^n patterns. Conjoin Ψ_σ for all patterns realized by V and $\neg\Psi_\sigma$ for all other patterns. Also conjoin the finite minor conditions expressing its affine dimension. Finally existentially quantify the coordinates of V .

The resulting sentence is a first-order sentence of ordered rings with integer coefficients. It is true in the given real-closed field. By the completeness of the theory of real-closed fields,

equivalently by Tarski's elimination theorem [11], it is true in the real algebraic numbers, which themselves form a real-closed field. Choose a witnessing configuration there.

A supporting affine functional has all its values nonpositive after subtracting its maximum. Its zero labels are exactly the labels on its maximizing face. Thus the all-nonpositive sign patterns recover all nonempty face label sets; adding the empty face gives the entire labeled face lattice. The construction preserves these patterns and their inclusions. $\square$

[write] The next two paragraphs follow the transfer theorem in manuscript 02; the family announced at their end is the lifting compression of [section 7](#).

This theorem is intentionally stronger than a statement about the numbers of faces: it preserves the entire affine covector system. It is also intentionally weaker than an exact coordinate representation. There is no order-preserving field map from **No** into $\mathbb{R}$ fixing $\mathbb{R}$, since a positive infinitesimal would have to become a positive real number smaller than every $1/m$. Nor does the theorem guarantee rational coordinates.

Consequently, such familiar finite combinatorial questions as which f -vectors occur do not acquire additional examples merely by allowing surreal coordinates. Any genuinely additional information must reside in scales, residue maps, simultaneous structures, or infinite families of observations. We turn to one such family.

The realization statement is classical transfer, not a new surreal realizability theorem. Likewise the general arc existence is an application of curve selection; [6] supplies a quantitative framework. The special contribution of the fixed-base setting is the elementary polynomial model with sharp degree and the linear finite-data path.

Example 18.2 (Polynomial arcs need not preserve nonlinear constraints). Take the finite surreal point

$$(x, y) = (\sqrt{1 - \varepsilon^2}, \varepsilon), \quad 0 < \varepsilon \ll 1.$$

It lies on the unit circle, has $y > 0$, and has standard part $(1, 0)$. There is no real polynomial arc $(X(t), Y(t))$ with $(X(0), Y(0)) = (1, 0)$, $X(t)^2 + Y(t)^2 = 1$, and $Y(t) > 0$ for small positive t . Indeed, if either polynomial is nonconstant, the coefficient of the largest even degree in $X^2 + Y^2$ is a positive sum of squares and cannot vanish. Both must therefore be constant, contradicting the endpoint and $Y(t) > 0$. An algebraic analytic arc exists, but a polynomial one does not. This example concerns the permitted extra polynomial constraints; it is not claimed as an obstruction arising from determinant signs alone.

19 A finite Puiseux model with the same residue

[write] [Theorem 19.1](#) is the bounded half of [theorem 18.1](#) (source 05) in manuscript 02's form: it keeps the full signed affine-hyperplane patterns and any finite list of real polynomial signs along a semialgebraic arc starting at the standard part, with convergent algebraic Puiseux germs. Both proofs are the same transfer-and-curve-selection argument (05 cites [5, 6], 02 cites [15, 6]). The circle example after [theorem 18.1](#) shows that such arcs cannot in general be taken polynomial. The finite-support specialization [theorem 41.2](#) of [Part III](#) is an effective counterpart for finite-support Hahn certificates.

The universal truncation theorem does not prevent a *different* well-chosen real one-parameter model from preserving any finite collection of incidence data. This is a useful distinction between an arbitrary coordinate truncation and a controlled semialgebraic specialization.

Theorem 19.1 (Finite incidence-and-residue realization). *Let $V \in O_K^{dn}$ be a finite bounded configuration and let $b = \text{st}(V) \in \mathbb{R}^{dn}$. Fix any finite collection of polynomials with real coefficients in the coordinates of V . There is a continuous semialgebraic arc $\gamma : [0, a) \rightarrow \mathbb{R}^{dn}$ such that*

- (a) $\gamma(0) = b$;
- (b) *for every sufficiently small $s > 0$, all the prescribed polynomial signs at $\gamma(s)$ equal those at V ;*
- (c) *for those s , the configuration $\gamma(s)$ has the same signed affine-hyperplane patterns and affine dimension as V .*

The coordinate germs of γ can be taken to be convergent algebraic Puiseux germs over $\mathbb{R}(s)$. In particular, they give a finite geometric model with the full face lattice of V and its prescribed standard-part limit.

Proof. Encode the prescribed polynomial signs and the complete finite list of affine sign patterns exactly as in Theorem 18.1, now keeping real parameters. Include the rank conditions. Quantifier elimination makes the resulting condition a semialgebraic set $S \subseteq \mathbb{R}^{dn}$, and the original configuration satisfies the same defining condition in K .

For every real $\delta > 0$, the field K satisfies

$$\exists z \in S(K) \quad \max_i |z_i - b_i| < \delta,$$

since $z = V$ is a witness. The inclusion $\mathbb{R} \subseteq K$ is elementary for ordered-ring formulas, by real-closed-field quantifier elimination. Hence the displayed assertion has a real witness. It follows that $b \in \bar{S}$ in the ordinary Euclidean topology.

If $b \in S$, the constant arc is sufficient. Otherwise apply the semialgebraic curve selection lemma to S at b . It gives a continuous semialgebraic arc with initial point b and positive-time values in S . This proves (a)–(c). A one-variable semialgebraic function has, after restricting to a sufficiently short interval, an algebraic Puiseux germ; its Puiseux expansion is convergent. Equivalently, after a common finite ramification $s = u^m$, all the coordinate germs are analytic at the endpoint. These are the standard curve-selection and algebraic-Puiseux facts from real algebraic geometry [15, 6]. $\square$

Here “algebraic” means algebraic over $\mathbb{R}(s)$, not that every coefficient is a real algebraic number. The limiting configuration may itself contain transcendental real coordinates. Theorem 19.1 is a synthesis of standard transfer and curve-selection tools, not a new curve selection lemma. It supplies neither explicit complexity bounds nor a coordinate embedding of the original surreal field.

Most importantly, its quantifier is *finite*. The chosen arc can preserve the final four-point hull and its residue, or any finite list of additional polynomial observations. It need not preserve the arbitrary infinite sequence of coordinate-truncation hulls constructed in Section 26. This is exactly how finite realizability and infinite scale complexity coexist.

20 Corrections to the earlier convexity draft

[[write] The draft corrected here, *Finite Convexity and Exact Lexicographic Optimization over the Surreals* (22 September 2026), was supplied privately to the authors of manuscripts 05 and 07; it is not part of ProveIt and could not be checked. The statements below are printed as standalone results.]

The earlier draft [10] contains a valid sufficiently-small comparison argument. Its Proposition 7.1, however, identifies the *entire* positive validity set with an interval. Its Theorem 9.1

and following explanation also require a distinction between a real domain and its scalar extension. Neither issue affects the proofs above.

20.1 The global validity set need not be an interval

On $P = [0, 1]$, take the objective hierarchy $c_0x = x$, $c_1x = -3x$, and $c_2x = 2x$. Lexicographic maximization selects $x = 1$. The compiled objective is

$$(1 - 3t + 2t^2)x = (1 - t)(1 - 2t)x.$$

It selects exactly the same maximizer for

$$t \in (0, 1/2) \cup (1, \infty), \quad (20.1)$$

not for a single interval. At $t = 1/2$ and $t = 1$ it is constant on the whole segment; between those points it selects $x = 0$.

Proposition 20.1 (Correct initial-radius statement). *For finitely many nonzero comparison polynomials with their required signs at 0^+ , the component of the simultaneous validity set adjacent to zero is $(0, \rho)$, where ρ is the least positive zero of one of the normalized nonzero comparison polynomials, or $+\infty$ when none has a positive zero. Over a real closed coefficient field, a finite ρ belongs to that field. The entire validity set may have additional components beyond ρ .*

Proof. Divide each polynomial by its lowest power of t ; its constant term is nonzero. Before its first positive zero its sign is constant, by the intermediate value property. At a positive zero the required strict sign fails, even if the polynomial merely touches zero and later has the same sign again. Taking the least positive zero over the finite list gives exactly the component adjacent to zero. Signs on subsequent root intervals can agree again, as (20.1) shows. $\square$

For a general ordered field the first zero may lie only in a real closure, so the endpoint may define a cut rather than an element of the original field. A sufficient algebraic bound such as Lemma 10.3 avoids this issue. We never use a global-interval assertion.

20.2 A real face and its scalar extension are different sets

Let $P_{\mathbb{R}} = \text{conv}_{\mathbb{R}}(v_1, \dots, v_N)$ with real vertices, and let $P_K = \text{conv}_K(v_1, \dots, v_N)$. Suppose a compiled objective makes exactly the labels in J optimal. Then

$$\arg \max_{x \in P_K} c \cdot x = \text{conv}_K\{v_j : j \in J\}, \quad (20.2)$$

not generally $\text{conv}_{\mathbb{R}}\{v_j : j \in J\}$. To prove (20.2), express x as a convex combination and subtract its objective value from the optimal vertex value. The result is a sum of nonnegative terms. Equality forces every positive coefficient to occur at an optimal vertex, exactly as over $\mathbb{R}$.

For example, maximizing the first coordinate on $[0, 1]_K^2$ gives the whole edge $\{1\} \times [0, 1]_K$, including $(1, \varepsilon)$ for a positive infinitesimal $\varepsilon \in K$. This point is not in the ordinary real edge. The correct shadow identity is

$$\text{st} \left(\arg \max_{P_K} c \cdot x \right) = \text{conv}_{\mathbb{R}}\{v_j : j \in J\},$$

by Proposition 13.1. The same optimal *vertex labels* and the same base-changed face are selected. If the old statement is read literally with optimization only over the subset $P_{\mathbb{R}} \subset \mathbf{No}^d$, then equality with the real face is correct; it is not the assertion about K -convex polytopes used elsewhere. This is a scalar-extension ambiguity, not a reason to discard the finite compiler.

21 Computation, certificates, and a formalization route

21.1 What the executable verifier actually does

The accompanying code `05-finite-scale-liftings-verify.py` [write] (delivered as `code/verify.py`) uses only Python's standard library and exact rational arithmetic. It accepts the finite examples encoded there as Laurent polynomials in a positive infinitesimal $\varepsilon = \omega^{-1}$. An input coefficient list is complete, not an oracle that may reveal further terms later. The procedure is:

1. find affine bases by exact determinants and compute barycentric coefficients by exact elimination;
2. subtract the affine gauge coefficientwise and retain exactly the rank-increasing coefficient rows;
3. build the polynomial model, every affine-basis slack, and every augmented maximal minor;
4. compare leading terms and signs before and after compression, compute the rational threshold, and verify the real specialization and several points on the linear shadow path.

The theorem proves validity on the whole segment; the sampled points are only additional implementation checks. Exact leading-term equality is checked on the finite determinant/slack list and on 60 additional rational dependence vectors per example. No finite test can replace the universal proof in Theorem 6.1.

The executed suite contains seven named cases, six members of the sharpness family, and forty randomized rational configurations using seed 20260930. All fifty-three cases passed, including the three-scale pivot list [1, 5, 9] and the validity-set counterexample. This gives 3,180 extra sampled dependence checks, in addition to the complete finite geometry test lists for each case. Machine-readable data are in `data/05-finite-scale-liftings-certificates.json`; the execution summary is in `data/05-finite-scale-liftings-verification_report.txt`. The program does not implement arbitrary surreal arithmetic, semialgebraic curve selection, or a Graver basis algorithm. The corresponding general results rest on the written proofs, not on a claimed implementation.

If the finite support has s coefficient rows and the affine-dependence dimension is q , an incremental elimination implementation can perform the pivot selection using $O(sq^2)$ field operations, after gauging. The supplied reference verifier recomputes some small ranks for simplicity; it is not presented as achieving that optimized complexity. Enumerating all affine bases involves at most $\binom{n}{d+1}$ candidates, and rational bit growth must be accounted for in any bit-complexity claim.

21.2 Why coefficient access alone does not make the bound computable

Proposition 21.1 (An explicit representation obstruction). *There is no algorithm that always computes m for rational-coefficient height series given by uniformly computable coefficient functions, even under the promise that each series actually has finite support and that the real base configuration is fixed.*

Proof. Use $A = (0, 1, 2, 3)$ and gauge the first two heights to zero. For a Turing machine e , let $x_e = \varepsilon^{T+1}$ if it halts for the first time at step T , and let $x_e = 0$ if it never halts. The

coefficient of ε^N in x_e is computable uniformly from e, N by simulating at most N steps. Every resulting series has at most one nonzero coefficient. Take heights $(0, 0, 1, x_e)$. Their dependence-image dimension is one if the machine never halts and two if it halts, since a nonzero positive infinitesimal is real-linearly independent of 1. An algorithm computing m from these coefficient functions would decide the halting problem. $\square$

Thus a complete finite support list and a computable coefficient function with a finite-support promise are very different input representations. The latter supplies no computable bound on the last possible nonzero coefficient. The rank theorem does not erase that distinction.

21.3 A finite certificate workflow

[write] This subsection and the next are from manuscript 02’s section “Algorithms, verification, and proof boundaries” (section 27 prints the rest). Its query-model bound [proposition 21.2](#) and 05’s [proposition 21.1](#) are different obstructions and both are kept: 02’s is an adversary argument in the query model (no algorithm that halts after finitely many coefficient queries computes the rank of every finite-support input), 05’s a reduction of the halting problem for uniformly computable coefficient functions. For a real base with a supplied finite rational array of height coefficients, the positive results suggest the following reproducible workflow. Compute affine bases and their barycentric slack vectors. Fix one basis to gauge away affine coefficient rows by (11.3). Scan the ordered row array, retaining each rank-increasing residual row. For every retained prefix, enumerate lower bases by lexicographic slack signs and deduplicate their zero label sets. Finally compute (11.5) and check the same signs after real rational specialization.

Every operation in this workflow is finite linear algebra and rational arithmetic. The output consists of selected row positions, cell label sets, and a positive rational sign threshold. Each claim can be rechecked without trusting a floating-point convex-hull package. The supplied program implements this workflow for test configurations. It does not purport to parse arbitrary surreal sign sequences or decide arbitrary normal-form supports.

21.4 A rank bound is not a coefficient-query bound

Proposition 21.2 (No finite-query extraction for unrestricted streams). *There is no algorithm which, given only exact coefficient queries to an arbitrary power-series vector, always halts and computes its complete comparison rank, even if the coefficient vectors belong to a fixed rational two-dimensional space and the input is promised to have finite support without a supplied degree bound.*

Proof. Let w^1, w^2 be independent rational vectors. Run a purported algorithm on the coefficient stream of $x = w^1$. If it always halts, it makes only finitely many coefficient queries. Choose a positive integer N larger than every queried index. The finite-support input $x' = w^1 + \varepsilon^N w^2$ gives exactly the same answers to all queries, including adaptive queries, so the algorithm follows the same computation and returns the same result. But the comparison ranks are 1 and 2, respectively, by Theorem 7.1. This is a contradiction. $\square$

Thus “at most q selected rows” is not “at most q coefficient inspections.” Effective treatment of infinite support needs an additional representation promise, a degree bound, or a method for deciding whether a new coefficient direction will ever occur. The obstruction is already present for ordinary finite-support series with an unknown cutoff; it is not solely a proper-class issue.

[write] Manuscript 05’s theorem-by-theorem Lean plan, a subsection here in the source, is printed in [section 83](#).

22 Further research questions and concrete routes

[write] Manuscript 02’s eight research directions are printed in [section 28](#). Four of them are close to questions here: 05’s question 1 and 02’s “Effective controlled specialization” both ask for effective bounds on arcs from a standard part; question 2 and 02’s “Beyond linear comparisons: realizability constraints” both ask when compressed minors come from actual points; question 4 and 02’s “Minimal geometric certificates versus full comparison rank” both ask for certificates smaller than the all-basis list; question 5 and 02’s “Formalization and representation-aware computation” both ask for representations in which the next rank increase can be certified. Each is printed where its source puts it, with this cross-reference.

The questions here are proposed projects, not assertions that every one is a previously published open problem. Their scopes exclude the finite-data and complete-type degree questions already answered above.

1. Moving-base determinant degenerations. For n bounded surreal points in dimension d , seek sharp complexity bounds for a real algebraic or Puiseux arc preserving all determinant signs and the labeled shadow. General quantitative curve selection already gives effective bounds; the task is to exploit the determinant structure rather than reprove mere existence. A useful first case is a planar configuration with one prescribed collinearity collapse and all other minors nonzero. Determine when polynomial coordinates suffice without adding extra nonlinear constraints.

2. More than one nonreal lifting coordinate. Replace one height by $r \geq 2$ surreal coordinates over a fixed real base. Maximal minors now involve products of heights. Compressing their coefficient vectors in an exterior-power representation is finite-dimensional, but the compressed data may fail the decomposability relations needed to come from actual points. Determine the sharp realizable compression bound already for $r = 2$, and identify the Plücker-type obstruction explicitly.

3. Singular fibers of the flag-to-secondary map. A complete dependence flag maps to a regular subdivision, and for integral A to a toric initial ideal. Generic first rows leave later flags invisible; thus the fibers need not be points or contractible. Describe the topology and incidence structure of fibers when the first row lies on a secondary or Gröbner wall. The coefficient-kernel description gives explicit semialgebraic conditions from which to start.

4. Certificates smaller than all bases or a Graver basis. The proofs give finite certificates, but the evident lists can be large. Can a certificate containing only a controlled set of supporting bases, redundancy witnesses, and essential lattice relations verify simultaneously the face lattice, the lower subdivision, and the initial ideal? Separate certificate size from the complexity of finding that certificate. A practical target is an output-sensitive checker rather than an efficient universal enumeration algorithm.

5. Effective infinite-support subclasses. Proposition [21.1](#) rules out uniform extraction from arbitrary coefficient programs. Identify normal-form languages with enough support

structure to certify that no later independent coefficient exists. Examples to investigate include rational generating functions and prescribed finite-dimensional coefficient recurrences. The desired result is an algorithm with a proved stopping criterion, not merely symbolic access to individual coefficients.

6. Relating flags to higher-rank algebraic degenerations. Compare the canonical real-dependence flag here with the higher-rank quasi-valuations in [4]. Determine precisely which information is lost when a matrix of weights is replaced by the flag modulo positive triangular and affine transformations. Equality of toric initial ideals is proved above; equality of valuation structures or multi-parameter families is a stronger requirement and should be tested separately.

7. Arithmetic constraints on a real replacement. With rational A and rational coefficient slices, the present construction gives rational specializations, but denominators may grow substantially. Optimize coefficient height and the size of a rational endpoint while retaining specified signs and a prescribed rational shadow. For an integral endpoint one must account for the shadow and allowed affine rescalings, rather than silently multiplying the entire path by a scalar. This is a concrete quantitative refinement of the existence theorem.

8. A small trusted formalization. Implement the coefficient-pivot and finite-slack certificate theorems first, with no surreal class machinery in the checker. Then connect them to the existing ordered-field geometry and normal forms. An especially useful milestone is a machine-checked version of the separation theorem: a family with linear finite combinatorial realizations but arbitrarily large minimum degree for the complete dependence type.

23 Conclusion

The real-base lifting problem admits an exact three-level answer. Arbitrary surreal support collapses to $m \leq n - d - 1$ selected coefficient slices when all real dependence leading terms are the invariant. The resulting complete sign type is an oriented partial flag, and its minimum polynomial germ degree is determined by m and, when a shadow is fixed, r_0 . Finite combinatorics is less demanding: a single real segment suffices, and in the integral case that segment may also preserve the entire toric initial ideal.

These results do not produce a new finite face lattice unavailable over the reals. They describe exactly what can be hidden behind a familiar face lattice, what standard part destroys, and which additional data restore it. The strongest claim of this manuscript is the proved sharp classification in this stated setting. Its possible novelty lies in that assembled theorem package and its implementation boundary; independent proof review and a broader priority audit remain appropriate.

24 Dependency and verification ledger

[write] Appendix A of manuscript 05. It records manuscript 05's results only; manuscript 02's proof-dependency audit is [table 5](#).

Result	Evidence and dependencies
Coefficient-pivot theorem	Full proof; Conway coefficient uniqueness, leading-sign rule, and finite-dimensional linear algebra.
Affine compression and flags	Full proofs from the pivot theorem and ordinary real linear algebra.
Exact polynomial degrees	Full constructions and rank lower bounds; no transfer or numerical approximation.
Finite face and slack certificates	Full determinant and barycentric proofs; finite rational examples checked.
Linear shadow realization	Full proof from standard part and finite sign stabilization; arbitrary finite test list.
Arbitrary degree separation	Explicit full-rank construction and exact lower bounds; finite family members checked.
Toric initial ideal	Full conformal-decomposition proof, with classical Graver-background attribution; no Graver implementation claimed.
General semialgebraic arcs	Application of cited real-closed-field transfer and curve selection, not a new general existence theorem.
Validity-set correction	Explicit quadratic counterexample and corrected initial-component proof; exact signs checked.
Coefficient-oracle obstruction	Direct reduction from the halting problem.
New Lean declarations	None supplied or claimed.

25 Source and reproduction notes

The fixed geometry source inspected was

```
Algebra/SurrealNumbers/Surreal/Algebra/Geometry.lean
```

at repository commit `5e0f4e05046bacda3c6ba2643f19b91aa632436b`, with blob `450fd4bbfc017e0da24c28e538af1cc02037ad5a`. The surrounding README was read on 30 September 2026; its fetched blob was `cf63fc5ff9ca9d8877e90f8e69ec34e6ded19d23`. These identify inspected sources, not a claim that the repository has stopped changing. The comparison covers these files, the reports guide, and targeted searches; it is not a complete audit of every manuscript in the repository.

The earlier companion draft was recovered from the research library under `article(20260923-000126).pdf`; the relevant statements are on its printed pages 5–6. It was consulted for comparison and correction, not imported as a theorem dependency. The current literature check included the September 2026 revision of [4]. No exhaustive priority search or independent referee report is represented by this record.

To reproduce the executable checks, run

```
python3 code/verify.py
```

from the package directory. To rebuild this document, run

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

A third run may be needed after a substantial change to the contents or cross-references. The bibliography is included below; no external `.bib` file is required. Standard TeX Live packages provide the fonts; no font files are distributed with this package.

[write] In this report: the verifier is shipped as `code/05-finite-scale-liftings-verify.py` and its outputs as `data/05-finite-scale-liftings-certificates.json` and `data/05-finite-scale-liftings-verification-report.txt`. The program writes its outputs under the delivered names, into `data/` beside its parent directory, so it must be rerun on a copy laid out as delivered; the report’s README gives the recipe. The manuscript’s own README and PDF are not shipped (this report and its README.md replace them); its SOURCES.md and PROOF_REVIEW.md are shipped as `05-finite-scale-liftings-SOURCES.md` and `05-finite-scale-liftings-PROOF_REVIEW.md`, and its checksum file `SHA256SUMS.txt` is not shipped. The research-library copy of the earlier draft named above is not in ProveIt.

Part II

Coordinate-truncation histories

Source and scope. [write] This Part prints Section 6 of manuscript 02 (named in [Part I](#), which also prints its abstract and status), followed by the parts of its Sections 9–11 and its two appendices that are not printed in [Part I](#): the recorded tests and the proof-dependency audit, its research directions, its conclusion, the exact recursion for the universal construction, and its reproducibility notes. Its operation, truncating the *coordinates* of a configuration at a common degree, is operation (iii) of 02’s introduction ([section 3](#)). It is not the common initial cut of lifting heights of [Part I](#), and the finiteness theorems of [Part I](#) do not apply to it. No other manuscript of the batch proves a result of this kind. Manuscript 05’s example “Finite truncation can destroy an exact equality” ([section 16.4](#)) and the examples of [Part III](#) showing that no truncation depth depends only on dimension and vertex count ([section 40](#)) are weaker instances of the phenomenon of [theorems 26.2](#) and [26.3](#).

Notation in this Part. [write] t is the variable of the power series u, f, g, H, J ; every sign is evaluated at $t = \varepsilon = \omega^{-1}$ (manuscript 02’s convention), and the same series are also real-analytic functions of a real t near 0 ([theorem 26.3](#)). $\tau = \omega^{-\omega}$. $\Delta = 1 - fg$ is an orientation determinant; $\rho \in \{-1, 1\}$ in [theorem 26.6](#) is a sign, not a threshold or a jet scale; H is a sign-generating series, not a height vector; M in [proposition 26.7](#) is a maximal order.

26 Four moving points can encode arbitrary histories

The preceding finiteness theorem is fundamentally linear: the base A is real and fixed, and every relevant test is a real linear functional of its heights. If the point coordinates move, orientation tests become polynomials in those coordinates. Truncation does not preserve exact multiplicative relations. The following examples exploit that difference.

For a power series $u(t) = \sum_{k \geq 0} u_k t^k$, write $u_{\leq N} = \sum_{k=0}^N u_k t^k$. A degree- N coordinate truncation means that this operation is applied to *every* coordinate at the same N . All signs are evaluated at the positive infinitesimal $t = \omega^{-1}$, or equivalently by the first nonzero coefficient.

26.1 A four-point orientation test

Lemma 26.1 (Triangle versus quadrilateral). *Let f, g be elements of an ordered field, with*

$$g > 0, \quad 2g - 2 + \Delta > 0, \quad \Delta = 1 - fg.$$

For the four points

$$A_0 = (0, 0), \quad B = (1, f), \quad C = (g, 1), \quad D = (0, 2), \quad (26.1)$$

the following hold. If $\Delta > 0$, they are the four vertices of a strictly convex quadrilateral in the order displayed. If $\Delta < 0$, B is strictly inside the triangle A_0CD . If $\Delta = 0$ and $g > 1$, B lies in the relative interior of the edge A_0C .

Proof. The four consecutive oriented triangle determinants are

$$\begin{aligned} [A_0, B, C] &= \Delta, \\ [B, C, D] &= 2g - 2 + \Delta, \\ [C, D, A_0] &= 2g, \\ [D, A_0, B] &= 2. \end{aligned} \quad (26.2)$$

Here $[X, Y, Z] = \det(Y - X, Z - X)$. For $\Delta > 0$ these signs give the strict convex ordering. This can also be checked without a polygon-order criterion: the line through each displayed consecutive pair has both other points strictly on its left, using the same four determinants.

For $\Delta < 0$, the edge tests for membership in the positively oriented triangle A_0CD are $-\Delta$, $2g - 2 + \Delta$, and 2 , all positive. Thus B is strictly inside that triangle. For $\Delta = 0$, the equality $fg = 1$ gives $B = C/g$; if $g > 1$, the scalar $1/g$ lies strictly between zero and one. $\square$

26.2 The simplest infinite alternation

Theorem 26.2 (Rational alternating truncations). *Let $g = 1 + t$ and $f = (1 + t)^{-1}$ in $\mathbb{R}((t)) \subseteq \mathbf{No}$, and let P_N be the convex hull of the common degree- N coordinate truncations of (26.1). For every $N \geq 1$,*

P_N is a quadrilateral when N is odd, and a triangle when N is even.

The full configuration has triangular hull, with B in the relative interior of A_0C . All points are finite, all these hulls have dimension two, and there are only four labeled generators.

Proof. The geometric-series identity gives

$$f_{\leq N} = \sum_{k=0}^N (-t)^k, \quad (1 + t)f_{\leq N} = 1 + (-1)^N t^{N+1}.$$

Consequently

$$\Delta_N = 1 - (1 + t)f_{\leq N} = (-1)^{N+1} t^{N+1}. \quad (26.3)$$

The other nonconstant side test is

$$2g - 2 + \Delta_N = 2t + (-1)^{N+1} t^{N+1} > 0.$$

Lemma 26.1 proves the parity statement. For the full coordinates $fg = 1$ and $g > 1$, so the final assertion about B follows from the same lemma. $\square$

The theorem already rules out a bound on the number of coordinate-truncation changes in terms of n and d . The alternating points are rational functions of t . They do not require arbitrary surreal exponents or an infinite-dimensional base configuration. The infinitude is in the list of truncation levels.

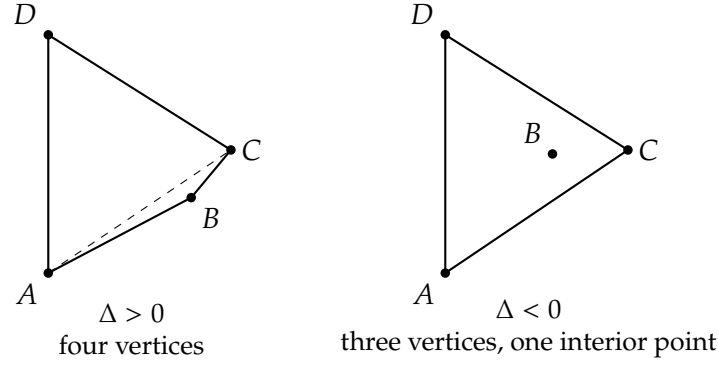


Figure 3: The two hull types detected by $1 - fg$. Here the schematic label A denotes A_0 . The drawing exaggerates the displacement of B from the diagonal; it is not a real-scale plot of infinitesimal coordinates.

26.3 A universal reciprocal-series construction

The alternating example can be generalized from a periodic pattern to an arbitrary prescribed pattern.

Theorem 26.3 (Universal binary coordinate-truncation histories). *Let $(\sigma_N)_{N \geq 1}$ be any sequence in $\{-1, 1\}$ with $\sigma_1 = 1$. There are power series $f, g \in \mathbb{Q}[[t]]$, convergent on a common real neighborhood of zero, such that*

$$fg = 1, \quad f = 1 - t + O(t^2), \quad g = 1 + t + O(t^2), \quad (26.4)$$

and, for every $N \geq 1$,

$$1 - f_{\leq N} g_{\leq N} = \sigma_N t^{N+1} + O(t^{N+2}). \quad (26.5)$$

The hull of the common degree- N truncations of (26.1) is therefore a quadrilateral for $\sigma_N = 1$ and a triangle for $\sigma_N = -1$. Every prescribed binary history from degree two onward occurs in this way.

Proof. Define

$$H(t) = \sum_{N \geq 1} \sigma_N t^{N-1}, \quad J(t) = H(t) + \frac{t^2}{4} H(t)^2. \quad (26.6)$$

Since $H(0) = J(0) = 1$, there is a unique formal square root of J with constant term 1. Put

$$\begin{aligned} g(t) &= 1 + \frac{t^2}{2} H(t) + t \sqrt{J(t)}, \\ f(t) &= 1 + \frac{t^2}{2} H(t) - t \sqrt{J(t)}. \end{aligned} \quad (26.7)$$

The coefficients are rational: the formal square-root recursion divides by 2, and all the coefficients of J are rational. Direct multiplication gives

$$fg = \left(1 + \frac{t^2}{2} H\right)^2 - t^2 \left(H + \frac{t^2}{4} H^2\right) = 1.$$

The constant and linear terms give (26.4), while addition gives the particularly useful identity

$$f + g = 2 + t^2 H. \quad (26.8)$$

Write $f = \sum_{k \geq 0} b_k t^k$ and $g = \sum_{k \geq 0} a_k t^k$, so $a_0 = b_0 = 1$. Through degree N , the product of the truncations agrees with the full product 1. At degree $N+1$, the only terms of the full product omitted by multiplying the two degree- N truncations are $a_{N+1}b_0 + a_0b_{N+1} = a_{N+1} + b_{N+1}$. By (26.8), this sum equals σ_N . The coefficient of degree $N+1$ in the truncated product is therefore $-\sigma_N$, proving (26.5).

It remains to verify convergence and the hull interpretation. The series H converges absolutely for $|t| < 1$. One can use a common quantitative neighborhood, independent of the sign sequence. For complex $|t| \leq 1/8$,

$$|H(t) - 1| \leq \frac{1}{7}, \quad |H(t)| \leq \frac{8}{7}, \quad |J(t) - 1| \leq \frac{1}{7} + \frac{1}{196} = \frac{29}{196} < 1.$$

The binomial series for $\sqrt{1 + (J - 1)}$ consequently converges near this closed disk and selects the root with constant term 1. The two series in (26.7) are convergent near zero. The construction thus works both analytically and formally, and its formal values at $t = \omega^{-1}$ are legitimate surreal normal forms.

For every $N \geq 1$, $g_{\leq N} = 1 + t + O(t^2)$ and $f_{\leq N} = 1 - t + O(t^2)$. The remaining side test in Lemma 26.1 is

$$2g_{\leq N} - 2 + \Delta_N = 2t + O(t^2) > 0.$$

Also $g_{\leq N} > 0$. The first nonzero term of Δ_N is exactly (26.5), so the lemma proves the prescribed hull type. $\square$

Remark 26.4 (Why the first bit is fixed). The normalization $f = 1 - t + O(t^2)$ and $g = 1 + t + O(t^2)$ forces $1 - f_{\leq 1}g_{\leq 1} = t^2$. Thus $\sigma_1 = 1$ is not an overlooked restriction: the first truncation is automatically a quadrilateral in this normalized family. All later bits are free. This is enough to realize every infinite binary pattern after one fixed initial stage.

Remark 26.5 (No contradiction with analytic nonoscillation). For each fixed analytic family, the full identity $fg = 1$ holds at every sufficiently small real t . The histories above are not the signs of one analytic function evaluated at a sequence of real arguments. Instead the polynomials $1 - f_{\leq N}g_{\leq N}$ themselves change with N . They approach the zero function while retaining prescribed leading signs. There is no accumulation of isolated zeros of a nonzero analytic function at an interior point.

26.4 The final hull can be selected independently

The full reciprocal configuration is degenerate along A_0C . A scale beyond all ordinary powers removes that degeneracy without constraining any finite truncation.

Theorem 26.6 (An independent uniform transfinite endpoint). *For the series in Theorem 26.3, set*

$$t = \omega^{-1}, \quad \tau = \omega^{-\omega}, \quad f^{(\rho)} = f - \rho\tau, \quad \rho \in \{-1, 1\}.$$

Use $f^{(\rho)}$ instead of f in (26.1). Every finite degree- N common coordinate truncation is unchanged. The full hull is a quadrilateral if $\rho = 1$ and a triangle with B strictly interior if $\rho = -1$. In both cases every three of the four full points are affinely independent. Thus the full configuration is uniform, while its finite coordinate-truncation history remains arbitrary.

Proof. The monomial τ is positive and smaller than t^m for every ordinary integer $m \geq 0$. Its exponent position is beyond every finite degree cut, so those cuts ignore it. The exact full determinant becomes

$$1 - gf^{(\rho)} = 1 - gf + \rho g\tau = \rho g\tau.$$

It has sign ρ . The remaining side test is $2g - 2 + \rho g \tau$, whose leading term is $2t > 0$. The other two oriented determinants are $2g > 0$ and $2 > 0$. Apply Lemma 26.1. The four determinants in (26.2) are all nonzero and enumerate all triples up to orientation, proving uniformity. $\square$

The theorem uses transfinite scale separation only to choose the endpoint. The arbitrary finite-degree history itself already uses ordinary convergent power series. For the simple alternating history, one may retain the rational $f = (1 + t)^{-1}$, $g = 1 + t$ of Theorem 26.2. Then the full uniform example lies in $\mathbb{Q}(t, \tau)^2$ coordinatewise. It is not necessary to use an arbitrary analytic series to obtain infinitely many changes with a uniform endpoint.

26.5 Where ordinary power-series stability returns

Proposition 26.7 (Finite determinacy for uniform ordinary series). *Let $V(t)$ be a configuration of $n \geq d + 1$ points with coordinates in $\mathbb{R}[[t]]$. Suppose every $(d + 1)$ -point affine determinant is nonzero. Let M be the maximum of the finite orders of these determinants. For every $N \geq M$, the common degree- N coordinate truncation has the same signs of all these determinants, and hence the same oriented matroid and face lattice, as the full configuration.*

Proof. All coordinate entries have nonnegative order. Truncating them at degree N changes each entry by a series of order at least $N + 1$. Expand a homogeneous affine determinant by multilinearity in its columns, including the constant homogeneous row. In every term of the difference between the full determinant and the truncated determinant, at least one entry is a truncation error and all remaining entries have nonnegative order. The determinant error therefore has order at least $N + 1$.

Each original nonzero determinant has order at most $M \leq N$, so its first nonzero coefficient survives unchanged. The sign pattern of the maximal homogeneous minors determines the oriented matroid of a uniform configuration and its supporting facets. In this uniform case one can also recover facets directly: a d -point candidate facet is supporting exactly when all determinants obtained by appending another point have one weak orientation, which here is strict. Hence the face lattice is unchanged. $\square$

The two hypotheses explain the boundary sharply. In the rational alternating example, a full affine determinant is identically zero, so it has no finite leading order to protect. In the transfinite uniform example, the previously zero determinant has order ω relative to the monomials $\omega^{-\gamma}$; no ordinary N reaches that order. Thus ordinary uniformity plus ordinary power-series support is a genuine stability regime, while full surreal uniformity alone is not.

26.6 A second truncation convention that must be excluded

Even with a fixed real base, deleting the *leading* part and retaining a tail is not covered by the lifting theorem. For example, take a square and place a height $u(t) = \sum_{k \geq 0} (-t)^k$ at one vertex, with the other three heights zero. The unique circuit evaluates to u up to a fixed nonzero real factor. The tail starting at degree N has leading sign $(-1)^N$, so the induced diagonal alternates. This is a tail history, not an initial-prefix history. The distinction is mathematical, not terminological.

27 Algorithms, verification, and proof boundaries

[write] Manuscript 02's finite certificate workflow and its query-model bound [proposition 21.2](#) are printed in [section 21](#), and its proposed Lean development in [section 83](#). The program is

shipped as `code/02-transfinite-scales-verify.py`; its outputs are shipped as `data/02-transfinite-scales-verification.json`, `data/02-transfinite-scales-hexagon_example.json` and `data/02-transfinite-scales-universal_history_example.json`, and the captured run as `data/02-transfinite-scales-verification_run.txt`.

27.1 Recorded exact-arithmetic tests

The accompanying `code/02-transfinite-scales-verify.py` uses Python’s standard-library `fractions.Fraction`. It requires no external Python package or network access. The recorded run passed **163,914 exact assertions**. These assertions cover the following finite cases.

Table 4: Scope of the supplied finite verification.

Family	What was checked
Sharp polygons	All nine sizes $4 \leq n \leq 12$: selected quotient rank, all prefix cell sets, refinement, and all basis-slack signs after the explicit rational specialization.
Redundant-row examples	Thirty seeded arrays of thirteen rational rows, including affine components and redundant coefficient directions: original and compressed prefix signs, cells, and real specialization.
Universal histories	All $2^7 = 128$ sign histories of length eight with first bit positive; twelve seeded histories of length thirty-two; one additional length-twelve example. Reciprocal jets, leading defect coefficients, and side-orientation signs were checked exactly.
Rational alternation	The full polynomial identity (26.3) for each $1 \leq N \leq 64$.

The count measures individual assertions, not independent theorems or statistical confidence. In particular, none of these tests proves the quantification over all real functionals, arbitrary infinite sign sequences, transfinite supports, or all dimensions. Those claims rest on the written proofs. The deterministic seeds are 20260930 for the compression tests and 31415926 for the longer history samples. The machine-readable output is `data/02-transfinite-scales-verification.json`; the hexagon and a prescribed-history coefficient example have separate JSON files.

The program uses a coefficient recursion equivalent to the closed construction (26.7); Appendix 30 derives it. This supplies a second algebraic view of the main negative example. There is no numerical approximation to ω in the program.

27.2 A proof dependency audit

The delicate assumptions are summarized in Table 5. These are boundaries of the results, not optional technicalities.

Table 5: Hypotheses whose omission would change or invalidate the claim.

Result	Essential boundary
Lifting compression	The base coordinates and comparison coefficients are real and fixed. Moving coordinates lead to nonlinear determinant expressions and are not covered.
History preservation	Cuts retain common initial normal-form segments. Independent coordinate cuts, sign-sequence cuts, and tail deletion are different operations.
Rank minimality	All real affine-dependence signs, including zero, are preserved. A finite circuit list or one final subdivision may require less information.
Real specialization	The explicit bound protects a finite family of functionals. It does not embed the entire higher-rank ordered comparison space into $\mathbb{R}$.
Refinement chains	Marked cell-label inclusion is used. Geometric containment alone can conceal whether an interior point is on the lower hull.
Universal histories	The degree-one bit is forced positive by the normalization. The freely prescribed history starts at degree two. There are four generators, not necessarily four extreme vertices.
Uniform endpoint	An added monomial beyond all ordinary powers chooses the endpoint. This is not an ordinary analytic correction with a finite Taylor order.
Ordinary stability	All relevant maximal determinants are nonzero and have finite orders, and coordinates have nonnegative orders.
Standard-part geometry	Coordinates are finite. The polar identity uses only finite polar points; a face image need not be a face.
Puiseux realization	Only finitely many polynomial observations and incidences are preserved. Curve selection and convergent algebraic Puiseux expansions are classical imported theorems.
Computations	Exact finite arithmetic tests selected examples. No Lean build, general surreal algorithm, or independent proof certification was performed.

28 Further research questions

The following are proposed research directions arising from this article. They are not assertions that each topic is absent from the literature, nor are they labeled as longstanding published open problems. The first three would most directly deepen the results proved here.

28.1 Which histories admit rational-function coordinates?

The universal construction permits arbitrary convergent reciprocal power series, while rational functions already give a periodic alternating history. Characterize the binary histories realizable by four points of the form (26.1) with $f, g \in \mathbb{Q}(t)$ and $fg = 1$. In the normalized reciprocal setting, the degree- $N + 1$ coefficient of the degree- N truncation defect equals that of

$f + g$. When nonzero it determines the history bit; a zero requires looking at higher degrees. For rational functions, the coefficients of $f + g$ satisfy a linear recurrence after finitely many indices. The problem therefore links geometric truncation histories to sign patterns of rational generating functions, with the additional constraint that the generating function is a reciprocal sum. A useful first target is a classification of eventually periodic histories and a precise description of the extra reciprocal constraint.

28.2 Stability in the presence of exact incidences

Proposition 26.7 treats configurations whose maximal determinants are all nonzero. The universal examples exploit a zero full determinant whose truncations have arbitrary leading signs. Find a necessary and sufficient condition, in a specified algebraic or analytic representation class, for the *zero* determinant relations to have eventually stable truncation signs. A condition merely on the final oriented matroid cannot suffice. Possible additional data include how an incidence ideal interacts with the coordinate truncation maps, and bounds on cancellation in products of truncated coordinates.

28.3 Effective controlled specialization

Theorem 19.1 is qualitative. Given rational polynomial incidence data, a compatible algebraic standard-part configuration, and a realizable sign pattern, obtain explicit bounds on a Puiseux arc's ramification, algebraic degree, and coefficient height. An especially useful version would preserve selected vanishing orders or a finite sequence of near-face optimality bands in addition to the final face lattice. Such a theorem must distinguish geometric sign complexity from the bit complexity of an exact coefficient representation. [write] Manuscript 05's question 1 (section 22) asks the same for determinant signs; theorem 41.2 of Part III gives an effective specialization for finite-support Hahn certificates.

28.4 Minimal geometric certificates versus full comparison rank

The rank p classifies how many coefficient layers are needed to preserve all real affine comparisons, whereas the strict history length r can be smaller. Determine an efficient canonical quotient of the comparison data that preserves just the marked subdivision history. How small can a slack or circuit certificate be compared with the all-basis list? What is the complexity of extracting it for a fixed dimension, and which families admit polynomial-size certificates without enumerating every candidate basis? A single final subdivision cannot determine p ; any proposed invariant must respect that information loss. [write] Manuscript 05's question 4 (section 22) asks the same for face lattices, lower subdivisions and toric initial ideals together.

28.5 A replacement for the false residue map on faces

The image of a face under standard part need not be a face. Develop a polyhedral object whose cells retain the fibers, infinitesimal widths, and near-optimality data discarded by ordinary residue. The identity in Proposition 14.2 suggests indexing such data by objective functionals and admissible infinitesimal error levels. The challenge is to obtain an intrinsic finite description when one exists, and to state precisely when transfinite scale labels are unavoidable. [write] Related: the questions of Part III on face specialization and oriented scale data and on joint residue data ask for structures of the same kind; proposition 14.2 is the single-objective case.

28.6 Beyond linear comparisons: realizability constraints

One can apply linear compression to a finite list of determinant values regarded as unrelated surreal scalars. That does *not* show that the compressed values are minors of an actual point configuration: Plücker relations and other polynomial dependencies may fail. Determine which rank-compression operations preserve realizability of valued or oriented matroid data. A useful theorem would identify a nontrivial class in which support compression can be lifted back to coordinate matrices without changing prescribed signs and selected valuation comparisons. [write] Manuscript 05's question 2 ([section 22](#)) asks this for two nonreal lifting coordinates over a fixed real base.

28.7 Infinite locally finite configurations

The proof of Theorem 7.1 depends on finite dimension of the coefficient quotient, not on a bound for support length. For infinite configurations the affine relation space is generally infinite dimensional. Which local finiteness, support, or uniform-rank assumptions imply local stabilization of coherent subdivisions? A satisfactory formulation should specify whether stabilization is pointwise on finite subconfigurations, uniform on bounded regions, or global. The four-point examples show that coordinatewise convergence alone is too weak even before infinitude is introduced.

28.8 Formalization and representation-aware computation

Carry out the layered Lean program in [section 83.2](#) and attach a checked theorem ledger. In parallel, classify effective representations for which the next rank-increasing coefficient can be found or its absence certified. Finite support with a known bound, rational functions, algebraic series, and general coefficient oracles have materially different capabilities. Proposition 21.2 shows that an implementation cannot hide those differences behind a single unrestricted "surreal input" interface. [write] Manuscript 02's layered Lean program is printed in [section 83](#); manuscript 05's question 5 ([section 22](#)) asks for such representations from the side of [proposition 21.1](#).

29 Conclusion

Finite surreal geometry has two distinct levels. At the level of a single finite incidence pattern, real-closed-field transfer gives an ordinary real algebraic model. At the level of a fixed real base with surreal lifting heights, finite-dimensional linear algebra still controls all normal-form prefix histories: at most $n - d - 1$ effective comparison layers are needed, and that bound is sharp.

Neither fact controls arbitrary coordinate truncation. Four bounded planar generators can encode any binary triangle/quadrilateral history from degree two onward, and a transfinite monomial can independently choose a uniform final endpoint. Thus the relevant additional structure of surreal polytopes is not a previously impossible finite face lattice. It is the organization of incidence data across scales, together with the precise operations used to expose or forget those scales.

30 An exact recursion for the universal construction

[write] Appendix A of manuscript 02.

The closed formula (26.7) is convenient for proving convergence. For exact finite calculations, the following recursion avoids formal square-root software. Write

$$g = \sum_{k \geq 0} a_k t^k, \quad f = \sum_{k \geq 0} b_k t^k, \quad a_0 = b_0 = 1, \quad a_1 = 1, \quad b_1 = -1.$$

For $n \geq 2$, assume the previous coefficients have been chosen and set

$$\begin{aligned} c_n &= - \sum_{k=1}^{n-1} a_k b_{n-k}, \\ M_n &= \sum_{k=2}^{n-1} a_k b_{n+1-k}, \\ a_n &= \frac{\sigma_n + c_n + M_n}{2}, \quad b_n = c_n - a_n. \end{aligned} \tag{30.1}$$

The equality $a_n + b_n = c_n$ forces the degree- n coefficient of fg to vanish. The coefficient of degree $n + 1$ in the product of the two degree- n truncations is

$$a_1 b_n + a_n b_1 + M_n = b_n - a_n + M_n = c_n - 2a_n + M_n = -\sigma_n.$$

Thus the leading truncation defect is σ_n . At the next step the condition of being reciprocal forces $a_{n+1} + b_{n+1} = \sigma_n$. Together with $a_2 + b_2 = 1 = \sigma_1$, this reproduces $f + g = 2 + t^2 H$.

Conversely, the identities $fg = 1$, $f + g = 2 + t^2 H$, and the linear terms $g = 1 + t + O(t^2)$, $f = 1 - t + O(t^2)$ select exactly the branch in (26.7). Hence the recursion computes that same pair. Every requested sign prefix determines the corresponding finite jets; an arbitrary continuation of the signs completes them to the convergent series of Theorem 26.3.

For example, $\sigma_2 = 1$ gives $a_2 = 1$, $b_2 = 0$, whereas $\sigma_2 = -1$ gives $a_2 = 0$, $b_2 = 1$. These yield, respectively, positive and negative leading degree-three defects. Later sign choices are independent and are obtained by the same rational recursion.

31 Reproducibility and source inventory

[write] Appendix B of manuscript 02; in this report its files are shipped under prefixed names. `code/verify.py` is `code/02-transfinite-scales-verify.py`; the three generated JSON files are `data/02-transfinite-scales-verification.json`, `data/02-transfinite-scales-hexagon_example.json` and `data/02-transfinite-scales-universal_history_example.json`; the Makefile is `code/02-transfinite-scales-Makefile`; the source and proof-audit notes are `02-transfinite-scales-sources.md` and `02-transfinite-scales-proof_audit.md`; the build record notes/`validation.json` is `data/02-transfinite-scales-validation.json` and describes the delivered PDF. The manuscript's `article.tex`, PDF and README are not shipped: this report replaces them. The commands below name the delivered layout; run them on a copy (the report's README gives the recipe).

The archive contains `article.tex`, its compiled PDF, `code/verify.py`, the three generated JSON data files, a Makefile, a README, and separate source and proof-audit notes. From the archive directory, run

```
python3 code/verify.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The verification script resolves its output paths relative to itself. The PDF build uses ordinary TeX packages and no shell escape, external bibliography processor, or network call. The Makefile performs an extra TeX pass for stable cross-references. No font files are distributed.

The repository geometry file inspected at the pinned commit has blob identifier

450fd4bbfc017e0da24c28e538af1cc02037ad5a.

The code-search index observed during the source check pointed to parent commit

5e0f4e05046bacda3c6ba2643f19b91aa632436b.

The actual geometry source was then read at the full pinned commit. This distinction is recorded to avoid presenting search-index freshness as a repository-wide proof audit. No original repository files were modified.

Part III

Affine residues and mixed-volume scales

Source and scope. [write] This Part prints, whole, manuscript 06 of batch 58: *Surreal Polytopes: Affine Residues, Mixed-Volume Scales, and Simultaneous Segment Models*, headed “ProveIt research program”, author line “Research article prepared for Vladimir Reshetnikov”, dated 30 September 2026 (archive `Surreal_Polytopes_Research_Package.zip`, main file `Surreal_Polytopes/article.tex`; delivered PDF of 27 pages, not shipped). It was written against ProveIt commit 6996fee43. Its program, recorded run, build file and provenance note are shipped as `code/06-affine-residues-verify.py`, `data/06-affine-residues-verification.json`, `code/06-affine-residues-Makefile` and `06-affine-residues-PROVENANCE.md`. Its README and checksum list SHA256SUMS are not shipped; they survive in the delivery archive of arrival commit b2b626d85. Every section, result, proof, example, remark, figure, table and research question of the manuscript is printed here, in the manuscript’s order, with three structural changes: its appendix is printed as the ordinary [section 45](#); its subsection “A modular Lean development” is printed in the formalization chapter ([section 83.3](#)); and its bibliography is printed as the list “References of Part III”. This Part is its own spine (the residue/volume spine of [section 1](#)); it shares no theorem with the other Parts beyond two foundations, both marked where they occur: the standard part of a finite hull ([lemma 34.2](#), proved in Part I as [proposition 13.1](#)) and finite real-closed-field transfer ([proposition 33.3](#), Part I’s [theorem 18.1](#)). The Part keeps its own definitions and hypotheses, fixed in [sections 32](#) and [33](#). In this Part “the article” and “the package” mean manuscript 06 and its delivery. Where manuscript 06 says “the report”, it means itself or, in [section 42](#), its recorded run. It never means this merged report.

Generality. Although manuscript 06 and this report have “surreal” in their titles, every theorem of this Part is proved for an *arbitrary* set-sized real closed ordered field $K \supseteq \mathbb{R}$ with its natural convex valuation (the standing hypothesis stated at the end of the first subsection below). The surreal field enters only as the source of examples: finitely many surreal coordinates lie in such a K ([section 33](#)), and ω^{-1} , $\omega^{-\omega}$ occur only in examples and in the audit. The word “surreal” in the titles is therefore not a hypothesis of any result here.

Notation in this Part. [write] The following symbols mean something different elsewhere in the report (section 1.3); the Part-local meaning is the one in force here.

- $\mathcal{O}$ is the ring of elements of K bounded by real constants and $\mathfrak{m}$ its maximal ideal: the same ring as $\mathcal{O}_K$ (Parts I–II) and $\mathcal{O}_K$ (Part IV). $\text{st} : \mathcal{O} \rightarrow \mathbb{R}$ is the standard part, as everywhere in the report.
- $v : K \rightarrow \Gamma \cup \{\infty\}$, $\Gamma = K^\times / \mathcal{O}^\times$, is ordered so that *larger values mean smaller elements*: $v(x) \geq v(y)$ iff $x/y \in \mathcal{O}$, and $v(x) > 0$ iff $x \in \mathfrak{m} \setminus \{0\}$. The value group need not embed in $\mathbb{R}$. (Part I’s leading exponent measures growth, in the opposite direction; Part V writes v and v .)
- The *affine residue* $\text{Res}(P)$ (definition 35.2) is a renormalised, dimension-preserving real polytope, the image of P in $(p_0 + L(P))/\mathfrak{m}L(P)$. It is *not* the standard part $\text{st}P$: the thin triangle of section 40 has $\text{st}P$ a segment but $\text{Res}(P)$ a triangle. Where this Part says “residue field”, “reduction” or “volume residue” (proposition 34.3), it means the standard part. The report reserves “affine residue” for Res .
- $L(P) = \text{span}_{\mathcal{O}}(P - P)$ is the *scale lattice*, a free $\mathcal{O}$ -module, not a discrete subgroup; it is unrelated to Part I’s L_h and L_A and to Part IV’s lattice-point count $L_P(n)$.
- $V(P_1, \dots, P_d)$ (printed V) is mixed volume, not Part I’s dependence space V or Part V’s coarse ring. Vol_d is determinant (triangulation) volume; Part IV writes vol_d for Euclidean volume of real polytopes.
- m is the number of bodies (not Part I’s comparison rank, not Part V’s m with $n = 2^m$); r, r_i are affine dimensions (not Part II’s history length or Part I’s shadow rank r_0); $q(B) = v \det W_B$ in lemma 38.2 (not $q = n - d - 1$); H is the grid bound $\binom{m-1}{d-1}$ in theorem 37.2 and a finitely generated group in lemma 41.1; S_i are inner segments and T_i probe segments (not Part IV’s trace S and implanted polytope T , not Part V’s slack matrix and triangle); ρ_g is the sign threshold of a specialized Laurent polynomial in section 41 (Part I’s threshold is η); s is the real specialization parameter and t^γ a formal Hahn monomial there (Part II’s t is a power-series variable evaluated at $t = \omega^{-1}$).
- $\tau_I, \tau(\alpha)$ and “the profile τ ” are the pure and full mixed-volume valuation profiles; they are *not* powers of the surreal $\tau = \omega^{-\omega}$ of the other Parts. Manuscript 06 writes η for $\omega^{-\omega}$ (in its examples, in section 41 and in its audit), and this Part keeps that letter so that τ means only a profile here; $\varepsilon = \omega^{-1}$ as in the source. Part I’s η is a different object (a sign threshold).
- Macro change only: manuscript 06’s `\Span` (plain span over the ring or field in context) is typeset with the report’s `\spanplain`; the printed output is unchanged.

Abstract of manuscript 06. Finite polytopes over the surreal numbers have no new finite face-lattice possibilities merely because the coordinates are surreal. Their genuinely non-Archimedean information lies instead in relative scales, cancellation, and degeneration. We develop an algebraic treatment of that information inside set-sized real closed subfields of the surreal field.

The principal result is a simultaneous inner-segment theorem: for every finite family of polytopes, a segment can be chosen inside each member so that all mixed-volume valuations involving distinct members are preserved at once. Consequently the pure mixed-volume valuation configurations are exactly the determinant-valuation configurations realizable over the given ordered field. A finite rational grid suffices for the segment choices. Applying the theorem to labelled copies of the bodies produces a representable valuated-matroid model

of the entire mixed-volume profile, and proves its exchange-convexity for arbitrary ordered value groups.

The underlying invariant is the translation-invariant module $L(P) = \text{span}_{\mathcal{O}}(P - P)$, where $\mathcal{O}$ is the ring of finite elements. A maximum vertex simplex gives an explicit basis of this module and a full-dimensional real affine residue, even when ordinary standard part collapses the original polytope. Mixed-volume valuations depend only on these modules and admit certificates using at most d^d determinants per coefficient after basis selection. Conversely, a full-dimensional module is determined by mixed-volume probes, with a $d + 1$ -measurement comparison test relative to a reference polytope.

Detailed proofs, higher-rank examples, finite-support specialization, exact computational audits, and a research agenda are provided. The classical ingredients and the unverified priority of the combined results are identified explicitly; no solution of the full real mixed-volume configuration problem or new Lean formalization is claimed.

Scope (title page of manuscript 06). Finite vertex sets; fixed finite dimension; the natural convex valuation; exact algebraic volume. No proper-class integration, Archimedean valuation hypothesis, or sequential compactness argument is used.

32 Research target, principal results, and status

32.1 Why the finite combinatorics is not the whole question

Replacing real coordinates by surreal coordinates does not, by itself, create new finite combinatorial types of polytopes. The incidence data of a fixed finite vertex configuration can be encoded in the first-order theory of real closed fields. That observation is useful, but it does not explain a triangle of height ω^{-1} , a determinant whose first nonzero term occurs at $\omega^{-\omega}$, or the mixed volume of two bodies thin in different directions.

The appropriate questions are more precise. Can one normalize a finite surreal polytope without destroying its affine dimension? Which finite algebraic data determine the scale of a mixed volume? Can all the scales of a family be realized simultaneously by much simpler objects? Can the resulting scale information be recovered from measurements? The article answers these questions under explicit hypotheses.

Throughout the main results, K is a set-sized real closed ordered field containing $\mathbb{R}$, $\mathcal{O}$ is its ring of elements bounded by real constants, and $v : K \rightarrow \Gamma \cup \{\infty\}$ is the corresponding valuation. The surreal application takes K inside **No**. Our convention is that larger valuations mean smaller quantities. The value group Γ need not embed in $\mathbb{R}$.

32.2 The principal theorem in a compact form

For nonempty finite polytopes $P_1, \dots, P_m \subset K^d$, put

$$\tau_I = v(V(P_{i_1}, \dots, P_{i_d})), \quad I = \{i_1 < \dots < i_d\} \subseteq [m].$$

We call this the *pure* profile: every body occurs at most once in an entry. The mixed-volume convention is $V(P, \dots, P) = \text{Vol}_d(P)$.

Theorem 32.1 (Simultaneous inner-segment model, preview). *For $m \geq d \geq 1$, there are segments $S_i \subseteq P_i$, possibly reduced to points, such that*

$$v(V(S_{i_1}, \dots, S_{i_d})) = \tau_I \quad \text{for every } I \in \binom{[m]}{d}.$$

The segments may be chosen with endpoints that are rational convex combinations of vertices of their respective polytopes.

The proof in Section 37 does more than choose a generic point informally. After selecting at most d vertex directions in each body, it gives a finite rational grid containing a successful simultaneous choice. If $S_i = [p_i, p_i + w_i]$, the segment formula becomes

$$\tau_I = v \det(w_{i_1}, \dots, w_{i_d}),$$

with $v(0) = \infty$. Thus every pure profile has a single matrix realization, not merely a different determinant witness for each entry.

Repetitions matter. Replacing a full-dimensional body by one segment cannot preserve its ordinary volume in dimension greater than one. For the *full* profile, which includes repetitions, we use either parallelotopes with at most d generators or d separately labelled segment copies of each body. The clone construction retains all repeated-body valuations and implies a higher-rank discrete exchange inequality (Theorem 38.4).

32.3 What the other theorems add

For a polytope P , define its *scale lattice*

$$L(P) = \text{span}_O(P - P).$$

The word lattice here means a finitely generated free O -module in a finite-dimensional K -space; it does not mean a discrete subgroup in the Euclidean sense. The following results describe the geometry behind the segment theorem.

Result	Content and location
Affine normalization	A maximum vertex simplex generates $L(P)$; normalization lies between a standard simplex and a cube. The residue preserves affine dimension (Section 35).
Mixed determinant formula	Every mixed-volume valuation is the minimum valuation of one-direction-per-body determinants; selected bases need at most d^d such determinants (Section 36).
Simultaneous simplification	Inner segments preserve all pure entries; parallelotopes preserve the full profile (Sections 36–37).
Discrete convexity	A clone realization gives the exchange axiom over any ordered value group, without a real-valued grading (Section 38).
Reconstruction	All segment probes characterize $L(P)$; a reference basis gives a finite $d + 1$ -measurement test (Section 39).
Finite certification	Finite-support higher-rank input admits a one-parameter model preserving specified finite certificates, but not the whole higher-rank order (Section 41).

32.4 Repository provenance and dependence

The inspected ProvelIt snapshot is commit 6996fee43cc97b6c16351def7507d59a95bf62f0, retrieved on 30 September 2026. The relevant development is Algebra/SurrealNumbers; its README records the constructed surreal ordered field, real-closedness, and a statement-level formalization ledger [17]. The report also draws a methodological connection to the finite

support certificates in *Reversion Beyond Archimedean Valuations* under Analysis/Transseries [18].

The inspection was targeted, not a complete audit of the repository. Search results were used for navigation; the principal source files were read at the immutable commit above. In particular, this article does not assume the correctness of every research claim in the repository. Its mathematical prerequisites are the classical real-closedness of the surreal field, elementary valuation algebra, and finite real polytope theory. The proofs below are independent of the repository’s unrefereed advanced analytic claims.

[[write] Re-verified at the write phase. The Transseries report cited here is at Analysis/Transseries/docs/series-and-transseries/Reversion_Beyond_Archimedean_Valuations/, with title and subtitle exactly as in the bibliography entry [18]; its finite positive-grading result, credited below as the antecedent of lemma 41.1, is the theorem “Finite supports admit a positive rational grading” there. The SurrealNumbers README still describes the constructed ordered field, its real-closedness and the formalization ledger. Nothing in this Part is formalized in Lean (see section 83.3).]

32.5 Literature boundary and novelty

Ordered-field polyhedral geometry is established mathematics; a modern formal treatment of face lattices over ordered fields is given by Allamigeon, Katz, and Strub [21]. Classical mixed-volume theory, including its determinant special case, is treated by Schneider [22]. Averkov, von Dichter, Richard, and Soprunov connect mixed volumes of zonoids to absolute Grassmannian coordinates and explicitly propose investigating a connection with tropical Grassmannians in their November 2024 version [23]. Our simultaneous segment theorem gives a concrete valued-field connection for *all finite polytopes*, with a precise realization field and no closure convention hidden in the claim.

The broader link between volume polynomials, Lorentzian polynomials, and M -convex tropicalizations is already part of the theory of Brändén and Huh [24]. Lattices, norms, buildings, and realizable valuated matroids also have an extensive theory; see, for example, Battistella, Kühn, Kuhrs, Ulirsch, and Vargas [25]. We do not claim to have invented these connections. The contribution developed here is the explicit package of simultaneous inner models, dimension-only determinant certificates, affine residues, reconstruction, and arbitrary-value-group proofs adapted to finite surreal geometry.

Remark 32.2 (Status of the claims). Every numbered theorem presented as a result is proved below, either from identified classical facts or by an explicit algebraic argument. Their correctness and the extent of their novelty have not been independently refereed, and an exhaustive priority search has not been completed. The computational supplement checks finite instances; it is not a Lean proof. No solution of the complete real mixed-volume configuration problem is claimed. These qualifications concern verification and priority, not a missing step being concealed in the stated theorems.

33 Set-sized foundations and finite geometry

33.1 Working inside the surreal field without a class-sized domain

The surreal numbers form a real closed ordered class field; standard references discuss this field and its universality properties [19]. For finitely many coordinate values, start with the set-sized field generated by those values and $\mathbb{R}$, then take its real closure inside $\mathbf{No}$. This

yields a suitable set-sized K containing all input. The elementary algebraic constructions used here remain in such a field.

Consequently a polytope in the article is the ordinary set

$$P = \text{conv}_K\{p_1, \dots, p_n\} = \left\{ \sum_{i=1}^n \lambda_i p_i : \lambda_i \in K_{\geq 0}, \sum_i \lambda_i = 1 \right\} \subset K^d.$$

This is not an assertion that the collection of *all* surreal points in the corresponding class convex hull is a set. Finite face data and volumes are compatible with enlarging K : their definitions use finite linear algebra and first-order formulas over the coefficient field.

33.2 Finite elements, residue, and order of magnitude

Define

$$\begin{aligned} \mathcal{O} &= \{x \in K : |x| \leq C \text{ for some } C \in \mathbb{R}_{>0}\}, \\ \mathfrak{m} &= \{x \in K : |x| < c \text{ for every } c \in \mathbb{R}_{>0}\}. \end{aligned}$$

The ring $\mathcal{O}$ is convex in K . For $x \neq 0$, either x is finite or $1/x$ is finite, so it is a valuation ring. Its nonunits are precisely $\mathfrak{m}$: a finite nonzero x is a unit exactly when $|x|$ is bounded below by a positive real constant.

Lemma 33.1 (Standard part). *For each $x \in \mathcal{O}$ there is a unique $r \in \mathbb{R}$ such that $x - r \in \mathfrak{m}$. The resulting map $\text{st} : \mathcal{O} \rightarrow \mathbb{R}$ is an order-preserving surjective ring homomorphism with kernel $\mathfrak{m}$.*

Proof. The real set $\{a \in \mathbb{R} : a \leq x\}$ is nonempty and bounded above. Let r be its supremum. For every positive real c , the definition of supremum gives $r - c < x < r + c$, allowing harmless weak inequalities first and then replacing c by $c/2$. Thus $x - r$ is infinitesimal. A nonzero real number cannot be infinitesimal, proving uniqueness.

Finite sums of infinitesimals are infinitesimal, and the product of a finite element with an infinitesimal is infinitesimal. Expanding $(r + \epsilon)(s + \delta)$ proves preservation of multiplication; addition is immediate. Constants give surjectivity, the kernel description follows from uniqueness, and positivity follows from convexity. $\square$

Write $\Gamma = K^\times / \mathcal{O}^\times$ additively and $v(x)$ for the class of $x \neq 0$. Order it so that

$$v(x) \geq v(y) \iff x/y \in \mathcal{O}.$$

Set $v(0) = \infty$, with $\gamma + \infty = \infty$. Positive real constants have value zero. A nonzero element belongs to $\mathfrak{m}$ exactly when its value is positive.

Lemma 33.2 (Finite positive sums and bounded comparison). *If $a_1, \dots, a_N \in K_{\geq 0}$, then*

$$v\left(\sum_{i=1}^N a_i\right) = \min_i v(a_i).$$

If $0 < c \leq C$ are real and $0 \leq a, b \in K$ satisfy $cb \leq a \leq Cb$, then $v(a) = v(b)$, including the case $a = b = 0$.

Proof. When some a_i is nonzero, choose a largest member M . The inequalities $M \leq \sum_i a_i \leq NM$ show that the quotient of the sum by M is a unit of $\mathcal{O}$. The same unit argument proves the second assertion. The all-zero case is immediate. $\square$

No assertion about infinite sums is involved. The absence of cancellation for a finite positive sum will replace all appeals to a real-valued valuation or an Archimedean growth parameter.

33.3 The precise finite transfer principle

Real closed fields satisfy quantifier elimination and have a complete first-order theory in the ordered-field language; semialgebraic geometry provides the usual framework [20]. In particular, $\mathbb{R} \subset K$ is an elementary extension. Only *finite-dimensional, finitely presented* instances are transferred here.

Proposition 33.3 (No new finite face types). *Every finite polytope over K has the face lattice of a polytope whose coordinates are real algebraic numbers. Any prescribed finite collection of rational-polynomial equalities, inequalities, and sign tests on the vertex coordinates can be preserved simultaneously when it is included in the realization formula.*

Proof. Fix the number n of labelled input points and their affine dimension. For a nonempty subset $S \subseteq [n]$, the assertion that exactly the labels in S lie on a supporting face is expressed by

$$\exists a \in K^d \exists c \in K : \quad a \cdot p_i = c \ (i \in S), \quad a \cdot p_i < c \ (i \notin S).$$

The full polytope corresponds to $S = [n]$; the empty face is added separately. Form the conjunction of this assertion or its negation for each of the finitely many subsets S . Include a rank formula for the affine dimension and the requested additional tests. Existentially quantify the vertex coordinates. The resulting sentence has rational coefficients and holds in K , hence in the real algebraic field, which is also real closed. Its realization has exactly the same labelled face sets and therefore the same inclusion lattice. $\square$

This does not preserve the natural valuation, infinitesimal relations to *every* real constant, or a whole infinite type. Nor does it assert a rational realization. The face-lattice result is the classical boundary condition for the rest of the article, rather than its proposed novel contribution.

[[write] This is the finite real-closed-field transfer that Part I states and proves for finite configurations as [theorem 18.1](#) (manuscripts 05 and 02). Manuscript 06's statement and proof are kept as printed because they are phrased for face lattices together with prescribed sign tests. The same principle appears again as Part IV's [proposition 48.1](#) and, strengthened to oriented matroids and extension complexity, as Part V's [theorem 74.3](#). The limitations in the preceding paragraph apply to every one of these statements.]

34 Algebraic volume and reduction to the residue field

34.1 Determinant volume, not surreal measure theory

For an affine d -simplex set

$$\text{Vol}_d \text{conv}(p_0, \dots, p_d) = \frac{1}{d!} |\det(p_1 - p_0, \dots, p_d - p_0)|.$$

A full-dimensional finite polytope admits a finite triangulation; summing these simplex volumes defines $\text{Vol}_d(P)$. Set the ambient d -volume to zero for a polytope of smaller affine dimension. This construction is independent of triangulation and has the ordinary finite additivity, monotonicity, translation invariance, and determinant covariance.

Here is a careful reason classical facts apply. For fixed vertex count and dimension there are finitely many candidate simplicial complexes on the input vertices. The assertion that a given complex triangulates the convex hull is a first-order statement: containment, covering, and intersection conditions are expressed by barycentric coordinates, with finitely many cells.

The candidate volume is a finite sum of absolute determinants. Existence of a triangulation and equality of the sums from any two candidates are true over $\mathbb{R}$, so they transfer to K . Equivalently, the graph of the fixed-vertex-count volume function is semialgebraic.

This argument is specific to *polytopes*. It does not claim that volumes of arbitrary semialgebraic sets, or arbitrary analytic integrals, are ordered-field-definable.

34.2 Mixed volume and its finite identities

Define mixed volume by polarization:

$$d! \, V(P_1, \dots, P_d) = \sum_{I \subseteq [d]} (-1)^{d-|I|} \text{Vol}_d \left(\sum_{i \in I} P_i \right), \quad (34.1)$$

where the empty Minkowski sum is $\{0\}$. For $d \geq 1$ its volume is zero. The classical real theory [22] and the preceding finite transfer argument give symmetry, Minkowski multilinearity, homogeneity for nonnegative scalars, independent translation invariance, nonnegativity, and monotonicity in every argument. They also give

$$d! \, V([0, u_1], \dots, [0, u_d]) = |\det(u_1, \dots, u_d)|. \quad (34.2)$$

For a matrix $A \in \text{GL}_d(K)$,

$$V(AP_1, \dots, AP_d) = |\det A| \, V(P_1, \dots, P_d). \quad (34.3)$$

For m bodies and a multi-index $\alpha \in \mathbb{N}^m$ of total degree d , use $P_i[\alpha_i]$ to denote α_i repeated arguments. The volume polynomial identity is

$$\text{Vol}_d \left(\sum_{i=1}^m t_i P_i \right) = \sum_{|\alpha|=d} \frac{d!}{\alpha_1! \cdots \alpha_m!} V(P_1[\alpha_1], \dots, P_m[\alpha_m]) t^\alpha \quad (t_i \in K_{\geq 0}). \quad (34.4)$$

The formula is a polynomial identity after the mixed coefficients have been defined. All transfers have bounded, finite vertex counts: a Minkowski sum is presented by finitely many sums of input vertices.

Remark 34.1 (A computational warning). Formula (34.1) is an exact definition but can suffer severe leading-term cancellation. It is not a safe way to infer a valuation by taking the minimum of the valuations of its signed terms. The positive determinant comparison in Section 36 provides a cancellation-safe alternative.

34.3 Standard part commutes with finite polytope volume

For a polytope whose vertices lie in $\mathcal{O}^d$, write $\text{st } P$ for its coordinatewise standard-part image.

Lemma 34.2 (Reduction of convex hulls and sums). *If $p_1, \dots, p_n \in \mathcal{O}^d$, then*

$$\text{st conv}_K(p_1, \dots, p_n) = \text{conv}_{\mathbb{R}}(\text{st } p_1, \dots, \text{st } p_n).$$

For bounded finite polytopes, $\text{st}(P + Q) = \text{st } P + \text{st } Q$.

Proof. Every convex coefficient belongs to $[0, 1] \subset \mathcal{O}$. Reducing a convex combination gives a real convex combination. Conversely a real convex combination of the reduced vertices is the reduction of the combination with the same real coefficients upstairs. The assertion about sums follows from additivity of standard part and independent choices of summands. $\square$

[[write] The first identity, $\text{st conv} = \text{conv st}$ for a finite bounded hull, is Part I's [proposition 13.1](#), where it is proved (manuscripts 05 and 02); Part IV's [lemma 47.2](#) states it once more. The Minkowski-sum clause, the proof above, and the volume and mixed-volume clauses built on it in [proposition 34.3](#) are manuscript 06's own and are printed only here.]

Proposition 34.3 (Volume and mixed-volume residue). *For finite polytopes with vertices in $\mathcal{O}^d$,*

$$\text{st Vol}_d(P) = \text{Vol}_d(\text{st } P), \quad (34.5)$$

$$\text{st } V(P_1, \dots, P_d) = V(\text{st } P_1, \dots, \text{st } P_d). \quad (34.6)$$

Dimension loss in the residue is allowed.

Proof. For fixed n, d , let f denote the semialgebraic function taking an n -tuple of points to the volume of its convex hull. Over $\mathbb{R}$, this function is continuous, including at lower-dimensional configurations: vertex convergence gives Hausdorff convergence, and volume is continuous on real compact convex sets [22].

Let $a = (\text{st } p_1, \dots, \text{st } p_n)$. For every positive real e , real continuity supplies a positive real δ for which

$$\|x - a\|_\infty < \delta \implies |f(x) - f(a)| < e.$$

This is a first-order assertion, because the graph of f is semialgebraic. It remains true in K . The tuple p satisfies its hypothesis for every such real δ , so $f(p) - f(a)$ is infinitesimal. Both quantities are finite, since all vertices fit in a real bounded box. This proves (34.5). Apply it to every sum in (34.1) and use Lemma 34.2 to obtain (34.6). $\square$

35 The scale lattice and canonical affine residue

35.1 A maximum vertex simplex is a module basis

Let P have affine dimension $r \geq 1$. Choose affine coordinates on its affine hull. Among the finitely many r -simplices formed by its input vertices, select one with largest absolute determinant, and label it $p_0, p_1, \dots, p_r$. Set

$$u_j = p_j - p_0, \quad B = (u_1, \dots, u_r).$$

A change of affine coordinates multiplies every determinant by the same nonzero factor, so maximality does not depend on that preliminary choice. The maximum exists by finite comparison; no compactness theorem is used.

Theorem 35.1 (Maximum-simplex normalization). *For the above choice,*

$$\Delta_r \subseteq \tilde{P} := B^{-1}(P - p_0) \subseteq [-1, 1]^r, \quad \Delta_r = \text{conv}(0, e_1, \dots, e_r). \quad (35.1)$$

Moreover

$$L(P) := \text{span}_{\mathcal{O}}(P - P) = \sum_{i=1}^n \mathcal{O}(p_i - p_0) = B\mathcal{O}^r. \quad (35.2)$$

Thus $L(P)$ is free of rank r .

Proof. The first containment holds because the selected simplex is in P . For an input vertex p , Cramer's rule expresses the j th coordinate of $B^{-1}(p - p_0)$ as

$$\frac{\det(u_1, \dots, u_{j-1}, p - p_0, u_{j+1}, \dots, u_r)}{\det B}.$$

The numerator is the determinant of another vertex simplex, possibly degenerate. Its absolute value is at most $|\det B|$. Therefore every coordinate lies in $[-1, 1]$; convexity gives the outer containment for all of P .

The vertex differences consequently lie in BO^r . Every difference of two convex combinations is an O -linear combination of these vertex differences, so $L(P) \subseteq BO^r$. Conversely each column of B is in $P - P$, and hence $BO^r \subseteq L(P)$. The columns are K -independent, so they are also an O -basis. $\square$

The equality with $\text{span}_O(P - P)$ proves independence from the chosen vertex presentation, including redundant points. Notice that freeness has been proved directly. No Noetherian hypothesis on O , and no general structure theorem for finitely generated modules over valuation rings, is needed.

35.2 An intrinsic affine residue

The coset $p_0 + L(P)$ is independent of the chosen point $p_0 \in P$. Reduce this affine space modulo $\mathfrak{m}L(P)$. Its translation space $L(P)/\mathfrak{m}L(P)$ is an r -dimensional real vector space.

Definition 35.2 (Affine residue polytope). The *affine residue* $\text{Res}(P)$ is the image of P in the affine quotient

$$(p_0 + L(P))/\mathfrak{m}L(P).$$

The quotient is an affine $\mathbb{R}$ -space, with no preferred origin.

Theorem 35.3 (Full-dimensional and coordinate-independent residue). *The affine residue of an r -dimensional finite polytope is a real r -dimensional polytope. In maximum-simplex coordinates it is represented by $\text{st } \tilde{P}$ and contains Δ_r . Different maximum-simplex choices give real affinely isomorphic representatives. Any invertible K -affine change of coordinates induces an affine isomorphism of the residue polytopes.*

Proof. Equation (35.2) identifies the affine quotient with $\mathbb{R}^r$, and Lemma 34.2 identifies the image of P with the convex hull of its reduced normalized vertices. Equation (35.1) shows that this image contains a real r -simplex.

For another basis B' and origin p'_0 , the change of normalized coordinates has linear part $A = B^{-1}B'$ and translation $b = B^{-1}(p'_0 - p_0)$. Since $BO^r = B'O^r$, both A and A^{-1} have entries in O ; also $b \in O^r$. Reducing gives an invertible real affine map, because $\text{st } A \text{ st } A^{-1} = I$. Finally a K -linear isomorphism T maps $L(P)$ onto $L(TP)$ and $\mathfrak{m}L(P)$ onto $\mathfrak{m}L(TP)$. Translations behave compatibly with the affine cosets. $\square$

For a point, $L(P) = 0$ and the affine residue is a point; all assertions extend with $r = 0$. The construction preserves dimension, not the full face lattice. Several vertices may coalesce in the residue, and a small face can disappear.

35.3 A leading-volume formula

For a full-dimensional $P \subset K^d$, let $D = |\det B|$ for its selected maximum simplex.

Corollary 35.4 (Volume scale and residue coefficient). *One has*

$$\frac{D}{d!} \leq \text{Vol}_d(P) \leq 2^d D, \quad (35.3)$$

$$v \text{Vol}_d(P) = vD, \quad (35.4)$$

$$\text{st} \left(\frac{\text{Vol}_d(P)}{D} \right) = \text{Vol}_d(\text{st } \tilde{P}) > 0. \quad (35.5)$$

Proof. Apply monotonicity of volume to (35.1) and then use determinant covariance. Lemma 33.2 gives the valuation identity. For the last statement, $\text{Vol}_d(P)/D = \text{Vol}_d(\tilde{P})$, and Proposition 34.3 applies. $\square$

The positive coefficient in (35.5) is expressed relative to the chosen determinant scale D . A valuation alone has no canonical leading coefficient without such a normalization or a chosen cross-section of the value group. The residue polytope supplies geometric information beyond the valuation.

36 Parallelotope models and finite determinant certificates

36.1 Dimension-only inner and outer models

For a positive-dimensional polytope P_i of affine dimension r_i , choose maximum-simplex directions $u_{i1}, \dots, u_{ir_i}$ as in Theorem 35.1, and put

$$Z_i = \sum_{j=1}^{r_i} [0, u_{ij}].$$

This is an r_i -dimensional parallelotope in its linear span. We use the term also for such a lower-dimensional object embedded in K^d .

Lemma 36.1 (Parallelotope sandwich). *After translating the selected origin p_{i0} to zero,*

$$\frac{1}{r_i} Z_i \subseteq P_i - p_{i0} \subseteq 2Z_i - \sum_{j=1}^{r_i} u_{ij}. \quad (36.1)$$

Proof. In normalized coordinates, $[0, 1/r_i]^{r_i} \subseteq \Delta_{r_i}$, since its coordinates are nonnegative and sum to at most one. The other inclusion is $\tilde{P}_i \subseteq [-1, 1]^{r_i} = 2[0, 1]^{r_i} - (1, \dots, 1)$. Apply the basis matrix. $\square$

Theorem 36.2 (Simultaneous full-profile parallelotope model). *For every multi-index $\alpha \in \mathbb{N}^m$ with $|\alpha| = d$, if every body occurring in that entry has positive affine dimension, then*

$$\begin{aligned} \left(\prod_{i:\alpha_i > 0} r_i^{-\alpha_i} \right) \text{V}(Z_1[\alpha_1], \dots, Z_m[\alpha_m]) &\leq \text{V}(P_1[\alpha_1], \dots, P_m[\alpha_m]) \\ &\leq 2^d \text{V}(Z_1[\alpha_1], \dots, Z_m[\alpha_m]). \end{aligned} \quad (36.2)$$

Consequently all entries have the same valuations for the two families. A point may be replaced by $Z_i = \{0\}$; any entry containing it is zero on both sides. Thus one choice of parallelotopes, each with at most d generators, preserves the entire mixed-volume valuation profile.

Proof. Apply monotonicity separately in all d argument slots of the mixed volume, using (36.1). Remove the independent translations and collect the positive scalar factors. The comparison constants are positive reals, so Lemma 33.2 gives equality of valuations, also when the compared mixed volumes vanish. An argument that is a point can be translated to zero, and multilinearity or polarization makes its mixed volume zero. $\square$

The same models work for every entry, including entries in which a body is repeated. There is no claim that their leading coefficients or face lattices agree with those of the input polytopes.

36.2 The determinant formula and explicit constants

For d bodies of positive affine dimensions $r_1, \dots, r_d$, define

$$M = \max_{1 \leq j_i \leq r_i} |\det(u_{1j_1}, \dots, u_{dj_d})|, \quad R_0 = \prod_{i=1}^d r_i. \quad (36.3)$$

When an argument is a point, use $M = 0$.

Theorem 36.3 (Finite mixed-determinant certificate). *For nonempty finite polytopes in K^d ,*

$$\frac{M}{d!} \leq V(P_1, \dots, P_d) \leq \frac{2^d}{d!} \sum_{j_1, \dots, j_d} |\det(u_{1j_1}, \dots, u_{dj_d})| \leq \frac{2^d R_0}{d!} M, \quad (36.4)$$

with the zero convention for a point argument. In particular,

$$v V(P_1, \dots, P_d) = \min_{j_1, \dots, j_d} v \det(u_{1j_1}, \dots, u_{dj_d}). \quad (36.5)$$

At most d^d determinants are required after the maximum-simplex bases have been selected.

Proof. Each segment $[p_{i0}, p_{i0} + u_{ij}]$ lies in P_i . Monotonicity and the segment formula give the lower bound for every determinant in (36.3), hence for their maximum.

By the upper half of Theorem 36.2, $V(P_1, \dots, P_d) \leq 2^d V(Z_1, \dots, Z_d)$. Expanding each zonotope by multilinearity and applying (34.2) gives the sum in (36.4). It contains R_0 terms, each at most M . If $M = 0$, the inequalities force the mixed volume to vanish. Otherwise the extreme comparison constants are positive real units and give (36.5). Finally $r_i \leq d$. $\square$

The lower constant is attained by independent segment arguments. No optimality is asserted for the upper constant. The bound counts the *final mixed determinants*, not the work needed to find a maximum simplex in a large vertex list. Brute-force basis selection can examine $\binom{n_i}{r_i+1}$ candidate simplices for body i .

36.3 An intrinsic determinant ideal

For finitely generated $\mathcal{O}$ -modules $L_1, \dots, L_d \subset K^d$, define

$$\mathfrak{D}(L_1, \dots, L_d) = \langle \det(w_1, \dots, w_d) : w_i \in L_i \rangle_{\mathcal{O}} \subset K.$$

This is an $\mathcal{O}$ -submodule of K , or fractional ideal in this usage; it need not be contained in $\mathcal{O}$.

Proposition 36.4 (Basis independence and the intrinsic formula). *For $L_i = L(P_i)$, the module $\mathfrak{D}(L_1, \dots, L_d)$ is generated by the finitely many determinants of the selected bases. It is either zero or a principal fractional ideal $\delta\mathcal{O}$, and*

$$v V(P_1, \dots, P_d) = v\delta \quad \text{when it is nonzero.}$$

Thus the mixed-volume valuation depends only on the scale lattices of its arguments. The minimum formula remains true if one uses all vertex differences instead of the selected bases.

Proof. Expand each w_i in its $\mathcal{O}$ -basis. Multilinearity expresses its determinant as an $\mathcal{O}$ -linear combination of the displayed finite basis determinants. Conversely each basis determinant is a generator of the intrinsic module. Among the finitely many nonzero determinants select one δ of least valuation. Every other determinant divided by δ lies in $\mathcal{O}$, so the generated module is $\delta\mathcal{O}$. Theorem 36.3 identifies its value with the mixed-volume value. Every vertex difference belongs to its corresponding module, while selected basis directions are vertex differences, proving the last assertion. $\square$

This is the decisive cancellation distinction. Individual determinants may have substantial cancellation and must be computed correctly. Once they are known, the mixed-volume scale is obtained by a finite minimum; there is no additional cancellation between those determinant scales.

37 Simultaneous inner segments and the valued Grassmannian

37.1 Avoiding finitely many leading-term cancellations

We first record the elementary finite-grid fact used to make the word “generic” effective.

Lemma 37.1 (A finite grid detects a nonzero polynomial). *Let $F \in \mathbb{R}[X_1, \dots, X_N]$ be nonzero and have degree at most H in each individual variable. If each $A_j \subset \mathbb{Q}$ contains $H + 1$ distinct numbers, some point of $A_1 \times \dots \times A_N$ is not a zero of F .*

Proof. For one variable this is the bound on the number of roots of a nonzero polynomial. Induct on N . If F vanished on the whole product grid, fixing the first $N - 1$ coordinates would give a polynomial in X_N with $H + 1$ roots, so all its coefficients would vanish on the smaller grid. The induction hypothesis would make every coefficient the zero polynomial, a contradiction. $\square$

Theorem 37.2 (Constructive simultaneous inner-segment theorem). *Let $m \geq d \geq 1$ and let $P_1, \dots, P_m \subset K^d$ be nonempty finite polytopes. Choose maximum-simplex bases in their positive-dimensional affine hulls and let $r_i = \dim P_i$. Put*

$$H = \binom{m-1}{d-1}.$$

There are rational coefficients

$$c_{ij} \in \left\{ \frac{1}{r_i(H+2)}, \frac{2}{r_i(H+2)}, \dots, \frac{H+1}{r_i(H+2)} \right\} \quad (r_i > 0) \quad (37.1)$$

such that the segments

$$S_i = [p_{i0}, p_{i0} + w_i], \quad w_i = \sum_{j=1}^{r_i} c_{ij} u_{ij},$$

with $w_i = 0$ for a point, satisfy

$$S_i \subseteq P_i, \quad v \mathbf{V}(S_i : i \in I) = v \mathbf{V}(P_i : i \in I) \quad \left(I \in \binom{[m]}{d} \right). \quad (37.2)$$

A successful choice exists in a grid of at most $(H + 1)^{md}$ points.

Proof. Introduce separate variables X_{ij} for each positive-dimensional body and set $w_i(X) = \sum_j X_{ij} u_{ij}$. For a d -subset $I = \{i_1 < \dots < i_d\}$, expand

$$D_I(X) = \det(w_{i_1}(X), \dots, w_{i_d}(X)) = \sum_{j_1, \dots, j_d} \det(u_{i_1 j_1}, \dots, u_{i_d j_d}) \prod_{k=1}^d X_{i_k j_k}.$$

Distinct choices $(j_1, \dots, j_d)$ produce distinct monomials, because the bodies in I are distinct labelled arguments.

Let $\tau_I = v \mathbf{V}(P_i : i \in I)$. If $\tau_I = \infty$, Theorem 36.3 says that every coefficient of D_I is zero; then no condition on the choice is necessary. Otherwise select a coefficient δ_I of valuation τ_I . All coefficients of D_I/δ_I lie in $\mathcal{O}$, and at least one is a unit. Reducing its coefficients gives a nonzero real polynomial

$$F_I(X) = \text{st}(D_I(X)/\delta_I) \in \mathbb{R}[X_{ij}].$$

Here standard part is applied coefficientwise, before any evaluation. Nonzeroness follows from distinctness of the monomials and from the chosen unit coefficient.

Take the product $F = \prod_{I: \tau_I < \infty} F_I$, using $F = 1$ if there are no such subsets. It is a nonzero real polynomial. Each variable X_{ij} occurs with degree at most one in F_I , and only when $i \in I$. There are at most H such subsets, so its degree in F is at most H . Lemma 37.1 applied to the grids in (37.1) gives a rational point c with $F(c) \neq 0$.

For each finite entry, evaluation at real c commutes with standard part, and $F_I(c) \neq 0$ says that $D_I(c)/\delta_I$ is a unit of $\mathcal{O}$. Hence $v D_I(c) = \tau_I$. Infinite entries remain zero identically. The segment formula gives all equalities in (37.2). Finally every selected coefficient is positive and

$$\sum_{j=1}^{r_i} c_{ij} \leq \frac{H+1}{H+2} < 1.$$

Thus $p_{i0} + w_i$ is a convex combination of the selected simplex vertices, and the whole segment lies in P_i . $\square$

The theorem proves the preview in Section 32. It applies even when no nonzero integer multiple of one positive valuation is cofinal in the value group. The only genericity argument occurs over the ordinary real residue field, and there it is replaced by a finite grid.

37.2 An exact equality of configuration images

Let $\mathcal{T}_{m,d}^{\text{pure}}(K)$ be the set of all vectors

$$(v \mathbf{V}(P_i : i \in I))_{I \in \binom{[m]}{d}}$$

obtained from nonempty finite polytopes in K^d . Define the determinant image

$$\mathcal{G}_{m,d}(K) = \left\{ (v \det W_I)_{I \in \binom{[m]}{d}} : W \in K^{d \times m} \right\},$$

where W_I is the submatrix with columns indexed by I in increasing order. Rank-deficient matrices contribute the all-infinite vector.

Corollary 37.3 (Pure profile realization). *One has the exact equality*

$$\boxed{\mathcal{T}_{m,d}^{\text{pure}}(K) = \mathcal{G}_{m,d}(K).} \quad (37.3)$$

After omitting the all-infinite vector and quotienting by common addition of a value, the right side is the valuation image of the K -realizable Grassmannian in Plücker coordinates.

Proof. Theorem 37.2 produces the matrix $W = (w_1, \dots, w_m)$ for every polytope profile. Conversely take $P_i = [0, w_i]$ and use (34.2); the factor $d!$ has valuation zero. $\square$

Several restrictions in this statement are essential. It is an equality of *valuation images over the specified ordered field K* , not a claim that every abstract valuated matroid is realizable. It does not replace K by an algebraically closed field, forget ordered-field realizability, or silently take a topological closure. The use of absolute determinants makes no difference to the value, since $v|a| = v(a)$. Thus (37.3) gives a concrete, field-specific link from polytope mixed-volume scales to the absolute and valued Grassmannian viewpoints discussed in [23].

37.3 Tropical Plücker constraints

Corollary 37.4 (A universal pure-profile relation). *Let $A \subseteq [m]$ have size $d - 2$, and let a, b, c, e be four distinct indices outside A . For any finite polytope family, the minimum of*

$$\tau_{Aab} + \tau_{Ace}, \quad \tau_{Aac} + \tau_{Abe}, \quad \tau_{Aae} + \tau_{Abc} \quad (37.4)$$

is attained at least twice, with the usual convention when all terms are infinite.

Proof. Represent the entire pure profile by one matrix, using Corollary 37.3. Its minors satisfy the three-term Plücker relation, with signs determined by the chosen ordering. In a valued field a sum of nonzero terms cannot vanish when exactly one term has strictly least valuation: dividing by that term gives a unit plus an infinitesimal. Apply this observation to the three products of minors. $\square$

For four planar bodies, the conclusion says that the minimum of $\tau_{12} + \tau_{34}$, $\tau_{13} + \tau_{24}$, and $\tau_{14} + \tau_{23}$ is attained at least twice. It is stronger than merely testing one mixed coefficient at a time. The same matrix realization supplies all determinantal relations simultaneously.

38 Full profiles, cloned segments, and higher-rank exchange

38.1 A colored determinant formula

Concatenate the maximum-simplex bases for $P_1, \dots, P_m$ into a matrix U , and give a column from body i the color i . For a d -element column subset B , let $\text{col}(B) \in \mathbb{N}^m$ record the counts of its colors. Put

$$\tau(\alpha) = v V(P_1[\alpha_1], \dots, P_m[\alpha_m]), \quad |\alpha| = d.$$

The zonotope expansion gives the precise multiplicity identity

$$V(Z_1[\alpha_1], \dots, Z_m[\alpha_m]) = \frac{\prod_i \alpha_i!}{d!} \sum_{B: \text{col}(B)=\alpha} |\det U_B|. \quad (38.1)$$

Indeed, choosing one generator in each ordered argument slot gives zero when the same column is chosen twice. Every surviving unordered column subset occurs $\prod_i \alpha_i!$ times. Consequently

$$\tau(\alpha) = \min_{B: \text{col}(B)=\alpha} v \det U_B, \quad (38.2)$$

where the minimum over no nonzero determinant is infinite.

This formula describes the profile as a colorwise minimum of determinant values. To prove its full exchange structure without invoking a real-valued tropicalization theorem, it is convenient to strengthen the realization.

Theorem 38.1 (Color-symmetric clone realization). *Make d labelled copies of each P_i . There exist vectors $w_{i,k} \in P_i - p_{i0}$, $1 \leq k \leq d$, such that for every d -element set B of distinct clone labels,*

$$v \det(w_{i,k} : (i, k) \in B) = \tau(\text{col}(B)). \quad (38.3)$$

Thus the determinant valuation depends only on the color counts, not on which clone labels are chosen. The vectors can be obtained from the finite rational grids of Theorem 37.2, applied to md labelled bodies.

Proof. Apply the simultaneous inner-segment theorem to the family in which each of the m bodies occurs as d separately labelled members. A pure mixed volume of this larger family is exactly the corresponding repeated-body mixed volume of the original family. Taking the segment direction vectors proves (38.3). $\square$

The number d of clones per body allows all multiplicities up to d . It is a statement about representing a finite profile, not about representing a polytope itself by the union of those segments.

38.2 A determinant exchange lemma

Lemma 38.2 (Valued basis exchange). *Let columns of a matrix over K be indexed by a finite set. For a d -element subset B , write $q(B) = v \det W_B$. If $q(I)$ and $q(J)$ are finite and $e \in I \setminus J$, there is $f \in J \setminus I$ for which*

$$q(I) + q(J) \geq q(I - e + f) + q(J - f + e). \quad (38.4)$$

Both terms on the right are finite for the selected f .

Proof. The determinant exchange identity expresses

$$\det W_I \det W_J = \sum_{f \in J \setminus I} \pm \det W_{I-e+f} \det W_{J-f+e}.$$

For completeness, expand the column w_e in the basis indexed by J using Cramer's rule, then use linearity in its slot in $\det W_I$ and multiply by $\det W_J$. Terms indexed by $f \in I \cap J$ have a repeated column in the first determinant and vanish. Reordering the columns accounts for the signs.

The left side is nonzero. The valuation of a finite sum is at least the minimum of its summands' valuations. Hence some nonzero summand has valuation at most $q(I) + q(J)$, which is exactly the assertion. $\square$

38.3 Exchange-convexity over an arbitrary value group

Definition 38.3. A function $f : \{\alpha \in \mathbb{N}^m : |\alpha| = d\} \rightarrow \Gamma \cup \{\infty\}$ will be called *exchange-convex* if, whenever $f(x), f(y) < \infty$ and $x_i > y_i$, there exists j with $x_j < y_j$ such that

$$f(x) + f(y) \geq f(x - e_i + e_j) + f(y + e_i - e_j).$$

This is the usual M -convex exchange axiom, stated with values in the ordered abelian group Γ rather than necessarily in $\mathbb{R}$.

Theorem 38.4 (Higher-rank mixed-volume exchange). *The full mixed-volume valuation profile τ is exchange-convex.*

Proof. Use the clone matrix from Theorem 38.1. For two multi-indices x, y with finite values, choose the clone subsets

$$I = \{(k, s) : 1 \leq s \leq x_k\}, \quad J = \{(k, s) : 1 \leq s \leq y_k\}.$$

They have size d . The initial-segment choice ensures that their common clones in each color are exactly the first $\min(x_k, y_k)$ labels.

If $x_i > y_i$, choose a clone e of color i in $I \setminus J$. Lemma 38.2 supplies $f \in J \setminus I$. Its color j satisfies $x_j < y_j$, precisely because the clone sets were nested within each color. The exchanged subsets have color counts $x - e_i + e_j$ and $y + e_i - e_j$. Substitute (38.3) into (38.4) to obtain the desired inequality. $\square$

This proof uses only finite determinants, reduction to an infinite residue field, and ordered-group comparisons. It needs no embedding $\Gamma \hookrightarrow \mathbb{R}$, no limiting process, and no assumption that a positive value has cofinal integer multiples. For real-valued settings, its conclusion lies within the established Lorentzian and M -convex framework [24]; the point here is an explicit matrix proof that continues unchanged at higher rank.

For example, taking $x = \alpha + e_i - e_j$ and $y = \alpha - e_i + e_j$ when both are admissible gives

$$2\tau(\alpha) \leq \tau(\alpha + e_i - e_j) + \tau(\alpha - e_i + e_j).$$

Thus the familiar midpoint inequality is only one consequence of the stronger exchange axiom. Infinite entries encode genuine vanishing, not a finite but unusually small mixed volume.

38.4 Exactly which entries are nonzero

Let $U_i = \text{span}_K(P_i - P_i)$. The following linear-algebra criterion makes the support of the full profile explicit.

Lemma 38.5 (Independent representatives of subspaces). *For subspaces $E_1, \dots, E_s$ of a finite-dimensional vector space over a field, there exist independent vectors $e_i \in E_i$ if and only if*

$$\dim \left(\sum_{i \in I} E_i \right) \geq |I| \quad (I \subseteq [s]).$$

Proof. Necessity is immediate. For sufficiency induct on s . If there is a nonempty proper subset I with equality, choose independent representatives for its subspaces by induction.

They form a basis of $W = \sum_{i \in I} E_i$. In the quotient by W , the remaining subspaces satisfy the dimension condition, since for $J \subseteq [s] \setminus I$,

$$\dim \left((W + \sum_{j \in J} E_j) / W \right) \geq |I| + |J| - |I| = |J|.$$

Choose independent representatives in the quotient and lift them.

If no such tight proper subset exists and $s > 1$, choose any nonzero $e_s \in E_s$. Every nonempty subset of the other indices spans dimension at least its size plus one. Quotienting by Ke_s decreases that dimension by at most one, so induction applies to the remaining $s - 1$ subspaces. The case $s = 1$ is immediate. $\square$

Corollary 38.6 (Support of the full profile). *For $|\alpha| = d$,*

$$\tau(\alpha) < \infty \iff \dim_K \left(\sum_{i \in J} U_i \right) \geq \sum_{i \in J} \alpha_i \text{ for all } J \subseteq [m]. \quad (38.5)$$

Proof. By the determinant formula, positivity of the mixed volume is equivalent to choosing independent representatives from the d slots, with U_i repeated α_i times. Lemma 38.5 applies. Its condition for all slot subsets is equivalent to (38.5): necessity follows by taking all slots of the colors in J ; sufficiency follows because an arbitrary slot subset uses no more than all α_i slots of each color it meets. $\square$

This is a finite subspace-rank condition. It separates the zero pattern, which uses only K -linear spans, from the nonzero scale values, which require the finer $\mathcal{O}$ -lattices.

39 Recovering a polytope's scale class from probes

39.1 All mixed responses classify the scale lattice

Fix full-dimensional polytopes $P, Q \subset K^d$. A *segment probe* is a tuple of $d - 1$ segments in K^d ; for $d = 1$ it is the empty tuple.

Theorem 39.1 (Scale-lattice classification). *The following are equivalent:*

(i) $L(P) = L(Q)$.

(ii) For every segment probe $T_1, \dots, T_{d-1}$,

$$v \, V(P, T_1, \dots, T_{d-1}) = v \, V(Q, T_1, \dots, T_{d-1}).$$

(iii) Replacing P by Q in any one slot of any mixed volume of finite polytopes leaves the valuation unchanged.

Proof. Proposition 36.4 gives (i) $\Rightarrow$ (iii), and segment arguments give (iii) $\Rightarrow$ (ii).

Assume (ii). Choose a maximum-simplex basis $B = (u_1, \dots, u_d)$ for P and write $D = \det B$. For each j , use as probe the segments in all directions u_k with $k \neq j$. For P the determinant minimum is vD : all basis choices except u_j give a repeated column, and u_j gives $\pm D$.

Let $C = (z_1, \dots, z_d)$ be a basis for $L(Q)$, and set $A = B^{-1}C$. By the determinant formula, the corresponding probe for Q has value

$$vD + \min_k v(A_{jk}).$$

Equality with vD forces every entry of row j of A to lie in $\mathcal{O}$. For all rows together, this gives $L(Q) \subseteq L(P)$. Repeating the argument with a basis of Q gives the reverse inclusion. $\square$

The probes range over directions in K^d , not merely real directions. Restricting to a smaller collection is a different question: hidden non-Archimedean shear data must not be discarded without proof.

39.2 A finite comparison test

Theorem 39.2 (A reference-basis $d + 1$ -measurement test). *Choose a maximum-simplex basis $B = (u_1, \dots, u_d)$ of P , and let $T^{(j)}$ be the $(d - 1)$ -tuple of segments $[0, u_k]$, $k \neq j$. For a full-dimensional Q , the equality $L(Q) = L(P)$ holds if and only if*

$$v \operatorname{Vol}_d(Q) = v \operatorname{Vol}_d(P), \quad (39.1)$$

$$v V(Q, T^{(j)}) \geq v \operatorname{Vol}_d(P) \quad (1 \leq j \leq d). \quad (39.2)$$

When these conditions hold, every inequality in (39.2) is an equality. Thus, relative to the known reference basis and scale of P , one volume value and d probe values suffice to test its entire scale class.

Proof. For a maximum-simplex basis C of Q , write $A = B^{-1}C$ as above. The threshold conditions say that all entries of A lie in $\mathcal{O}$. The volume condition and Corollary 35.4 say that $v \det A = 0$. Hence $\det A$ is a unit and

$$A^{-1} = \frac{\operatorname{adj}(A)}{\det A}$$

has entries in $\mathcal{O}$. Therefore $A \in \operatorname{GL}_d(\mathcal{O})$ and the lattices are equal. Conversely lattice equality makes A and its inverse integral, so its determinant is a unit and every row has some unit entry. The volume equality and all probe equalities follow. $\square$

No minimality of the number $d + 1$ is asserted. It is a uniform explicit upper bound for a comparison test, not a reconstruction algorithm from unlabelled real measurements.

39.3 A geometric interpretation of equal lattices

Proposition 39.3 (Bounded real distortion). *For full-dimensional $P, Q \subset K^d$, $L(P) = L(Q)$ if and only if there are $a \in P$, $b \in Q$ and positive real constants c, C such that*

$$c(P - a) \subseteq Q - b \subseteq C(P - a). \quad (39.3)$$

Proof. For a maximum-simplex normalization of a d -polytope, center at the centroid $h = (1/(d + 1), \dots, 1/(d + 1))$ of Δ_d . With $\delta = 1/(d(d + 1))$, the simplex contains $h + [-\delta, \delta]^d$, while the normalized polytope minus h is contained in $[-2, 2]^d$. Thus its centered shape lies between two real scalar multiples of the cube in its $\mathcal{O}$ -basis.

If $L(P) = L(Q)$, the basis change belongs to $\operatorname{GL}_d(\mathcal{O})$. All entries of it and its inverse are bounded by a common positive real number. Matrix multiplication therefore sends a cube into a real scalar multiple of a cube in both directions. Combining the two cube comparisons gives (39.3).

Conversely take differences of points in the two inclusions. The resulting inclusions of difference sets imply $cL(P) \subseteq L(Q) \subseteq CL(P)$. Positive real constants are units of $\mathcal{O}$, so both outer modules equal $L(P)$. $\square$

The equivalence class detected by all mixed-volume valuations is exactly this bounded-distortion class. The affine residue then records a real shape inside the class, and finer infinitesimal data may distinguish polytopes even when their residue shapes agree.

40 Examples, counterexamples, and what the invariants forget

40.1 A triangle whose ordinary shadow is only a segment

Let $\varepsilon = \omega^{-1} > 0$ and

$$P = \text{conv}\{(0, 0), (1, 0), (0, \varepsilon)\}.$$

Its ordinary standard part is the horizontal unit segment, but

$$L(P) = \mathcal{O}e_1 \oplus \mathcal{O}(\varepsilon e_2), \quad B = \begin{pmatrix} 1 & 0 \\ 0 & \varepsilon \end{pmatrix}, \quad B^{-1}P = \Delta_2.$$

Hence $\text{Res}(P)$ is a real triangle, not a segment. Its area is $\text{Vol}_2(P) = \varepsilon/2$, its volume value is $v\varepsilon$, and its normalized residue area is $1/2$. The rescaling has not removed the infinitesimal information: that information is retained in the embedded scale lattice and its determinant.

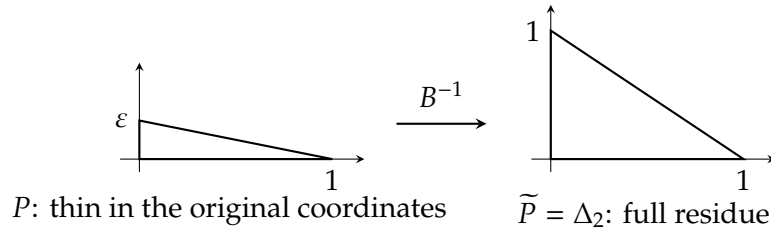


Figure 4: Affine normalization of an infinitesimally thin triangle. The left vertical scale is exaggerated; the picture is schematic, not a real-scale rendering of an infinitesimal height.

40.2 Cancellation can lie beyond every fixed power scale

Take $\eta = \omega^{-\omega}$ and $u = (1, 1)$, $w = (1, 1 + \eta)$. Then

$$\det(u, w) = \eta.$$

Keeping only the standard part of each coordinate makes both columns $(1, 1)$ and destroys the determinant. Moreover $0 < \eta < \varepsilon^N$ for every positive integer N , so no fixed finite order in powers of ε reveals it. The correct finite certificate is the determinant itself, computed with enough exact coefficient information to resolve its cancellation.

Even in rank one there is no truncation depth depending only on dimension and vertex count. For each integer $N > 0$, the three-vertex triangle with height ε^N has nonzero area $\varepsilon^N/2$. A rule discarding all terms above a fixed power cutoff fails on some such triangle.

[[write] These two observations are weaker instances of Part II's results on coordinate truncation. There, [theorem 26.2](#) gives four labelled points with coordinates in $\mathbb{R}((\varepsilon))$ whose degree- N truncated hulls alternate between a quadrilateral and a triangle for every N . [theorem 26.3](#) realizes *every* prescribed binary history in the same way. So no truncation depth works, and moreover the truncated combinatorics need not stabilize at all. Part I's [section 16.4](#) records the corresponding failure for an exact algebraic equality. The examples here are kept because they concern a determinant (a scale) rather than a face lattice.]

40.3 Identical individual scales do not determine a mixed scale

Use the rank-two convention

$$v\varepsilon = (0, 1), \quad v\eta = (1, 0) \quad \text{in lexicographically ordered } \mathbb{Z}^2.$$

Define three parallelograms

$$\begin{aligned} P &= [0, e_1] + [0, \varepsilon e_2], \\ Q &= [0, e_1 + \varepsilon e_2] + [0, \eta e_2], \\ Q' &= [0, \eta e_1] + [0, e_2]. \end{aligned}$$

Their individual areas satisfy

$$\text{Vol}_2(P) = \varepsilon, \quad \text{Vol}_2(Q) = \text{Vol}_2(Q') = \eta.$$

Each is affinely a square, and each has an affine residue affinely isomorphic to a square. Nevertheless expansion into segments gives

$$\begin{aligned} V(P, Q) &= \frac{1}{2}(\varepsilon + \eta + \varepsilon + 0) = \varepsilon + \frac{1}{2}\eta, \\ V(P, Q') &= \frac{1}{2}(0 + 1 + \varepsilon\eta + 0) = \frac{1}{2}(1 + \varepsilon\eta). \end{aligned}$$

The mixed values are respectively $(0, 1)$ and $(0, 0)$. Thus individual volume values and separate affine residue isomorphism classes do not determine a family's mixed profile. The *relative positions* of its scale lattices in the common ambient vector space matter.

40.4 A four-body Plücker example

Let

$$\begin{aligned} P_1 &= [0, e_1] + [0, \eta e_2], & P_2 &= [0, e_1 + \varepsilon e_2] + [0, \eta e_2], \\ P_3 &= [0, e_2] + [0, \eta e_1], & P_4 &= [0, e_2 + \varepsilon e_1] + [0, \eta e_1]. \end{aligned}$$

All four bodies have positive area. Their pure mixed values are

$$\tau_{12} = \tau_{34} = v\varepsilon, \quad \tau_{13} = \tau_{14} = \tau_{23} = \tau_{24} = 0.$$

([write] Here $\tau_{ab} = v V(P_a, P_b)$ is an entry of the pure profile of [corollary 37.4](#); the η in the generators is the surreal $\omega^{-\omega}$, written τ elsewhere in the report.) In this example the first displayed generator of each body already gives a simultaneous inner-segment model. The three Plücker sums are $2v\varepsilon, 0, 0$, so the minimum is attained exactly twice. The example makes visible the difference between two nearly parallel pairs and the transverse cross-pairs.

40.5 The affine residue need not retain all faces

Let

$$P_\varepsilon = \text{conv}\{(0, 0), (1, 0), (1, 1), (0, 1), (-\varepsilon, 1/2)\}.$$

It is a pentagon over K , of area $1 + \varepsilon/2$. Its scale lattice is $\mathcal{O}^2$, and its affine residue is the square $[0, 1]^2$. Thus a full-dimensional residue can lose a vertex even though it loses no dimension. Maximum-simplex normalization does not repair this loss of fine face information; it is deliberately a first residue invariant.

[[write] This is the same phenomenon as Part I's lost-face example (14.2) (manuscript 02). There a pentagon has a vertex infinitesimally below the bottom edge of the rectangle $[-1, 1] \times [0, 1]$, and that vertex's standard part is not a face. Here the vertex $(-\varepsilon, 1/2)$ lies infinitesimally outside an edge of the square. The loss occurs for the standard part and for the affine residue alike: the scale lattice here is already $\mathcal{O}^2$, so $\text{Res}(P_\varepsilon)$ is affinely isomorphic to $\text{st } P_\varepsilon = [0, 1]^2$. Part I's [proposition 14.2](#) shows how an infinitesimal optimality band recovers the whole limiting face; see the re-scoped [question 43.6](#).]

The real triangle Δ_2 and the real unit square also have the same scale lattice $\mathcal{O}^2$, but areas $1/2$ and 1 . The common lattice fixes the value, not the leading coefficient. This is why a determinant minimum is an exact scale formula rather than an exact mixed-volume formula.

40.6 Why the pure/full distinction cannot be dropped

For $d > 1$, a full-dimensional P has $\text{Vol}_d(P) > 0$, while every segment S has $\text{Vol}_d(S) = 0$. Therefore no single-segment replacement can preserve the repeated entry $V(P[d])$. The simultaneous theorem is formulated for *distinct labels* for this reason. The clone theorem solves the repeated-entry issue by providing independent segment choices for different copies of the same body.

40.7 An exact characterization of full profiles

The clone theorem also has a converse that is useful when formulating further realization problems.

Corollary 40.1 (Full profiles as color-symmetric determinant data). *An array $t(\alpha) \in \Gamma \cup \{\infty\}$, $|\alpha| = d$, is the full mixed-volume valuation profile of finite polytopes in K^d if and only if there is a $d \times md$ matrix over K , with d columns of each of m colors, such that every d -element column subset B satisfies*

$$v \det W_B = t(\text{col}(B)).$$

Proof. Necessity is Theorem 38.1. For sufficiency, make Z_i the zonotope generated by the d columns of color i . For any α , formula (38.1) expresses the corresponding mixed volume as a positive real factor times a finite sum of absolute determinants. All eligible determinants have value $t(\alpha)$, or all are zero when $t(\alpha) = \infty$. There is at least one eligible column subset for every $|\alpha| = d$, since every color has d labels. The positive-sum lemma gives the required value. $\square$

This is a realizability characterization with matrix witnesses, not a claim that the exchange axiom alone is sufficient. Removing the matrix witnesses in favor of a usable intrinsic list of conditions is one of the research questions below.

41 Finite higher-rank certificates and real specialization

41.1 The finite-support input model

The preceding results apply to arbitrary finite tuples of elements of K ; they are not an algorithm for comparing arbitrary surreal expressions. For an effective specialization theorem, restrict the coordinates to finite sums

$$f = \sum_{\gamma \in A} a_\gamma t^\gamma, \quad a_\gamma \in \mathbb{R},$$

where A is a finite subset of a totally ordered abelian group. Rational functions in such sums are also allowed when their denominators are certified nonzero. The order is the Hahn leading-term order: the least exponent determines the sign and value of a nonzero sum.

A finite certificate may contain coordinate expressions, determinants, nonzero denominators, polynomial differences encoding comparisons, and products needed for specified additive comparisons of valuations. Expand these expressions exactly first; cancellation within an expression must already have been resolved. For comparisons of absolute determinants, one may use the polynomial difference of their squares.

41.2 Finite order data admit an integer-valued model

Lemma 41.1 (Finite positive grading). *Let H be a finitely generated subgroup of a totally ordered abelian group, and let $D \subset H_{>0}$ be finite. There exists a group homomorphism $\lambda : H \rightarrow \mathbb{Z}$ with $\lambda(\delta) > 0$ for every $\delta \in D$.*

Proof. The group H is finitely generated and torsion-free, so choose an isomorphism $H \cong \mathbb{Z}^s$. No nontrivial nonnegative integer combination of members of D can be zero, because all its summands are positive in H .

The origin is not in the real convex hull of the corresponding integer vectors. Otherwise the finite linear system $\sum a_\delta \delta = 0$, $\sum a_\delta = 1$, $a_\delta \geq 0$ would have a real solution and hence a rational solution: a nonempty polytope defined over $\mathbb{Q}$ has a rational vertex, or a rational point in its minimal supporting face, by Gaussian elimination. Clearing denominators would give a forbidden nonnegative integer relation.

Strictly separate this finite convex hull from zero in $\mathbb{R}^s$. The separating linear functional is positive on all members of D . Approximate its coefficients rationally closely enough to preserve these finitely many strict inequalities, and clear denominators. The resulting integer linear functional defines λ . $\square$

The finite positive-grading principle has a close antecedent in the repository's support-certificate discussion [18]; the application here is to determinant and polytope sign certificates. It is not asserted for an infinite support, and it is not asserted to preserve the full order of H .

Theorem 41.2 (One-parameter preservation of a finite certificate). *Let $\mathcal{F}$ be a finite collection of finite-support Hahn polynomials. There is an integer-valued homomorphism on the group generated by their exponents such that the substitution*

$$t^\gamma \longmapsto s^{\lambda(\gamma)} \tag{41.1}$$

preserves, for every member of $\mathcal{F}$, zero versus nonzero, sign, leading coefficient, and the ordered list of its exponents. It also preserves the order of the valuations of the nonzero members of $\mathcal{F}$. The resulting expressions are ordinary one-variable Laurent polynomials. Finite rational expressions are covered by including their nonzero numerators and denominators in the certificate.

Proof. Let A be the union of all supports occurring in $\mathcal{F}$. Apply Lemma 41.1 to the positive differences $\beta - \alpha$ of all distinct pairs $\alpha, \beta \in A$ with $\alpha < \beta$. The resulting homomorphism is strictly order-preserving on the finite set A . Thus no two terms within a certificate polynomial acquire the same exponent, and the term ordering, leading coefficient, sign, and nonzeroness are unchanged. Zero expressions remain zero because the substitution is a group-algebra homomorphism. Comparing leading exponents across different members gives the assertion about their valuations. Including denominators ensures that specialization of each rational expression is defined in $\mathbb{R}(s)$. $\square$

An equality of exponent sums is automatically preserved by a group homomorphism. To preserve a specified *inequality* between such sums, include the corresponding products or monomials among the finitely many certificate expressions before applying the theorem. This prevents an unwarranted claim that every order comparison in the infinite group has been retained.

[[write] [Theorem 41.2](#) is the effective, finite-support counterpart of Part I's [theorem 19.1](#) (manuscript 02). That theorem takes an arbitrary finite bounded configuration over K and a finite list of polynomial sign conditions. By real-closed-field transfer and semialgebraic curve selection it gives a convergent algebraic Puiseux arc over $\mathbb{R}(s)$ that also keeps the prescribed standard-part limit, but it is not effective. The theorem here assumes finite-support Hahn input. In exchange it gives an explicit integer grading, ordinary Laurent polynomials with the same leading coefficients and exponent order, and the explicit threshold ρ_g below. Part IV's [theorem 50.1](#) is a third one-parameter statement, preserving a whole real trace by algebraic germs. None of the three preserves the entire higher-rank order (see the example below).]

41.3 A certified small positive real parameter

Suppose a nonzero specialized expression is

$$g(s) = c_0 s^{m_0} + c_1 s^{m_1} + \cdots + c_k s^{m_k}, \quad m_0 < m_1 < \cdots < m_k \in \mathbb{Z}.$$

For $k > 0$ set

$$\rho_g = \min \left(1, \frac{|c_0|}{2 \sum_{j=1}^k |c_j|} \right),$$

and use $\rho_g = 1$ for a single monomial. If $0 < s < \rho_g$, then

$$|s^{-m_0} g(s) - c_0| \leq s \sum_{j=1}^k |c_j| < |c_0|/2,$$

so $g(s)$ has the sign of its leading coefficient. A common parameter smaller than all the finitely many thresholds preserves all certificate signs simultaneously. For rational input coefficients, a rational choice of s gives a rational finite configuration satisfying those tests.

This result does not contradict non-Archimedean behavior. It reproduces a *specified finite collection* of inequalities and equalities. It does not reproduce the assertion $0 < \eta < \varepsilon^N$ for every integer N .

Example 41.3 (Why an entire rank-two order cannot be compressed). For lexicographic $\mathbb{Z}^2$ with $v\eta = (1, 0)$ and $v\varepsilon = (0, 1)$, an order-preserving homomorphism to $\mathbb{Z}$ would have $a = \lambda(1, 0) > N\lambda(0, 1) = Nb$ for every $N > 0$, with $b > 0$. No two integers a, b satisfy this. In contrast, finitely many comparisons are preserved by $\lambda(a, b) = Ma + b$ for a sufficiently large integer M . The required M depends on the certificate, not only on the geometric dimension.

[[write] This is the same phenomenon as Part I's [section 16.5](#) (manuscript 05): the scales ω^{-1} and $\omega^{-\omega}$ satisfy $\omega^{-\omega} < \omega^{-N}$ for every N , so replacing them by t^a, t^b with fixed positive real exponents in a rank-one monomial group cannot preserve all these comparisons. There, as here, the finite theorems preserve only a specified finite list of comparisons.]

41.4 An explicit finite workflow

A proof-producing implementation for finite-support coordinates can use the following stages. First establish affine ranks and select maximum vertex simplices by exact determinant

comparisons. Second compute the mixed determinant lists and their least valuations. Third, when a pure matrix model is requested, enumerate the rational grids in Theorem 37.2 and retain a choice passing every determinant value test. Fourth collect all the resulting finite polynomial tests, construct a positive grading, and produce a real-parameter sign bound.

The certificate consists of the selected simplex indices, exact comparison expressions, basis determinants, chosen rational coefficients, and the finite grading inequalities. Its verification is finite algebra. The theorem gives a terminating exhaustive-grid method when exact coefficient and order operations are available. It does not give a polynomial-time algorithm, a universal truncation depth, or a decision procedure for unrestricted surreal-number notation.

42 Exact verification and a formalization route

42.1 What the supplied program actually checks

The accompanying code/06-affine-residues-verify.py uses only the Python standard library. Its coefficient ring is

$$\mathbb{Q}[\eta, \eta^{-1}, \varepsilon, \varepsilon^{-1}],$$

with rational coefficients and lexicographically ordered exponent pairs. Finite sums in this ring embed into the surreal example $\eta = \omega^{-\omega}$, $\varepsilon = \omega^{-1}$. All sign tests use exact leading terms; no floating-point arithmetic is used. ([write] The program and its recorded output name $\eta = \omega^{-\omega}$ eta, for example "exponent_order": "Z^2 lexicographic, eta=(1,0), epsilon=(0,1)" and the key lambda_eta.)

For planar polytopes, the program computes convex hulls by exact orientation tests, areas by the shoelace formula, and mixed areas by polarizing the area of the pairwise-sum convex hull. It then compares these independently computed mixed areas with the maximum-simplex mixed-determinant certificate. Thus the audit does not define the mixed area by the very minimum formula it is checking.

The included report (data/06-affine-residues-verification.json) records the following run, using seed 20260930:

Audit category	Count
Random full-dimensional polygon pairs	160
Normalized residue checks from those pairs	320
Scale-lattice/probe equivalence comparisons	320
Exact common-affine-map covariance checks	24
Additional lower-dimensional or zero cases	4
Explicit rank-two mixed-area examples	2
Four-body simultaneous inner-segment families	20
Four-body tropical Plücker checks	20
Three-color families with six labelled clones	12
Individual M -convex exchange tests	396

The program also checks the thin triangle, the cancellation determinant, the pentagon with square residue, and a finite compression certificate. Its successful inner-segment choices are searched randomly within the proved rational grids, with a fixed finite audit budget; that convenience is distinguished in the source from the exhaustive-grid existence proof. The run used at most four trials for a four-body family and at most three for a six-clone family.

To reproduce the default report, run

```
python3 code/06-affine-residues-verify.py \
  --output /path/to/scratch/verification.json
```

[write] Manuscript 06 printed `python3 verify.py`. Without `-output` the program writes `verification.json` next to itself, that is, into `code/`; pass an output path outside the report directory, as above, so that the shipped record is never overwritten. The delivered `code/06-affine-residues-Makefile` still names the delivery paths `verify.py` and `article.tex`.) The output, shipped as `data/06-affine-residues-verification.json`, contains the exact example expressions, counts, and specialization data. For more instances, the command-line options `--pairs`, `--seed`, and `--output` are available. These are finite-instance tests, not a formal proof of the universal statements and not an exhaustive comparison with the literature.

42.2 A modular Lean development

[write] The formalization roadmap of manuscript 06 is printed in [section 83](#), together with its statement that the package contains no Lean source and reports no new Lean build.

43 Further research questions and topics

The following questions are proposed directions arising from the proved results. Their formulation here is not a claim that every item is a previously published open problem or that no partial answer exists in related literature. Each question isolates something the current theorems do not settle.

Question 43.1 (Intrinsic full-profile realization). Give a practical intrinsic characterization of the arrays that admit the color-symmetric matrix realization of [Corollary 40.1](#). Which additional conditions, beyond support rank constraints and M -convex exchange, express realizability over an ordered field? A useful first target is a complete analysis for three or four colors in dimension three, keeping the realization field explicit.

Question 43.2 (Leading coefficients above a fixed scale profile). Fix a realizable valuation profile τ . Describe the possible positive leading coefficients of the mixed volumes after choosing a compatible system of determinant scales. The simultaneous segment theorem determines the scales but not these coefficients. Can the coefficient problem be organized by real residue configurations attached to the lowest-valued minors, rather than by the entire original coordinate series?

Question 43.3 (Joint residue data for several lattices). A single affine residue is full-dimensional, but separate residue shapes do not determine mixed scales. Construct a canonical finite joint reduction for a family of embedded scale lattices that retains their relative position and supports leading-coefficient calculations. For higher-rank valuations, determine when a finite flag of successive reductions is sufficient and how its length depends on the family.

[write] Related material elsewhere in the report, which leaves the question open. Part II's research direction [section 28.5](#) (manuscript 02) asks for a polyhedral object that keeps the fibres, infinitesimal widths and near-optimality data discarded by the standard part; a joint reduction of several scale lattices would be one such object. Finite flags of successive reductions are proved to suffice in two narrower settings. For the rows of one halfspace system, Part IV's [theorem 49.2](#) needs at most $d + 1$ separated real levels per row. For the heights of a lifting of a fixed real configuration, Part I's [theorem 6.1](#) needs at most $n - d - 1$ real layers. Neither is a joint reduction of several scale lattices.]

Question 43.4 (Efficient deterministic inner-segment selection). The rational grid gives an explicit finite existence bound, but exhaustive search is generally large. Find a deterministic selection algorithm with substantially better complexity for the structured product of normalized determinant polynomials in the proof. Separate arithmetic-operation complexity from bit complexity of exponents and coefficients, and compare with the randomized exact search used in the supplement.

Question 43.5 (Minimal probe systems). Theorem 39.2 gives $d + 1$ measurements relative to a reference basis. Determine optimal bounds under various access models: known reference lattice, unknown basis, restricted directions, and uniform tests independent of the reference polytope. The distinction between arbitrary K -directions and real directions should be explicit.

Question 43.6 (Face specialization and oriented scale data). Characterize when reduction maps faces of P onto faces of $\text{Res}(P)$, and when vertices or entire face intervals disappear. Enrich the lattice and residue data by signed leading minors, and compare the resulting structure with valuated oriented-matroid and regular-subdivision data. A first useful result would be a finite certificate guaranteeing that all faces survive a chosen normalization.

[[write] *Re-scoped: partly answered in this report.* Part I's [proposition 14.2](#) (manuscript 02) shows, for every real functional c , that the standard part of a suitable infinitesimal optimality band $\{x \in P : c \cdot x \geq M - \eta\}$ is exactly the maximizing face of $\text{st } P$. So every exposed face of the reduction is recovered from near-optimal points of P , even when the exact maximizing face of P reduces to a nonface ((14.2), and the pentagon in [section 40](#)). Applied to the bounded normalized polytope $\tilde{P} = B^{-1}(P - p_0)$, whose standard part represents $\text{Res}(P)$ ([theorem 35.3](#)), the same identity describes the faces of the affine residue. Still open: when the faces of P themselves map onto faces of $\text{Res}(P)$, the signed-minor enrichment, the comparison with valuated oriented matroids, and the finite survival certificate. Part II's research direction [section 28.5](#) (manuscript 02) asks for the same kind of replacement for the false face map, and cites the same band identity as its starting point.]

Question 43.7 (Adaptive cancellation certificates). For coordinates supplied as truncatable infinite Hahn series, find hypotheses under which determinant-leading terms can be certified by an adaptive terminating procedure. The height- ε^N examples exclude a bound depending only on vertex count and dimension. The appropriate parameters may involve support geometry, coefficient dependencies, or an input-specific separation certificate rather than a valuation cutoff.

[[write] Part II's [theorem 26.3](#) (manuscript 02) sharpens the obstruction. It gives convergent rational power series coordinates for which the sign of the truncated defect $1 - f_{\leq N} g_{\leq N}$, which decides the truncated hull type, follows any prescribed pattern in N . So a procedure that stops once successive truncations agree for a while can be fooled. The question itself is not answered there.]

Question 43.8 (Dual geometric invariants). Develop mixed-volume and residue invariants involving polar bodies, widths, and successive directional scales. Simple cube comparisons already control coarse polar lattices in suitably centered situations; the harder target is joint leading-coefficient and incidence information under duality, especially for nonsymmetric bodies and families with different scale lattices.

Question 43.9 (Beyond finite vertex sets). Extend the theory to selected definable convex bodies or controlled limits of polytopes. A satisfactory formulation must first specify the volume object and its coefficient field: the finite determinant definition does not automatically extend to arbitrary semialgebraic volume. Determine which additional hypotheses retain

a finite inner-model theorem and which introduce genuinely infinite residue or support phenomena.

Question 43.10 (Formal certificates in the repository). Implement and independently check the maximum-simplex module theorem, the mixed determinant squeeze, and the simultaneous grid theorem as separate generic Lean modules. Connect finite-support computational certificates to those statements without assuming unrestricted comparison of surreal expressions. The concrete goal is an auditable end-to-end example in which a higher-rank mixed-volume profile is verified from vertex data and then instantiated in the constructed surreal field.

44 Conclusion

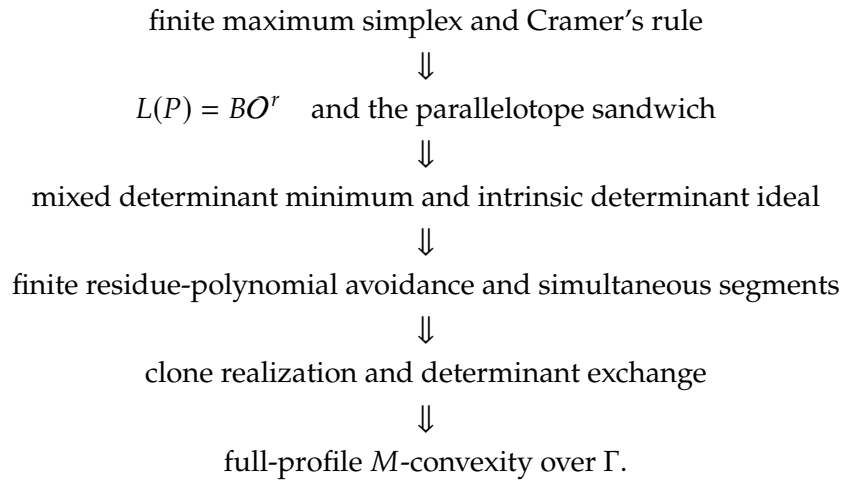
The finite face lattice alone does not distinguish real from surreal polytope geometry. The first meaningful extra layer is the embedded scale lattice; a maximum vertex simplex makes it explicit and supplies a full-dimensional affine residue. The scale of every mixed volume is then a finite determinant minimum.

The simultaneous inner-segment theorem adds a global compatibility fact: all pure mixed-volume scales of a finite family come from one matrix. The clone construction extends this compatibility to repeated entries and produces the full exchange axiom over arbitrary ordered value groups. These results give exact simplification and reconstruction statements, while the examples identify what they deliberately forget: residue coefficients, fine faces, and unrestricted higher-order information.

The resulting research program is finite and algebraic at its core. It can use the surreal field to accommodate very large hierarchies of scales without relying on class-sized measure theory or on a nonexistent universal real-valued parameter. Its next substantive challenge is to recover the coefficient and incidence geometry above those scale data, not merely to reprove the ordinary finite combinatorics over a larger field.

45 Proof dependencies and audit boundaries (Appendix A of manuscript 06)

The main scale results have the following dependency chain:



The affine residue and its leading-volume identity use the separate standard-part volume lemma. The probe classification uses the determinant ideal and the adjugate formula, not the

genericity argument. Finite one-parameter specialization is an additional effective result for finite-support coordinates; it is not needed for any existence theorem about arbitrary finite surreal coordinates.

Potential failure mode	How the statements handle it
Proper-class domain	All inputs and constructions are placed in a set-sized real closed subfield of $\mathbf{No}$.
Lower-dimensional body	Its module has its affine rank; a point argument makes the mixed volume zero. Support is decided by Corollary 38.6.
Infinite or infinitesimal size	Maximum-simplex coordinates put the body in a real bounded cube without changing its affine rank.
Cancellation	Determinants are computed exactly. Generic segment selection avoids the finitely many leading cancellations simultaneously.
Higher-rank value group	Proofs use finite comparisons and real residue polynomials, never cofinal multiples of a positive value.
Repeated arguments	One segment per original body is insufficient; parallelotope models or separately labelled clones are used.
Unjustified transfer	Only fixed finite polytope formulas and their semialgebraic determinant volumes are transferred.
Overstated effectiveness	Algorithms assume exact finite-support arithmetic. Arbitrary surreal comparison and polynomial-time bounds are not claimed.
Overstated novelty	Classical ingredients are credited. The combined results are research proofs supplied for review, not certified historical priority claims.

Part IV

Real traces and lattice-point counting

Source and scope. [write] This Part prints, whole, manuscript 03 of batch 58: *Infinitesimal Boundary Geometry of Surreal Polytopes: Lexicographic traces, one-parameter descent, and Ehrhart universality*, headed “Surreal geometry / lattice enumeration”, author line “Research manuscript prepared for Vladimir Reshetnikov” (the PDF metadata of the delivered file reads “Research manuscript prepared with ChatGPT”), dated September 30, 2026 (archive surreal_polytope_research.zip, main file surreal_polytopes/surreal_polytopes.tex; delivered PDF of 28 pages on Letter paper, not shipped). It was written against ProveIt commit 6996fee43. Every section, result, proof, example, remark, figure, table and research question of the manuscript is printed here, in the manuscript’s order, with three structural changes: its two appendices are printed as the ordinary sections 60 and 61; its subsection “A realistic formalization sequence” is printed in the formalization chapter (section 83.4); and its bibliography is printed as the list “References of Part IV”. This Part is its own spine (section 1): it shares no theorem with the other Parts beyond two foundations, both marked where they occur. These are the shadow of a finite hull (lemma 47.2, proved in Part I as proposition 13.1) and finite real-closed-field transfer (proposition 48.1; Part I’s theorem 18.1 and Part V’s theorem 74.3). The Part keeps its own definitions and hypotheses, fixed in section 47. In this Part “the article” and “this manuscript” mean manuscript 03. As in Part III, every theorem is stated for an arbitrary set-sized real closed field $K \supseteq \mathbb{R}$ with no rank or valuation hypothesis; the surreal field supplies the examples and, through bounded omnific

integers, the reason for counting ordinary lattice points.

Notation in this Part. [write] The following symbols mean something different elsewhere in the report (section 1.3); the Part-local meaning is the one in force here. No symbol of manuscript 03 has been renamed.

- O_K and $\mathfrak{m}_K$ are the finite elements and the infinitesimals: the same ring and ideal as O_K (Parts I–II) and $O, \mathfrak{m}$ (Parts III and V). st is the standard part, and the *shadow* $Q = \text{st}(P)$ is the standard part of the polytope, as in Part I.
- The *real trace* $S = \text{Tr}_{\mathbb{R}}(P) = P \cap \mathbb{R}^d$ (printed $\text{Tr}_{\mathbb{R}}$) is the set of real points of P . It is *not* the shadow: it can be empty or non-closed while the shadow is a full polytope (example 47.4); in general $\text{int } Q \subseteq S \subseteq Q$ only for a full-dimensional shadow. Nor is it Part V’s matrix trace tr , or Part III’s affine residue Res .
- $L_P(n) = \#(nP \cap \mathbb{Z}^d)$ counts lattice points; it is unrelated to Part III’s scale lattice $L(P)$ and to Part I’s L_h . $\mathbf{Oz}$ is the ring of omnific integers.
- ε and δ denote an *arbitrary* positive infinitesimal of K , not necessarily ω^{-1} ; t is a real parameter tending to 0^+ , or the positive infinitesimal of the ordered field $\mathbb{R}(t)$ (not Part II’s $t = \omega^{-1}$); $\eta > 0$ in theorem 50.1 is a *real* cutoff for that parameter (not Part I’s sign threshold, not $\tau = \omega^{-\omega}$).
- $\rho_1 \gg \cdots \gg \rho_r$ is a separated basis of positive elements of a real subspace of K (lemma 49.1); it is not Part V’s monomial ρ or Part III’s threshold ρ_g .
- m is the number of rows in theorem 49.2, a period in the proof of theorem 52.1, and a nonsquare integer in section 57 (not Part III’s number of bodies); S, T, G, Q, B, E, H are the trace, an implanted real polytope, a face, the shadow, a bounding box, a realization space or generating function, and a halfspace intersection or generating function (not Part V’s slack matrix S or triangle T); $A(Q)$ is the lattice-normalized boundary measure and μ_F the normalized facet measure; V_i are coefficient spaces (not Part I’s dependence space); $\beta(P)$ is the surface coefficient.
- vol_d is Euclidean volume of real sets; on real polytopes it agrees with Part III’s determinant volume Vol_d .
- No valuation is used in this Part (see the remark after theorem 50.1).

Abstract of manuscript 03. A finite surreal polytope has an ordinary realizable face lattice, but its infinitesimal boundary placement can carry arithmetic information invisible in that lattice and in its standard-part shadow. We give a finite lexicographic description of its real trace, with at most $d + 1$ levels per inequality in dimension d . Combining this description with semialgebraic curve selection gives a one-parameter algebraic realization preserving the entire real trace, the shadow, the labeled face lattice, and every ordinary lattice-counting value. We then construct a face-implantation operation that imports arbitrary lower-dimensional real-polytope counting functions into a fixed rational shadow. For a lattice shadow, we prove an explicit second asymptotic coefficient and determine its full attainable interval. If the shadow is simple, every value in this interval is attained without changing the labeled face lattice or any vertex standard part. Finally, a quadrilateral with the four integral vertex shadows of the unit square has counting function $(n + 1)^2 - \lceil \alpha n \rceil$ for any irrational $0 < \alpha < 1$; its generating function has the unit circle as a natural boundary and is not D-finite.

Status and scope (title page of manuscript 03). This manuscript supplies mathematical proofs and exact finite checks. It has not been independently refereed or formalized in Lean. Its classical inputs are explicitly attributed; priority for the assembled results is not certified. No named published open conjecture is claimed solved. All counting statements concern ordinary integer dilations of polytopes bounded by a real box.

46 The research questions and their answers

The relevant starting point in ProveIt is its construction of the surreal field and its omnific-integer subring, together with its distinction between research manuscripts and mapped formal theorems [26, 27]. The particularly useful bridge is that an omnific integer bounded in absolute value by a real number is an ordinary integer. The repository contains an explicit Lean declaration of this statement [28]. Consequently, finite polytopes in bounded surreal space lead to a precise, ordinary lattice-enumeration problem. Their coordinates may be non-Archimedean; the lattice points being counted are not.

The organizing question is not whether arbitrary finite combinatorics becomes possible over the surreals. Real-closed-field transfer already rules that out for finite face lattices. Rather, we ask what information survives when one retains the real numbers and their integer lattice while moving supporting hyperplanes through infinitesimal distances.

46.1 Four questions posed and resolved here

Does the shadow determine lattice enumeration? No. Even a surreal quadrilateral with exactly the four integral vertex shadows of the unit square can have non-quasipolynomial counts. Its generating function can have infinitely many analytic obstructions in the strongest possible form: every point of the unit circle is a singular boundary point. The construction and proof appear in Section 52.

Can arbitrarily deep surreal scales be necessary for the real trace? Not for a finite system in fixed dimension. An individual affine inequality in d coordinates needs at most $d + 1$ lexicographic levels on real inputs. A stronger global theorem replaces the whole polytope by one algebraic parameter while preserving its trace and finite combinatorics simultaneously; see Sections 49–50. This is a statement about a specified finite geometric object, not about collapsing the surreal field or its full value group.

How much counting behavior can be hidden on a face? Every real polytope lying in that face can be implanted. If G is a proper face of a rational polytope Q and $T \subseteq G$ is any real polytope, we construct P with shadow Q and

$$L_P(n) = L_Q(n) - L_G(n) + L_T(n) \quad (n \geq 1).$$

The last summand need not be an Ehrhart quasipolynomial. This isolates a universality mechanism rather than merely an isolated counterexample.

What remains of the first two Ehrhart coefficients? For a full-dimensional lattice shadow Q , the leading coefficient is always $\text{vol}_d(Q)$. The next coefficient is the sum of the normalized measures of the portions of facets retained in the real trace, minus half the normalized

boundary measure of Q . Its complete attainable interval is

$$\left[-\frac{1}{2} \sum_F \mu_F(F), \frac{1}{2} \sum_F \mu_F(F) \right].$$

For a *simple* lattice shadow every value is attainable by perturbing its existing facets, with no change in the labeled face lattice and no change in any vertex standard part. Sections 53 and 54 prove these statements.

46.2 What is classical, and what is being assembled

The real-closed-field transfer principle and semialgebraic curve selection are classical [30, 31, 32]. Lexicographic inequalities are also established objects in optimization [33]. Classical Ehrhart theory and reciprocity apply to rational polytopes [34]. Irrationality does not by itself preclude polynomial enumeration: period collapse already occurs for irrational triangles [35]. Finally, the natural-boundary step below is an application of the classical Pólya–Carlson theorem, not a new complex-analytic theorem [36].

The contributions developed here are the finite trace normal form and its trace-preserving descent application, the face-implantation construction, the surface-coefficient formula and its sharp realization theorem, and the explicit connection of these statements to bounded surreal and omnific geometry. Their proofs are given in full, except for the clearly named classical inputs. These results may overlap consequences of existing non-Archimedean convexity or real-algebraic geometry; the literature checks recorded here are not an exhaustive priority search.

47 Fields, finite polytopes, shadows, and traces

47.1 A set-sized setting

Throughout, K is a set-sized real closed ordered field containing $\mathbb{R}$ as an ordered subfield. When an infinitesimal is needed we suppose that $0 < \varepsilon < r$ for every $r \in \mathbb{R}_{>0}$. Define

$$\mathcal{O}_K = \{x \in K : |x| \leq r \text{ for some } r \in \mathbb{R}_{>0}\}, \quad \mathfrak{m}_K = \{x \in K : |x| < r \text{ for every } r \in \mathbb{R}_{>0}\}.$$

Completeness of $\mathbb{R}$ gives a unique standard part $\text{st}(x) \in \mathbb{R}$ for $x \in \mathcal{O}_K$, characterized by $x - \text{st}(x) \in \mathfrak{m}_K$. To see existence, take the supremum of the real numbers below x ; the defining property of the supremum excludes a positive real distance in either direction. Uniqueness follows because no nonzero real number is infinitesimal. Directly from this characterization,

$$\text{st}(x + y) = \text{st}(x) + \text{st}(y), \quad \text{st}(xy) = \text{st}(x) \text{st}(y),$$

and $x \geq 0$ implies $\text{st}(x) \geq 0$. Thus $\mathcal{O}_K$ is a convex subring, $\mathfrak{m}_K$ is its maximal ideal, and its residue field is $\mathbb{R}$.

A finite tuple of surreal coordinates can be placed in such a field: take inside **No** the real closure of the field generated by $\mathbb{R}$ and that tuple. This uses the classical real closedness of the surreal field [29]. All constructions below take place in this set-sized field. No application of first-order model theory treats the proper class **No** as a set-sized structure, and no saturation hypothesis is used.

Definition 47.1. A *finite K -polytope* is the convex hull of finitely many points of K^d , with convex combinations using nonnegative coefficients in K . It is *real-bounded* if it is contained in $[-M, M]^d$ for some $M \in \mathbb{R}_{>0}$. For such a nonempty polytope P , its *shadow* and *real trace* are

$$Q = \text{st}(P) = \{\text{st}(x) : x \in P\} \subseteq \mathbb{R}^d, \quad S = \text{Tr}_{\mathbb{R}}(P) = P \cap \mathbb{R}^d.$$

Standard part is applied coordinatewise.

Finite-dimensional polyhedral linear algebra works over real closed fields. In particular a finite convex hull has a finite halfspace representation, allowing pairs of opposite inequalities to encode its affine hull, and a nonempty bounded intersection of finitely many closed halfspaces is a finite convex hull. These statements can be obtained from finite elimination, or from their fixed-size first-order real versions by transfer [30]. We never invoke topological compactness of a bounded subset of K^d .

Lemma 47.2 (The shadow of a finite convex hull). *If $P = \text{conv}_K(v_1, \dots, v_N)$ is real-bounded, then*

$$\text{st}(P) = \text{conv}_{\mathbb{R}}(\text{st}(v_1), \dots, \text{st}(v_N)).$$

In particular, its shadow is an ordinary closed real polytope.

Proof. If $x = \sum_j \lambda_j v_j$ with $\lambda_j \geq 0$ and $\sum_j \lambda_j = 1$, then every λ_j is finite and its standard part is nonnegative. Taking standard parts produces a real convex combination. Conversely, a real convex combination of the shadow vertices is the standard part of the same real combination of their lifts. $\square$

[[write] This is Part I's [proposition 13.1](#), where it is proved (manuscripts 05 and 02), and the first clause of Part III's [lemma 34.2](#). Manuscript 03's statement and proof are kept because the rest of this Part cites them.]

Lemma 47.3 (Interior points survive). *If $Q = \text{st}(P)$ has dimension d , then*

$$\text{int } Q \subseteq S \subseteq Q, \quad \text{cl}_{\mathbb{R}} S = Q.$$

Proof. The second inclusion is immediate because $\text{st}(x) = x$ for real x . For $x \in \text{int } Q$, choose a real d -simplex in Q containing x in its interior. Its vertices $q_0, \dots, q_d$ have lifts $p_0, \dots, p_d \in P$ by Lemma 47.2. The matrix whose columns are $(1, p_j)$ has determinant with nonzero standard part. It is therefore invertible over K . The solution of

$$\sum_{j=0}^d \lambda_j p_j = x, \quad \sum_{j=0}^d \lambda_j = 1$$

has standard part equal to the strictly positive barycentric coordinates of x in the real simplex. Hence every $\lambda_j > 0$, so $x \in P$. Taking closures gives the last assertion. $\square$

Example 47.4 (A shadow is not a trace). The segment $\{(x, \varepsilon) : 0 \leq x \leq 1\}$ has shadow $[0, 1] \times \{0\}$ and empty real trace. Thus the dimension assumption in Lemma 47.3 cannot be dropped. Even for full-dimensional shadows the trace need not be closed: $[\varepsilon, 1 - \varepsilon]$ has trace $(0, 1)$ and shadow $[0, 1]$.

47.2 Ordinary and omnific lattice points

Conway's omnific integers are the surreal normal forms with no negative exponents and with an ordinary integer constant coefficient [29]. A nonconstant such form has a nonzero leading term of positive exponent, so it is not bounded in absolute value by a real number. It follows that

$$\mathbf{Oz} \cap \mathbf{O}_{\mathbf{No}} = \mathbb{Z}. \quad (47.1)$$

This is also the content of the repository theorem `omnific_bounded_iff`; Section 56 records the exact path and the verification boundary [28].

[[write] `omnific_bounded_iff` is the one Lean declaration that any Part of this report uses as an input; the other Parts cite Lean declarations at most as background. It states that an omnific integer whose surreal value is bounded in absolute value by a real number is the cast of an ordinary integer. Re-verified at the write phase: it is at line 65 of `Algebra/SurrealNumbers/Surreal/Foundations/OmnificUnits.lean`, in namespace `Surreal.Foundations.SignSequence`, next to `omnific_isFinite_iff` (line 55) and `omnific_eq_intConstant_of_finite` (line 42), and the file is unchanged since the pin. Manuscript 03 also proves (47.1) directly from normal forms, as above. *None of this Part's own results is formalized*: the Lean statement covers only the bridge (47.1), not the trace, descent, implantation, counting or surface-coefficient theorems.]

For any real-bounded K -polytope and an *ordinary* positive integer n , define $L_P(n) = \#(nP \cap \mathbb{Z}^d)$. In the surreal case,

$$L_P(n) = \#(nP \cap \mathbf{Oz}^d) = \#(nP \cap \mathbb{Z}^d) = \#(nS \cap \mathbb{Z}^d). \quad (47.2)$$

The omnific and integer sets agree coordinatewise by (47.1); the last equality holds over any K because $x/n \in \mathbb{R}^d$ for real x . The counts are finite, and we set $L_P(0) = 1$ for nonempty P . For a real polytope or bounded real set T , write $L_T(n) = \#(nT \cap \mathbb{Z}^d)$ when the ambient space is understood.

Equation (47.2) is a reduction to the *real trace*, not to the shadow. It also explains why the integer-part structure does not add new bounded lattice points. Infinite omnific dilations, infinite-coordinate polytopes, and counts indexed by a proper class are outside our scope.

48 The classical limit: no new finite face lattices

Proposition 48.1 (Finite combinatorial transfer). *Every finite polytope over a real closed ordered field has a realization over the real algebraic numbers with the same labeled vertex–facet incidences, and consequently the same face lattice. Real boundedness is not required.*

Proof. Choose a finite vertex list and a finite inequality description, including affine-hull equations when necessary. With their sizes fixed, the following requirements form a finite first-order formula in the language of ordered fields: the convex hull equals the halfspace intersection; the listed points are pairwise distinct vertices; and each specified vertex–row slack is zero or strictly positive according to the original incidence data. Convex-hull membership is expressed with finitely many existential barycentric coordinates. Extremality is expressed by exclusion from the convex hull of the other listed vertices. H/V equality is expressed by quantifying a point of the fixed ambient dimension. None of these formulas names a nonrational parameter: all coordinates and coefficients are variables.

The formula has a solution in the given real closed field. Completeness of the theory of real closed fields gives a solution in the real algebraic numbers [30]. Every face of a polytope

is an intersection of its supporting facets, so the preserved incidence data determine the labeled face lattice. Equivalently one can transfer the finite determinant sign pattern of a homogeneous vertex configuration. $\square$

This observation prevents a misleading research claim: finite surreal polytopes do not create previously unrealizable finite face lattices. It does not say that all face lattices have rational realizations, that given coordinates descend, or that standard parts and valuations are preserved. Most importantly, quantification over $\mathbb{Z}$ is not part of the pure ordered-field language. Proposition 48.1 gives no transfer theorem for lattice-counting functions.

[[write] Proposition 48.1 is the finite real-closed-field transfer that Part I proves as theorem 18.1 (manuscripts 05 and 02); Part III's proposition 33.3 is another instance. Part V's theorem 74.3 (manuscript 07) strengthens it for polytopes over a real closed field: a real-algebraic realization can keep the oriented matroid and the linear and semidefinite extension complexities as well. The limitations just stated apply to all of these: none transfers standard parts, valuations or lattice counts.]

The rest of this article retains exactly the data discarded by that proposition: the distinguished real subfield, its standard-part map, and the boundary trace on the ordinary integer lattice.

49 A finite lexicographic normal form for real traces

For positive $a, b \in K$, write $a \gg b$ if b/a is infinitesimal. For a finite tuple of real numbers, lexicographic nonnegativity means either that every entry is zero or that its first nonzero entry is positive.

Lemma 49.1 (Separated real basis). *Let $V \subseteq K$ be a nonzero real vector space of finite dimension r . There is a basis of positive elements*

$$\rho_1 \gg \rho_2 \gg \cdots \gg \rho_r > 0$$

such that, for all $c_1, \dots, c_r \in \mathbb{R}$,

$$\operatorname{sgn} \left(\sum_{j=1}^r c_j \rho_j \right) = \operatorname{sgn}(c_k)$$

when k is the first index with $c_k \neq 0$.

Proof. Choose a real basis $v_1, \dots, v_r$ and let ρ_1 be the largest of $|v_1|, \dots, |v_r|$. It belongs to V , is positive, and every v/ρ_1 with $v \in V$ is finite: the coefficients of a fixed finite real linear combination give a real bound. Therefore

$$c_1 : V \longrightarrow \mathbb{R}, \quad c_1(v) = \operatorname{st}(v/\rho_1)$$

is real linear and takes ρ_1 to 1. Its kernel has dimension $r - 1$, and every element of the kernel is infinitesimal relative to ρ_1 . Moreover,

$$V = \mathbb{R}\rho_1 \oplus \ker c_1.$$

Apply the same argument inside the kernel. Induction gives an exact basis expansion after r steps. Each later basis element divided by an earlier one is infinitesimal. After dividing a nonzero combination by its first active ρ_k , its leading coefficient is a nonzero real number and the remaining finite sum is infinitesimal. This proves the sign rule. $\square$

This argument uses exact finite-dimensional linear algebra. It does not truncate a Hahn series, assume that its support is finite, or require that the value group have finite rank.

Theorem 49.2 (Lexicographic trace theorem). *Let*

$$P = \{x \in K^d : b_i - a_i \cdot x \geq 0, 1 \leq i \leq m\}$$

be any finite halfspace system. For every nonzero row there exist real affine forms $\ell_{i1}, \dots, \ell_{ir_i}$, where $r_i \leq d + 1$, such that

$$P \cap \mathbb{R}^d = \{x \in \mathbb{R}^d : (\ell_{i1}(x), \dots, \ell_{ir_i}(x)) \geq_{\text{lex}} 0 \text{ for all } i\}. \quad (49.1)$$

More precisely, there are positive separated scales ρ_{ik} with

$$b_i - a_i \cdot x = \sum_{k=1}^{r_i} \rho_{ik} \ell_{ik}(x) \quad (x \in K^d). \quad (49.2)$$

Zero rows may be deleted.

Proof. Apply Lemma 49.1 to

$$V_i = \text{span}_{\mathbb{R}} \{b_i, a_{i1}, \dots, a_{id}\}.$$

Its dimension is at most $d + 1$. Expand each coefficient in the separated basis and collect the real affine forms multiplying the basis elements. This gives the exact identity (49.2). When x is real, all $\ell_{ik}(x)$ are real, so the sign rule is precisely (49.1). $\square$

Corollary 49.3 (Finite semilinear structure). *The real trace of a real-bounded finite K -polytope is a convex semilinear set. It is a finite disjoint union of relatively open real polytopes.*

Proof. A lexicographic condition is a finite union of conditions of the form

$$\ell_1 = \dots = \ell_{k-1} = 0, \quad \ell_k > 0,$$

together with the all-zero condition. Thus it is semilinear. Subdivide a real bounding box by all the affine hyperplanes in the description and by its boundary hyperplanes. Each relatively open cell has a constant truth value for the trace formula. The selected cells give the stated finite disjoint union; boundedness makes their closures polytopes. Convexity follows from convexity of P and the real subfield. $\square$

Example 49.4 (Why leading standard parts are insufficient). The inequality

$$1 - y + \varepsilon(x - \alpha) \geq 0$$

on real (x, y) first tests $1 - y$. Only on the line $y = 1$ does it test $x - \alpha$. Taking the standard part of its coefficients loses the second test and incorrectly keeps the entire line segment on that boundary.

49.1 Polynomial compression, without a combinatorial guarantee

Corollary 49.5 (A degree- d one-infinitesimal trace description). *Suppose P is real-bounded and its shadow is full-dimensional. Fix a real bounding box B containing P . For any positive infinitesimal δ , set*

$$P_{\text{flat}} = \left\{ x \in B_K : \sum_{k=1}^{r_i} \delta^{k-1} \ell_{ik}(x) \geq 0 \text{ for all } i \right\}.$$

Then P_{flat} is a nonempty real-bounded polytope with the same real trace and shadow as P . Its row coefficients are polynomials in δ of degree at most d . In the surreal setting all its ordinary counting values agree with those of P .

Proof. On real inputs powers of δ implement exactly the same lexicographic sign rule. Thus the traces agree. The common trace contains $\text{int } Q$ by Lemma 47.3, so P_{flat} is nonempty and its shadow is full-dimensional. Applying that lemma to both polytopes shows that both shadows are the closure of the common trace. Equation (47.2) proves the counting claim. $\square$

This compression need not preserve the original face lattice. Different rows may interact through their relative infinitesimal scales at nonreal points. The next theorem preserves those interactions by selecting an algebraic curve in an appropriate realization space rather than replacing each row independently.

50 One-parameter algebraic descent preserving the entire trace

An *algebraic semialgebraic germ* below is a germ at $t = 0^+$ of a real semialgebraic function algebraic over $\mathbb{R}(t)$. It can be chosen Nash on a sufficiently small positive interval. The coefficients of its defining polynomial may be arbitrary real numbers; “algebraic” here does not mean “defined over $\mathbb{Q}$.”

Theorem 50.1 (Trace-preserving algebraic descent). *Let $P \subseteq K^d$ be a nonempty real-bounded finite polytope, with its distinct vertices labeled $v_1, \dots, v_N$. There exist $\eta > 0$ and a family of ordinary real polytopes P_t , $0 < t < \eta$, whose labeled vertices and inequality coefficients are algebraic semialgebraic germs, such that:*

- (i) *every P_t has the same labeled vertex–inequality incidences and face lattice as P ;*
- (ii) *its labeled vertices $v_j(t)$ tend to $\text{st}(v_j)$;*
- (iii) *for every fixed real point x , membership $x \in P_t$ is eventually constant and equals membership $x \in \text{Tr}_{\mathbb{R}}(P)$.*

Evaluating these germs at a positive infinitesimal gives a polytope P^ in a real closure of $\mathbb{R}(t)$, embedded in a real closed extension, with exactly the same real trace, shadow, and labeled face lattice as P . All ordinary lattice counts agree. For every fixed ordinary $n \geq 1$,*

$$L_{P_t}(n) = L_P(n)$$

for all sufficiently small positive real t , where the cutoff may depend on n . No full-dimensional-shadow hypothesis is needed.

Proof. We give the finite encoding because preserving all real points at once is stronger than preserving a finite determinant sign list.

Step 1: encode every row with adjacent ratios. Use the exact expansion (49.2) and divide row i by its positive ρ_{i1} . Introduce one variable u_{ik} for each adjacent ratio, and write

$$f_i(u, x) = \ell_{i1}(x) + u_{i1}\ell_{i2}(x) + u_{i1}u_{i2}\ell_{i3}(x) + \cdots + (u_{i1} \cdots u_{i,r_i-1})\ell_{ir_i}(x). \quad (50.1)$$

The original ratio tuple $u_{ik}^0 = \rho_{i,k+1}/\rho_{ik}$ consists of positive infinitesimals. Substituting it recovers the original normalized rows exactly. Choose a closed real box B containing P strictly inside its boundary.

Step 2: construct a semialgebraic realization space. Let $z = (z_1, \dots, z_N)$ be variable points. Let E be the set of real pairs (u, z) satisfying the following finite first-order conditions: $0 < u_{ik} < 1$; all $z_j \in B$; the polyhedron

$$H(u) = \{x \in B : f_i(u, x) \geq 0 \text{ for every } i\}$$

equals $\text{conv}(z_1, \dots, z_N)$; the z_j are distinct and each lies outside the convex hull of the others; and each slack $f_i(u, z_j)$ is zero or strictly positive according to the original row-vertex table. The box inequalities can be included in this table as well.

H/V equality and exclusion from a finite convex hull have first-order formulas with finitely many barycentric variables. Quantifier elimination therefore makes E a semialgebraic set over $\mathbb{R}$ [30]. The same formula interpreted over K contains the original point (u^0, v) .

Step 3: approach the standard-part endpoint over the reals. Put $a = (0, \text{st}(v_1), \dots, \text{st}(v_N))$. For every real $r > 0$, the point (u^0, v) lies in the radius- r ball about a . Hence the existential sentence asserting a point of E in that ball is true over K and, by elementary transfer over the named real parameters, over $\mathbb{R}$. Therefore $a \in \text{cl } E$.

When there are ratio variables, $a \notin E$ because they must be positive. Semialgebraic Nash curve selection gives a curve

$$(u(t), z(t)) \in E, \quad (u(t), z(t)) \longrightarrow a \quad (t \downarrow 0),$$

with algebraic semialgebraic coordinates [31, 32]. If there are no ratio variables, all rows are positive multiples of real rows, the polytope and its vertices are already real, and the constant family suffices. Define $P_t = H(u(t))$.

Step 4: preserve geometry and the whole trace. By the definition of E , the labeled vertices and all their row incidences are unchanged for every sufficiently small positive t . These data preserve the face lattice, including when the shadow loses dimension. The limit of the vertex list gives (ii).

Fix any real x . For a given row, suppose k is the first index with $\ell_{ik}(x) \neq 0$. Divide (50.1) by the positive product $u_{i1}(t) \cdots u_{i,k-1}(t)$. Its value tends to the nonzero real number $\ell_{ik}(x)$, since every later term contains a product of ratios tending to zero. If all the forms vanish, the row vanishes identically. Thus each row eventually has exactly its original lexicographic truth value at x . There are only finitely many rows, which proves (iii). Points outside the fixed box are excluded throughout.

Step 5: evaluate at an infinitesimal. The ordered field $\mathbb{R}(t)$ with t positive infinitesimal embeds by $t \mapsto \delta$ into any real closed extension containing such a δ . Here δ is transcendental over the real closed field $\mathbb{R}$. The embedding extends to a real closure; the selected real algebraic germs are elements of that ordered real closure. Semialgebraic eventual identities

and inequalities are preserved by this evaluation. In particular the defining formula for E remains true, and the row argument now gives exact membership agreement for every real x . Vertex limits become standard parts, proving equality of shadows by Lemma 47.2. Equation (47.2) gives the counting claim.

Finally $P_t \subseteq B$ for all small t . For fixed n there are only finitely many candidates in $nB \cap \mathbb{Z}^d$. Apply (iii) to their quotients by n and take the minimum of finitely many positive cutoffs. This proves the last assertion without asserting a cutoff uniform in n . $\square$

Remark 50.2 (What descent does not preserve). The theorem does not preserve arbitrary valuations of all rational expressions, the original field embedding, or the infinite hierarchy of scales outside the finite description. It supplies algebraic germs, not necessarily polynomial coordinate functions or an effective degree bound. The finite degree bound of Corollary 49.5 concerns a weaker, trace-only construction. Confusing these two conclusions would erase a substantive distinction.

Remark 50.3 ([write] Descent and Part V). The polytope P^* of theorem 50.1 lies in a real closure of $\mathbb{R}(t)$, whose natural value group is $\mathbb{Q}$, of rank one. There every positive infinitesimal has an order-unit value, so Part V's flat ideal of beyond-all-orders elements (its jet kernel) is zero (theorem 72.2, manuscript 07; theorem 72.7, manuscript 04). Finite coordinate tuples of P^* are determined by their power jets, and the invisible-complexity phenomena of Part V cannot occur for P^* . This does not conflict with Part V. As the preceding remark says, descent keeps the real trace, the shadow, the labeled face lattice and the lattice counts, and it expressly does not keep valuations, the field embedding or the hierarchy of scales. Manuscript 03 uses no valuation at all, and claims nothing about the jets or the extension complexity of P .

Corollary 50.4. *Every ordinary lattice-counting sequence of a real-bounded finite surreal polytope is already the sequence of a polytope over a real closure of a one-infinitesimal extension $\mathbb{R}(t)$, with the same labeled finite combinatorics and the same shadow.*

This is a reduction of the field needed for a finite observation. It does not turn the sequence into an ordinary Ehrhart quasipolynomial: the next sections show why one infinitesimal is already sufficient for substantial arithmetic complexity.

51 Face implantation and boundary universality

Let $Q \subseteq \mathbb{R}^d$ be a full-dimensional rational polytope and G a nonempty proper face. Choose a real affine function s_G nonnegative on Q whose zero set there is exactly G . Such a function can be obtained by adding the nonnegative facet slacks of the facets whose intersection is G . For a real polytope $T \subseteq G$, choose finitely many real affine functions $h_1, \dots, h_r$ on $\mathbb{R}^d$ such that

$$T = \{x \in G : h_j(x) \geq 0 \text{ for all } j\}.$$

An inequality description on $\text{aff } G$ extends to the ambient space by linear algebra. Lower-dimensional T is allowed.

Theorem 51.1 (Face implantation). *For a positive infinitesimal ε , set*

$$P_\varepsilon = \{x \in Q_K : s_G(x) + \varepsilon h_j(x) \geq 0 \text{ for } 1 \leq j \leq r\}. \quad (51.1)$$

Then P_ε is a nonempty real-bounded polytope with

$$\text{st}(P_\varepsilon) = Q, \quad \text{Tr}_{\mathbb{R}}(P_\varepsilon) = (Q \setminus G) \cup T. \quad (51.2)$$

For every ordinary $n \geq 1$,

$$L_{P_\varepsilon}(n) = L_Q(n) - L_G(n) + L_T(n). \quad (51.3)$$

The empty choice $T = \emptyset$ is also possible by taking the single function $h_1 = -1$.

Proof. For a real $x \in Q \setminus G$, the positive real number $s_G(x)$ dominates each infinitesimal $\varepsilon h_j(x)$, so all the added inequalities hold. For real $x \in G$, they hold exactly when all $h_j(x) \geq 0$. This proves the trace identity. In particular the trace contains $\text{int } Q$, so P_ε is nonempty. It is a bounded halfspace intersection, hence a polytope.

Its shadow is contained in Q because $P_\varepsilon \subseteq Q_K$. It contains $\text{int } Q$ and is closed by Lemma 47.2; hence it contains Q . The shadow identity follows. For $n > 0$, dilation is injective and the trace identity is a disjoint replacement of G by T . Counting lattice points gives (51.3). The argument is unchanged when T is empty. $\square$

Corollary 51.2 (A fixed-cube universality formula). *For every real polytope $T \subseteq [0, 1]^{d-1}$, there is a real-bounded K -polytope with unit-cube shadow and*

$$L_P(n) = n(n+1)^{d-1} + L_T(n) \quad (n \geq 1). \quad (51.4)$$

Proof. Implant $T \times \{1\}$ on the top facet of $[0, 1]^d$. Then $L_Q(n) = (n+1)^d$ and $L_G(n) = (n+1)^{d-1}$. $\square$

A boundary perturbation can therefore contain the entire counting sequence of an arbitrary real polytope of one lower dimension, modulo the explicit polynomial in (51.4). Rationality of the shadow alone cannot control quasipolynomiality, generating-function rationality, or recurrence properties of the resulting sequence. This is a structural explanation of the counterexamples, not a claim that irregular enumeration of irrational polytopes is a newly discovered phenomenon.

Proposition 51.3 (Localization by face dimension). *If the implanted face has dimension k , then*

$$L_{P_\varepsilon}(n) - L_Q(n) = O(n^k).$$

Thus implantation on a face of codimension c cannot change terms of order strictly higher than n^{d-c} .

Proof. Both nG and nT lie in a fixed k -dimensional affine space dilated by n , and in a ball of radius $O(n)$ within it. Their distinct integer points are at least unit distance apart. Packing disjoint balls of fixed radius in that affine space gives $O(n^k)$ points. Apply (51.3). For $k = 0$ the assertion simply says $O(1)$. $\square$

52 An integral-shadow quadrilateral with a natural boundary

The general implantation formula already gives a pentagonal example by cutting a thin wedge from a square. A more economical construction perturbs one existing facet and uses only four vertices.

Theorem 52.1 (The irrational boundary quadrilateral). *Fix an irrational real number $0 < \alpha < 1$ and a positive infinitesimal ε . Define*

$$P_{\alpha, \varepsilon} = \{(x, y) \in K^2 : 0 \leq x \leq 1, \quad 0 \leq y \leq 1 + \varepsilon(x - \alpha)\}. \quad (52.1)$$

It is a quadrilateral with cyclically ordered vertices

$$(0, 0), \quad (1, 0), \quad (1, 1 + \varepsilon(1 - \alpha)), \quad (0, 1 - \varepsilon\alpha). \quad (52.2)$$

Its shadow is $[0, 1]^2$, every vertex shadow is integral, and its labeled face lattice is that of the square. Nevertheless,

$$L_{P_{\alpha,\varepsilon}}(n) = (n+1)^2 - \lceil \alpha n \rceil \quad (n \geq 0). \quad (52.3)$$

For $n \geq 1$ this is

$$L_{P_{\alpha,\varepsilon}}(n) = n^2 + (2 - \alpha)n + \{\alpha n\}. \quad (52.4)$$

The sequence is not eventually quasipolynomial. Its ordinary generating function has the unit circle as a natural boundary; it is not D -finite, and the sequence is not P -recursive.

Proof. The upper bounds on the two vertical sides in (52.1) are positive. The four intersections in (52.2) are the four vertices, with the stated incidences. Their standard parts prove the shadow assertion. On real points the upper inequality tests $1 - y$ first and $x - \alpha$ only when $y = 1$. Thus

$$\mathrm{Tr}_{\mathbb{R}}(P_{\alpha,\varepsilon}) = ([0, 1] \times [0, 1)) \cup ([\alpha, 1] \times \{1\}). \quad (52.5)$$

For a positive ordinary n , the admissible integer columns are $x = 0, 1, \dots, n$. The rows $y = 0, 1, \dots, n - 1$ occur in every column. At $y = n$, the last inequality is $\varepsilon(x - \alpha n) \geq 0$, so the top row retains exactly $n + 1 - \lceil \alpha n \rceil$ points. There are no higher integer rows. Hence

$$L(n) = n(n+1) + n + 1 - \lceil \alpha n \rceil.$$

The convention at $n = 0$ agrees with (52.3). Since αn is nonintegral for $n > 0$, expanding the ceiling gives (52.4).

Suppose the sequence were eventually quasipolynomial with period m . On each progression $n \equiv r \pmod{m}$ it would eventually equal a polynomial $p_r(n)$. By (52.4), the polynomial $p_r(n) - n^2 - (2 - \alpha)n$ is bounded on that progression and so is constant. Consequently $\{\alpha(n+m)\} = \{\alpha n\}$ for sufficiently large n in the progression. This would imply $\alpha m \in \mathbb{Z}$, a contradiction. The argument even excludes constituent polynomials with arbitrary real coefficients.

Write $a_n = \lceil \alpha n \rceil$ for $n \geq 0$, and $w_n = a_{n+1} - a_n$. Then $w_n \in \{0, 1\}$ and

$$\frac{1}{N} \sum_{n=0}^{N-1} w_n = \frac{a_N}{N} \longrightarrow \alpha. \quad (52.6)$$

If $E(z) = \sum_{n \geq 0} L(n)z^n$ were rational, so would be $H(z) = \sum_{n \geq 0} a_n z^n$, because

$$\sum_{n \geq 0} (n+1)^2 z^n = \frac{1+z}{(1-z)^3}.$$

Since $a_0 = 0$, the series of the increments is

$$W(z) = \sum_{n \geq 0} w_n z^n = \frac{1-z}{z} H(z),$$

which would also be rational. The coefficients of a rational series satisfy an eventual fixed-length linear recurrence with constant coefficients. For a finite-valued sequence, the finitely many possible recurrence state vectors evolve deterministically. A state must repeat, after which the sequence is periodic. An eventually periodic 0–1 sequence has a rational limiting mean, contradicting (52.6). Thus E is not rational.

Its coefficients are integers and grow quadratically, so its radius of convergence is exactly 1. The Pólya–Carlson theorem now implies that the unit circle is a natural boundary [36,

Theorem 1.1]. An analytic D-finite germ satisfies a linear differential equation with polynomial coefficients. Away from the finitely many zeros of its leading coefficient, solutions analytically continue locally along paths. Such a germ cannot have an entire circle as a natural boundary. Hence E is not D-finite. The coefficient correspondence between D-finite series and P-recursive sequences gives the final assertion [38]. $\square$

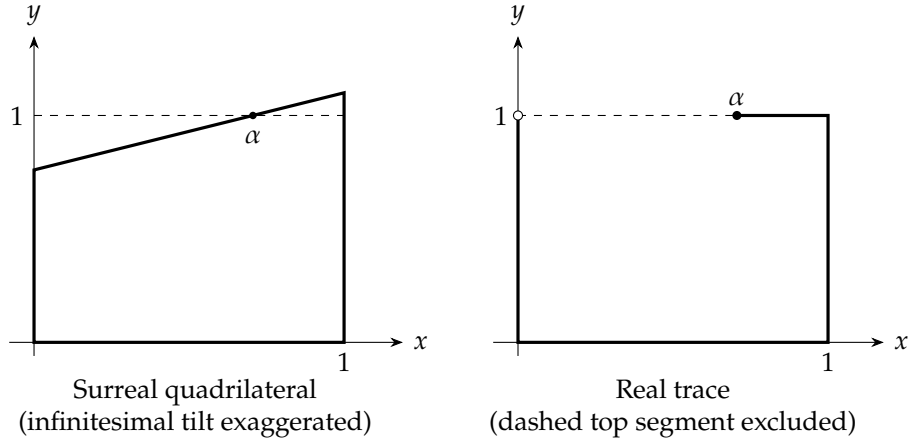


Figure 5: A four-vertex perturbation with a square shadow. The trace removes $[0, \alpha) \times \{1\}$, including the upper-left corner, while retaining the rest of the square. The left drawing is schematic, not a numerical representation of an infinitesimal.

Corollary 52.2 (Algebraic input already suffices). *Taking $\alpha = 1/\sqrt{2}$ gives the phenomenon over a real closure of $\mathbb{Q}(\sqrt{2})(\varepsilon)$, with coefficients linear in one infinitesimal. Thus neither real-algebraic flag coefficients, integral vertex shadows, nor simplicity of the shadow guarantees a quasipolynomial counting function.*

Corollary 52.3 (Every dimension at least two). *For every $d \geq 2$ there is a d -polytope with the labeled combinatorics and integral vertex shadows of the unit d -cube whose counting series has the unit circle as a natural boundary and whose sequence is not P-recursive.*

Proof. Take $P_{\alpha,\varepsilon} \times [0, 1]^{d-2}$. Its counts are

$$(n+1)^{d-2}((n+1)^2 - \lceil \alpha n \rceil).$$

If these counts satisfied a polynomial-coefficient recurrence, substitution would give one for $L_{P_{\alpha,\varepsilon}}(n)$: replace each coefficient of the shifted product by that coefficient times $(n+j+1)^{d-2}$. Nonzero polynomial coefficients remain nonzero. This contradicts Theorem 52.1. The series is therefore nonrational; its integer coefficients have polynomial growth and radius 1, so Pólya–Carlson applies again. Non-quasipolynomiality follows as well, since every quasipolynomial sequence is P-recursive. $\square$

52.1 Why a sufficiently small real number is not a uniform substitute

Replace ε in (52.1) by a fixed small positive real t . Ordinary Euclidean area is

$$\text{vol}_2(P_{\alpha,t}) = \int_0^1 (1 + t(x - \alpha)) dx = 1 + t \left(\frac{1}{2} - \alpha \right). \quad (52.7)$$

The elementary lattice-cell estimate proved in Lemma 53.1 below gives

$$\lim_{n \rightarrow \infty} \frac{L_{P_{\alpha,t}}(n)}{n^2} = 1 + t \left(\frac{1}{2} - \alpha \right) \neq 1.$$

In contrast the surreal count has leading coefficient 1. Hence no fixed positive real t reproduces *all* counting values. For each fixed n , a sufficiently small t does reproduce the value, as guaranteed by Theorem 50.1.

For $\alpha = 1/\sqrt{2} > 1/2$ the discrepancy is especially explicit:

$$\lim_{n \rightarrow \infty} \frac{L_{P_{\alpha,t}}(n) - n^2}{n} = -\infty \quad (t > 0 \text{ fixed and small}),$$

whereas taking the eventual $t \downarrow 0$ value first yields $2 - 1/\sqrt{2}$ after $n \rightarrow \infty$. The issue is the order of quantifiers, not a numerical accuracy requirement.

53 The universal surface coefficient

We now describe what always survives from the first two Ehrhart coefficients. The shadow in this section is a full-dimensional *lattice* polytope $Q \subseteq \mathbb{R}^d$, with $d \geq 2$. A facet F has direction lattice

$$\Lambda_F = \text{lin}(F - F) \cap \mathbb{Z}^d.$$

Let μ_F be $(d-1)$ -dimensional measure on $\text{aff } F$, normalized so that a fundamental parallelepiped of Λ_F has measure one. Define

$$A(Q) = \sum_{F \text{ facet of } Q} \mu_F(F). \quad (53.1)$$

For the real trace S of P , put

$$R_F = \text{cl}_{\text{aff } F}(S \cap \text{relint } F). \quad (53.2)$$

An empty or lower-dimensional R_F has μ_F -measure zero.

53.1 The elementary counting estimates

Lemma 53.1 (Volume estimate for real polytopes). *For a full-dimensional bounded real polytope $T \subseteq \mathbb{R}^k$ and a full-rank lattice Λ , with lattice-normalized measure μ ,*

$$\#(nT \cap \Lambda) = \mu(T)n^k + O(n^{k-1}).$$

The estimate also holds for a bounded convex semilinear set whose closure is full-dimensional. A bounded subset of a fixed j -dimensional affine space has at most $O(n^j)$ points of any fixed ambient integer lattice in its n -dilate.

Proof. Choose a half-open fundamental parallelepiped of Λ . Its translates tile the space and each contains exactly one lattice point under a fixed consistent boundary convention. Associate such a tile with every lattice point of nT . The symmetric difference between their union and nT lies in a neighborhood of the boundary of fixed width depending only on the tile. A polytope has finitely many facets. The volume of that neighborhood is $O(n^{k-1})$, by bounding a fixed-width neighborhood of each dilated facet. Comparing the two volumes proves the first estimate. No rationality of the vertices is used.

A bounded convex semilinear set has a closure that is a polytope: its finite cell decomposition has finitely many closure vertices, and the convex closure is their convex hull. The cells missing from the set but lying in its closure have lower dimension, since a convex set contains the relative interior of its closure. They contribute $O(n^{k-1})$ lattice points. That bound, and the last assertion of the lemma, follow by placing disjoint balls of fixed radius around the lattice points within the affine space containing them. Distinct integer points have distance at least one, and the dilated set lies in a ball of radius $O(n)$ in that space. $\square$

Lemma 53.2 (Classical interior coefficient). *For a full-dimensional lattice polytope Q and $d \geq 2$,*

$$\#(n \operatorname{int} Q \cap \mathbb{Z}^d) = \operatorname{vol}_d(Q)n^d - \frac{A(Q)}{2}n^{d-1} + O(n^{d-2}). \quad (53.3)$$

Proof. We use the classical Ehrhart polynomial and reciprocity [34]. Write $L_Q(n) = c_d n^d + c_{d-1} n^{d-1} + O(n^{d-2})$. Lemma 53.1 gives $c_d = \operatorname{vol}_d(Q)$, while reciprocity gives $L_{\operatorname{int} Q}(n) = (-1)^d L_Q(-n)$. Their difference, which counts the boundary, has leading term $2c_{d-1}n^{d-1}$.

On the other hand the boundary is the union of its facets. Since every facet contains a lattice vertex, its affine lattice on dilation is a translate of Λ_F . Lemma 53.1 gives its leading count $\mu_F(F)n^{d-1}$. Intersections of facets have dimension at most $d-2$ and contribute $O(n^{d-2})$. Thus the boundary count has leading term $A(Q)n^{d-1}$, proving $c_{d-1} = A(Q)/2$ and then (53.3). $\square$

53.2 The coefficient formula

Theorem 53.3 (Boundary-trace surface coefficient). *Let $P \subseteq K^d$ be a real-bounded polytope whose shadow Q is a full-dimensional lattice polytope, with $d \geq 2$. Then*

$$L_P(n) = \operatorname{vol}_d(Q)n^d + \beta(P)n^{d-1} + O(n^{d-2}), \quad (53.4)$$

where

$$\beta(P) = \sum_F \left(\mu_F(R_F) - \frac{1}{2} \mu_F(F) \right). \quad (53.5)$$

In particular

$$-\frac{A(Q)}{2} \leq \beta(P) \leq \frac{A(Q)}{2}. \quad (53.6)$$

Proof. By Lemma 47.3, every real interior point of Q belongs to the trace and no trace point lies outside Q . Decompose S into its interior part, its intersections with the relative interiors of the facets, and its intersection with the union of the faces of codimension at least two. The last part has $O(n^{d-2})$ lattice points after dilation, by Lemma 53.1.

By Corollary 49.3, each $S \cap \operatorname{relint} F$ is bounded, convex, and semilinear. Its closure R_F is a real polytope, possibly empty or lower-dimensional. Choose a lattice vertex $z_F \in F$. The lattice in $n \operatorname{aff} F$ is

$$\mathbb{Z}^d \cap n \operatorname{aff} F = n z_F + \Lambda_F.$$

Translating by $n z_F$ and using the normalized measure, the preceding lemma gives

$$\#(n(S \cap \operatorname{relint} F) \cap \mathbb{Z}^d) = \mu_F(R_F)n^{d-1} + O(n^{d-2}).$$

If R_F has smaller dimension, the same statement follows from the packing bound with zero leading measure. Adding these finitely many facet contributions to (53.3) proves (53.4)–(53.5). Since $R_F \subseteq F$, each of its measures lies between zero and $\mu_F(F)$, proving the bounds. $\square$

Example 53.4 (The quadrilateral revisited). For the unit square, $A(Q) = 4$. Three facets are retained up to sets of one-dimensional measure zero; the retained portion of the top facet has length $1 - \alpha$. Therefore

$$\beta(P_{\alpha,\varepsilon}) = (4 - \alpha) - 2 = 2 - \alpha,$$

in agreement with (52.4). The remaining term is the bounded nonperiodic sequence $\{\alpha n\}$. Even in dimension two the $O(1)$ remainder need not approach a limit.

Remark 53.5 (Why the shadow is assumed lattice). If Q is only rational, a dilated facet can meet the integer lattice on some residue classes of n and miss it on others. A single constant surface coefficient need not exist even for $P = Q$. Theorem 53.3 is therefore not stated for arbitrary rational shadows. One can formulate residue-class versions after tracking affine-lattice congruences, but that additional bookkeeping is separate from the theorem proved here. For any full-dimensional real shadow the simpler inclusions $\text{int } Q \subseteq S \subseteq Q$ still give $L_P(n) = \text{vol}_d(Q)n^d + O(n^{d-1})$ by Lemma 53.1.

54 The full attainable interval, with fixed combinatorics

The bounds in (53.6) are sharp not merely at their endpoints: there are no gaps anywhere between them. More strongly, simplicity of the shadow permits this freedom while preserving its entire labeled face lattice.

Theorem 54.1 (Complete surface-coefficient range). *Fix a full-dimensional lattice polytope $Q \subseteq \mathbb{R}^d$ with $d \geq 2$. Among real-bounded K -polytopes with shadow Q , the values of $\beta(P)$ are exactly*

$$[-A(Q)/2, A(Q)/2].$$

Every value can be attained using one positive infinitesimal and inequality coefficients affine in that infinitesimal.

Proof. The upper and lower bounds were proved in Theorem 53.3. Fix $0 \leq \theta \leq 1$. For each facet F , choose a point in its relative interior and contract F toward that point by the real factor $\theta^{1/(d-1)}$. This gives a real polytope $T_F \subseteq F$ with

$$\mu_F(T_F) = \theta \mu_F(F).$$

At $\theta = 0$ it is a point. Choose affine functions h_{Fj} cutting out T_F inside F , and let s_F be the original nonnegative facet slack. Define

$$P = \{x \in Q_K : s_F(x) + \varepsilon h_{Fj}(x) \geq 0 \text{ for all } F, j\}.$$

Every real interior point of Q belongs to P . Its shadow is Q by the same closed-shadow argument as in Theorem 51.1. For a real point in relint F , all other facet slacks are strictly positive, so only the inequalities associated with F matter. Its trace there is exactly $T_F \cap \text{relint } F$, whose closure has measure $\theta \mu_F(F)$. Consequently

$$\beta(P) = (\theta - 1/2)A(Q).$$

As θ ranges through $[0, 1]$, this gives the full claimed interval. All rows are affine in the single parameter ε . $\square$

54.1 Stability of a simple polytope under infinitesimal facet motion

A d -polytope is simple if exactly d facets meet at each vertex. Write an irredundant facet representation of a fixed full-dimensional simple real polytope as

$$Q = \{x \in \mathbb{R}^d : s_F(x) = b_F - a_F \cdot x \geq 0 \text{ for every facet } F\}.$$

For arbitrary real affine forms $h_F = c_F + g_F \cdot x$, perturb only these existing inequalities:

$$P_\varepsilon = \{x \in K^d : s_F(x) + \varepsilon h_F(x) \geq 0 \text{ for all } F\}. \quad (54.1)$$

Unlike (51.1), this construction is *not* intersected with Q_K .

Lemma 54.2 (Simple labeled stability). *The polytope in (54.1) is nonempty, full-dimensional, and real-bounded. It has the same labeled face lattice as Q . Every vertex has the standard part of its correspondingly labeled vertex in Q , and its shadow is Q .*

Proof. First we establish boundedness; this must precede taking standard parts of hypothetical vertices. Since Q is bounded, its recession cone $\{u : a_F \cdot u \leq 0 \text{ for all } F\}$ is zero. Hence for every real unit vector u ,

$$f(u) = \max_F a_F \cdot u > 0.$$

By ordinary real compactness of the unit sphere, its minimum is a real $c > 0$. The first-order polynomial statement

$$\forall u \left(\sum u_j^2 = 1 \implies \bigvee_F a_F \cdot u \geq c \right)$$

transfers to K . Choose real bounds for all the finitely many coefficients of g_F . For unit $u \in K^d$, the error $|\varepsilon g_F \cdot u|$ is less than $c/2$, so some perturbed normal $A_F = a_F - \varepsilon g_F$ satisfies $A_F \cdot u \geq c/2$. Choose a real $B > 0$ exceeding every $|b_F + \varepsilon c_F|$. For nonzero feasible x , apply this to $u = x/\|x\|$; a positive square root exists because K is real closed. Feasibility then gives

$$\frac{c}{2} \|x\| \leq A_F \cdot x \leq b_F + \varepsilon c_F < B.$$

Thus $\|x\| < 2B/c$. The polyhedron is real-bounded. An ordinary interior point of Q has strictly positive real original slacks and therefore strictly positive perturbed slacks, proving nonemptiness and full dimension.

Let v be a vertex of Q , and $I(v)$ its d incident facets. The corresponding original normal matrix is nonsingular. The perturbed matrix has the same nonzero determinant standard part, so its system has a unique solution v_ε with $\text{st}(v_\varepsilon) = v$. At every other facet the original slack at v is strictly positive; hence its perturbed slack at v_ε is positive. The point v_ε is feasible and is a vertex, being the intersection of d independent active hyperplanes.

Conversely, a vertex w of the perturbed full-dimensional polytope has d independent active rows. It is finite by the preceding bound. Taking standard parts shows that $\text{st}(w) \in Q$ lies on all of those d distinct original facets. In a simple polytope a point on d distinct facets must be a vertex: the facets through any positive-dimensional face are fewer than d . One way to see this is to choose a vertex of that face; its d incident facet normals are independent, and the intersection of any d of them is that vertex alone. Thus these are exactly the incident facets of an original vertex, and the corresponding original matrix is nonsingular. Its unique perturbed solution is the already constructed v_ε . This proves that there are no other vertices.

At the constructed vertices the zero slacks are exactly the original zero slacks. The vertex-facet table, and hence the labeled face lattice, is unchanged. The vertex standard parts and Lemma 47.2 give the shadow assertion. $\square$

Theorem 54.3 (Full range with fixed labeled simple combinatorics). *Let $Q \subseteq \mathbb{R}^d$, $d \geq 2$, be a full-dimensional simple lattice polytope. Every*

$$b \in [-A(Q)/2, A(Q)/2]$$

is realized as $\beta(P)$ by a real-bounded K -polytope P satisfying all of the following simultaneously: its shadow is Q ; its labeled face lattice equals that of Q ; its labeled vertex standard parts are exactly the original integral vertices of Q ; and it uses one affine-in- ε perturbation of each original facet inequality, with no added facets.

Proof. Put $\theta = 1/2 + b/A(Q) \in [0, 1]$. For every facet F , choose a real affine function h_F for which

$$\mu_F(F \cap \{h_F \geq 0\}) = \theta \mu_F(F). \quad (54.2)$$

Such a choice exists: a nonconstant linear functional on the $(d-1)$ -dimensional facet has sublevel sections whose volume varies continuously from zero to full measure as the level traverses its range. Continuity follows since the moving level hyperplane has measure zero and the facet is bounded. For the endpoints use $h_F = -1$ or $h_F = 1$.

Form P_ε as in (54.1). Lemma 54.2 proves all the geometric assertions. At a real point x on relint F , exactly the original F -slack is zero; the others are strictly positive. Consequently trace membership there is precisely $h_F(x) \geq 0$. The closure of that retained part has the measure in (54.2); deleting the boundary of F does not change it. Theorem 53.3 therefore gives

$$\beta(P_\varepsilon) = \sum_F (\theta - 1/2) \mu_F(F) = (\theta - 1/2) A(Q) = b.$$

□

Theorem 54.3 says more than that one can choose open or closed facets. Intermediate real portions of a facet, selected by an infinitesimal tilt, continuously interpolate its contribution. Neither the finite face lattice nor the full labeled list of vertex standard parts records that selection. Simplicity is used only in the combinatorial stability lemma; Theorem 54.1 establishes the interval without it.

54.2 Dimension one behaves differently

A real-bounded interval with a positive-length lattice shadow $[a, b]$, with $a, b \in \mathbb{Z}$, retains all points of (a, b) and independently retains or excludes its two endpoints. Hence for $n \geq 1$ its count is

$$(b - a)n - 1 + e_a + e_b, \quad e_a, e_b \in \{0, 1\}.$$

Its constant term is one of $-1, 0, 1$, not an arbitrary value in an interval. There is no positive-dimensional facet along which a threshold can slide. This explains the $d \geq 2$ condition in the continuous-range theorems.

55 A rationality criterion that restores Ehrhart behavior

The preceding examples identify a failure of a natural extension of Ehrhart theory. There is nevertheless a useful sufficient condition that replaces rational vertex coordinates by rational data in the trace flags.

Definition 55.1. A real-bounded polytope has a *rational flag presentation* if its real trace is described by finitely many lexicographic nonnegativity conditions on real affine forms all of whose coefficients are rational, together with a rational bounding box. This is a property of a possible presentation, not a claim about the canonicity of the basis in Lemma 49.1.

Theorem 55.2 (Rational flags imply positive-dilation quasipolynomiality). *If P has a rational flag presentation, then $L_P(n)$ is quasipolynomial for positive integers n . Its ordinary generating function, with the convention $L_P(0) = 1$, is rational.*

Proof. Use the hyperplane arrangement of the rational affine forms, including the rational box boundary. The selected cells form a finite disjoint union of relatively open rational polytopes $C_1, \dots, C_s$. Each is the relative interior of its closure, a rational polytope. For $n > 0$ the dilation map is injective and the dilated cells remain disjoint, so

$$L_P(n) = \sum_{j=1}^s \#(nC_j \cap \mathbb{Z}^d).$$

Classical rational Ehrhart theory and reciprocity imply that the count of the relative interior of a rational polytope is quasipolynomial on positive integers [34]. This statement includes lower-dimensional rational affine hulls and their residue-class restrictions. A finite sum of quasipolynomial is quasipolynomial, with period dividing a common multiple of their periods. Splitting into residue classes shows that its positive-index generating function is rational. Changing or supplying the single coefficient at $n = 0$ preserves rationality. $\square$

Example 55.3 (Why $n = 0$ needs a convention). Let $P = [\varepsilon, 1 - \varepsilon]$. Its trace is $(0, 1)$, described by the rational strict inequalities $x > 0$ and $1 - x > 0$. Then

$$L_P(n) = n - 1 \quad (n \geq 1), \quad L_P(0) = 1.$$

The positive-dilation polynomial does not take the conventional value at zero. Any theorem using relatively open trace cells must either treat $n = 0$ separately or adopt a different valuation convention there.

Remark 55.4 (Not a necessary condition). The theorem is a sufficient criterion, not an equivalence. Irrational real polytopes can have polynomial counting functions; the examples of Cristofaro-Gardiner, Li, and Stanley show that arithmetic cancellations and period collapse must be taken seriously [35]. Conversely, the choice $\alpha = 1/\sqrt{2}$ in Theorem 52.1 shows that replacing “rational” by “real algebraic” in the hypothesis is insufficient. An exact classification would have to detect cancellation among boundary pieces rather than inspect each coefficient separately.

55.1 Three notions of rationality that must not be conflated

A polytope may have rational or integral *vertex shadows*; its trace may admit a description by *rational flags*; and its counting *generating function* may be rational. These are different properties. The quadrilateral has integral vertex shadows but irrational trace data and a nonrational generating function. Rational flags imply a rational generating function by Theorem 55.2. The manuscript does not assert any converse from generating-function rationality to a specific kind of geometric presentation.

The stronger descent theorem does not conflict with this distinction. Its one-parameter algebraic coefficients can contain arbitrary real constants. Eliminating extra infinitesimal scales is not the same as eliminating irrational real thresholds.

55.2 A classical analytic dichotomy for every bounded trace count

The analytic consequence of Pólya–Carlson is more general than the quadrilateral example. It supplies a useful complete dichotomy once the geometric rationality question has been answered.

Corollary 55.5 (Analytic classification of bounded counts). *Let P be any nonempty real-bounded finite K -polytope, without a dimension assumption on its shadow, and let $E_P(z) = \sum_{n \geq 0} L_P(n)z^n$. The following are equivalent:*

- (i) E_P is rational;
- (ii) E_P is D-finite;
- (iii) $L_P(n)$ is P-recursive;
- (iv) $L_P(n)$ is eventually quasipolynomial.

If these conditions fail, the unit circle is a natural boundary of E_P . This is a direct application of classical results, not a new analytic classification theorem.

Proof. A real bounding box gives $0 \leq L_P(n) = O(n^d)$, and the values are integers. If only finitely many are nonzero then E_P is a polynomial. Otherwise integrality and polynomial growth imply radius of convergence exactly one. Pólya–Carlson says that a nonrational such series has the unit circle as a natural boundary [36, 37]. The finite-singularity argument in Theorem 52.1 then excludes D-finiteness. Conversely a rational series is D-finite. The standard coefficient correspondence identifies D-finiteness with P-recursiveness [38].

In the rational case, the integer-coefficient form of Pólya–Carlson also says that every pole is a root of unity [36, Theorem 1.1]. Partial fractions therefore express the coefficients, after a finite initial segment, as finite sums of terms $n^j \zeta^n$ with ζ a root of unity. They are polynomial on residue classes modulo a common order. Conversely, splitting an eventually quasipolynomial sequence into its residue classes gives a rational series. $\square$

[[write] The root-of-unity clause is classically the Fatou–Kronecker argument, not part of the natural-boundary dichotomy proper. By Fatou’s lemma a rational power series with integer coefficients is $A(z)/C(z)$ with $A, C \in \mathbb{Z}[z]$ and $C(0) = 1$. Radius of convergence 1 puts every root of the reversed polynomial, which is monic with integer coefficients, in the closed unit disc. Kronecker’s theorem then makes each nonzero root a root of unity. Manuscript 03 cites this through the integer-coefficient statement it attributes to [36, Theorem 1.1] and does not prove it; the equivalence of (i) and (iv) rests on it.]

Thus for the present unweighted, one-variable counting problem there is no intermediate nonrational D-finite case. The genuine remaining task is to determine from the boundary geometry which side of the dichotomy occurs.

56 Repository integration and the exact proof boundary

56.1 Snapshot and source use

The repository tree inspected for this work had commit

6996fee43cc97b6c16351def7507d59a95bf62f0.

The root and surreal-project readmes were consulted for orientation. The surreal report guide and the omnific source below were then fetched at this explicit commit. Code-search results may refer to older indexed commits; they were used as locators, not as proof of the contents of the pinned snapshot.

The key source is

```
Algebra/SurrealNumbers/Surreal/Foundations/OmnificUnits.lean
```

Its namespace is `Surreal.Foundations.SignSequence`. The three relevant declarations are

```
omnific_eq_intConstant_of_finite
omnific_isFinite_iff
omnific_bounded_iff
```

The last states that an omnific integer has a real bound on its absolute value exactly when it is an ordinary integer cast [28]. This is the bridge in (47.1).

The source was read; this work did *not* rebuild the repository, check its complete transitive axiom dependencies, or independently audit its surreal-field implementation. The mathematical article can be read independently using the standard normal-form argument given in Section 47. None of its new trace, descent, implantation, or surface-coefficient theorems is being represented as an existing Lean theorem.

56.2 A realistic formalization sequence

[write] The formalization roadmap of manuscript 03 is printed in [section 83](#).

56.3 A theorem dependency map

Result	Principal input	Status in this package
Bounded omnific reduction	Normal form; repository boundedness theorem	Source inspected; direct mathematical argument
Lexicographic trace	Standard part; finite real linear algebra	Full proof, no Lean certificate
Algebraic descent	Trace expansion; real-closed transfer; curve selection	Full application proof with classical inputs
Face implantation	Dominance of a real slack over an infinitesimal	Full constructive proof
Surface coefficient	Trace cells; lattice-cell estimate; Ehrhart reciprocity	Full application proof
Fixed-combinatorics range	Simple facet stability; continuous section volume	Full constructive proof
Quadrilateral count	Exact lexicographic membership; finite row count	Full proof and exact finite checks
Natural boundary	Nonrationality; Pólya–Carlson	Classical analytic input explicitly used

57 Exact computational checks and reproducibility

The accompanying code/03-boundary-geometry-verify.py uses only the Python standard library. All assertions use integers, rational numbers, exact arithmetic in $\mathbb{Q}(\sqrt{2})$, or polynomials in a formal positive infinitesimal ordered by their first nonzero coefficient. No floating-point arithmetic is used to resolve a boundary comparison.

57.1 What was actually checked

For $\alpha = 1/\sqrt{2}$, the program verifies the 16 vertex–facet slacks of the quadrilateral, its four positive cyclic orientation determinants, and the exact shoelace identity

$$2 \operatorname{vol}_2(P_{\alpha,\varepsilon}) = 2 + \varepsilon(1 - \sqrt{2}).$$

It independently enumerates integer candidates in an enlarged bounding box for every $1 \leq n \leq 150$: 1,205,575 candidate membership tests in total. Every result agrees with (52.3). The ceiling formula and its 0–1 increments are checked for $1 \leq n \leq 10,000$.

For rational thresholds $\alpha = p/q$ with $2 \leq q \leq 12$ and $1 \leq p < q$, it checks 3,960 threshold/dilation cases, including vanishing third forward differences on each tested progression of step q . For the three-dimensional face implantation of

$$T = [1/\sqrt{2}, 1] \times [1/\sqrt{3}, 1]$$

on the top of the unit cube, it checks all $1 \leq n \leq 35$ by 443,555 candidate tests against the exact formula

$$n(n+1)^2 + (n+1 - \lceil n/\sqrt{2} \rceil)(n+1 - \lceil n/\sqrt{3} \rceil).$$

Finally it verifies real specialization at $t_n = 1/(4(n+1))$ for $1 \leq n \leq 100$, using 353,700 exact $\mathbb{Q}(\sqrt{2})$ sign comparisons. The specialization parameter deliberately depends on n .

For nonsquare positive integer m and $n > 0$, the program evaluates

$$\lceil n/\sqrt{m} \rceil = \operatorname{isqrt}(\lfloor n^2/m \rfloor) + 1$$

and verifies the defining inequalities $m(c-1)^2 < n^2 < mc^2$. Thus the tests near the irrational cut do not depend on a decimal approximation.

n	$\lceil n/\sqrt{2} \rceil$	$L_{P_{1/\sqrt{2},\varepsilon}}(n)$	$(n+1)^2$
0	0	1	1
1	1	3	4
2	2	7	9
3	3	13	16
4	3	22	25
5	4	32	36
6	5	44	49
7	5	59	64
8	6	75	81
9	7	93	100
10	8	113	121

The successful test report is in `data/03-boundary-geometry-verification.json`; `data/03-boundary-geometry-counts.csv` records values for $0 \leq n \leq 100$. To reproduce the checks and compile the article, run

```
python3 code/03-boundary-geometry-verify.py --out-dir /path/to/scratch
latexmk -pdf -interaction=nonstopmode article.tex
```

A double invocation of `pdflatex`, with an additional pass when cross-references change, can replace `latexmk`. All figures are contained in the TeX source; no external image assets are needed.

[[write] Delivery names replaced. Manuscript 03 printed `verify.py`, `verification.json`, `counts.csv` and `python3 verify.py`; the shipped names are those above. Without `-out-dir` the program writes `verification.json` and `counts.csv` next to itself, that is, into `code/`. The report's recorded outputs are under `data/` and should never be overwritten, so pass an output directory outside the report. The program writes the CSV with Python's `csv` module, so its lines end in CRLF on every platform. The shipped `data/03-boundary-geometry-counts.csv` keeps those bytes: 102 lines (a header and $n = 0, \dots, 100$), each ending in CRLF, and marked `-text` for Git. Manuscript 03's own TeX file `surreal_polytopes.tex` is not shipped: its text is this Part of the report's `article.tex`, which the second command builds.]

57.2 What finite checks do not establish

Agreement through 10,000 does not prove nonperiodicity or exclude a large period. Those conclusions use the irrationality argument in Theorem 52.1. The checks do not establish the general trace, descent, or surface-coefficient theorems. They do not prove Pólya–Carlson, certify a natural boundary, or substitute for a proof assistant. Their role is to catch arithmetic, incidence, dilation, and endpoint mistakes in the explicit constructions.

58 Further research questions

The following are proposed research problems arising from this article. Unless an external source is explicitly attached to a question, it is not being represented as an established named open problem in the literature.

Question 58.1 (An intrinsic classification of traces). Which bounded convex semilinear sets $S \subseteq \mathbb{R}^d$ are the real traces of finite polytopes over a real closed extension? Can one characterize them without choosing a halfspace presentation over the extension?

Theorem 49.2 provides a necessary finite lexicographic description. Every finite conjunction of such descriptions has an infinitesimal halfspace realization, but an arbitrary convex semilinear set is initially given by a Boolean combination, not necessarily such a conjunction. The useful target is a geometric characterization, including compatibility of the selected portions on intersecting faces and a criterion for prescribing the shadow when it loses dimension.

Question 58.2 (Minimal trace depth and description size). What is the least maximum lexicographic depth among all descriptions of a fixed trace, and how does it trade off against the number of rows?

The $d + 1$ bound proved here is a rowwise coefficient-space bound. It does not prove that $d + 1$ is an optimal invariant of traces after changing their presentations. A normal form with canonical depth would be useful for certificate design and for comparing independent constructions.

Question 58.3 (Effective algebraic descent). Given exact algebraic coefficients and a finite certificate for a lexicographic presentation and incidence data, what degree, coefficient height, and complexity bounds suffice for the descent curve in Theorem 50.1?

Quantitative curve selection supplies relevant general machinery [31]. The open task here is to exploit the special multi-affine ratio structure in (50.1), account for the realization-space quantifiers, and produce bounds suitable for an executable certificate. The degree- d trace-only compression is not already such a bound for incidence-preserving descent.

[[write] Related, not an answer: for coordinates given as finite-support Hahn polynomials, Part III's [theorem 41.2](#) (manuscript 06) specializes to ordinary Laurent polynomials in one parameter, with an explicit integer grading and explicit sign thresholds. It keeps any specified *finite* certificate. Whether such a certificate can be chosen to keep the whole real trace, and so give an effective descent with degree bounds, is not proved in this report. Part I's [theorem 19.1](#) is the non-effective real-algebraic analogue for finite sign data.]

Question 58.4 (Surface freedom for nonsimple fixed combinatorics). For a nonsimple lattice polytope Q , which values of β can be realized while preserving its labeled face lattice and all vertex shadows?

Theorem [54.1](#) gives the full interval when combinatorics is allowed to change, and Theorem [54.3](#) gives it with fixed combinatorics when Q is simple. At a nonsimple vertex, preserving several dependent facet intersections imposes compatibility equations on the perturbations. Determining whether these constraints shrink the interval, and in which examples, is a sharply delimited next problem.

Question 58.5 (The next asymptotic layer). Which conditions ensure a coefficient of n^{d-2} , and can it be expressed using traces on ridges and second-level facet data?

The surface term always exists for lattice shadows, but the bounded fractional-part remainder in dimension two already prevents a universal constant next term. A suitable theorem may require rational flags, averaging in n , residue-class restrictions, or a discrepancy term rather than an ordinary coefficient. Face implantation provides test cases with defects localized at each codimension.

Question 58.6 (Rationality and cancellation). Within traces defined by real-algebraic flags, can one decide when the counting series is rational, or characterize when irrational boundary contributions cancel to give a quasipolynomial?

Rational flags give a sufficient condition; irrational period collapse prevents a naive converse [\[35\]](#). The face-implantation identity reduces many examples to lower-dimensional real-polytope enumeration, offering both obstructions and constructive test families.

Question 58.7 (Weighted and multivariate counting). What replaces Corollary [55.5](#) for algebraic or real weights, several dilation parameters, or generating functions recording individual lattice coordinates?

The unweighted one-variable dichotomy is already resolved by that corollary, including lower-dimensional shadows. Arbitrary real weights remove the integer-coefficient hypothesis, while several variables require an appropriate notion of boundary and multivariate D-finiteness. Existing multivariate Pólya–Carlson results provide relevant analytic input [\[37\]](#); the task is to identify geometric hypotheses that let these results be applied to the proposed weighted or multivariate enumeration.

Question 58.8 (Reciprocity with boundary data). Is there a useful reciprocity formula for trace counts stated in terms of constructible weights on faces rather than just open versus closed polytopes?

A trace may retain a proper region of a facet and may make additional choices on its boundary. Ordinary closed-polytope reciprocity cannot simply be reused without recording those choices. The rational cell decomposition in Theorem [55.2](#) provides a finite starting point, but an intrinsic formula should be independent of the chosen subdivision.

Question 58.9 (Symmetry-constrained perturbations). What parts of the surface-coefficient interval remain attainable under central symmetry or under a prescribed finite affine symmetry group?

The independent facet choices in Theorem 54.3 need not respect a group action. Averaging perturbations preserves some algebraic symmetries but may change the selected boundary regions. Orbitwise constraints and self-paired facets should be analyzed explicitly rather than inferred from the unconstrained theorem.

Question 58.10 (Finite enriched combinatorial invariants). What finite augmentation of the face lattice and vertex shadows suffices to reconstruct the real trace and its lattice counts?

The lexicographic affine flags are one sufficient description, but not an intrinsic minimal invariant. Ordinary oriented-matroid signs alone forget the real threshold data, while unrestricted valuation information is more than the counting sequence needs. An invariant adapted to boundary flags may bridge these extremes.

Question 58.11 (Other integer parts and unbounded geometry). How do the conclusions change for a different integer part of an ordered field, or when coordinates and dilation parameters are allowed to be infinite?

The present reduction depends on the exact bounded-element property (47.1). Infinite dilations do not automatically yield finite sets of lattice points, so a new counting invariant must be specified before an analogue of Ehrhart theory can be meaningfully formulated. Ordinary cardinality, polynomial evaluation in an infinite argument, and surreal-valued numerosity are not interchangeable definitions.

Question 58.12 (A checked geometry pipeline). Can the exact row-expansion, incidence, and finite-count certificates be assembled into a Lean-checked pipeline with an explicitly small trust boundary?

The immediate milestone is a checked proof of the quadrilateral count and the bounded omnific reduction, followed by trace equivalence. A general symbolic discovery procedure for arbitrary surreal inputs is a much stronger demand and is not implied by the finite certificate theorems.

59 Conclusion

Finite surreal convex geometry has a classical combinatorial boundary: its face lattices are already realizable over the real algebraic numbers. Its arithmetic boundary is different. The real trace, not the standard-part shadow, determines ordinary and bounded omnific lattice counts. That trace is finitely lexicographic, and a single algebraic parameter can preserve it together with the entire finite incidence structure.

Inside this finite framework, infinitesimal boundary motion is expressive. It implants arbitrary lower-dimensional real-polytope counting functions, changes the surface coefficient through its whole allowable interval, and can do so without changing a simple lattice shadow or any of its labeled vertex standard parts. A four-vertex, one-infinitesimal example already escapes quasipolynomial and P-recursive enumeration. These statements give a concrete research program and an exact formalization target without requiring genuinely class-sized convexity or a new notion of infinite lattice cardinality.

60 Additional scope checks and potential failure modes (Appendix A of manuscript 03)

Finite generators versus topological compactness. “Polytope” means a finite convex hull, equivalently a bounded finite halfspace system in the setting used here. Proofs over K use linear algebra or first-order transfer. The only compactness argument in the simple-stability proof takes place on the *real* unit sphere and is then converted to a polynomial statement before transfer.

The interior-lifting hypothesis. A full-dimensional shadow ensures that its interior points belong to the trace. It is not enough that P itself have positive dimension, and even full dimension of P need not imply full dimension of its shadow. The one-parameter descent theorem deliberately avoids relying on this lemma and therefore allows rank loss in the shadow.

A representation-dependent flag is not an invariant. The separated basis exists for every finite coefficient space, but it is not canonically chosen. The trace theorem asserts existence of a bounded-depth presentation. It does not assert uniqueness of the scales or optimality among all presentations.

Zero standard part does not mean zero slack. When a leading slack vanishes at a real point, its infinitesimal terms can still decide membership. Dropping these terms is exactly the error exposed by (52.5). Conversely, a strictly positive real slack cannot be overturned by a finite sum of infinitesimal corrections.

One parameter is not a uniform real approximation. An infinitesimal evaluation and a fixed positive real evaluation have different quantifier meanings. Descent gives eventual agreement for each fixed real point and each fixed finite lattice query. Equation (52.7) shows why a single real evaluation generally cannot answer all dilations.

Real-algebraic is not rational. All defining constants of the explicit quadrilateral can lie in $\mathbb{Q}(\sqrt{2})$, yet its count is not quasipolynomial. The rational flag theorem requires rational coefficients for the real affine tests, not merely algebraic coefficients or integral shadow vertices.

No general reciprocity theorem is smuggled in. Classical Ehrhart reciprocity is used for the lattice shadow in Lemma 53.2 and for rational cells in Theorem 55.2. It is not assumed for an arbitrary irrational trace or for an arbitrary surreal polytope.

The analytic step has an external dependency. The arithmetic proof establishes nonrationality directly. The assertion of a natural boundary uses Pólya–Carlson, and the non-P-recursive conclusion uses its analytic consequence together with the standard D-finite correspondence. Finite experimental evidence is not used as a substitute for any of these implications.

61 Reproducibility and source ledger (Appendix B of manuscript 03)

The archive contains the TeX source, compiled PDF, standard-library verification program, its JSON report, a CSV of example counts, and a README. The bibliography gives stable DOIs

or arXiv identifiers for the classical inputs and commit-pinned paths for the two repository documents used substantively. The source review was performed on September 30, 2026. No repository files were modified or committed as part of this work.

[[write] In this report the delivered verification program is shipped as `code/03-boundary-geometry-verify.py`, its JSON report as `data/03-boundary-geometry-verification.json`, and the CSV as `data/03-boundary-geometry-counts.csv` (CRLF line endings, as delivered). The TeX source is printed as this Part, not shipped as a file. The compiled PDF and the member README are not shipped; they survive only in the delivery archive `surreal_polytope_research.zip` of arrival commit `b2b626d85`.]

The source `OmniFicUnits.lean` was returned with blob identifier

`c9ce3771e76d4b21ff76814682f37f63cc4664f8`.

This identifies the inspected file; it is not a claim that the entire repository was locally compiled. ([write] The blob of this file at the report's placement is still `c9ce3771`.) The article imports no unreviewed research-report theorem as a hidden prerequisite: its surreal-specific bounded-integer fact is also justified directly, and its remaining external inputs are the classical mathematical results named in the text.

Part V

Extension complexity beyond all orders

Source and scope. [write] This Part is built from two batch-58 manuscripts. Its base, printed whole and in its own order, is manuscript 07, *Invisible Complexity in Surreal Polytopes*, subtitle *Explicit polygons with identical all-order jets and identical initial slack matrices*, author line “Research manuscript prepared for Vladimir Reshetnikov”, dated 30 September 2026 (archive `surreal_polytopes.zip`, main file `surreal_polytopes.tex`; repository pin `6996fee43cc97b6c16351def7507d59a95bf62f0`; its delivered PDF, which is not shipped, has 26 pages according to its README). Into it are merged the results of manuscript 04, *Beyond-All-Orders Extension Complexity in Surreal Polygon Geometry*, subtitle *Flat deformations, classical face lattices, and a sharp-order complexity gap*, author line “Research manuscript prepared for Vladimir Reshetnikov / Developed with ChatGPT; independent review and formalization pending”, dated September 30, 2026 (archive `surreal_polytopes_beyond_all_orders.zip`, main file `article.tex`; the same pin `6996fee43`; delivered PDF of 26 pages, not shipped). The two manuscripts answer the same question: explicit polygons whose coordinates agree to all orders of a chosen infinitesimal with a polygon of small extension complexity, but whose own extension complexity is much larger. Manuscript 07 is the base because it covers every $n \geq 4$ (04: $n = 2^m$), linear and semidefinite lifts, residue descent for every convex valuation ring (04: standard part and nonnegative rank only), every fixed dimension, and a sharper parameter count. Manuscript 04 contributes a second construction on one circle with a two-sided square-root order, a field- and scale-general classification of when all-order jets lose information, a finite-order rank-one analogue, failure of lifting coherent finite-order factorizations, algebraic finite-accuracy approximants, a rational-chart generalization and an in-principle decision procedure. Every item of 04 is printed where it belongs and marked “source 04”; items proved by both manuscripts are printed once and name both. Labels of 07 carry the prefix `poly:ic:`, labels of 04 the prefix `poly:bao:`. This Part keeps its own definitions and hypotheses: the finite-element ring $\mathcal{O}$, standard part `st` and convex hulls are those of [Part I](#), restated here in the words of 07 and 04. The formalization roadmap of 07 is printed in [section 83](#); nothing in this Part is formalized.

Notation in this Part. [write] The valuation is written v throughout this Part, with the convention of 04: $v(x) > v(y)$ precisely when x/y is infinitesimal, so a *larger* value means a *smaller* element and $v(\omega^{-\gamma}) = \gamma$. (Manuscript 07 writes v without defining it; manuscript 04 writes v ; we renamed 04's v to v . Manuscript 04 also uses v_i for monomials in the proof of [theorem 74.1](#); those are renamed μ'_i .) The natural value group is G (07's letter; 04's Γ in its Section 3 is renamed G); Γ , Γ_2 and Γ_n denote the specific groups ([72.1](#)), [corollary 72.4](#) and ([70.11](#)). The flat ideal of a positive infinitesimal scale x is written with one symbol, $\mathfrak{i}_x = \bigcap_{m \geq 1} x^m \mathcal{O}$ (07's $\mathfrak{j}$); manuscript 04's $\mathcal{I}_\varepsilon$ is printed as $\mathfrak{j}_\varepsilon$. The jet scale is always the subscript of $\mathfrak{j}$: a general positive infinitesimal is s in statements from 07 and ε in statements from 04 ([section 63.4](#) and [theorems 72.7](#) and [76.1](#)); in 04's explicit family it is specialized to $\varepsilon = \omega^{-1}$, the report's ε . Manuscript 07's main construction uses the fixed scale $\rho = \omega^{-\omega}$; following the report-wide convention it is printed as $\tau = \omega^{-\omega}$ wherever it denotes that surreal. Where 07 re-chooses the scale, the letter ρ is kept as a general parameter: in [corollary 72.4](#) ($\rho = \omega^{-1}$) and in [theorem 74.2](#) (a chosen infinitesimal); the auxiliary bound called τ in the proof of [theorem 74.2](#) is renamed σ , and the tolerance set $\mathcal{T} \ni \tau$ of 04's [theorem 74.2](#) is renamed $\mathcal{D} \ni \delta$ (07's letters). The two quoted abstracts keep their own letters: ρ in 07's abstract is this Part's τ , and P_n in 04's abstract is this Part's P_n° . Tempting false readings, with the true one beside them:

- P_n, Q_n (07) are polygons in the triangle T ; manuscript 04's cyclic polygon on the unit circle, also called P_n there, is printed as P_n° with vertices p_i° , and its regular reference polygon is R_n with vertices r_i . R_N in ([66.1](#)) is 07's reflection map, not a polygon.
- st is the standard part. $\mathcal{O}/\mathfrak{j}_s$ is a different quotient and is not $\mathbb{R}$; the coarse residue field of V_τ ([proposition 72.10](#)) is a third one. Manuscript 04's "algebraic finite-accuracy shadows" are real-algebraic approximants, not standard parts; we call them *approximants* ([corollary 74.6](#)).
- tr is the matrix trace (07), not Part IV's real trace $\text{Tr}_{\mathbb{R}}$.
- q_i are 07's easy vertices; in 04's text q_i are distinct ordinary primes, entering only through $\sqrt{q_i}$ and $\mu_i = \omega^{-\omega\sqrt{q_i}}$. a_n (07) is the real root of $X^{2n+1} - 2$; a_i (04) is $\tan(\theta_i/2)$.
- η_ℓ (07) are perturbation monomials and $\eta = \eta_1$; in 04, η is a flat bound ([proposition 71.1](#)) or a positive rational tolerance ([corollary 74.6](#)).
- D : 07's orientation determinant $D(X, Y, Z)$ of points and $D = h(h+1)/2$ in [theorem 67.4](#); 04's $D(a, b, c)$ is the orientation determinant in circle parameters ([70.9](#)). H_N (07) is a parabola hull; $H(V)$ (04) is a $3 \times n$ point matrix.
- m : an ordinary integer exponent in 07 ($\mathfrak{j}_s, <_{\text{pow}}$) and in 04 $n = 2^m$; M is 04's exponent letter. $\mathfrak{m}$ is the maximal ideal of $\mathcal{O}$. T is 07's triangle, but also a nonnegative factor matrix in 04 ($S = TU$) and in the proof of [theorem 73.5](#); $U(n)$ is 07's easy bound, U also a factor matrix. S is always a slack matrix.
- K is a set-sized real closed field; K_0, K_n and F_n are specific fields of 07 and 04; k is a residue field in [section 73](#) and $k = n - 2$ in 07's construction.

Across the report: ρ here is not Part III's threshold ρ_g nor Part IV's separated scales $\rho_1 \gg \dots \gg \rho_r$; V_τ is not Part I's dependence space nor Part III's mixed volume.

Abstract of manuscript 07. Can the real shadow, the complete expansion at a chosen infinitesimal scale, or the leading terms of a slack matrix determine the exact complexity of a surreal polytope? We give explicit negative answers for both linear and positive-semidefinite

lifts. For every ordinary finite $n \geq 4$ we construct two strictly convex, labeled surreal n -gons Q_n, P_n inside the same real triangle. They have the same real shadow and oriented matroid; their vertex coordinates and canonical slack matrices agree modulo every power of a specified infinitesimal ρ ; and corresponding nonzero slack entries have identical valuations and leading coefficients. Nevertheless,

$$\text{xc}(Q_n) \leq 2\lceil \log_2(n-2) \rceil + 2, \quad \text{xc}(P_n)(\text{xc}(P_n) - 1) \geq 2n,$$

and $\text{sxc}(P_n) \geq (2n)^{1/4}$, while $\text{sxc}(Q_n) \leq \text{xc}(Q_n)$. Every coordinate is a sum of at most three Conway monomials, and all examples fit one set-sized real closed subfield of valuation rank three.

The proof combines a binary reflection lift with explicit algebraically independent, beyond-all-powers perturbations. We supply a non-Archimedean proof of the generic semidefinite lower bound using finite definability, rather than rational exposing objectives, which can fail over the surreals. We also prove residue descent for nonnegative and semidefinite ranks, identify the exact kernel of power-jet observations, establish a sharp rank-one/rank-two boundary for all-order ambiguity, and extend the construction to arbitrary set-sized precision requirements.

These are proved statements within the manuscript, not a claim to have settled a previously named classical conjecture. The generic lower bounds, reflection method, cone-factorization correspondence and real-closed-field transfer are classical. The proposed contribution is their explicit multiscale separation package. Priority and correctness have not been independently certified; no Lean formalization is claimed.

Keywords (manuscript 07): surreal polytope; extension complexity; nonnegative rank; semidefinite rank; convex valuation ring; infinitesimal perturbation; real closed field; formal jets.

Abstract of manuscript 04. Infinitesimal coordinates do not create new finite polytopal face lattices: real-closed-field transfer already prevents this. They can, however, conceal large changes in extension complexity beneath every power of a prescribed infinitesimal. We construct explicit cyclic surreal n -gons P_n , for $n = 2^m$, and regular n -gons R_n on the same unit circle such that their labeled vertex coordinates agree modulo $\varepsilon^M \mathcal{O}$ for every ordinary integer $M \geq 1$, where $\varepsilon = \omega^{-1}$. Nevertheless,

$$m \leq \text{xc}(R_n) \leq 2m, \quad \lceil \sqrt{n} \rceil \leq \text{xc}(P_n) \leq \min\{n, \lceil 24\sqrt{n} \rceil\}.$$

The polygons also have identical standard parts, leading affine Plücker data, and leading slack matrices; every finite linear objective has an optimal-value difference smaller than every power of ε . The construction uses algebraically independent Conway monomials and a rational parametrization of the circle. A general theorem identifies the exact boundary: over any real closed $K \supseteq \mathbb{R}$, unbounded such gaps in all-order jet fibers occur precisely when $\bigcap_{M \geq 1} \varepsilon^M \mathcal{O} \neq 0$, equivalently when $v(\varepsilon)$ is not an order unit of the natural value group. We give full proofs of these statements, a finite-order rank-one analogue, a rational-chart generalization, and an explicit account of why finite real-algebraic transfer is compatible with the obstruction. The classical extension-complexity bounds are credited separately from the proposed flat-geometric synthesis.

Status (manuscript 04). This is a research manuscript, not a claim of a refereed breakthrough or a Lean-checked result. The flat-fiber theorems and explicit all-order examples are offered as proposed contributions. Their priority has not been independently certified. No named published conjecture is claimed solved. The auxiliary exact checks verify identities and finite examples, not the asymptotic lower bounds.

[write] Manuscript 07 has no separate status paragraph; its status is the last paragraph of its abstract above and its table in [section 62](#). In 04’s abstract the valuation symbol is printed as v (04: v).

62 The questions, the answer, and the research boundary

A polytope can be geometrically almost indistinguishable from another and still require a much larger exact optimization model. Surreal coefficients make it possible to ask a stronger version: can the difference remain invisible not merely at a small numerical tolerance, but at *every* power of a prescribed infinitesimal? The answer is yes, even with the same number of vertices, the same full-dimensional real shadow and the same leading slack data.

Here a linear extended formulation is an exact description of a polytope as an affine image of a polytope in a possibly larger space. Its size is the number of facet inequalities; equations do not count. A semidefinite lift uses an affine slice of the cone of positive-semidefinite symmetric matrices. Its size is the matrix order, not the number of scalar entries. Both notions are exact and allow arbitrary coefficients in the ambient real closed field.

The concrete questions answered below are the following.

- (1) Does equality of all power jets of labeled vertices force comparable linear or semidefinite extension complexity? No: the ratios are unbounded.
- (2) Does the zero pattern together with every leading valuation and leading coefficient of a slack matrix determine these complexities? No, even for rank-three slack matrices of polygons.
- (3) Does taking a real or more general ordered residue lose complexity in a controlled direction? Yes: both complexities can only decrease.
- (4) Is non-Archimedean order alone sufficient for nontrivial all-order ambiguity? No: the chosen scale must fail to be an order unit in the value group. Valuation rank two is necessary and sufficient for examples with one layer of such ambiguity.

These are questions formulated and answered here. They are not presented as a list of previously published open conjectures.

62.1 The question of manuscript 04

[write] The following is the opening of manuscript 04 (its Section 1, “The question and the main result”); its explicit family and main theorem are printed in [section 70](#).

A polytope can be complicated in two different ways. Its vertices and faces can have an intricate incidence pattern, or its particular coordinates can make it difficult to express as the projection of a simpler polytope. The first notion is combinatorial. The second is measured by *extension complexity*: the minimum number of facets of a higher-dimensional polytope that projects onto the given one.

For a polygon, every face lattice is a cycle. This leaves considerable room for coordinate-dependent behavior. Regular polygons have logarithmic extension complexity, whereas suitable generic polygons require at least square-root complexity. These are classical facts, not discoveries of this manuscript [43, 44]. Kwan, Sauermann, and Zhao proved that every *cyclic* n -gon—one with all vertices on a circle—has extension complexity at most $24\sqrt{n}$ [53]. Here “cyclic polygon” does not refer to the higher-dimensional cyclic polytopes on the moment curve.

The question investigated here is more refined:

Can a polygon be indistinguishable from a regular polygon at every order of a fixed infinitesimal scale, while retaining square-root rather than logarithmic extension complexity?

The answer is affirmative in the surreal field, and its exact valuation theoretic boundary can be identified. The interesting phenomenon is not a new abstract face lattice. It is loss of information about an exact existential problem under an all-order asymptotic observation map.

[write] Question (1) of 07 and the question of 04 are the same question for two reference polygons: a logarithmic-size polygon in a triangle (07) and a regular polygon on the unit circle (04). Question (4) of 07 and 04’s “exact valuation theoretic boundary” are the same order-unit theorem ([theorem 72.2](#), with its two-sided consequences [corollary 72.4](#) and [theorem 72.7](#)).

62.2 Relationship to ProveIt and the previous convexity report

The selected repository documentation describes a surreal field development, real closedness, surcomplex algebraic closedness, finite geometry, and a large collection of research reports. It explicitly separates research claims from statements mapped to Lean declarations [39, 40]. The repository snapshot recorded during this investigation is 6996fee43cc97b6c16351def7507d59a95bf62f0, dated 30 September 2026. The audit consulted the root README, the surreal README, the report catalogue, and selected directory metadata; it was not an exhaustive theorem-by-theorem audit or an independent rebuild.

[[write] The draft corrected here, *Finite Convexity and Exact Lexicographic Optimization over the Surreals* (22 September 2026), was supplied privately to the authors of manuscripts 05 and 07; it is not part of ProveIt and could not be checked. The statements below are printed as standalone results.]

The earlier draft *Finite Convexity and Exact Lexicographic Optimization over the Surreals* already treats finite Farkas theory, linear-programming duality, finite convexity theorems and exact lexicographic compilation [41]. That draft was located in the supplied research library; its presence in the maintained repository tree was not established in this audit. The present paper does not reclassify that material as new. Instead, it studies *the size of exact lifts and the information lost by scale reduction*.

62.3 Relation to ProveIt (source 04)

The inspected snapshot of VladimirReshetnikov/ProveIt contains a surreal-number library and a collection of research reports that explicitly distinguish manuscript arguments, finite computational tests, and mapped Lean declarations [54, 55]. Its computer-algebra report also separates exact denotation from approximation and from decidable equality [56]. The present work develops a geometric consequence of that separation: even complete information at every power of one selected scale need not determine a finite exact optimization invariant.

We use the established ordered-field and normal-form foundations. We do not assume the correctness of the repository’s unreviewed new research claims. The source snapshot and the limited inspection scope are recorded in [section 80](#). The current paper supplies no Lean development and makes no assertion about having rebuilt the repository.

[write] Both descriptions were re-checked against the current tree. The surreal README (Algebra/SurrealNumbers/README.md, lines 1–27 and 147) describes the report collection,

the real closed surreal field, the algebraically closed surcomplex numbers, finite geometry among the formalized topics, and the ledger that maps statements to Lean declarations; the computer-algebra report’s README (Algebra/SurrealNumbers/docs/foundations-and-computation/computer-algebra/README.md, lines 24–28) separates “exact denotation from effective coefficient access from decidable equality/order from certified approximation”. The Lean module `Surreal.Surcomplex.RegularPolygon` (Algebra/SurrealNumbers/Surreal/Surcomplex/RegularPolygon.lean: `vertex`, `vertex_radius`, `side_length`, `area_eq`) formalizes regular polygons at surreal radius, which is related to 04’s regular R_n , but neither manuscript uses it; nothing in this Part is formalized.

62.4 What is classical and what is proposed here

Fiorini, Rothvoß and Tiwary established generic polygon lower bounds and small extended formulations for special polygons [43]. Reflection relations are an established tool for compact formulations [44]. Gouveia, Robinson and Thomas proved the generic semidefinite lower bound [47]. The slack-factorization interpretation originates with Yannakakis and its cone generalization [45, 46]; normalization and lower-semicontinuity methods for positive ranks are also established [48].

Component	Status in this manuscript
Generic rank bounds	Classical mechanism, reproved over arbitrary real closed fields; the semidefinite proof avoids an Archimedean step.
Binary reflection lift	Explicit specialization of the classical reflection-relation method.
Residue descent	Valuation-ring formulation with algebraic proofs; not claimed as a new closure theorem.
Explicit P_n, Q_n	Proposed separation package: finite monomial support, all-order agreement, identical initial slacks, and diverging lift sizes.
Rank and precision boundary	Consequences proved here, with the exact scale and set-size hypotheses stated.
Formal verification	Finite exact computational checks only; no machine-checked proof of the manuscript.

Searches of the selected documentation and primary literature did not locate this exact explicit separation package. This is a limited novelty check, not evidence that every related formulation is absent from the literature. All uses of “new” should therefore be understood as *candidate new formulations and consequences*, subject to expert review.

62.5 What is new here, and what is imported (source 04)

The proposed contribution is the combination of exact circular geometry, explicit nonreal algebraic independence, and agreement to *all* orders, together with a necessary-and-sufficient condition for the same phenomenon in arbitrary real closed extensions of $\mathbb{R}$. The resulting failure of reconstruction persists even after one supplies all leading Plücker coordinates, all leading slack entries, and every finite-direction linear optimization value to all orders.

The following ingredients are established mathematics: real closedness of the surreal field and uniqueness of Conway normal forms [52]; quantifier elimination for real closed fields [50, 51]; Yannakakis’ slack-factorization theorem [45]; reflection constructions for regular

polygons [44]; and the polygon bounds just cited. We prove the needed elementary extensions and calculations rather than treating the phrase “by transfer” as a substitute for specifying the finite formula.

63 Fields, polytopes, lifts, and power-jet information

63.1 Set-sized fields first

Unless otherwise stated, K is a set-sized real closed ordered field. For standard parts we assume $K \supseteq \mathbb{R}$. A finite surreal configuration, and any finite collection of putative lift certificates, lie in a set-sized real closed subfield of **No**: generate an ordered field from their coefficients and take its relative real closure in **No**. Thus every first-order argument below is made in a genuine set-sized field.

We use the classical real closedness and normal-form arithmetic of **No** [42]. In particular,

$$\omega^a \omega^b = \omega^{a+b}, \quad a < b \implies \omega^a \ll \omega^b, \quad (63.1)$$

where $x \ll y$ means $|x| < y/m$ for every positive ordinary integer m . A nonzero finite sum of distinct Conway monomials cannot vanish. All arithmetic inside an exponent is surreal *field* arithmetic; no unstated ordinal-arithmetic convention is used.

Every n, m, r, k in this paper is an ordinary finite integer. No result concerns a polytope with an infinite number of vertices. A statement about lifts “over **No**” means a statement about finite coefficient certificates; it does not apply first-order model theory directly to a proper class.

63.2 Finite convexity and the two lift sizes

For $v_1, \dots, v_N \in K^d$ define

$$\text{conv}_K(v_1, \dots, v_N) = \left\{ \sum_j \lambda_j v_j : \lambda_j \geq 0, \sum_j \lambda_j = 1 \right\}.$$

Dimension and facets are relative to the affine hull. A vertex is an extreme point. The finite vertex–facet correspondence, linear-programming duality and the finite Farkas alternative hold over real closed fields. They follow from their finite algebraic proofs or from real-closed-field transfer. No order-topological compactness is assumed.

We allow size zero for a point (a constant affine image of the zero-dimensional cone). For a positive-dimensional polytope P , let $\text{xc}_K(P)$ be the smallest number of facets of a polytope E admitting an affine surjection onto P . The usual definition allowing polyhedral extensions gives the same number. Indeed, a nonnegative factorization of a slack matrix gives a bounded extension after zero factor columns are removed: each remaining nonnegative auxiliary coordinate is bounded by a bounded slack divided by a fixed positive entry. This also explains why unbounded lifts cannot circumvent the lower bound proved later.

Let $\mathcal{S}^k(K)$ denote the symmetric $k \times k$ matrices. Write $\mathcal{S}_+^k(K)$ for the positive-semidefinite cone; equivalently, every principal minor is nonnegative. Define $\text{sx}_K(P)$ as the least k for which an affine slice of $\mathcal{S}_+^k(K)$ has an affine image equal to P . The slice itself need not be bounded.

Definition 63.1 (Two factorization ranks). For a nonnegative matrix $M \in K^{f \times N}$, $\text{rank}_+^K(M)$ is the least r such that $M = UV$ with $U \in K_{\geq 0}^{f \times r}$ and $V \in K_{\geq 0}^{r \times N}$. Its semidefinite rank $\text{rank}_{\text{psd}}^K(M)$ is the least k admitting $A_i, B_j \in \mathcal{S}_+^k(K)$ with $M_{ij} = \text{tr}(A_i B_j)$.

A facet–vertex slack matrix of $P = \{x : Ax \leq b\} = \text{conv}(v_j)$ has entries $S_{ij} = b_i - A_i v_j$. The classical factorization theorems imply

$$\text{xc}_K(P) = \text{rank}_+^K(S), \quad \text{sx}_K(P) = \text{rank}_{\text{psd}}^K(S), \quad \text{sx}_K(P) \leq \text{xc}_K(P). \quad (63.2)$$

We import the real factorization theorem [45, 46]. Its extension to K needs no analytic limiting argument: for each fixed finite dimension, vertex count, facet count and proposed rank, existence of a factorization and existence of a lift are first-order statements in ordered-field arithmetic. Positive semidefiniteness is a finite list of principal-minor inequalities. The universal equivalence over $\mathbb{R}$ therefore holds over every real closed field. Quantifier elimination for these fields is the standard algebraic input; see also its formalized constructive treatment in [49].

[write] Manuscript 04 proves the linear half of (63.2) directly over K , checking the field and boundedness issues instead of transferring; it is printed as [proposition 63.6](#) in [section 63.6](#) below.

63.3 Finite elements, standard part, and all power jets

Suppose now that $K \supseteq \mathbb{R}$. Its finite-element ring and infinitesimal ideal are

$$\mathcal{O} = \{x \in K : |x| \leq c \text{ for some } c \in \mathbb{R}_{>0}\}, \quad \mathfrak{m} = \{x : |x| < 1/m \text{ for every } m \geq 1\}.$$

Completeness of the *ordinary real coefficient field* identifies $\mathcal{O}/\mathfrak{m}$ with $\mathbb{R}$; write the residue map as st . This does not assert completeness of K .

For a positive infinitesimal s , define

$$\mathfrak{j}_s = \bigcap_{m \geq 1} s^m \mathcal{O}. \quad (63.3)$$

Equality of all s -power jets means equality in every quotient $\mathcal{O}/s^m \mathcal{O}$, or equivalently that the difference lies in $\mathfrak{j}_s$. For a vector, $\mathfrak{j}_s^M$ denotes the Cartesian power of the ideal. These are quotients of the *valuation ring*, not of the field K . In a field the ideal generated by a nonzero s is the whole field.

[write] Manuscript 04 (its Section 3) fixes a positive infinitesimal $\varepsilon \in K$ and sets

$$\mathfrak{j}_\varepsilon = \bigcap_{M \geq 1} \varepsilon^M \mathcal{O}, \quad (63.4)$$

the same ideal for $s = \varepsilon$ (04 writes $\mathcal{I}_\varepsilon$). Here and throughout 04's text M ranges over ordinary finite integers.

For positive infinitesimals a, b we write

$$a <_{\text{pow}} b \iff 0 < a < b^m \text{ for every } m \geq 1.$$

This notation means “smaller than every power,” not merely $a < b$.

Lemma 63.2 (The power-flat ideal; sources 07 and 04). *For every positive infinitesimal s ,*

$$\mathfrak{j}_s = \{x \in K : |x| < s^m \text{ for every } m \geq 1\}.$$

It is a prime ideal of $\mathcal{O}$ and the kernel of

$$\mathcal{O} \longrightarrow \varprojlim_m \mathcal{O}/s^m \mathcal{O}.$$

Moreover $s^{-h} \mathfrak{j}_s = \mathfrak{j}_s$ for each integer $h \geq 0$. [source 04] For $x \in \mathcal{O}$, membership $x \in \mathfrak{j}_s$ is also equivalent to $v(x) > Mv(s)$ for every $M \geq 1$, with $v(0) = +\infty$; and $\mathfrak{j}_s$ is a convex prime ideal of $\mathcal{O}$.

Proof. If $x \in s^{m+1}\mathcal{O}$, write $x = s^{m+1}u$ with u finite. Then $|su| < 1$, so $|x| < s^m$. Conversely, $|x| < s^m$ makes x/s^m finite. The ideal and kernel assertions follow from the intersection definition. If $x, y \in \mathcal{O} \setminus \mathfrak{j}_s$, there are m, n with $|x| \geq s^m$ and $|y| \geq s^n$; hence $xy \notin \mathfrak{j}_s$. Finally, shifting every exponent m by a fixed integer does not change the defining all-orders condition. $\square$

Second proof (source 04), with the valuation and convexity clauses. [write] (Written for $s = \varepsilon$, as in 04.) If $x = \varepsilon^{M+1}u$ with u finite, then $\varepsilon|u| < 1$, and hence $|x| < \varepsilon^M$. Conversely, $|x| < \varepsilon^M$ implies $x/\varepsilon^M \in \mathcal{O}$. Membership in $\varepsilon^M\mathcal{O}$ means $v(x) \geq Mv(\varepsilon)$, and using $M + 1$ instead of M turns the intersection of these weak inequalities into the asserted family of strict ones.

The intersection in (63.4) is an ideal. It is convex because $|y| \leq |x|$ and $x \in \mathfrak{j}_\varepsilon$ imply $|y| < \varepsilon^M$ for every M . If $a, b \notin \mathfrak{j}_\varepsilon$, choose $r, s \geq 1$ with $|a| \geq \varepsilon^r$ and $|b| \geq \varepsilon^s$. Then $|ab| \geq \varepsilon^{r+s}$, so $ab \notin \mathfrak{j}_\varepsilon$. This proves primeness. $\square$

In particular, a nonzero error can disappear in the *entire* s -adic completion. This is failure of separatedness, not slow convergence. Exact equality tests in K still distinguish such an error from zero.

63.4 The all-order observation map (source 04)

[write] This subsection prints the rest of the first part of 04's Section 3 ("The all-order observation kernel"); its second part, on the valuation boundary, is printed with the same theorem of 07 in [section 72](#). In this subsection ε is an arbitrary positive infinitesimal of K .

Consequently $\mathcal{O}/\mathfrak{j}_\varepsilon$ is an ordered integral domain. It is generally not $\mathbb{R}$ and need not be a field. In fact $\varepsilon \notin \mathfrak{j}_\varepsilon$, so the class of ε is a nonzero element in the maximal ideal $\mathfrak{m}/\mathfrak{j}_\varepsilon$. It is important not to conflate this quotient with standard part.

Definition 63.3 (All-order jet data). The all-order observation map is the ring homomorphism

$$J_\varepsilon : \mathcal{O} \longrightarrow \varprojlim_{M \geq 1} \mathcal{O}/\varepsilon^M\mathcal{O}, \quad x \longmapsto (x \bmod \varepsilon^M\mathcal{O})_M. \quad (63.5)$$

Two finite coordinate arrays have the same ε -jet data if their images under this map agree coordinatewise.

The kernel of J_ε is exactly $\mathfrak{j}_\varepsilon$. This language does not assume that an arbitrary surreal has a convergent or formal expansion in integer powers of ε . The individual rings $\mathcal{O}/\varepsilon^M\mathcal{O}$ may have nilpotents; they are not being treated as ordered fields.

[write] The map J_ε (capital italic J) is 04's jet map; its kernel is the flat ideal $\mathfrak{j}_\varepsilon$ (fraktur j). The map of [lemma 63.2](#) is the same map for the scale s .

Lemma 63.4 (Regular rational expressions preserve flat agreement; source 04). *Let $b \in \mathbb{R}^s$, $z - b \in \mathfrak{j}_\varepsilon^s$, and $f \in \mathbb{R}(X_1, \dots, X_s)$ admit a presentation $f = g/h$ with $h(b) \neq 0$. Then $f(z)$ is finite and*

$$f(z) - f(b) \in \mathfrak{j}_\varepsilon.$$

Proof. The ideal property gives $g(z) - g(b), h(z) - h(b) \in \mathfrak{j}_\varepsilon$. Since $\mathfrak{j}_\varepsilon \subseteq \mathfrak{m}$, the standard part of $h(z)$ is $h(b) \neq 0$. Consequently $h(z)$ and $h(b)$ are units of $\mathcal{O}$. The difference

$$\frac{g(z)}{h(z)} - \frac{g(b)}{h(b)} = \frac{(g(z) - g(b))h(b) - g(b)(h(z) - h(b))}{h(z)h(b)}$$

belongs to $\mathfrak{j}_\varepsilon$. $\square$

The condition on the denominator is essential. Division by a sufficiently small flat quantity can reveal information that J_ε discards.

[write] In lemma 63.4 the letter s is a number of variables (04's usage), not a scale. The lemma is the real-base-point case, at an arbitrary scale, of the rational-function clause of 07's lemma 65.1, which is stated at the scale τ with base points in $\mathbb{R}(\tau)$.

63.5 Conventions of manuscript 04

[write] Manuscript 04's Section 2 ("Fields, polytopes, and size conventions") restates the setting above in its own words; it is printed here in full.

A set-sized field is enough (source 04). Unless specifically stated otherwise, K is a *set-sized* real closed ordered field containing the usual $\mathbb{R}$ as an ordered subfield. A finite list of surreal coordinates is contained in such a field: take the field generated by $\mathbb{R}$ and those coordinates and then its real algebraic closure inside the real closed surreal field. This is a set. Accordingly, every use of first-order logic below takes place in a set-sized structure.

The notation $\mathbf{x}_{\mathbb{N}_0}$ means that a finite extension is allowed arbitrary surreal coefficients. It will follow from scalar-extension invariance in corollary 74.7 that the answer is already attained in any real closed subfield containing the input data. No first-order model is made out of a proper class.

For finite $V = (v_1, \dots, v_n) \subset K^d$, define

$$\text{conv}_K(V) = \left\{ \sum_{i=1}^n \lambda_i v_i : \lambda_i \in K_{\geq 0}, \sum_i \lambda_i = 1 \right\}.$$

A polytope is such a convex hull. Throughout, polytopes are nonempty. Faces are the subsets on which a linear functional achieves its maximum, together with the empty face. A linear functional on a polytope attains its maximum at a listed vertex because

$$c \cdot \sum_i \lambda_i v_i \leq \max_i c \cdot v_i.$$

This is finite algebra. We never invoke topological compactness of a closed bounded subset of a non-Archimedean field.

The usual finite halfspace description, vertex criterion, and linear Farkas lemma hold over real closed fields. One may prove them by finite linear elimination. Alternatively, each fixed-size statement is a first-order assertion in the language of ordered fields and transfers from $\mathbb{R}$ by Tarski's theorem [50]. We use these finite facts in the slack-factorization argument below.

Finite elements and standard part (source 04). Define

$$\mathcal{O} = \{x \in K : |x| \leq N \text{ for some } N \in \mathbb{N}\}, \quad \mathfrak{m} = \{x \in K : |x| < 1/N \text{ for all } N \geq 1\}.$$

The finite elements form a convex valuation ring; $\mathfrak{m}$ is its maximal ideal. Its units are exactly the finite elements with finite reciprocal. Every $x \in \mathcal{O}$ is infinitely close to a unique real number: take the supremum in $\mathbb{R}$ of the reals not exceeding x . This defines the ring homomorphism $\text{st} : \mathcal{O} \rightarrow \mathbb{R}$ with kernel $\mathfrak{m}$. Nonnegative elements have nonnegative standard part, and st fixes $\mathbb{R}$.

[write] This is the same ring $\mathcal{O}$ as 07's (a bound by a positive real is a bound by an integer).

Proposition 63.5 (Standard part of a finite convex hull; source 04). *If all $v_i \in \mathcal{O}^d$, then*

$$\text{st}(\text{conv}_K(v_1, \dots, v_n)) = \text{conv}_{\mathbb{R}}(\text{st}(v_1), \dots, \text{st}(v_n)).$$

Proof. A convex coefficient belongs to $[0, 1] \subset \mathcal{O}$. Applying st to a convex combination gives a real convex combination. Conversely, use any real convex coefficients as elements of K . These two observations give the two inclusions. $\square$

The proposition does not assert preservation of the face lattice or of extension complexity. Our examples retain the former but not the latter.

[write] This is Part I's [proposition 13.1](#) and 07's (65.6); all three proofs are the same argument.

63.6 Slack factorization over a real closed field (source 04)

[write] From 04's Section 4 ("Extension complexity over ordered fields"). Its scalar-extension proposition is printed with 07's [corollary 74.7](#); its parameter count after 07's [theorem 67.1](#); its reflection lemma and cyclic upper bound in [section 70](#).

For a full-dimensional polytope $P \subset K^d$ with vertices v_j and irredundant facet inequalities $a_i \cdot x \leq b_i$, its slack matrix is

$$S_{ij} = b_i - a_i \cdot v_j \geq 0.$$

The nonnegative rank $\text{rank}_+^K(S)$ is the least r for which $S = TU$, with $T \in K_{\geq 0}^{f \times r}$ and $U \in K_{\geq 0}^{r \times n}$. Positive rescaling of a row or column does not change it. [write] (rank_+^K is 07's rank_+^K of [definition 63.1](#).)

Proposition 63.6 (Yannakakis over a real closed field; source 04, the linear half of 07's (63.2)). *If P has positive dimension, then*

$$\text{xc}_K(P) = \text{rank}_+^K(S).$$

Here extension complexity counts facets of a bounded polytope in its affine hull, and allows an affine projection.

Proof. This is the classical factorization theorem [45]; we check the field and boundedness issues. Suppose $S = TU$ and discard any zero columns of T . Consider

$$Q = \{(x, y) : Ax + Ty = b, y \geq 0\}.$$

Every point of Q projects into P . Each vertex v_j lifts using the j th column of U , and convex combinations of these lifts show that all of P is the projection. The set Q is bounded: x ranges over P , and for each column ℓ choose i with $T_{i\ell} > 0$. Then

$$0 \leq y_\ell \leq \frac{b_i - a_i \cdot x}{T_{i\ell}},$$

whose right-hand side is bounded on the finite convex hull P . Thus Q is a polytope with at most r facets in its affine hull.

Conversely, let Q have r facet slacks $g_\ell \geq 0$ and project onto P . Pull back the i th facet slack of P to a nonnegative affine function on Q . The affine Farkas lemma writes it as $c_i + \sum_\ell T_{i\ell} g_\ell$, with all coefficients nonnegative. There is a point of Q above that facet of P , where the function is zero; hence $c_i = 0$. Choose one lift w_j of each vertex v_j and set $U_{\ell j} = g_\ell(w_j)$. Evaluation gives $S = TU$. $\square$

For a matrix with fixed dimensions and an ordinary integer r , introduce the finite formula

$$\text{NF}_r(S) : \exists(T_{i\ell}, U_{\ell j}) \left[T_{i\ell}, U_{\ell j} \geq 0 \wedge \bigwedge_{i,j} S_{ij} = \sum_{\ell=1}^r T_{i\ell} U_{\ell j} \right]. \quad (63.6)$$

All conjunctions and sums here are finite. This is the precise first-order content of a nonnegative-rank upper bound.

64 The explicit separation theorem

[write] In this section and in [sections 65 to 69](#) and [72](#) manuscript 07's scale $\rho = \omega^{-\omega}$ is printed as τ (see "Notation in this Part"). A second construction, on one circle, from manuscript 04 is printed in [section 70](#).

Set

$$\varepsilon = \omega^{-1}, \quad \tau = \omega^{-\omega}. \quad (64.1)$$

For $n \geq 4$, put $k = n - 2$ and $u_j = j/(k + 1)$, $1 \leq j \leq k$. Define

$$A = (\tau, 1 - 2\tau), \quad B = (1 - 2\tau, \tau), \quad (64.2)$$

$$C_j = (\tau u_j, \tau(1 - u_j)^2). \quad (64.3)$$

The easy configuration, in cyclic order, is

$$q_1 = A, \quad q_{j+1} = C_j \ (1 \leq j \leq k), \quad q_n = B. \quad (64.4)$$

Let a_n be the positive real root of $X^{2n+1} - 2$, and set

$$\eta_\ell = \omega^{-\omega^2 a_n^{\ell-1}} \quad (1 \leq \ell \leq 2n), \quad p_i = q_i + (\eta_{2i-1}, \eta_{2i}). \quad (64.5)$$

Finally,

$$Q_n = \text{conv}(q_1, \dots, q_n), \quad P_n = \text{conv}(p_1, \dots, p_n), \quad T = \text{conv}((0, 0), (1, 0), (0, 1)).$$

A subscript on conv or T specifies the coefficient field when needed.

Theorem 64.1 (Explicit all-order separation). *The configurations (64.4)–(64.5) have the following properties for every ordinary finite $n \geq 4$.*

- (i) P_n, Q_n are strictly convex n -gons contained in the strict interior of T . Every increasing labeled triple has positive orientation in both. Their real shadows are the same full-dimensional triangle: $\text{st}(P_n) = \text{st}(Q_n) = T_{\mathbb{R}}$.

- (ii) The configurations have identical τ -power jets: $p_i - q_i \in \mathfrak{i}_\tau^2$. In fact, writing $\eta = \eta_1$,

$$P_n \subseteq Q_n + [-\eta, \eta]^2, \quad Q_n \subseteq P_n + [-\eta, \eta]^2, \quad \eta \prec_{\text{pow}} \tau.$$

Also $T \subseteq P_n + [-3\tau, 3\tau]^2$ and $T \subseteq Q_n + [-2\tau, 2\tau]^2$, while $3\tau < \varepsilon^m$ for every $m \geq 1$.

- (iii) For the canonical edge slacks

$$S(v)_{ij} = \det(v_{i+1} - v_i, v_j - v_i), \quad v_{n+1} = v_1,$$

$S(p) - S(q)$ is entrywise in $\mathfrak{i}_\tau$. The two slack matrices have the same zero pattern, the same valuation of each entry, and the same leading coefficient of each nonzero entry. Both have ordinary rank three.

(iv) With $U(n) = 2\lceil \log_2(n-2) \rceil + 2$,

$$\text{xc}(Q_n) \leq U(n), \quad \text{sxc}(Q_n) \leq U(n), \quad (64.6)$$

$$r = \text{xc}(P_n) \implies r(r-1) \geq 2n, \quad (64.7)$$

$$h = \text{sxc}(P_n) \implies \left(\frac{h(h+1)}{2} \right)^2 \geq 2n. \quad (64.8)$$

In particular $\text{xc}(P_n) \geq \sqrt{2n}$ and $\text{sxc}(P_n) \geq (2n)^{1/4}$.

(v) The bounds remain valid over every real closed ordered extension containing these coordinates, including the finite-certificate interpretation over all of **No**. Every coordinate has at most three Conway monomials. All the examples, for all finite n , lie in one set-sized real closed field of valuation rank three.

Consequently neither the labeled order type, the full system of power jets, nor the initial slack matrix controls exact lift complexity up to any uniform multiplicative constant. The ratios $\text{xc}(P_n)/\text{xc}(Q_n)$ and $\text{sxc}(P_n)/\text{sxc}(Q_n)$ both tend to infinity along these examples. The first ratio is at least of order $\sqrt{n}/\log n$; the second is at least of order $n^{1/4}/\log n$.

[write] Manuscript 07 proves no upper bound for its hard polygon beyond the trivial $\text{xc}(P_n) \leq n$, so its ratio is bounded below only. The same-circle construction of manuscript 04 (theorem 70.1) has a two-sided order $\Theta(\sqrt{n}/\log n)$, because its hard polygon is cyclic and the Kwan–Sauermann–Zhao bound applies; it covers only $n = 2^m$ and linear lifts.

64.1 Proof architecture

The easy polygon is a tiny affine image of equally spaced points on a parabola, with two additional vertices near the other corners of T . Binary reflection gives a logarithmic-size extension. The hard polygon adds one algebraically independent monomial to each coordinate. These monomials are smaller than every power of τ , so they change no τ -power jet or slack leading term, but they force large algebraic parameter counts in any exact lift. The following four sections give the individual proofs; Section 72 proves the common-field assertion.

65 Geometry of the cap and stability beyond all powers

65.1 Four exact orientation identities

For three planar points write $D(X, Y, Z) = \det(Y - X, Z - X)$. Temporarily let $C(u) = (\tau u, \tau(1-u)^2)$. Direct expansion gives

$$D(C(u), C(v), C(w)) = \tau^2(v-u)(w-u)(w-v), \quad (65.1)$$

$$D(A, C(u), C(v)) = \tau(v-u)[1 + \tau(uv - u - v - 1)], \quad (65.2)$$

$$D(C(u), C(v), B) = \tau(v-u)[2 - u - v + \tau(uv + 2u + 2v - 4)], \quad (65.3)$$

$$D(A, C(u), B) = (1 - 3\tau)[1 - \tau(u^2 - u + 2)]. \quad (65.4)$$

For $0 < u < v < w < 1$ every displayed expression has positive leading coefficient. These four cases exhaust the increasing triples of (64.4). Hence every cyclic edge has all other vertices strictly on its left, and every point is an exposed vertex. This proves strict convexity and the stated cyclic order without an appeal to a picture.

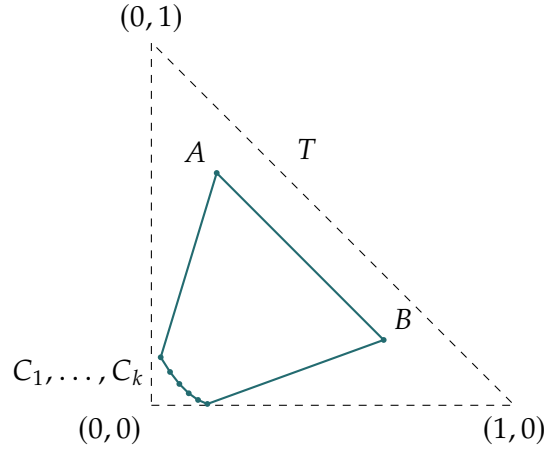


Figure 6: A real specialization of the easy construction, drawn with $\tau = 0.18$ and $k = 6$ to make the cap visible. This schematic does not model the surreal scale inequalities; those are proved algebraically. The hard perturbations would occur at a separate, still smaller scale. [write] (Source 07; its ρ is printed as τ .)

All coordinates are positive. At A, B the coordinate sum is $1 - \tau$. At $C(u)$ it is $\tau(1 - u + u^2) < 1$. Therefore Q_n lies in the strict interior of T .

65.2 A stability lemma with an explicit scope

Lemma 65.1 (Flat perturbations of finite algebraic observations). *Let $q \in \mathbb{R}(\tau)^M$ have finite coordinates, and let $e \in \mathfrak{j}_\tau^M$. If $f \in \mathbb{R}(\tau)[X_1, \dots, X_M]$, then*

$$f(q + e) - f(q) \in \mathfrak{j}_\tau.$$

If $f(q) \neq 0$, the two values have the same sign and the same Conway leading monomial and coefficient. If F is a rational function over $\mathbb{R}(\tau)$ whose denominator is nonzero at q , its denominator is also nonzero at $q + e$, and $F(q + e) - F(q) \in \mathfrak{j}_\tau$.

Proof. Every nonzero $c \in \mathbb{R}(\tau)$ has the form $\tau^h u$ for some $h \in \mathbb{Z}$, where u is finite with nonzero real standard part. This follows by factoring the lowest nonzero power of τ from the numerator and denominator. Thus multiplication by c preserves $\mathfrak{j}_\tau$.

Expanding $f(q + e) - f(q)$ produces a finite sum of terms, each containing an e -factor. All other coordinate factors are finite; the coefficient factors preserve $\mathfrak{j}_\tau$. The sum is therefore flat. If $f(q) = \tau^h u \neq 0$, the error divided by τ^h is still flat and in particular infinitesimal. It cannot change $\text{st}(u)$ or its sign.

For rational functions, apply the polynomial assertion to numerator and denominator. The two denominators have the same nonzero leading term, and their inverses grow at most like a fixed negative power of τ . Cross-multiplication then shows that the difference of the quotients is flat. $\square$

The lemma controls a specified family of observations. It does not say that exact algebra cannot detect the perturbation. For example,

$$2q_{1,1} + q_{1,2} - 1 = 0, \quad 2p_{1,1} + p_{1,2} - 1 = 2\eta_1 + \eta_2 > 0. \quad (65.5)$$

The right side is invisible in every τ -power quotient, but an exact zero test distinguishes it immediately. Thus this is an information-loss theorem for specified reductions, not a lower bound for an unrestricted symbolic equality oracle.

65.3 Perturbed convexity and the full-dimensional shadow

For the scales in (64.5), $a_n > 1$ and

$$0 < \eta_{2n} < \cdots < \eta_1 = \omega^{-\omega^2} < \tau^m \quad (m \geq 1).$$

Apply Lemma 65.1 to the orientation determinants and to the strict inequalities $x > 0$, $y > 0$, $1 - x - y > 0$. All their reference values are nonzero elements of $\mathbb{R}(\tau)$, with the required signs. It follows that P_n is also a strictly convex n -gon in the strict interior of T , with the same labeled order type.

For completeness, standard part commutes with a finite convex hull of finite points:

$$\text{st}(\text{conv}_K(v_1, \dots, v_N)) = \text{conv}_{\mathbb{R}}(\text{st}(v_1), \dots, \text{st}(v_N)). \quad (65.6)$$

Indeed, convex coefficients lie in $[0, 1] \subseteq O$, so one inclusion follows by taking their standard parts. Conversely, real convex coefficients themselves are admissible lifts. [write] (This is proposition 63.5 of source 04 and Part I's proposition 13.1.) Here

$$\text{st}(A) = (0, 1), \quad \text{st}(B) = (1, 0), \quad \text{st}(C_j) = (0, 0),$$

and the perturbations have zero standard part. This proves the common triangular shadow.

Matching the same convex coefficients on p_i and q_i proves the two inclusions with error box $[-\eta_1, \eta_1]^2$ in Theorem 64.1. Each vertex of T is within 2τ in the maximum norm of a reference vertex: use A, B for the nonzero corners and any C_j for the origin. Interpolating these three matches with real or K -valued convex weights proves $T \subseteq Q_n + [-2\tau, 2\tau]^2$. The perturbed inclusion follows because $2\tau + \eta_1 < 3\tau$. Finally, $3\tau < \varepsilon^m$ for each finite m , as $\omega - m$ is positive infinite. This proves parts (i) and (ii) of Theorem 64.1.

66 An explicit logarithmic-size reflection lift

The following elementary recurrence is the source of the small upper bound. Its mechanism is an affine instance of the reflection-relation method [44]. We give the whole construction because it is a short, exact certificate over any ordered field.

Lemma 66.1 (A two-inequality reflection step). *Let*

$$H_N = \text{conv}\{(j, j^2) : j = 0, \dots, N\}, \quad M = \lfloor N/2 \rfloor.$$

If H_M is the affine image of a polytope with r inequalities, then H_N has an extension with at most $r + 2$ inequalities.

Proof. The affine involution

$$R_N(x, y) = (N - x, y + N^2 - 2Nx) = (x, y) + (N - 2x)(1, N) \quad (66.1)$$

sends (j, j^2) to $(N - j, (N - j)^2)$. The two index sets $\{0, \dots, M\}$ and $\{N - M, \dots, N\}$ cover $\{0, \dots, N\}$, so $H_N = \text{conv}(H_M \cup R_N(H_M))$.

Suppose the given extension uses a variable z with affine output $(x(z), y(z))$. Introduce one scalar λ and impose exactly

$$0 \leq \lambda \leq N - 2x(z). \quad (66.2)$$

The upper bound is nonnegative because $0 \leq x(z) \leq M$. Project the new extension by

$$(z, \lambda) \mapsto (x(z) + \lambda, y(z) + N\lambda).$$

For fixed z , this traces the segment joining $(x(z), y(z))$ and its image under R_N . Thus every output lies in the required convex hull. Conversely, the new feasible set is convex, and its image contains both endpoint sets, so its image contains their convex hull. The extension remains bounded: z was bounded and $0 \leq \lambda \leq N$. $\square$

[write] Manuscript 04's reflection lemma (lemma 70.2, a Euclidean reflection in a hyperplane through the origin) is the same two-inequality step for a different involution; it is printed with 04's regular polygons in section 70.

Corollary 66.2 (Parabola formulation). *For every $N \geq 1$,*

$$\text{xc}(H_N) \leq 2\lceil \log_2(N+1) \rceil.$$

More explicitly, follow the chain $N, \lfloor N/2 \rfloor, \dots, 0$ in reverse, beginning at $(0, 0)$, and apply (66.2) at each nonzero stage. The number of stages is $\lceil \log_2(N+1) \rceil$.

Proof. The starting point requires no inequalities. Each nonzero stage adds two. The binary length identity gives the stated count. $\square$

Lemma 66.3 (Adding one vertex). *If a positive-dimensional polytope P has an extension with r facets, then $\text{conv}(P \cup \{a\})$ has one with at most $r + 1$ facets.*

Proof. Let E be an extension and $\pi : E \rightarrow P$ its affine output. Form a pyramid over E in one extra dimension. The base and one pyramid facet over each facet of E give $r + 1$ facets. Extend π affinely so that the apex is sent to a . The image is precisely $\text{conv}(P \cup \{a\})$. $\square$

The cap $\text{conv}(C_1, \dots, C_k)$ is an invertible affine image of H_{k-1} : first replace j by $(j+1)/(k+1)$, then replace (u, u^2) by $(\tau u, \tau(1 - 2u + u^2))$. For $k \geq 2$ this cap has positive dimension. Adding A and B and applying Corollary 66.2 and lemma 66.3 yields

$$\text{xc}(Q_n) \leq 2\lceil \log_2 k \rceil + 2 = U(n).$$

A nonnegative orthant is a diagonal positive-semidefinite cone, so the same upper bound holds for $\text{sx}(Q_n)$. No approximation of the parabola or of the output polygon has been used.

67 Generic lower bounds over real closed fields

This section isolates the only complexity-lower-bound input. It is a field-general version of the classical algebraic parameter-counting method [43, 47]. Its hypotheses concern algebraic independence of coordinates, not independence of the inequalities of an unknown lift. All auxiliary coefficients of that lift remain unrestricted.

67.1 A linear parameter count

Theorem 67.1 (Generic linear lower bound). *Let $F \subseteq K$ be an ordered subfield, and let $P \subset K^d$ be a full-dimensional polytope with N vertices. Suppose its dN labeled vertex coordinates are algebraically independent over F . Then every linear extension of P with r facets satisfies*

$$dN \leq r(r-1). \tag{67.1}$$

Proof. Choose a bounded extension E and put $q = \dim E$. A bounded q -dimensional polytope has at least $q + 1$ facets, so $q \leq r - 1$. One elementary reason is that at most q inequality normals either have a common nonzero kernel direction or, if they form a basis, admit a direction making every scalar product negative; either case gives an unbounded ray from a feasible point.

The affine output $E \rightarrow P$ has rank d . Complete its d coordinate functions to an affine coordinate system on $\text{aff}(E)$. In these coordinates the output is projection onto the first d entries, so no separate output parameters need to be counted. Write the r facet inequalities using $q + 1$ coefficients each. Divide each row by the absolute value of one of its nonzero coefficients. One coefficient in every row is now 1 or -1 , both in F , leaving at most rq scalar parameters.

Let L be the field generated over F by these parameters. At each vertex of E , q linearly independent active equations determine the coordinates. Gaussian elimination puts all those coordinates in L . Every vertex v of P is the image of a vertex of E : expose v by a K -linear objective and take a vertex of the resulting nonempty face of E . Here there is no requirement that the objective have coefficients in F or in $\mathbb{R}$.

Therefore all dN coordinates of P lie in L , and

$$dN \leq \text{trdeg}_F L \leq rq \leq r(r - 1).$$

The equality between bounded-extension size and the usual formulation size was explained in [Section 63](#), so allowing unbounded polyhedral extensions does not evade the result. $\square$

[write] Manuscript 04 proves the same parameter count in a form that does not assume full independence: it bounds the complexity by the transcendence degree s of the coordinate field. That form is needed for 04's circle construction, whose $2n$ coordinates have transcendence degree only n ([remark 70.6](#)); it is printed next. [Theorem 67.1](#) supersedes it in the count: the proof above never uses independence before its last line, so it shows $\text{trdeg}_F F(\text{vertex coordinates}) \leq r(r - 1)$ for every full-dimensional P with an r -facet extension, which is sharper than 04's $s \leq r^2$.

Proposition 67.2 (Parameter count; source 04). *Let $F \subseteq K$ be any subfield, and let $P \subset K^d$ be a full-dimensional polytope. If the field generated over F by its vertex coordinates has transcendence degree s , then*

$$\text{xc}_K(P) \geq \lceil \sqrt{s} \rceil. \quad (67.2)$$

Proof. This is the parameter-count method in [43], with its base field made explicit. If $d = 0$, then $s = 0$ and the conclusion is immediate. Otherwise, let an e -dimensional polytope Q with r facets project onto P . Choose affine coordinates on the affine hull of Q in which the projection is onto the first d coordinates. Write $Q = \{z : Bz \leq c\}$. There are $r(e + 1)$ entries in (B, c) . Every vertex of Q is the unique solution of e independent tight facet equations, so its coordinates belong to $F(B, c)$ by Cramer's rule. Every vertex of P is the image of a vertex of Q : maximize an exposing functional and take a vertex of the resulting nonempty face. Hence

$$s \leq \text{trdeg}_F F(B, c) \leq r(e + 1).$$

A bounded full-dimensional e -polytope has $r \geq e + 1$, giving $s \leq r^2$. $\square$

The coordinate change does not require adding a separate parameter budget for the projection. All its effects are already encoded in the rewritten facet coefficients (B, c) . The proposition does not require $F = \mathbb{Q}$. We will use $F = \mathbb{R}$ for the explicit surreal family and $F = \mathbb{R}_{\text{alg}}$ for the one-parameter existence theorem.

[write] In [proposition 67.2](#) the letters s (a transcendence degree), Q , B and c are local to 04's proof. Its Q is not 07's easy polygon.

67.2 Finite definability replaces rational exposing objectives

Lemma 67.3 (Finite definable sets are algebraic). *If a finite subset $X \subset K^d$ is definable in the language of ordered fields with parameters from a subfield F , then every coordinate of every point of X is algebraic over F .*

Proof. A coordinate projection of X is finite and F -definable. By quantifier elimination it is described by a Boolean combination of finitely many polynomial sign conditions with coefficients in F . Discard identically zero polynomials. If a point a is transcendental over F , none of these polynomials vanishes at a . Their finitely many real roots partition K into intervals on which all signs are constant. The point a lies in one such open interval, so the defining formula would contain that whole interval, contradicting finiteness. Apply this argument to each coordinate projection. $\square$

For a polytope, its finite vertex set is definable using only its membership predicate. In particular,

$$x \in P \wedge \forall y, z \in P (2x = y + z \implies y = z) \quad (67.3)$$

defines the vertices. The midpoint characterization of extremality holds over any ordered field: a nontrivial finite convex representation permits a small positive transfer of weight in both directions.

Theorem 67.4 (Generic semidefinite lower bound). *Under the hypotheses of [Theorem 67.1](#), any semidefinite lift of P of matrix order h satisfies, with $D = h(h+1)/2$,*

$$dN \leq D^2 \leq h^4. \quad (67.4)$$

Proof. Let the lift be an affine slice of $\mathcal{S}_+^h(K)$ with affine output onto P . The affine space defining the slice may be parametrized injectively by ℓ variables, with $\ell \leq D$. Since the output has full-dimensional image, its linear part on that affine space has rank d . Make an affine change of these ℓ variables so that the first d variables are exactly the output coordinates. The lift is then

$$P = \left\{ x \in K^d : \exists z \in K^{\ell-d}, \quad G_0 + \sum_{i=1}^d x_i G_i + \sum_{j=1}^{\ell-d} z_j G_{d+j} \succeq 0 \right\}. \quad (67.5)$$

The matrices need not have bounded entries.

In fact $\ell \leq D - 1$. If $\ell = D$, the defining affine space is all of $\mathcal{S}_+^h(K)$. An affine image of the entire positive-semidefinite cone is a translated cone. A bounded translated cone is a point, contradicting $\dim P = d \geq 1$. Thus the pencil has at most D coefficient matrices, each with D independent entries. Let L be generated over F by those at most D^2 entries.

The set (67.5) is L -definable because positive semidefiniteness is given by principal minors. Formula (67.3) makes its vertex set finite and L -definable. By [Lemma 67.3](#), all dN vertex coordinates are algebraic over L . Algebraic independence over F now gives

$$dN \leq \text{trdeg}_F L \leq D^2 \leq h^4.$$

This argument does not require Slater duality, an attained optimum in the lift, or a rational exposing objective. $\square$

67.3 Why the Archimedean proof needs care

The real proof in [47, Theorem 2.1] uses a rational objective uniquely exposing a chosen vertex. Such objectives exist for real polytopes but can fail over $K \supsetneq \mathbb{R}$. The following exact example isolates the issue without challenging the real theorem.

Example 67.5 (A vertex invisible to all real objectives). Let $0 < s < 1/m$ for every $m \geq 1$ and set

$$P = \text{conv}((0, 0), (-1, s), (1, -2s), (0, -1)).$$

The origin is a vertex. It uniquely maximizes $c_x x + c_y y$ exactly when

$$c_y > 0, \quad s < \frac{c_x}{c_y} < 2s.$$

The vector $(3s/2, 1)$ satisfies this condition, but no nonzero vector in $\mathbb{R}^2$ does. Its real shadow is the full-dimensional triangle with vertices $(-1, 0), (1, 0), (0, -1)$; the standard part of the origin lies in the relative interior of its upper edge.

Proof. Strictly beating the other three points is equivalent to $-c_x + sc_y < 0$, $c_x - 2sc_y < 0$, and $-c_y < 0$. These are exactly the conditions stated. A positive real number cannot lie between s and $2s$, while a nonpositive one does not satisfy the left inequality. $\square$

Thus one must not transport the rational-objective step unchanged into a non-Archimedean field. [Lemma 67.3](#) supplies an algebraic replacement. It also shows why a single surreal coefficient carrying a long normal form cannot secretly encode arbitrarily many independent coordinates for a *semialgebraic* lift: finite definability only recovers elements algebraic over the field of its scalar parameters.

[write] The proofs of [theorem 67.1](#) and [proposition 67.2](#) use a K -linear exposing objective, not a real one, and are therefore unaffected by [example 67.5](#).

68 Explicit independence and the hard lower bounds

A genericity argument alone would only assert the existence of a suitable perturbation. Here the independence has a finite, exact normal-form proof.

Lemma 68.1 (A monomial independence certificate). *For each fixed n , the $(2n + 1)$ -tuple*

$$\tau, \eta_1, \dots, \eta_{2n}$$

is algebraically independent over $\mathbb{R}$. Consequently, the $2n$ coordinates of $p_1, \dots, p_n$ are algebraically independent over $\mathbb{R}(\tau)$, and in particular over $\mathbb{R}$.

Proof. Eisenstein's criterion at 2 makes $X^{2n+1} - 2$ irreducible over $\mathbb{Q}$. Hence $1, a_n, \dots, a_n^{2n-1}$ are linearly independent over $\mathbb{Q}$. Consider a polynomial in formal variables $R, E_1, \dots, E_{2n}$ with real coefficients. A monomial $R^b \prod_\ell E_\ell^{m_\ell}$ evaluates to

$$\omega^{-b\omega - \omega^2 \sum_\ell m_\ell a_n^{\ell-1}}.$$

Two distinct exponent tuples cannot give the same surreal exponent. Indeed, equality would give

$$(b - b')\omega + \omega^2 \sum_\ell (m_\ell - m'_\ell) a_n^{\ell-1} = 0.$$

Divide by ω^2 . The second summand is real and the first is an infinitesimal real multiple of ω^{-1} . Thus each must vanish. The independence of the powers of a_n then gives $m_\ell = m'_\ell$ for every ℓ , and also $b = b'$.

A nonzero polynomial therefore becomes a finite sum of distinct Conway monomials with nonzero real coefficients. It cannot vanish. This proves the first assertion. Clearing denominators shows that the η_ℓ are independent over $\mathbb{R}(\tau)$. Finally, coordinatewise translation by $q_i \in \mathbb{R}(\tau)^2$ is a polynomial automorphism over $\mathbb{R}(\tau)$, so it preserves algebraic independence. $\square$

[write] Manuscript 04's independence certificate (lemma 70.5, rationally independent exponents give algebraically independent Conway monomials) is the general form of the step used here; 04 applies it with the exponents $\omega\sqrt{q_i}$, q_i distinct primes, where 07 uses $\omega^2 a_n^{\ell-1}$ and Eisenstein's criterion.

Applying Theorems 67.1 and 67.4 with $d = 2$, $N = n$, and $F = \mathbb{R}$ proves (64.7) and (64.8). The argument allows the coefficients of a proposed lift to lie in any larger real closed field. Enlarging that field does not create an algebraic relation among the fixed independent vertex coordinates. Thus no choice of extra surreal scales in the lift can defeat the parameter count.

The word *explicit* here has a precise meaning. All coordinates are specified by rational arithmetic, the unique positive root of one given integer polynomial, and finitely many named Conway monomials. It does not assert an algorithm for arbitrary surreal comparison. Nor does it supply a small rational numerical realization with the same provable lower bounds: those are different effectivity questions.

68.1 Numerical sizes of the proved bounds

Table 6 lists bounds, not computed exact complexities. The semidefinite lower column uses the stronger inequality $D^2 \geq 2n$, not only $h^4 \geq 2n$. The easy column bounds *both* lift sizes of Q_n .

n	Easy upper bound	Hard LP lower bound	Hard PSD lower bound
258	18	24	7
4 098	26	92	13
65 538	34	363	27
1 048 578	42	1 449	54
16 777 218	50	5 794	108

Table 6: Consequences of the proved inequalities. The large examples are symbolically defined families, not explicitly enumerated in the finite verification script. A lower bound smaller than the easy upper bound does not establish a gap at that particular n . [write] (Source 07. For example, the row $n = 258$ is a linear gap, $24 > 18$, but not a semidefinite one, $7 < 18$.)

Remark 68.2 (Where the linear gap starts). [write] Recomputing from the stated formulas $U(n) = 2\lceil \log_2(n-2) \rceil + 2$ and " $r(r-1) \geq 2n$ " (least such r), the first $n \geq 4$ at which the proved hard linear lower bound exceeds the easy upper bound is $n = 121$: there $U(121) = 16$ and the least r with $r(r-1) \geq 242$ is 17. The gap is not yet monotone: for $131 \leq n \leq 153$ the bound $U(n) = 18$ is not exceeded, and from $n = 154$ on (checked to $n = 2 \cdot 10^5$, and for all larger n because $\sqrt{2n} > 2 \log_2 n + 4$ there) every n has a strict linear gap. The semidefinite columns of table 6 give a strict gap only from the row $n = 1\,048\,578$ on ($54 > 42$). All five rows reproduce exactly. These are bounds, as the caption says, not computed complexities. (Computation by the write phase, not by source 07.)

69 Identical initial slack matrices and identical jets

69.1 Canonical normalization matters

A facet slack row can be multiplied by any positive scalar without changing its factorization ranks. Its valuation and leading coefficient do change under such a multiplication. A comparison of initial slack matrices is therefore meaningful only after a normalization has been fixed. For our cyclically labeled polygons we use the same determinant convention on both sides:

$$S(v)_{ij} = D(v_i, v_{i+1}, v_j), \quad i, j \in \{1, \dots, n\}, \quad v_{n+1} = v_1. \quad (69.1)$$

The strict orientation statements already proved show that this is a facet–vertex slack matrix. The zeros occur exactly at $j = i, i + 1$, with cyclic indices.

Theorem 69.1 (Initial-slack and jet agreement). *For P_n, Q_n the matrices (69.1) satisfy*

$$S(p)_{ij} - S(q)_{ij} \in \mathfrak{i}_\tau \quad \text{for every } i, j.$$

For each nonzero corresponding pair,

$$v(S(p)_{ij}) = v(S(q)_{ij}), \quad \text{lc}(S(p)_{ij}) = \text{lc}(S(q)_{ij}), \quad (69.2)$$

and even $S(p)_{ij}/S(q)_{ij} - 1 \in \mathfrak{i}_\tau$. The only nonzero reference valuations are $0, v(\tau), 2v(\tau)$. Both matrices have ordinary rank three, but their nonnegative and semidefinite ranks have the separations of Theorem 64.1.

Proof. Each slack is a polynomial of degree two in the coordinates. Its perturbation is flat by Lemma 65.1. Reference entries are polynomials of degree at most two in τ , because every reference coordinate is affine in τ . The orientation formulas show that every nonzero one has a positive leading real coefficient. Thus its valuation is one of $0, v(\tau), 2v(\tau)$, and a flat error can change neither that valuation nor its leading coefficient. Dividing by the reference entry, which has growth of a fixed power of τ , preserves flatness. The incident zeros remain zero by the determinant identity itself, not by a perturbation estimate.

Every row is an affine function of the vertex coordinates, so the ordinary rank is at most three. The submatrix on the first three rows and columns has the pattern

$$\begin{pmatrix} 0 & 0 & a \\ b & 0 & 0 \\ c & d & 0 \end{pmatrix}, \quad a, b, d > 0.$$

Its determinant is $abd > 0$, including when $n = 4$. Thus the rank is exactly three. The final assertion follows from (63.2) and the bounds already proved. $\square$

If $s \neq 0$ has normal form beginning $c\omega^{-\gamma}$, write $\text{in}_v(s) = c\omega^{-\gamma}$ and put $\text{in}_v(0) = 0$. Then the theorem says that $\text{in}_v(S(p)) = \text{in}_v(S(q))$ entrywise. The result is stronger than coincidence of tropical valuations or of a Boolean support matrix: it also preserves the actual real coefficients of those leading monomials.

[write] The same-circle family of source 04 has the analogous properties with all nonzero slack valuations equal to 0 (section 71, (71.1)).

69.2 What the negative answer does and does not cover

The theorem rules out a function of the *initial slack matrix alone* that recovers either exact factorization rank for all these inputs. Together with (64.6)–(64.8), it rules out a uniform multiplicative estimate from that information: the same input data occur with an unbounded ratio of possible complexities.

The all-orders assertion is stronger in a different direction. The labeled vertex tuple, the labeled slack tuple, and every fixed polynomial observation over $\mathbb{R}(\tau)$ have the same images modulo all powers of τ whenever the observation has finite values; the rational-function version is given in Lemma 65.1. These statements do not preserve all exact polynomial *zero predicates*: (65.5) is a counterexample. There is no contradiction between a generic hard configuration and an algebraically special easy one.

Nor does this assert a failure of all tropical methods. A method that retains extra initial forms of polynomial relations, higher coarsenings, or additional exact certificates has strictly more information than the entrywise initial slack matrix used here. The precise insufficiency statement must not be broadened beyond its input model.

70 A second construction: regular and cyclic polygons on one circle (source 04)

[write] This section prints manuscript 04's explicit family and main theorem (its Subsection 1.1), the regular-polygon and cyclic bounds (from its Section 4) and its Section 5 proper ("The explicit cyclic construction"). It is the second construction of this Part, beside 07's theorem 64.1. Its hard polygon, called P_n in 04, is printed as P_n° with vertices p_i° to keep it apart from 07's P_n ; in 04's text q_i are distinct ordinary primes; the flat ideal $\mathcal{I}_\varepsilon$ of 04 is printed $\mathfrak{j}_\varepsilon$ and its valuation v as ν .

70.1 An explicit family

Let $n = 2^m$ with $m \geq 2$, and put

$$\theta_i = \frac{(2i+1)\pi}{n}, \quad a_i = \tan \frac{\theta_i}{2}, \quad 0 \leq i < n. \quad (70.1)$$

These a_i are ordinary real algebraic numbers. No vertex has angle π , so they are all defined. The trigonometric notation specifies ordinary constants only; no trigonometric function on infinite surreals is used. Define the rational circle parametrization

$$\varphi(z) = \left(\frac{1-z^2}{1+z^2}, \frac{2z}{1+z^2} \right). \quad (70.2)$$

Let q_i be pairwise distinct ordinary primes. In **No**, set

$$\varepsilon = \omega^{-1}, \quad \mu_i = \omega^{-\omega \sqrt{q_i}}, \quad r_i = \varphi(a_i), \quad p_i^\circ = \varphi(a_i + \mu_i), \quad (70.3)$$

and let $R_n = \text{conv}(r_0, \dots, r_{n-1})$ and $P_n^\circ = \text{conv}(p_0^\circ, \dots, p_{n-1}^\circ)$. Every expression ω^α in this manuscript denotes a *Conway monomial*; it is not an unqualified use of an analytic exponential with base ω .

Theorem 70.1 (Explicit all-order cyclic gap; source 04). *The polygons in (70.3) have the following properties.*

- (i) R_n is regular, P_n° has exactly n vertices in the same cyclic labeling, and every vertex of both polygons lies on $x^2 + y^2 = 1$.
- (ii) For every ordinary $M \geq 1$ and every i ,

$$\|p_i^\circ - r_i\|_\infty < \varepsilon^M.$$

Consequently their coordinates have the same image in every $O/\varepsilon^M O$, and their standard parts agree.

- (iii) All 3×3 affine orientation determinants have the same nonzero leading real coefficient and valuation for the two configurations. Their determinant-normalized slack matrices have the same exact zero pattern and the same leading entries.
- (iv) Their extension complexities satisfy

$$m \leq \text{xc}(R_n) \leq 2m, \quad \lceil \sqrt{n} \rceil \leq \text{xc}(P_n^\circ) \leq \min\{n, \lceil 24\sqrt{n} \rceil\}. \quad (70.4)$$

In particular,

$$\frac{\text{xc}(P_n^\circ)}{\text{xc}(R_n)} = \Theta\left(\frac{\sqrt{n}}{\log n}\right).$$

The proof is given in [section 70.6](#). The square-root lower bound comes from transcendence degree, using the classical parameter-count method of Fiorini–Rothvoß–Tiwary [43]. The square-root upper bound is the theorem of Kwan–Sauermaann–Zhao transferred to a real closed field [53]. Thus the order of the gap is sharp within the cyclic family relative to a regular reference polygon. No improved universal bound for ordinary polygons is asserted.

[write] Comparison with 07’s [theorem 64.1](#): 07 separates both linear and semidefinite lift sizes for every $n \geq 4$ inside a fixed triangle, with a one-sided lower bound for the hard polygon; 04 separates linear lift sizes for $n = 2^m$ on a fixed circle, with the two-sided order above. The jet scale is $\tau = \omega^{-\omega}$ in 07 and $\varepsilon = \omega^{-1}$ in 04; 04’s standard parts are the regular polygon itself, not a triangle.

70.2 A direct logarithmic extension for a regular polygon

The following standard reflection construction [44] also supplies the constant in (70.4).

Lemma 70.2 (Adding a reflected copy; source 04). *Let $Q \subset K^d$ lie in $u \cdot x \geq 0$, with $u \neq 0$, and let $\sigma(x) = x - 2(u \cdot x)u/(u \cdot u)$. Then*

$$\text{xc}_K(\text{conv}(Q \cup \sigma Q)) \leq \text{xc}_K(Q) + 2.$$

Proof. Take an extension E of Q with projection π . Adjoin a variable λ and the two inequalities

$$0 \leq \lambda \leq \frac{2u \cdot \pi(z)}{u \cdot u}.$$

Project the resulting bounded polytope by $(z, \lambda) \mapsto \pi(z) - \lambda u$. Its image is convex and consists of segments joining x to $\sigma(x)$ for $x \in Q$. It contains both Q and σQ , and every such segment belongs to their convex hull. Thus the image is exactly the claimed polytope. $\square$

[write] In [lemma 70.2](#), Q is any polytope and σ a hyperplane reflection; 07’s two-inequality step [lemma 66.1](#) is the same device for the affine involution R_N of the parabola.

Corollary 70.3 (Source 04). *For $n = 2^m$, $m \leq \text{xc}_K(R_n) \leq 2m$.*

Proof. Starting with the vertex at angle π/n , suppose that the first t vertices at angles $(2j+1)\pi/n$, $0 \leq j < t$, have been obtained, where $t \leq n/2$. Reflect their convex hull in the line at angle $2t\pi/n$. The reflected vertex indices are $2t - 1 - j$, so this doubles the list to the first $2t$ vertices. The original t vertices lie strictly on one side of the reflecting line, because their angle differences from it are between $-\pi$ and 0 . Applying [lemma 70.2](#) for $t = 1, 2, \dots, 2^{m-1}$, where t doubles at each step, uses $2m$ inequalities. A point has zero facets in its affine hull.

For the lower bound, a polytope with r facets has at most 2^r vertices: a vertex is determined by its set of tight facets. Each vertex of a projection is the image of some vertex of the original polytope. Thus $n \leq 2^r$ and $r \geq m$. $\square$

[write] The lower bound $m \leq \text{xc}$ is a general counting bound; 07 states no lower bound for its easy polygon Q_n .

70.3 The cyclic upper bound and its transfer

Proposition 70.4 (Transferred cyclic bound; source 04). *Every cyclic n -gon over a real closed ordered field satisfies*

$$\text{xc}_K(P) \leq \min\{n, \lceil 24\sqrt{n} \rceil\}.$$

Proof. Over $\mathbb{R}$, the $24\sqrt{n}$ bound is [53, Theorem 1.3]. We explain why it applies in K . Fix the ordinary integers n and $r = \lceil 24\sqrt{n} \rceil$. Translate and rescale a circle to the unit circle; a positive square root of its squared radius exists in K . For vertices in counterclockwise order set

$$S_{ij} = \det \begin{pmatrix} 1 & p_{i,1} & p_{i,2} \\ 1 & p_{i+1,1} & p_{i+1,2} \\ 1 & p_{j,1} & p_{j,2} \end{pmatrix}, \quad i+1 \pmod{n}. \quad (70.5)$$

The circle equations, distinctness of the vertices, and positivity of all nonincident S_{ij} are polynomial conditions. They specify the desired cyclic polygon and its facet slacks. The sentence saying that every such configuration satisfies $\text{NF}_r(S)$ is first-order, with rational coefficients. It holds in $\mathbb{R}$ by the cited theorem and [proposition 63.6](#), and hence in every real closed K . Finally a polygon itself has n facets, giving the other upper bound. $\square$

The integer r is chosen externally for each fixed n . No quantification over a set of integers inside the ordered field, and no assertion about randomness in a non-Archimedean field, is involved.

[write] In (70.5) the p_i are the vertices of an arbitrary cyclic polygon; the matrix is the same edge-normalized slack as 07's (69.1), written as an affine 3×3 determinant.

70.4 Rational identities on the circle

For every z in an ordered field, $1 + z^2 > 0$ and

$$\varphi_1(z)^2 + \varphi_2(z)^2 = 1, \quad 1 + \varphi_1(z) = \frac{2}{1 + z^2}, \quad z = \frac{\varphi_2(z)}{1 + \varphi_1(z)}. \quad (70.6)$$

Thus the coordinate field of a list of points $\varphi(z_i)$ is exactly the field of its parameters z_i .

Two further identities make all scale assertions explicit:

$$\varphi_1(s) - \varphi_1(a) = -\frac{2(s-a)(s+a)}{(1+s^2)(1+a^2)}, \quad (70.7)$$

$$\varphi_2(s) - \varphi_2(a) = \frac{2(s-a)(1-as)}{(1+s^2)(1+a^2)}. \quad (70.8)$$

The affine orientation determinant is

$$D(a, b, c) := \det \begin{pmatrix} 1 & \varphi_1(a) & \varphi_2(a) \\ 1 & \varphi_1(b) & \varphi_2(b) \\ 1 & \varphi_1(c) & \varphi_2(c) \end{pmatrix} = \frac{4(b-a)(c-a)(c-b)}{(1+a^2)(1+b^2)(1+c^2)}. \quad (70.9)$$

For instance, multiplying the three rows by their denominators turns the numerator determinant into one with rows $(1+a^2, 1-a^2, 2a)$; elementary column operations reduce it to the Vandermonde determinant, with the factor and sign in (70.9). These are rational identities, not asymptotic expansions.

[write] In (70.7)–(70.8) the letter s is a free variable, not a scale.

70.5 Algebraic independence below every observed scale

Lemma 70.5 (Independent monomials; source 04). *If $\alpha_1, \dots, \alpha_s \in \mathbf{No}$ are linearly independent over $\mathbb{Q}$, then the Conway monomials $\omega^{-\alpha_1}, \dots, \omega^{-\alpha_s}$ are algebraically independent over $\mathbb{R}$.*

Proof. Substitution into a nonzero polynomial gives a finite sum

$$\sum_{e \in E} c_e \omega^{-\sum_i e_i \alpha_i}, \quad c_e \in \mathbb{R} \setminus \{0\}, \quad e \in \mathbb{N}^s,$$

where E is finite. Distinct exponent vectors e give distinct surreal exponents, by rational linear independence. After sorting these exponents, this is a nonempty finite Conway normal form. Uniqueness of normal forms [52] says that it is nonzero. $\square$

The numbers $\sqrt{q_1}, \dots, \sqrt{q_s}$ for distinct primes are linearly independent over $\mathbb{Q}$; in fact 1 together with these numbers is linearly independent. One elementary proof works in the multiquadratic field generated by the prime square roots. The independent prime square classes give automorphisms that change the sign of any one square root and fix the others. Subtracting a putative relation from its image under one such automorphism kills every term except the chosen coefficient. The independence of the prime square classes follows from unique factorization in $\mathbb{Z}$.

Since multiplication by the nonzero surreal ω preserves rational linear independence, lemma 70.5 applies to $\alpha_i = \omega\sqrt{q_i}$. Also $\omega\sqrt{q_i} > M$ for every ordinary M , and hence

$$0 < \mu_i = \omega^{-\omega\sqrt{q_i}} < \omega^{-M} = \varepsilon^M. \quad (70.10)$$

We have therefore exhibited finitely many algebraically independent nonzero elements of the flat ideal, not merely assumed their existence.

70.6 Proof of the main theorem

Proof of theorem 70.1. Choose a set-sized real closed field containing $\mathbb{R}$, ε , and the finitely many μ_i . All subsequent operations occur there. The ordinary half-angle identities show that r_i is the regular polygon vertex $(\cos \theta_i, \sin \theta_i)$. Equation (70.6) puts each p_i° on the same unit circle.

By (70.10) and lemma 63.4, or directly by (70.7)–(70.8), both coordinates of $p_i^\circ - r_i$ belong to j_ε . Their maximum absolute value is still in j_ε . This proves part (ii). The standard parts r_i are distinct, so the p_i° are distinct.

Every affine orientation determinant of three distinct r_i is nonzero: a line intersects a circle in at most two points, as is also clear from (70.9). The corresponding determinant for P_n° differs

from it by an element of $\mathbf{j}_\varepsilon$. It consequently has the same sign, valuation zero, and nonzero leading real coefficient. In particular the edge slacks (70.5) are positive at every nonincident vertex and zero exactly at the two incident vertices. Each p_i° is uniquely exposed by adding its two adjacent edge slack functionals and minimizing the sum. All p_i° are vertices, the edges occur in the same cyclic order, and part (iii) follows as well.

Put $z_i = a_i + \mu_i$. Translation by constants in $\mathbb{R}$ preserves algebraic independence, so the z_i are algebraically independent over $\mathbb{R}$. The inverse identity in (70.6) gives

$$\mathbb{R}(p_{i,1}^\circ, p_{i,2}^\circ : 0 \leq i < n) = \mathbb{R}(z_0, \dots, z_{n-1}).$$

Its transcendence degree over $\mathbb{R}$ is exactly n . Applying [proposition 67.2](#) gives the lower bound for P_n° ; applying [proposition 70.4](#) gives its upper bound. The two bounds for R_n come from [corollary 70.3](#). Scalar-extension invariance permits the notation $\text{xc}_{\mathbf{No}}$ throughout.

Finally the ratio lies between $\sqrt{n}/(2m)$ and $\lceil 24\sqrt{n} \rceil/m$. Since $m = \log_2 n$, these have the same asymptotic order. $\square$

Remark 70.6 (The circle consumes one parameter per vertex; source 04). The coordinate transcendence degree here is n , not $2n$; every vertex satisfies $x_i^2 + y_i^2 = 1$. Applying the unconstrained generic-polygon lower bound $\sqrt{2n}$ without accounting for these relations would be an error. The square-root order remains sharp because the cyclic upper bound applies.

Remark 70.7 (The circle count against 07's parameter count). [write] This is the count $s = n$ that 04's [proposition 67.2](#) needs and 07's [theorem 67.1](#) does not provide as stated (the $2n$ coordinates of P_n° are not independent). The proof of [theorem 67.1](#), which bounds the transcendence degree of the coordinate field by $r(r-1)$, gives with $s = n$ the slightly sharper $\text{xc}(P_n^\circ)(\text{xc}(P_n^\circ) - 1) \geq n$. This observation combines the two sources and is made by the write phase; neither manuscript states it. The order $\sqrt{n}$ is unchanged.

70.7 Only two valuation ranks are needed

The construction does not require the full hierarchy of surreal scales. Let

$$K_n = \text{the real closure inside } \mathbf{No} \text{ of } \mathbb{R}(\omega^{-1}, \mu_0, \dots, \mu_{n-1}).$$

Before real closure the values belong to the additive group generated by 1 and $\omega\sqrt{q_i}$. This follows directly for polynomials from normal-form uniqueness, and for rational functions by subtraction of values. After real closure the values belong to

$$\Gamma_n = \mathbb{Q} + \omega \text{span}_{\mathbb{Q}}\{\sqrt{q_0}, \dots, \sqrt{q_{n-1}}\}. \quad (70.11)$$

Indeed, if $x \neq 0$ is algebraic and $\sum_j b_j x^j = 0$, at least two nonzero summands have the same least value; otherwise the uniquely least one could not cancel. Thus $(j-k)v(x) = v(b_k) - v(b_j)$, so its value lies in the divisible hull of the base value group. Conversely, real closure supplies positive roots and hence all rational multiples of base values. This proves equality with (70.11).

The low component $\mathbb{Q}$ is a convex subgroup, and the quotient is an Archimedean subgroup of $\mathbb{R}$. Both components are nonzero. The natural valuation therefore has rank two. Its element $v(\varepsilon) = 1$ lies in the low component and is not an order unit. The all-order effect is already present in this small ordered-field setting.

Remark 70.8 (One rank-two field for every n). [write] Put $\mu_q = \omega^{-\omega\sqrt{q}}$ for each ordinary prime q (04 indexes the same monomials by vertex, $\mu_i = \mu_{q_i}$). The real closure L inside $\mathbf{No}$ of $\mathbb{R}(\omega^{-1}, \mu_q : q \text{ prime})$ has value group $\mathbb{Q} + \omega \text{span}_{\mathbb{Q}}\{\sqrt{q} : q \text{ prime}\}$ and so is one rank-two field for every n : it contains every K_n , whatever distinct primes are chosen, and when the q_i are the

first n primes the fields K_n form a chain whose union is L . Manuscript 04 states a field K_n for each n . The proof is 04's: the exponents 1 and $\omega\sqrt{q}$ (q prime) are linearly independent over $\mathbb{Q}$, because 1, $\sqrt{q}$ are and ω is infinite, so the least-value argument above gives the divisible hull; its convex subgroups are $0 \subsetneq \mathbb{Q} \subsetneq$ the whole group, with Archimedean quotients.

71 What the asymptotic data cannot recover (source 04)

[write] Manuscript 04's Section 7, for its same-circle family. Its bounded-factor normalization (04's Proposition 7.1) is printed with 07's residue theorem as [theorem 73.2](#), together with the paragraph that follows it in 04. The corresponding statements for 07's family are [theorem 69.1](#) and [lemma 65.1](#).

71.1 Leading Plücker data and slack entries

For a labeled planar configuration V , put

$$H(V) = \begin{pmatrix} 1 & \cdots & 1 \\ v_{0,1} & \cdots & v_{n-1,1} \\ v_{0,2} & \cdots & v_{n-1,2} \end{pmatrix}.$$

Its maximal minors are the affine orientation determinants. In our examples all minors indexed by three distinct columns are nonzero and have valuation zero. Their leading coefficients are exactly the minors of $H(R_n)$. Thus the signs, valuations, and leading coefficients of all these Plücker coordinates are identical for P_n° and R_n .

The edge-normalized slack matrix (70.5) obeys the stronger entrywise identity

$$S(P_n^\circ) - S(R_n) \in \mathfrak{j}_\varepsilon^{n \times n}. \quad (71.1)$$

All nonzero entries have valuation zero and the same positive leading coefficient; the zero pattern is exactly $j = i$ or $j = i + 1$ in both matrices. It follows from [proposition 63.6](#) that this identical leading slack data is compatible with nonnegative ranks of order $\log n$ and $\sqrt{n}$.

These statements concern *entries of the chosen slack matrices and maximal minors of the point matrices*. They do not say that every polynomial identity, or every minor of every derived matrix, is preserved. A polynomial that vanishes at the regular configuration may acquire a nonzero flat value. Such exact equalities are precisely information that an all-order observation can lose.

71.2 Standard part cannot increase nonnegative rank

There is also a useful positive statement, proved without taking limits of possibly unbounded factors. [write] It is printed as the second proof of 07's [theorem 73.2](#) (label `poly:bao:prop:normalization`), of which it is the case $V = \mathcal{O}$, $k = \mathbb{R}$.

71.3 All finite linear optimization values agree to all orders

For a finite vertex list V and $c \in K^d$, define the support function

$$h_V(c) = \max_i c \cdot v_i.$$

This is well-defined without any topological hypothesis.

Proposition 71.1 (Flat stability of bounded observations; source 04). *Suppose $V, W \in (\mathcal{O}^d)^n$ and $v_i - w_i \in \mathfrak{j}_\varepsilon^d$ for each i . Then for every $c \in \mathcal{O}^d$,*

$$h_V(c) - h_W(c) \in \mathfrak{j}_\varepsilon.$$

There is also $\eta \in \mathfrak{j}_\varepsilon \cap K_{\geq 0}$ such that every point of $\text{conv}_K(V)$ is within η in the maximum norm of a point of $\text{conv}_K(W)$, and conversely.

Proof. For finite lists of scalars, $|\max_i \alpha_i - \max_i \beta_i| \leq \max_i |\alpha_i - \beta_i|$. Apply this to $\alpha_i = c \cdot v_i$ and $\beta_i = c \cdot w_i$. Each difference is in $\mathfrak{j}_\varepsilon$, and so is the finite maximum of their absolute values. For the second statement set $\eta = \max_i \|v_i - w_i\|_\infty \in \mathfrak{j}_\varepsilon$. To $\sum_i \lambda_i v_i$ associate $\sum_i \lambda_i w_i$ with the same convex coefficients. The norm of the difference is at most η . Interchanging V, W proves the other direction. $\square$

In particular every finite-direction linear program on P_n° and R_n has the same all-order optimal-value data. This does not mean that the optimal faces always agree, nor does it cover objectives with coefficients as large as reciprocals of the discarded flat perturbations.

[write] The second clause is the analogue of 07's [theorem 64.1](#) (ii) (Hausdorff-type inclusions with error box $[-\eta, \eta]^2$); in [proposition 71.1](#) the scale ε is arbitrary.

The signed area is a polynomial in the vertex coordinates and therefore also agrees modulo $\mathfrak{j}_\varepsilon$. Squared side lengths do as well. For a side whose length has positive real standard part, the difference of lengths is the difference of squared lengths divided by the sum of the lengths; that denominator is a unit in $\mathcal{O}$. Thus the perimeters of our two polygons agree modulo $\mathfrak{j}_\varepsilon$, while the circumradius is exactly one for both. None of these metric observations determines their extension complexities.

71.4 Coherent low-rank factorizations at every finite order

The obstruction can be stated as an explicit failure of lifting a coherent family of approximate solutions to an exact one.

Theorem 71.2 (All-order factorization does not imply exact factorization; source 04). *For $n = 2^m$ with $m \geq 10$, the matrix $S(P_n^\circ)$ has nonnegative rank strictly greater than $2m$, but there are fixed nonnegative real matrices T_0, U_0 of inner dimension at most $2m$ such that*

$$S(P_n^\circ) - T_0 U_0 \in \varepsilon^M \mathcal{O}^{n \times n} \quad \text{for every } M \geq 1.$$

In particular their reductions give compatible algebraic factorizations of every finite jet of $S(P_n^\circ)$, induced by nonnegative lifts, without an exact nonnegative factorization of that inner dimension over K .

Proof. By [corollary 70.3](#) and [proposition 63.6](#), choose a nonnegative real factorization $S(R_n) = T_0 U_0$ of inner dimension at most $2m$. Then use (71.1). The lower bound $\text{rank}_+^K(S(P_n^\circ)) \geq \lceil 2^{m/2} \rceil > 2m$ for $m \geq 10$ excludes the exact factorization. $\square$

Remark 71.3 (The hypothesis $m \geq 10$ is conservative). [write] The inequality $\lceil 2^{m/2} \rceil > 2m$ already holds at $m = 9$ ($\lceil \sqrt{512} \rceil = 23 > 18$); it fails for $2 \leq m \leq 8$ (at $m = 8$, $16 = 16$). The theorem is therefore true for $m \geq 9$ by the same proof. This is a conservative hypothesis, not an error. (Checked by the write phase.)

No positivity structure is being imposed on the possibly nonreduced rings $\mathcal{O}/\varepsilon^M \mathcal{O}$. The factors there are simply the reductions of actual nonnegative real factors. Their compatibility is especially strong: the same pair T_0, U_0 works for every M . The remaining error is not zero;

it lies in the nonzero kernel j_ε . This is a nonseparatedness phenomenon, not an appeal to a failure of numerical precision.

[write] This theorem bears on 07's question on lifting a residue factorization ([question 82.4](#)): coherent finite-order solutions, even one fixed pair for all orders, are not a sufficient lifting criterion. Manuscript 07 makes the same point in words after that question.

Corollary 71.4 (An information-theoretic obstruction; source 04). *There is no function of the labeled all-order vertex jets alone that returns extension complexity for all cyclic surreal polygons. The same is true after adjoining their standard parts, leading Plücker data, leading slack entries, and all finite-direction support-function jets.*

Proof. For sufficiently large n , [theorem 70.1](#) and [proposition 71.1](#) give two inputs with identical such data but different answers. $\square$

This is not a computability claim about every representation of surreal numbers. Exact finite expressions for the μ_i , for example, contain information that the observation map intentionally discards.

[write] (The corollary is unlabelled in 04; the label `poly:bao:cor:information` is added here. 07's counterpart is the discussion after [theorem 69.1](#), which also covers semidefinite rank.)

72 A common field and a sharp valuation-rank boundary

72.1 All examples in one rank-three field

Let $\mathbb{R}_{\text{alg}}$ be the real algebraic numbers and define the divisible ordered subgroup of the additive surreal field

$$\Gamma = \{c\omega^2 + b\omega + d : c \in \mathbb{R}_{\text{alg}}, b, d \in \mathbb{Q}\}. \quad (72.1)$$

Its order is lexicographic by the coefficients c, b, d . Let

$$K_0 = \mathbb{R}(\omega^{-\gamma} : \gamma \in \Gamma) \subseteq \mathbf{No}, \quad K = \{x \in \mathbf{No} : x \text{ is algebraic over } K_0\}.$$

The latter is the relative real closure, with the inherited order.

Proposition 72.1 (Fixed field and fixed rank). *The field K is set-sized and real closed, with real residue field and natural value group exactly Γ . In particular its valuation rank is three. It contains every P_n, Q_n , together with ε and τ , for all ordinary finite $n \geq 4$.*

Proof. The set of monomial generators is a set; rational expressions in finitely many such generators and real coefficients form a set. Nonzero polynomials over K_0 have finitely many roots, so the collection of their surreal roots is also set-sized. It is a field. Positive square roots and roots of odd-degree polynomials over it exist in $\mathbf{No}$ and remain algebraic over K_0 , proving real closedness.

Finite sums and quotients of monomials from Γ have valuations in Γ , and every $\gamma \in \Gamma$ is realized by a generator. If a nonzero x is algebraic over K_0 , choose a polynomial relation $\sum_i a_i x^i = 0$. At least two nonzero summands have the same least valuation; otherwise the uniquely dominant summand could not cancel. Thus

$$v(a_i) + iv(x) = v(a_j) + jv(x) \quad \text{for some } i \neq j.$$

Since Γ is divisible, this puts $v(x)$ in Γ . Therefore algebraic closure inside the ordered field adds no values. The residue field is $\mathbb{R}$, as $K \subseteq \mathbf{No}$ contains all real constants. The convex subgroup chain is exactly

$$0 \subsetneq \mathbb{Q} \subsetneq \mathbb{Q}\omega + \mathbb{Q} \subsetneq \Gamma;$$

each successive quotient is Archimedean. This gives valuation rank three. Every $a_n^{\ell-1}$ is real algebraic, so every exponent in (64.1) and (64.5) belongs to Γ . $\square$

This avoids any infinite-summation issue. The proof does not need a full Hahn-series field, a support-completion theorem, or a changing universe as n increases. The exponent group has countably infinite rational rank but only three Archimedean layers. These are different invariants.

72.2 The exact all-orders kernel condition

For a nontrivial ordered abelian group, a positive element γ is an *order unit* if every positive group element is at most $m\gamma$ for some positive ordinary integer m .

The valuation boundary (source 04). The natural value group is denoted by $G = v(K^\times)$, ordered so that $v(x) > v(y)$ precisely when x/y is infinitesimal. A positive $\gamma \in G$ is an *order unit* if for every $\beta \in G$ there is an integer $M \geq 1$ with $\beta \leq M\gamma$. [write] (04 writes Γ and v ; renamed G and v . The two definitions of order unit agree.)

Theorem 72.2 (When all power jets lose information; sources 07 and 04). *Let $K \supseteq \mathbb{R}$ be real closed, with natural value group G , and let s be a positive infinitesimal. Then*

$$j_s \neq 0 \iff v(s) \text{ is not an order unit in } G. \quad (72.2)$$

Thus all s -power jets determine a finite coordinate tuple exactly if and only if $v(s)$ is an order unit.

Proof. By Lemma 63.2, a nonzero $x \in j_s$ has $v(x) > mv(s)$ for every m : use the bound with $m+1$ if a strict valuation comparison is needed. Conversely, any value $\beta > mv(s)$ for every m is represented by a nonzero field element x with $|x| < s^m$ for every m . This is exactly the negation of the order-unit condition. Equality of all coordinate jets is equivalent to each coordinate difference belonging to this kernel. $\square$

Second proof (source 04; its Proposition 3.3, “Exact nonseparatedness criterion”, stated for $s = \varepsilon$).

A nonzero flat element has value larger than every integer multiple of $v(\varepsilon)$, by lemma 63.2. Conversely, a value with this property is realized by a nonzero element of K by the definition of the value group; that element is finite and belongs to j_ε . $\square$

In a rank-one valued ordered field, the value group is Archimedean and every positive value is an order unit. Thus $j_\varepsilon = 0$. A field of higher rank can have $j_\varepsilon = 0$ for one choice of ε and $j_\varepsilon \neq 0$ for another. The distinction is relative to the observation scale, not just to the rank of the field. [write] (Source 04.)

Example 72.3 (The basic surreal flat element; source 04). For $\varepsilon = \omega^{-1}$, the nonzero element $\omega^{-\omega}$ is smaller than $\varepsilon^M = \omega^{-M}$ for every ordinary M . Thus J_ε is not injective. Equality of all its observations is not equality in $\mathbf{No}$.

Corollary 72.4 (Rank two is the sharp existence threshold). *No valued field of natural valuation rank one has distinct labeled polytopes with identical all-order power jets at a positive infinitesimal scale. There is, however, a single set-sized real closed field of valuation rank two containing pairs of polygons with all the jet, initial-slack, shadow and complexity-separation properties above.*

Proof. A rank-one value group is Archimedean, so every positive element is an order unit. Then [Theorem 72.2](#) makes the jet kernel zero.

For the converse, repeat the construction with

$$\rho = \omega^{-1}, \quad \eta_\ell = \omega^{-\omega a_n^{\ell-1}}, \quad \Gamma_2 = \mathbb{R}_{\text{alg}}\omega + \mathbb{Q}.$$

[write] (That is, with the scale τ of (64.1) replaced by $\rho = \omega^{-1}$; here ρ is 07's general scale letter.) The same orientation, reflection and independence proofs apply. The exponents of ρ and the η_ℓ are independent in exactly the way used in [Lemma 68.1](#). The common field is the relative real closure of $\mathbb{R}(\omega^{-\gamma} : \gamma \in \Gamma_2)$, and the proof of [Proposition 72.1](#) now gives rank two. The separate assertion that the entire polygon is flat-close to T at a *larger* scale ε is omitted: its distance to T is $O(\rho)$. All-order agreement *between the two polygons* remains unchanged. $\square$

The scale condition is sharper than a condition on the whole field. A rank-two field can contain a scale whose value is itself an order unit; power jets at that scale are separated. Our examples deliberately choose a lower layer, leaving room for a nonzero higher-layer error.

Remark 72.5 (Rank three in 07, rank two in 04). [write] Manuscript 07's "valuation rank three" ([theorem 64.1](#) (v), [proposition 72.1](#)) comes only from keeping $\varepsilon = \omega^{-1}$ as a third layer, below the layers of τ and of the η_ℓ , so that both polygons are also flat-close to the triangle T at the larger scale ε ; [corollary 72.4](#) drops exactly that assertion and works in rank two. Manuscript 04's construction needs only rank two from the start: its jet scale is $\varepsilon = \omega^{-1}$ itself, in the low layer $\mathbb{Q}$ of Γ_n ((70.11)), and the perturbations μ_i live in the upper layer. The two statements are consistent, and both fit a single rank-two field for all n ([corollary 72.4](#); [remark 70.8](#)). 04's [corollary 72.9](#) below is 07's paragraph just printed, made explicit: in one rank-two field one scale has a nonzero flat ideal and another has none.

72.3 An exact criterion for hidden complexity (source 04)

[write] Manuscript 04's Section 6. In this subsection ε is an arbitrary positive infinitesimal, as in 04, except in [corollary 72.9](#).

The explicit prime-monomial construction gives independence even over all of $\mathbb{R}$. Surprisingly, the existence of an arbitrary single nonzero flat element suffices for the qualitative and quantitative gap.

Lemma 72.6 (One infinitesimal parameter and generic real coefficients; source 04). *Let $\delta \in K \setminus \mathbb{R}$, and let $F \subseteq \mathbb{R}$ be a subfield. Assume δ is transcendental over $\mathbb{R}$, $b_i \in F$, and $c_1, \dots, c_s \in \mathbb{R}$ are algebraically independent over F . Then*

$$b_1 + \delta c_1, \dots, b_s + \delta c_s$$

are algebraically independent over F .

Proof. For a nonzero $f \in F[X_1, \dots, X_s]$ of total degree D , regard $f(b_1 + Tc_1, \dots, b_s + Tc_s)$ as a polynomial in T with real coefficients. Its leading coefficient is $f_D(c_1, \dots, c_s)$, where f_D is the highest homogeneous part of f . It is nonzero by algebraic independence of the c_i . Substitution $T = \delta$ therefore cannot give zero, since δ is transcendental over $\mathbb{R}$. $\square$

Every nonzero infinitesimal in an ordered extension of $\mathbb{R}$ is transcendental over $\mathbb{R}$: a real closed field has no proper ordered algebraic extension. This observation supplies the hypothesis on δ .

Theorem 72.7 (Flat-ideal classification; source 04). *Let $K \supseteq \mathbb{R}$ be real closed and let $\varepsilon > 0$ be infinitesimal. The following conditions are equivalent.*

- (a) $j_\varepsilon \neq 0$.
- (b) $v(\varepsilon)$ is not an order unit of the natural value group.
- (c) For every $n = 2^m$, $m \geq 2$, there is a cyclic n -gon $P_n^\circ \subset K^2$ on the unit circle, with the same labeled ε -jet data as the regular polygon R_n , and

$$\lceil \sqrt{n} \rceil \leq \text{xc}_K(P_n^\circ) \leq \min\{n, \lceil 24\sqrt{n} \rceil\}.$$

- (d) The ratios $\text{xc}_K(P)/\text{xc}_K(R_n)$ are unbounded among cyclic n -gons P with the same labeled ε -jet data as R_n , with $n = 2^m$ allowed to vary.

Proof. The equivalence of (a) and (b) is [theorem 72.2](#). Assume (a) and choose $0 < \delta \in j_\varepsilon$. Let $F = \mathbb{R}_{\text{alg}}$, the field of real algebraic numbers. For each n , choose real numbers $c_0, \dots, c_{n-1}$ algebraically independent over F , and set

$$z_i = a_i + \delta c_i, \quad p_i^\circ = \varphi(z_i).$$

Such finite independent lists exist because $\mathbb{R}$ has infinite transcendence degree over $\mathbb{R}_{\text{alg}}$. The a_i from (70.1) belong to F , so [lemma 72.6](#) makes the z_i algebraically independent over F . The rational inverse (70.6) makes the vertex-coordinate field have transcendence degree n over F . [Proposition 67.2](#) and [proposition 70.4](#) give the desired bounds. Because $\delta c_i \in j_\varepsilon$, the rational-flat lemma and the strict orientation signs of the base polygon show that these are cyclic n -gons with the prescribed jet data. This proves (c).

Condition (c), together with $\text{xc}_K(R_n) \leq 2 \log_2 n$, implies (d). If (a) fails, equality of jet data is literal coordinate equality by (63.5); hence every polygon occurring in (d) equals its labeled regular reference and every ratio is one. Thus (d) implies (a). $\square$

The theorem is not a criterion for extension complexity of an individual polygon. It is a complete criterion for whether *unbounded complexity gaps can occur within the specified all-order observation fibers*. In the sufficiency proof the lower bound uses the real algebraic base field, not all of $\mathbb{R}$; this distinction is essential because the construction uses generic real coefficients.

Remark 72.8 (One order-unit theorem in both sources). [write] The equivalence (a) $\Leftrightarrow$ (b) is [theorem 72.2](#), printed once above for both sources. 04's (a) $\Leftrightarrow$ (c) $\Leftrightarrow$ (d) (gaps occur iff $j_\varepsilon \neq 0$ iff $v(\varepsilon)$ is not an order unit) and 07's [corollary 72.4](#) (rank one never, rank two already suffices) are the same order-unit theorem read in two directions: 04 fixes an arbitrary field and scale and characterizes the scales, 07 asks which valuation ranks admit some scale. Since a rank-one value group has only order units, 04's criterion implies the first half of 07's corollary; 04's condition (c), applied in one fixed rank-two field with $j_\varepsilon \neq 0$ (for instance the field L of [remark 70.8](#)), gives its second half for linear lifts of cyclic polygons. 07's own proof of the second half also covers semidefinite lifts and the slack and shadow properties.

Corollary 72.9 (Two scales in one rank-two field; source 04). *In the field K_n of (70.11), ω^{-1} has a nonzero flat ideal, while μ_0 has zero flat ideal.*

Proof. The value of ω^{-1} lies in the lower convex component and is bounded above by $v(\mu_0)$ after any integer multiplication. The value $v(\mu_0) = \omega\sqrt{q_0}$, on the other hand, is an order unit of Γ_n : every leading real coefficient in the upper component is bounded by an integer multiple of $\sqrt{q_0}$, and a further increment also dominates any lower component. Apply [theorem 72.2](#). $\square$

Thus changing the observation scale can restore injectivity without changing the field. The obstruction should not be described simply as “higher rank makes all asymptotic information incomplete.”

72.4 Coarsening explains the lost information

Define the convex subring

$$V_\tau = \bigcup_{m \geq 0} \tau^{-m} \mathcal{O}. \quad (72.3)$$

It consists of elements whose growth is bounded by some finite negative power of τ , up to a real finite factor.

Proposition 72.10 (The coarse residue behind all-order agreement). *The ring V_τ is a convex valuation ring of K , its maximal ideal is $\mathfrak{j}_\tau$, and its residue field is real closed. The field $\mathbb{R}(\tau)$ embeds into that residue field, and p_i and q_i have identical residues for every i .*

Proof. The union in (72.3) is a convex ring. If x does not belong to it, then x^{-1} is smaller than every power of τ , so $x^{-1} \in \mathfrak{j}_\tau \subseteq V_\tau$. Hence it is a valuation ring. An element $x \in V_\tau$ is a nonunit precisely when it is smaller than every power of τ ; this identifies its maximal ideal with $\mathfrak{j}_\tau$. The general real-closed residue assertion is proved in [Lemma 73.1](#) below. Every nonzero element of $\mathbb{R}(\tau)$ has valuation a finite integer multiple of $v(\tau)$ and is therefore a unit of V_τ , proving injectivity on that field. Finally, $p_i - q_i \in \mathfrak{j}_\tau^2$. $\square$

Thus the hard polygon becomes the easy configuration after a suitable coarse residue, while ordinary standard part takes both to T . The next section proves that the direction of the resulting complexity inequality is forced in every such residue reduction.

[write] For 04’s family the analogous coarse ring is $\bigcup_{m \geq 0} \varepsilon^{-m} \mathcal{O}$, with maximal ideal $\mathfrak{j}_\varepsilon$; manuscript 04 uses instead the ordered quotient $\mathcal{O} \rightarrow \mathcal{O}/\mathfrak{j}_\varepsilon$ (see the paragraph after [theorem 73.2](#)). The coarse residue is not the standard part, and neither is $\mathcal{O}/\mathfrak{j}_\varepsilon$.

73 Residue descent for positive ranks and polytope lifts

The main example shows that residue reduction can erase a large amount of complexity. It cannot create complexity. We prove this for an arbitrary convex valuation ring, not just the finite-element ring associated with $\mathbb{R}$. These are algebraic counterparts of familiar factor-normalization and rank-closure arguments [48]; novelty is not claimed for the underlying lower-semicontinuity principle.

Let V be a convex valuation ring in a real closed ordered field K , with maximal ideal $\mathfrak{n}$ and ordered residue field $k = V/\mathfrak{n}$. Write $\bar{x}$ for the residue of $x \in V$. A bar over a vector or matrix is entrywise reduction.

Lemma 73.1 (Real closedness of the residue). *The ordered residue field k is real closed.*

Proof. A positive $\bar{a} \in k$ has a positive lift $a \in V$. Its positive square root in K is bounded by $1 + a \in V$, hence belongs to V and reduces to a square root of $\bar{a}$.

Lift a monic odd-degree polynomial over k coefficientwise to one over V . It has a root $x \in K$. Every such root satisfies

$$|x| \leq 1 + \sum_{i < d} |a_i|,$$

where the a_i are its nonleading coefficients: outside this bound the leading power dominates the sum of all others. Convexity puts x in V , and its residue is a root of the original polynomial. The square-root and odd-degree-root characterization now proves real closedness. $\square$

73.1 Nonnegative factors can be made integral

Theorem 73.2 (Nonnegative rank under residue; sources 07 and 04). *If M is a finite nonnegative matrix over V , then*

$$\text{rank}_+^k(\bar{M}) \leq \text{rank}_+^K(M).$$

More strongly, every length- r nonnegative factorization of M can be rescaled to one of length at most r whose factors lie in V .

Proof. Write $M = UW$ with nonnegative factors, and remove zero columns of U together with the matching rows of W . For each remaining factor index α , set

$$s_\alpha = \sum_i U_{i\alpha} > 0, \quad U'_{i\alpha} = U_{i\alpha}/s_\alpha, \quad W'_{\alpha j} = s_\alpha W_{\alpha j}.$$

Then $M = U'W'$, $0 \leq U'_{i\alpha} \leq 1$, and

$$0 \leq W'_{\alpha j} \leq \sum_\beta W'_{\beta j} = \sum_i M_{ij}.$$

Convexity of the valuation ring implies that all entries of U' , W' belong to it. Reducing the finite matrix identity gives a nonnegative factorization of $\bar{M}$ of the same or smaller length. $\square$

Remark 73.3 (Manuscript 04's form: bounded factor normalization). [write] Manuscript 04 proves the case $V = \mathcal{O}$, $k = \mathbb{R}$ (standard part) as its Proposition 7.1 ("Bounded factor normalization"), with a different normalization (column maxima instead of column sums) and one extra conclusion. Its statement: *Let $S \in \mathcal{O}_{\geq 0}^{f \times n}$. Then $\text{rank}_+^{\mathbb{R}}(\text{st } S) \leq \text{rank}_+^K(S)$. Moreover, any nonnegative factorization $S = TU$ can be replaced, after omitting zero columns, by one in which every entry of both factors is finite and every column of the first factor has maximum one.* Its proof follows.

Second proof (source 04), for $V = \mathcal{O}$. For a nonzero column ℓ of T , let $a_\ell = \max_i T_{i\ell} > 0$ and replace

$$T_{i\ell} \mapsto T'_{i\ell} = T_{i\ell}/a_\ell, \quad U_{\ell j} \mapsto U'_{\ell j} = a_\ell U_{\ell j}.$$

The product is unchanged and $0 \leq T'_{i\ell} \leq 1$. Choose i_ℓ with $T'_{i_\ell \ell} = 1$. Nonnegativity of all other summands gives

$$0 \leq U'_{\ell j} \leq S_{i_\ell j}.$$

Thus both factors are finite. Applying st entrywise produces a nonnegative real factorization of the same inner dimension. $\square$

The same normalization permits reduction through the ordered quotient $\mathcal{O} \rightarrow \mathcal{O}/\mathfrak{j}_\varepsilon$. For the matrices in (71.1), the reduction is literally the real matrix $S(R_n)$ embedded in this quotient. Its nonnegative rank there equals its real nonnegative rank: a factorization over the quotient would also be one over the real closure of its fraction field, where [corollary 74.7](#) applies. Hence passage to the quotient can strictly lower the rank by the stated asymptotic factor, but cannot raise it for a finite matrix with nonnegative factors. [write] (Source 04.)

The normalization is necessary. A factorization such as $1 = \omega \cdot \omega^{-1}$ cannot simply be reduced term by term in the finite-element ring: one factor is not in the ring. Nonnegativity allows normalization without cancellations, which is why the proof works.

73.2 Semidefinite normalization

Theorem 73.4 (Semidefinite rank under residue). *For every finite nonnegative matrix M over V ,*

$$\text{rank}_{\text{psd}}^k(\bar{M}) \leq \text{rank}_{\text{psd}}^K(M).$$

Any size- h factorization can be replaced, after a possible compression, by positive-semidefinite factors of size at most h with entries in V .

Proof. Take $M_{ij} = \text{tr}(A_i B_j)$ with $A_i, B_j \geq 0$, and put $A = \sum_i A_i$. If $A = 0$, then every $A_i = 0$ and $M = 0$, which is immediate. Otherwise compress to the range of A using an orthogonal basis over K . For positive-semidefinite matrices,

$$\ker A = \bigcap_i \ker A_i :$$

if the quadratic form of their sum vanishes, each nonnegative summand vanishes, and a zero quadratic value for a positive-semidefinite matrix puts the vector in its kernel. Thus the A_i have no components on the removed subspace. Compressing each B_j preserves the traces and positive semidefiniteness.

On the resulting space A is positive definite. Real closedness supplies its positive square root and inverse. Define

$$C_i = A^{-1/2} A_i A^{-1/2}, \quad D_j = A^{1/2} B_j A^{1/2}.$$

Then $C_i, D_j \geq 0$, $\sum_i C_i = I$, and

$$\text{tr}(C_i D_j) = M_{ij}, \quad \text{tr}(D_j) = \sum_i M_{ij}.$$

Every diagonal entry of C_i lies between 0 and 1. Every diagonal entry of D_j lies between 0 and $\text{tr}(D_j) \in V$. If $H \geq 0$, its quadratic forms on $e_a \pm e_b$ give

$$2|H_{ab}| \leq H_{aa} + H_{bb}.$$

Hence all off-diagonal entries of the C_i, D_j also lie in V . Reducing preserves every nonnegative principal minor, so the reduced matrices are positive semidefinite over k . Reduction also preserves the trace identities, yielding the desired factorization. $\square$

All ingredients in this proof are finite linear algebra over a real closed field. The normalization matrices themselves need not lie in V ; only the resulting factors must. No extraction of convergent subsequences or compactness in a surreal topology is involved.

73.3 Polytope shadows cannot have larger lift size

For $P = \text{conv}_K(v_1, \dots, v_N)$ with $v_j \in V^d$, define $\bar{P} = \{\bar{x} : x \in P\}$. Every convex coefficient belongs to V . Conversely, nonnegative residue coefficients summing to one can be lifted to nonnegative elements of V and normalized by their sum, which is a unit with residue one. Therefore

$$\bar{P} = \text{conv}_k(\bar{v}_1, \dots, \bar{v}_N). \tag{73.1}$$

Theorem 73.5 (Residue descent of extension complexity). *Let $P \subset K^d$ be full-dimensional with a finite vertex set in V^d . If $\bar{P}$ is positive-dimensional, then*

$$\text{xc}_k(\bar{P}) \leq \text{xc}_K(P), \quad \text{sxc}_k(\bar{P}) \leq \text{sxc}_K(P). \quad (73.2)$$

The dimension of $\bar{P}$ may be smaller than d . For a point shadow the same conclusions hold with the convention that the lift size of a point is zero.

Proof. Take a relative facet inequality $\bar{a}_i \cdot x \leq \bar{b}_i$ of $\bar{P}$, extending its linear functional from $\text{aff}(\bar{P})$ to k^d . Lift the coefficients of $\bar{a}_i$ to a vector $a_i \in V^d$, and set

$$\beta_i = \max_j a_i \cdot v_j.$$

It belongs to V , and $\bar{\beta}_i = \bar{b}_i$ because the residue inequality is tight. The row

$$M_{ij} = \beta_i - a_i \cdot v_j$$

is a finite nonnegative row over V . Its reduction is the slack of the residue points in that facet inequality.

Let S be a facet–vertex slack matrix of P . Finite linear-programming duality, applied to the tight valid inequality $a_i \cdot x \leq \beta_i$, expresses its slack row as a nonnegative combination of the rows of S . The constant term matches exactly because the inequality is tight. Thus $M = TS$ for some $T \geq 0$ over K . Left multiplication by a nonnegative matrix cannot increase either factorization rank: for the semidefinite rank, replace row factors by the corresponding nonnegative sums. Consequently,

$$\text{rank}_+^K(M) \leq \text{xc}_K(P), \quad \text{rank}_{\text{psd}}^K(M) \leq \text{sxc}_K(P).$$

Apply [Theorems 73.2](#) and [73.4](#) to M .

The columns of $\bar{M}$ include every vertex of $\bar{P}$, possibly with repetitions and extra nonvertex points. Removing those extra columns cannot increase a factorization rank. Adding them back also cannot increase it, because each is a convex combination of vertex slack columns. Hence $\bar{M}$ has exactly the positive ranks of a usual slack matrix of $\bar{P}$. Equation (63.2) proves (73.2). The point case is immediate from the stated convention. $\square$

Corollary 73.6 (Real standard part). *If $K \supseteq \mathbb{R}$ and a full-dimensional polytope has finite coordinates, then neither linear nor semidefinite extension complexity increases under standard part, even if its dimension drops.*

[write] For 04’s family the standard part of P_n° is R_n itself, so [corollary 73.6](#) gives $\text{xc}(R_n) \leq \text{xc}(P_n^\circ)$, consistent with (70.4); for 07’s family the standard part is the triangle $T_{\mathbb{R}}$.

Remark 73.7 (Why facetwise reduction is not a proof). Reducing a chosen list of facet inequalities of P may erase an entire row or produce redundant inequalities. Likewise, a vertex can reduce to a nonvertex, as in [Example 67.5](#). The proof above starts instead with facets of the *residue polytope* and lifts each to a tight inequality of P . This avoids any unsupported claim that standard part induces a face-lattice homomorphism.

74 Prescribed precision and finite conservativity

74.1 Every fixed precision can fail in rank one (source 04)

[write] Manuscript 04’s Section 8 (“Finite precision and arbitrary observation scales”); its second theorem is printed with 07’s [theorem 74.2](#) below.

The all-order ambiguity is absent in rank one, but no *fixed finite* order is sufficient without additional restrictions.

Theorem 74.1 (Explicit finite-order gaps in rank one; source 04). *For every integer $M \geq 1$ and $n = 2^m$, $m \geq 2$, there is a rank-one real closed ordered subfield $K \subset \mathbf{No}$ containing $\mathbb{R}$ and $\varepsilon = \omega^{-1}$, and a cyclic n -gon $P_{n,M}^\circ \subset K^2$, such that*

$$\|p_i^\circ - r_i\|_\infty < \varepsilon^M, \quad \lceil \sqrt{n} \rceil \leq \text{xc}_K(P_{n,M}^\circ) \leq \min\{n, \lceil 24\sqrt{n} \rceil\}.$$

The all-order map J_ε on this K is nevertheless injective.

Proof. Use

$$\mu'_i = \omega^{-(M+1+\sqrt{q_i})}, \quad p_i^\circ = \varphi(a_i + \mu'_i),$$

and take the real closure of $\mathbb{R}(\varepsilon, \mu'_0, \dots, \mu'_{n-1})$ inside $\mathbf{No}$. The real exponents $M+1+\sqrt{q_i}$ are rationally independent: a relation among them would be a relation among 1 and the distinct prime square roots. Thus the μ'_i are algebraically independent over $\mathbb{R}$, giving the lower bound exactly as before. They are smaller than ε^{M+1} ; the finite coefficients in (70.7)–(70.8) then give the strict coordinate bound by ε^M . Cyclicity gives the upper bound.

The value group of the rational-function field lies in the additive subgroup of $\mathbb{R}$ generated by 1 and these real exponents. The least-value cancellation argument following (70.11) shows that the real closure still has value group in their rational span. It is nontrivial and Archimedean, hence of rank one. Now use [theorem 72.2](#) to conclude that $j_\varepsilon = 0$ and J_ε is injective. $\square$

[write] In this proof 04's monomials v_i are printed μ'_i , since v is the valuation in this Part.

This theorem distinguishes two assertions that should not be merged. A particular finite truncation can omit important information even in a separated setting. Equality of *all* truncations forces equality in a separated setting; its failure in [theorem 70.1](#) needs a nonzero flat ideal.

74.2 Beyond an arbitrary set of tolerances

The two named scales in [Theorem 64.1](#) are not essential. The surreal set-cut property permits a version that escapes every member of any prescribed *set* of positive tolerances.

[write] There is also a proper-class version of the choice of scale, while the constructed polygons and every proof still use only set-sized data (source 04's introduction to its version, part (b) below). In part (b) 04's tolerance set $\mathcal{T} \ni \tau$ is printed $\mathcal{D} \ni \delta$.

Theorem 74.2 (Set-sized precision escape; sources 07 and 04). *Two versions are proved.*

(a) [source 07] *Let $F \subseteq \mathbf{No}$ be a set-sized subfield containing $\mathbb{R}$, let $\mathcal{D} \subseteq \mathbf{No}_{>0}$ be a set, and fix $n \geq 4$. There exist surreal n -gons P, Q and a positive infinitesimal ρ such that:*

- (i) P, Q lie in the strict interior of T , have shadow $T_\mathbb{R}$, and have the same labeled order type;
- (ii) for every $\delta \in \mathcal{D}$, $T \subseteq P + [-\delta, \delta]^2$ and $T \subseteq Q + [-\delta, \delta]^2$;
- (iii) their labeled coordinates and canonical slacks agree modulo every power of ρ , and their corresponding nonzero slacks have the same leading terms;
- (iv) $\text{xc}(Q), \text{sx}(Q) \leq U(n)$, while the coordinates of P are algebraically independent over F and therefore satisfy the lower bounds (64.7)–(64.8).

The statement concerns existence over all of $\mathbf{No}$; each resulting pair and each finite certificate still live in a set-sized real closed field.

- (b) [source 04: set-indexed tolerance strengthening] Let $\mathcal{D}$ be any set of positive surreal numbers. There are cyclic polygons P_n° , $n = 2^m$, on the unit circle with the bounds (70.4), such that

$$\|p_i^\circ - r_i\|_\infty < \delta \quad \text{for every } \delta \in \mathcal{D} \text{ and every vertex } i.$$

The exponent scale used in the construction may be chosen once for all n .

Proof of (a) (source 07). Choose $\rho > 0$ below every $1/m$ and every $\delta/4$ with $\delta \in \mathcal{D}$. The right-hand constraints form a set, so the surreal cut property supplies such a ρ . Construct Q from (64.2)–(64.4) [write] (with ρ in place of τ). Choose $\sigma > 0$ below every ρ^m .

Every interval $(0, \sigma)$ in $\mathbf{No}$ is a proper class: the injection $x \mapsto \sigma/(1+x)$ maps the positive surreal class into it. On the other hand, the elements algebraic over a set-sized field form a set. Choose successively $2n$ elements $e_1, \dots, e_{2n}$ in $(0, \sigma)$, each transcendental over $F(\rho)$ and all earlier choices. This is a finite sequence of choices; no class-length choice principle is needed. The resulting tuple is algebraically independent over $F(\rho)$.

Put $p_i = q_i + (e_{2i-1}, e_{2i})$. Every perturbation lies in i_ρ , so the stability and slack proofs apply. Translation over $F(\rho)$ preserves algebraic independence. The reflection lift of Q and the generic lower bounds of P give (iv). Finally, $T \subseteq P + [-3\rho, 3\rho]^2$ and $3\rho < \delta$ for every prescribed δ , proving the tolerance assertion. $\square$

Proof of (b) (source 04). For each $\delta \in \mathcal{D}$, let $c_\delta \omega^{b_\delta}$ be its leading Conway term, with $c_\delta > 0$. Choose a positive surreal Λ larger than every element of the set

$$\{N : N \geq 1\} \cup \{|b_\delta| + N : \delta \in \mathcal{D}, N \geq 1\}.$$

Such a surreal exists by the set-sized cut construction. Put $\mu_i = \omega^{-\Lambda\sqrt{q_i}}$ and $p_i^\circ = \varphi(a_i + \mu_i)$. For each δ , the leading exponent of μ_i/δ is $-\Lambda\sqrt{q_i} - b_\delta$, which is less than $-N$ for every ordinary N . In particular μ_i/δ is infinitesimal. The difference identities (70.7)–(70.8), whose remaining factors are finite, give the stated strict tolerance bound.

Multiplication by nonzero Λ preserves rational independence of the $\sqrt{q_i}$, so the monomials are algebraically independent over $\mathbb{R}$. The proof of theorem 70.1 gives the same cyclic complexity bounds. The choice of Λ was independent of n . $\square$

The set qualification cannot be removed. No nonzero positive surreal is smaller than *every* positive surreal, since half of it is another positive surreal. The theorem is not a construction inside an arbitrary preassigned small subfield, which might not have enough transcendence or coinital scales.

The requirement that $\mathcal{D}$ be a set is indispensable. Taking all positive surreals as tolerances would force every coordinate difference to be zero. The theorem does not say that a fixed set-sized subfield contains polygons below all of its own positive tolerances; the newly chosen scale can lie outside that subfield. [write] (Source 04, on part (b).)

[write] The two parts differ in what they control: (a) fixes n and makes two polygons flat-close to each other at a chosen ρ and close to the triangle below every tolerance; (b) makes one cyclic polygon for every $n = 2^m$ close to the regular polygon below every tolerance, with one scale Λ for all n , which (a) does not assert.

74.3 Finite algebraic polytope data remain conservative

Non-Archimedean geometry in this paper does not create new finite oriented-matroid realizability types. The phenomenon belongs to infinite scale requirements and the information discarded by residue, not to a failure of real-closed-field transfer.

[write] This subsection also prints manuscript 04's Section 9 ("Finite algebraic transfer and its precise limits"), which opens: "The negative results concern an infinitary observation scheme. Every fixed finite semialgebraic question still behaves classically."

Theorem 74.3 (Simultaneous finite conservativity; sources 07 and 04). *For any positive-dimensional finite polytope over a real closed field, there is a polytope with real algebraic coordinates having the same labeled oriented matroid and exactly the same linear and semidefinite extension complexities. Its real extension complexities have those same values. Numerical coordinates and infinite scale requirements need not be preserved.*

Proof. Fix the finite dimension, vertex count and facet count of the given polytope, and choose vertex and facet descriptions. They can be asserted to describe the same full-dimensional polytope by an ordered-field formula. The signs of the finitely many homogeneous determinants encode the labeled oriented matroid. Facet incidence and nonredundancy can also be expressed by finite sign and rank conditions.

Write r and h for the two exact complexities. By (63.2), exact nonnegative rank r is the conjunction of existence of a length- r nonnegative factorization and nonexistence of one of length $r - 1$. Exact semidefinite rank h is expressed in the same way with positive-semidefinite factors. These are first-order conditions; the nonexistence clauses are allowed to be universal.

Quantify the unknown vertex and facet coefficients, keeping only rational constants in the formula. The resulting sentence is true in the original real closed field, hence also in the real algebraic numbers by completeness of real closed fields. This gives the required algebraic realization. Finally, an inclusion of real closed fields is elementary, so its ranks do not change on passing from the real algebraic numbers to $\mathbb{R}$. $\square$

Remark 74.4 (Manuscript 04's form, including the point case). [write] Manuscript 04 states the linear half as its Theorem 9.1 ("Real-algebraic realization with exact extension complexity"): *Let P be a finite-dimensional polytope over a real closed ordered field, with finitely many labeled vertices. There is a polytope P^{alg} with coordinates in $\mathbb{R}_{\text{alg}}$ having the same labeled face lattice and the same exact extension complexity. The result applies in particular to finite surreal polytopes.* The labeled face lattice is determined by the labeled oriented matroid, so for positive dimension this is a special case of theorem 74.3; 04 also covers the zero-dimensional case, a point, where 07's statement assumes positive dimension. 04's proof, which writes the face-lattice conditions as explicit formulas (74.1), follows.

Second proof (source 04), including the point case. A zero-dimensional polytope is a point; its face lattice and extension complexity zero are realized by the origin. Otherwise work in the affine hull so that the dimension is $d \geq 1$. Let there be n vertices, f facets, and let $k = \text{xc}(P)$. These are fixed ordinary integers. Write variables $V = (v_1, \dots, v_n)$, A , and b for a vertex list and an f -row halfspace description.

For each nonempty proper subset $I \subset \{1, \dots, n\}$, the assertion that I is exactly the set of vertices of an exposed face is the finite formula

$$E_I(V) : \quad \exists(c, \beta) \left[\bigwedge_{i \in I} c \cdot v_i = \beta \wedge \bigwedge_{j \notin I} c \cdot v_j < \beta \right]. \quad (74.1)$$

There are only finitely many subsets I . Impose E_I or $\neg E_I$ according to the face lattice of P , and impose a nonzero affine d -minor for full dimension. Singletons in this pattern ensure that all listed points are vertices.

The halfspace and convex-hull descriptions agree if

$$\forall x \left[Ax \leq b \iff \exists \lambda_1, \dots, \lambda_n \geq 0 \left(\sum_j \lambda_j = 1 \wedge x = \sum_j \lambda_j v_j \right) \right].$$

Require each row to expose its prescribed facet, and define $S_{ij} = b_i - a_i \cdot v_j$. Finally impose

$$\text{NF}_k(S) \wedge \neg \text{NF}_{k-1}(S).$$

These conditions, preceded by existential quantifiers for V, A, b , form one finite first-order sentence with rational coefficients. It is true in the original real closed field. Completeness of the theory of real closed fields makes it true in $\mathbb{R}_{\text{alg}}$ [50]. The resulting polytope has the prescribed faces by (74.1), and exact extension complexity k by [proposition 63.6](#). $\square$

Thus surreal coordinates introduce neither new finite polytopal face lattices nor new attainable pairs consisting of a labeled face lattice and an exact extension-complexity value. The new feature is how different realizable values can coexist inside an asymptotic fiber, not an extension-complexity number impossible over the reals. [write] (Source 04.)

Remark 74.5 (Descent and transfer go in different directions). [write] 07's residue descent ([theorem 73.5](#), [corollary 73.6](#)) changes the polytope, replacing it by its residue, and complexity can only drop. 04's real-algebraic transfer ([theorem 74.3](#), [corollary 74.6](#)) also changes the polytope, replacing it by a real-algebraic realization, and the exact complexity is kept. The two maps are different (a residue map versus a choice of algebraic witness by transfer) and the statements do not conflict: the approximant of [corollary 74.6](#) is not the standard part of P_n° (which is R_n , of logarithmic complexity), and a real-algebraic realization with exact complexity $\text{xc}(P_n^\circ)$ is not a residue of P_n° .

Corollary 74.6 (Algebraic finite-accuracy approximants; source 04). *Fix $n = 2^m$ and a positive rational η . There is a cyclic polygon with real algebraic coordinates within η of the corresponding regular vertices and with extension complexity exactly $\text{xc}(P_n^\circ)$ from [theorem 70.1](#). In particular it has extension complexity at least $\lceil \sqrt{n} \rceil$.*

Proof. In the formula of [theorem 74.3](#), additionally impose the unit-circle equations, the cyclic slack signs, and $|p_{i,j} - r_{i,j}| < \eta$. The $r_{i,j}$ lie in $\mathbb{R}_{\text{alg}}$, so these are allowed parameters. The explicit surreal configuration is a witness, and $\mathbb{R}_{\text{alg}}$ is an elementary substructure of the real closed field containing it. Transfer gives the requested algebraic witness. $\square$

[write] Manuscript 04 titles this corollary “Algebraic finite-accuracy shadows”. The polygon it produces is a real-algebraic approximant of the regular polygon, chosen by transfer; it is not a standard part (the standard part of P_n° is R_n) and not a residue in the sense of [section 73](#). We therefore say “approximant”.

The transcendence-degree certificate itself is not preserved by this specialization: all coordinates of the resulting polygon are algebraic. What is preserved is the finite statement $\neg \text{NF}_{k-1}(S)$, whose truth was established using the transcendental witness. This is a useful way to separate a proof device from the property it certifies.

Why there is no contradiction (source 04). The assertion $0 < \mu < \varepsilon^M$ for *every ordinary* M is an infinite family of formulas, not one formula obtained by quantifying over ordinary integers in the language of ordered fields. Nor can $\mathbb{R}$ contain a positive real ε smaller than every positive rational. Consequently [theorem 74.3](#) neither preserves the entire hierarchy of infinitesimals nor embeds the coefficient field into $\mathbb{R}$.

The theorem is a finite realization statement. It is not a statement that one may substitute a single small real number for all surreal monomials and preserve every possible equality, inequality, and complexity bound simultaneously.

Corollary 74.7 (Scalar extension does not simplify fixed real data; sources 07 and 04). *If a polytope is defined over a real closed subfield $F \subseteq K$, its linear and semidefinite extension complexities are identical over F and over K . In particular, a fixed real polytope does not acquire a smaller exact lift merely by allowing surreal coefficients.*

Proof. For each proposed factorization rank, existence is a first-order existential statement with parameters in F . Elementarity transfers its truth in either direction. Take the least successful finite rank. $\square$

Remark 74.8 (Manuscript 04's form: scalar-extension invariance). [write] Manuscript 04 states the linear half as its Proposition 4.2 ("Scalar-extension invariance"): *If $K \subseteq L$ are real closed ordered fields and S has entries in $K_{\geq 0}$, then $\text{rank}_+^K(S) = \text{rank}_+^L(S)$. The extension complexity of a polytope defined over K is likewise unchanged by extension to L . For a finite surreal input, it is unchanged by allowing the entire surreal field as the coefficient domain.* It adds the matrix form and the clause on the whole class **No**, which is how 04 justifies writing xc_{No} . Its proof follows.

Second proof (source 04). An inclusion of real closed ordered fields is elementary by quantifier elimination [50, 51]. Apply this to (63.6) for each r . The polytope assertion follows from [proposition 63.6](#). A putative factorization with surreal entries uses only finitely many coefficients; these and K lie in a set-sized real closed extension, where the same argument applies. $\square$

The corollary and the main example concern different situations. Scalar extension keeps a polytope's exact coordinates unchanged. Our perturbation changes them by nonzero elements of a jet kernel. Neither operation can be substituted for the other.

For a fixed n , one may preserve any *finite* subcollection of the scale inequalities along with the two exact complexities in a real algebraic realization of the pair. One cannot preserve the requirement $0 < \eta < \rho^m$ for every m with positive ordinary real $\rho < 1$ and η [write] (here ρ and η are real stand-ins for the scale τ and the perturbation η_1). That requirement is an infinite scheme, not one of the finite ordered-field sentences used in the proof. The conclusion also does not supply rational-coordinate realizations: $\mathbb{Q}$ is not real closed.

75 The same separation in every fixed dimension

The construction is not confined to a two-dimensional shadow. Iterated pyramids preserve the invisible perturbations while replacing the triangle by a simplex of any fixed dimension.

Lemma 75.1 (Faces do not increase lift size). *If F is a nonempty face of a polytope P , then $\text{xc}(F) \leq \text{xc}(P)$ and $\text{sxc}(F) \leq \text{sxc}(P)$.*

Proof. Pull a supporting equality for F back to an extension of P . For a polyhedral extension this gives a face of the extension, which has at most as many relative facets. For a semidefinite

lift it simply adds an affine equality to the defining slice without changing the matrix order. The resulting images are exactly F . $\square$

Corollary 75.2 (Fixed-dimensional simplex shadows). *For every fixed $d \geq 2$ and every $n \geq 4$ there are d -dimensional polytopes $\widehat{P}_{n,d}, \widehat{Q}_{n,d}$ with $n + d - 2$ vertices, the same labeled face structure, and standard shadow the standard d -simplex, such that their coordinates agree to every order in τ . They admit slack matrices with identical entrywise leading terms and identical all-order jets, while*

$$\text{xc}(\widehat{Q}_{n,d}), \text{sxc}(\widehat{Q}_{n,d}) \leq U(n) + d - 2,$$

and both hard lower bounds from [Theorem 64.1](#) hold with $\widehat{P}_{n,d}$ in place of P_n . All these examples remain in the same rank-three field.

Proof. Apply $d - 2$ ordinary pyramids to each polygon, using the same successive apices at the new standard coordinate vectors. Each pyramid adds one vertex and at most one inequality to an extension. The original polygon is a face, so [Lemma 75.1](#) preserves its lower bounds. The standard part of an iterated pyramid over the standard triangle is the standard d -simplex.

For a pyramid, put the base at last coordinate zero and the apex at the last unit vector. From base inequalities $b_i - a_i \cdot x \geq 0$, the side inequalities are $b_i(1 - z) - a_i \cdot x \geq 0$ and the base inequality is $z \geq 0$. Ordering old vertices before the apex and side facets before the base gives a slack matrix

$$\begin{pmatrix} S & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus common zero patterns, initial entries and flat differences persist at every step. The new vertex coordinates are ordinary constants, so no larger coefficient field is needed. $\square$

Strict interior containment in the simplex is not asserted in this higher-dimensional corollary: the pyramid apices are simplex vertices. The shadow is nevertheless full-dimensional and fixed, and the gaps remain unbounded as the ordinary finite vertex count grows.

76 A rational-chart generalization (source 04)

[write] Manuscript 04's Section 10. It generalizes the lower-bound half of 04's circle construction to rationally parametrized families in any dimension, beside 07's pyramid extension [corollary 75.2](#). Here ε is an arbitrary positive infinitesimal and s a number of parameters.

The circle is a particularly clean realization space because it supplies one exact free parameter per vertex and a sharp-order upper bound. The lower-bound construction extends to rationally parametrized families.

Theorem 76.1 (Generic configurations in a flat rational-chart fiber; source 04). *Let $K \supseteq \mathbb{R}$ be real closed with $\mathfrak{j}_\varepsilon \neq 0$. Let $F \subseteq \mathbb{R}$ have at least s algebraically independent real elements over it. Suppose*

$$\Phi \in F(Z_1, \dots, Z_s)^{dn}$$

is regular at $b \in F^s$, its value $\Phi(b)$ is the labeled vertex list of a full-dimensional real d -polytope P_0 with exactly n vertices, and Φ has a rational left inverse defined over F and regular at $\Phi(b)$. Then there exists a K -polytope P with the same labeled vertex jets as P_0 , exactly n vertices, and

$$\text{xc}_K(P) \geq \lceil \sqrt{s} \rceil.$$

Every rational identity satisfied identically by the parametrization is still exact at P . Every strict affine determinant sign of the base configuration is preserved. In particular, if the base configuration is in affine general position, its labeled oriented matroid and face lattice are preserved.

Proof. Choose $0 < \delta \in \mathfrak{j}_\varepsilon$ and $c_1, \dots, c_s \in \mathbb{R}$ algebraically independent over F . Put $z = b + \delta c$ and take $\Phi(z)$ as the vertex list. Its coordinates agree with $\Phi(b)$ modulo $\mathfrak{j}_\varepsilon$ by [lemma 63.4](#). Each real vertex of P_0 has an exposing real functional with a strictly positive gap from all other vertices. Infinitesimal perturbation preserves these finitely many gaps, so all n new points remain vertices. A nonzero affine d -minor likewise preserves full dimension.

By [lemma 72.6](#), the parameters $z_1, \dots, z_s$ are algebraically independent over F . The rational map and its rational left inverse identify their generated field with the vertex-coordinate field, so this field has transcendence degree s over F . Apply [proposition 67.2](#). Rational identities remain identities under substitution; regularity ensures the denominators used here do not vanish. Every originally nonzero determinant retains its sign because its change is flat. In affine general position this covers all maximal minors and hence the full oriented-matroid data and its face lattice. $\square$

A base configuration with zero maximal minors needs separate care. Flat perturbation can turn such a zero into a nonzero flat determinant, so preserving all originally *strict* signs does not generally preserve a nongeneric face lattice. The cyclic planar case avoids this problem because every triple of distinct circle points is noncollinear.

The theorem suggests a systematic strategy: choose a rational realization chart with many recoverable parameters, choose a low-complexity base point, and move generically in an invisible flat direction. The dimension of the chart gives a lower bound; a separate geometric theorem is needed for any matching upper bound. One must not infer a universal upper bound from the chart argument alone.

[write] As in [remark 70.7](#), the proof of 07's [theorem 67.1](#) would give $\text{xc}_K(P)(\text{xc}_K(P) - 1) \geq s$ here. 07's semidefinite count [theorem 67.4](#) is likewise stated only for fully independent coordinates; 04 states no semidefinite version.

77 Exact computation versus discarded information (source 04)

[write] Manuscript 04's Section 11. Its F_n is a field of 04, not 07's subfield letter F .

77.1 A decidable coefficient core for the explicit examples

The explicit family need not be described by an oracle for arbitrary surreals. Consider the ordered subfield

$$F_n = \mathbb{R}_{\text{alg}}(\varepsilon, \mu_0, \dots, \mu_{n-1}) \subset \mathbf{No}$$

with the monomials from (70.3). The generators are algebraically independent over $\mathbb{R}_{\text{alg}}$: the values 1 and $\omega\sqrt{q_i}$ are rationally independent, and the proof of [lemma 70.5](#) applies.

A polynomial is a finite sum of terms

$$c_{k,e} \varepsilon^k \mu_0^{e_0} \cdots \mu_{n-1}^{e_{n-1}}, \quad c_{k,e} \in \mathbb{R}_{\text{alg}}.$$

After combining equal exponent vectors, the leading term is the one with least value

$$k + \omega \sum_i e_i \sqrt{q_i}.$$

Comparison of these values is lexicographic comparison of

$$\left(\sum_i e_i \sqrt{q_i}, k \right), \tag{77.1}$$

and comparison in the first coordinate is exact comparison of ordinary real algebraic numbers. Distinct exponent vectors cannot have the same pair. The sign of the whole polynomial is therefore the sign of its unique leading coefficient. A rational function is compared by the signs of numerator and denominator. This gives exact field operations, zero testing, and ordering for this finite rational-monomial core.

Proposition 77.1 (In-principle decidability for the explicit input family; source 04). *For a specified $n = 2^m$, the exact value $\text{xc}_{\mathbf{No}}(P_n^\circ)$ is computable from the finite expression (70.3). No effective presentation of the entire surreal field is required.*

Proof. The constants a_i are effectively real algebraic: for $n = 2^m$ they can be computed by the usual real half-angle radical formulas, with signs fixed by their specified angles. The coordinates and edge slacks of P_n° therefore lie effectively in F_n . Apply an effective real-closed-field quantifier-elimination procedure [50, 51] to $\text{NF}_r(S(P_n^\circ))$, for successive integers $3 \leq r \leq \min\{n, \lceil 24\sqrt{n} \rceil\}$. Treat the finitely many coefficients as parameters during elimination and evaluate the resulting polynomial sign conditions by the exact ordered-field procedure above. The first successful r is the nonnegative rank over a real closure of F_n . By [corollary 74.7](#) it is also the nonnegative rank over $\mathbf{No}$, and [proposition 63.6](#) identifies it with extension complexity. $\square$

This proposition asserts termination, not practicality. It does not supply a good running-time bound, a compact minimal extension, or an implemented quantifier-elimination backend. It explains why the information-theoretic obstruction is compatible with an effective exact representation: the expression retains the independent flat generators, whereas all-order ε -jet data erase them.

77.2 What the accompanying program actually checks

The archive includes `code/04-beyond-all-orders-verify.py` and its recorded output `data/04-beyond-all-orders-verification.json`. The program uses exact SymPy arithmetic for the circle and inverse identities, both difference formulas, the orientation determinant, reflection interpolation, rational cyclic polygon slack signs and rank, bounded factor normalization, and the finite index-doubling scheme for regular polygons. [write] (Delivered as `code/verify.py` and `data/verification.json`; shipped in this report under the prefixed names printed here. Run in place, the script would write an unprefixed `data/verification.json` one level above its own directory; rerun it on a copy with the delivered layout, as the report README describes.)

The recorded run used Python 3.13.5 and SymPy 1.14.0 and passed **114 of 114** exact finite checks. The rational test polygons are only test cases for the algebra and sign conventions. They are not claimed to instantiate the transcendence-degree lower bound. No optimization solver was used to certify a minimum nonnegative rank; no computer search is offered as evidence of algebraic independence or of any statement quantified over all orders. The proof of the main lower bound is instead the exact normal-form and parameter-count argument in [section 70](#).

For orientation, [table 7](#) lists theorem bounds, not computed extension complexities. Large values of n make the distinction between low-order observations and exact formulation size conspicuous even though the vertex perturbations are flat for every n .

m	$n = 2^m$	$\text{xc}(R_n) \leq$	$\text{xc}(P_n^\circ) \geq$	$\text{xc}(P_n^\circ)/\text{xc}(R_n) \geq$
8	256	16	16	1
10	1 024	20	32	8/5
12	4 096	24	64	8/3
16	65 536	32	256	8
20	1 048 576	40	1 024	128/5
24	16 777 216	48	4 096	256/3

Table 7: Bounds supplied by the proofs. No exact minimum extension complexity is numerically computed here. The row $m = 8$ alone does not certify a strict gap. [write] (Source 04; its P_n is printed P_n° . All six rows reproduce from (70.4). The last column divides the lower bound for P_n° by the upper bound $2m$ for R_n ; compare 07’s table 6.)

78 Reproducibility and final conclusions

[write] The formalization roadmap of manuscript 07 is printed in section 83 (section 83.5); in 07 it stood between its research questions and this section. Manuscript 04 has no roadmap section. Nothing in this Part is formalized.

78.1 What the executable checks establish

The companion code/07-invisible-complexity-verify.py uses rational arithmetic and SymPy, without floating-point optimization. The recorded run checks:

1. the four symbolic orientation identities and three reflection identities;
2. 132 088 positive triple orientations and 32 384 slack entries in reference and rationally perturbed polygons for $4 \leq n \leq 36$;
3. all 4 686 candidate active bases of the binary reflection lifts for $1 \leq N \leq 31$, with the feasible vertices projecting exactly onto $\{(j, j^2) : 0 \leq j \leq N\}$;
4. 636 symbolic canonical slack entries for $4 \leq n \leq 12$, including their zero pattern and positive lowest coefficients;
5. sample factor-normalization identities and the integer rounding used in Table 6.

The reflection index identity is additionally checked for $1 \leq N \leq 256$. The rational perturbations in these tests are deliberately *not* claimed to be algebraically independent. The all-orders theorem, the surreal independence proof and the universal lower bounds are not validated by substituting very small real numbers.

The run used Python 3.13.5 and SymPy 1.14.0. The JSON output records the counts and exact scope. Running the script again regenerates it. The LaTeX source compiles directly with `latexmk -pdf`; it does not need external image assets or downloaded fonts.

[write] File names: the script was delivered as `verification/verify.py`, its JSON output as `verification/results.json` and its notes as `verification/README.md`; they are shipped here as `code/07-invisible-complexity-verify.py`, `data/07-invisible-complexity-results.json` and `07-invisible-complexity-verification-README.md` (whose text still says “Run `python verification/verify.py`”). The script writes `results.json` next to itself, so running it in place would add an unprefixed `code/results.json`; rerun it on a copy. The delivered LaTeX source `surreal_polytopes.tex` and its 26-page PDF are not shipped; this Part is their merged print, built with the report’s `article.tex`.

78.2 One fully expanded small lift

For H_7 , the binary chain is $0, 1, 3, 7$. The complete formulation is

$$\begin{aligned} 0 &\leq \lambda_1 \leq 1, \\ 0 &\leq \lambda_2 \leq 3 - 2\lambda_1, \\ 0 &\leq \lambda_3 \leq 7 - 2\lambda_1 - 2\lambda_2, \end{aligned}$$

with output

$$x = \lambda_1 + \lambda_2 + \lambda_3, \quad y = \lambda_1 + 3\lambda_2 + 7\lambda_3.$$

This uses six inequalities to produce the eight parabola vertices. For the cap with $k = 8$, apply the affine map

$$\Phi_k(x, y) = \left(\frac{\tau(x+1)}{k+1}, \frac{\tau(y-2kx+k^2)}{(k+1)^2} \right).$$

Two pyramid steps then add A and B . This example illustrates that the logarithmic upper bound is an actual formulation compiler, not only a counting argument.

78.3 Conclusion

The central distinction is between an exact finite polytope and its scale-reduced information. A surreal polygon can retain a large exact linear or semidefinite lift complexity after its labeled coordinates have become indistinguishable, to every power of a chosen scale, from a polygon with a logarithmic-size lift. Even the complete entrywise initial slack matrix fails to recover what was lost. This happens with uniformly finite normal-form support and inside one fixed small real closed field.

The positive counterpart is equally important: every convex residue reduction can only lower these lift sizes. The obstruction is not a failure of real-closed-field algebra, but the nonzero kernel of the chosen power-jet map. Valuation rank two already permits that kernel; a third layer allows both polygons themselves to be flat-close to a common full-dimensional real simplex at a larger scale.

The manuscript supplies explicit constructions, complete arguments for its stated results relative to identified classical inputs, and finite verification artifacts. Independent review and formalization remain the next checks on the proposed contribution, rather than conclusions assumed from the act of generating the article.

78.4 Conclusion of manuscript 04

Finite surreal polytopes obey the classical finite realizability theory, but their exact formulation sizes need not be visible at a chosen scale. A regular polygon and an explicit cyclic perturbation can agree in every power jet, every leading affine Plücker coordinate, every leading slack entry, and every finite-direction linear optimal value, while their extension complexities have different asymptotic orders.

The exact obstruction is the nonzero kernel $\bigcap_M \varepsilon^M \mathcal{O}$. Its presence is equivalent to unbounded cyclic complexity gaps within all-order jet fibers; its absence makes the full jet map injective. This supplies both a negative theorem about reconstruction and a positive boundary theorem identifying when that particular loss of information cannot occur. The main lesson is not that surreal geometry is disconnected from classical geometry. It is that exact algebraic relations and asymptotic observations are different kinds of data, even for finite polygons on a fixed circle.

79 Source and dependency audit (Appendix A of manuscript 07)

The repository was accessed on 30 September 2026. The branch metadata reported the following commit and timestamp:

```
6996fee43cc97b6c16351def7507d59a95bf62f0
2026-09-30T16:07:39Z
```

The selected documentation paths were `README.md`, `Algebra/SurrealNumbers/README.md`, and `Algebra/SurrealNumbers/docs/README.md`. A large tree response was truncated, so it is not treated as an exhaustive inventory. No claim of novelty relies on an assertion that every repository file was searched.

The earlier finite-convexity draft was retrieved by its title and abstract from the research library. Its role here is to mark previously covered topics. None of its unreviewed theorems is a necessary new assumption: the finite linear-programming facts used above are standard ordered-field facts, and the manuscript specifies where the classical factorization and real-closed-field results enter. [write] (The draft is private and not part of ProvelIt; see the note in [section 62.](#))

The external audit used the primary papers [43, 44, 47, 46, 48], the classical surreal reference [42], and the real-algebraic formalization reference [49]. Searches also combined “surreal polytopes,” “surreal extension complexity,” “nonnegative rank Hahn,” and related valuation terms. The search did not establish priority. In particular, the paper does not advertise the generic lower bounds, reflection relations, slack factorization, or rank lower-semicontinuity as original discoveries.

[write] The three documentation paths exist in the current tree, and the pin and its timestamp agree with the repository history. The delivered `SOURCE_AUDIT.md` of 07 is shipped as `07-invisible-complexity-SOURCE_AUDIT.md`; its file list names the delivered layout.

80 Repository, literature, and verification audit (Appendix A of manuscript 04)

80.1 Pinned repository scope

The repository was inspected through its GitHub connection at the snapshot

```
6996fee43cc97b6c16351def7507d59a95bf62f0.
```

The directly read sources relevant to this article were:

1. `Algebra/SurrealNumbers/README.md`, for the stated foundational library and the distinction between formalization and research drafts.
2. `Algebra/SurrealNumbers/docs/README.md`, for the report guide and its verification boundaries.
3. Under `Algebra/SurrealNumbers/docs/`, the report `foundations-and-computation/computer-algebra/README.md`, for the distinction between exact representations, coefficient access, and approximation.

The tree was inspected and a scoped indexed search for the phrase “extension complexity” returned no results. This is not an assertion that every repository file was read or that no related observation exists anywhere in the repository. No change was made to the repository.

[write] Re-checked in the current tree: `rg -il "extension complexity"` over the whole repository, `.lake` excluded, finds no file outside this report's own directory (and `rg -il "extended formulation"` likewise none). This report, and in particular this Part, is now the repository's treatment of extension complexity. The three paths above exist; the computer-algebra README is described accurately (see [section 62](#)).

80.2 Literature dependence and novelty boundary

The crucial external inputs are recorded in [table 8](#). The exact all-order flat-fiber formulation and prime-monomial cyclic construction were not found in the sources inspected here. This limited search is not a certification of originality. The constituent methods are intentionally exposed so that an expert can assess whether the synthesis is new, implicit in earlier work, or best viewed as a useful corollary.

Table 8: Dependence and claim status (source 04).

Ingredient or conclusion	Source or argument	Status
Real closed fields; finite transfer	Tarski; Collins [50, 51]	Classical input
Conway monomials and normal forms	Standard surreal theory, as in [52]	Classical input
Slack-factorization equality	Yannakakis [45]; field check in proposition 63.6	Classical theorem, adapted proof
Regular polygon upper bound	Reflection method [44]; direct proof in corollary 70.3	Classical bound
Transcendence-degree lower bound	Parameter-count method [43]; arbitrary base in proposition 67.2	Classical method
Cyclic square-root upper bound	Kwan–Sauermann–Zhao [53]; transfer in proposition 70.4	Imported theorem
Explicit all-order cyclic gap	Theorem 70.1	Proposed synthesis with proof
Flat-ideal classification	Theorem 72.7	Proposed structural contribution
Invisible exact factorization obstruction	Theorem 71.2	Consequence proved here
Finite-accuracy and set-tolerance versions	Theorems 74.1 and 74.2	Consequences proved here
Exact finite algebraic shadow [write] (approximant)	Theorem 74.3	Standard transfer mechanism, explicit formulation
Exact finite checks	<code>code/04-beyond-all-orders-verify.py</code> ; 114/114 passed	Symbolic and finite, not formal proof
Lean verification of this article	No files or build supplied	Not claimed

80.3 Reproduction

The LaTeX source is standalone and uses an internal bibliography. From the archive root one may run

```
python code/04-beyond-all-orders-verify.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```


`pdflatex -interaction=nonstopmode -halt-on-error article.tex`

The delivered PDF was compiled from this source. The program exits with an error if any check fails and rewrites its JSON report only on success. A Python environment with SymPy is needed; no internet access or proprietary solver is required for the finite checks. The exact quantifier-elimination algorithm described in [proposition 77.1](#) is not implemented by this script.

[write] Delivered as python code/verify.py, writing data/verification.json; the shipped script is code/04-beyond-all-orders-verify.py and its recorded output data/04-beyond-all-orders-verification.json (with data/04-beyond-all-orders-requirements.txt and code/04-beyond-all-orders-build.sh, which still names article.tex and build/). Run in place, the script writes an unprefixd data/verification.json; use a copy with the delivered layout. The delivered article.tex and its 26-page PDF are not shipped: the report's article.tex is the merged report, which the pdflatex lines above now build, and this Part is 04's merged print.

81 Notation and possible misreadings (Appendix B of manuscript 04)

[write] Printed with this Part's symbols (j_ε for 04's I_ε , P_n° for 04's P_n); see also "Notation in this Part" at the opening of [Part V](#).

Notation	Meaning and restriction
K	A set-sized real closed ordered field; it contains $\mathbb{R}$ in the flat-ideal theorems, but need not do so in the general scalar-extension or effective-core arguments.
No	The surreal class field, used only through finite input lists and set-sized subfields when applying first-order logic.
$\mathbb{R}_{\text{alg}}$	The real algebraic numbers, not all reals.
ω^α	Conway monomial, with normal-form multiplication and ordering.
$\mathcal{O}, \mathfrak{m}$	Finite elements and ordinary infinitesimals relative to the embedded real field.
st	Ordinary real standard part; not the quotient by j_ε .
j_ε	$\bigcap_{M \geq 1} \varepsilon^M \mathcal{O}$, the all-order flat ideal.
J_ε	The compatible list of reductions modulo $\varepsilon^M \mathcal{O}$; no integer-power expansion is assumed to exist.
xc_K	Minimum facet count of a bounded extension polytope over K , allowing affine projection and measuring facets in the affine hull.
rank_+^K	Nonnegative factorization rank over the specified field.
Cyclic polygon	A planar polygon with vertices on one circle; not a higher-dimensional moment-curve polytope.
Leading slack data	Entrywise values and leading coefficients for the specific determinant-normalized slack matrix, not all its minors.
All-order agreement	Equality under J_ε , not equality in K unless $j_\varepsilon = 0$.

82 Further research questions and concrete next steps

[write] The ten questions of manuscript 07 (its Section 13) and the nine of manuscript 04 (its Section 12, "Further research questions"), with duplicates merged and both sources named. Three pairs are merged: the exact orders of the hard families, lifting a reduced factorization,

and the formalization path. 04's remaining questions follow 07's in their own subsections, as 04 prints them. Where a result of the other source bears on a question, a [write] note re-scopes it.

The following questions are proposed by the present results. They are a research program, not a claim that each has already appeared as a named open problem. In particular, the existence and valuation-rank threshold questions settled above are not listed again as unresolved.

The questions below are proposed directions growing directly out of the proved results. They are not presented as an exhaustive catalogue of published open problems, and no priority claim is attached to their formulation. [write] (Source 04.)

82.1 Questions of manuscript 07, with the matching questions of manuscript 04

Question 82.1 (Exact orders for the explicit hard families; sources 07 and 04). [source 07] What are the asymptotic orders of $\text{xc}(P_n)$ and $\text{sx}(P_n)$ for the particular monomial construction (64.5)? [source 04, "Exact complexity inside a fixed jet fiber"] For the explicit prime-monomial polygon P_n° , determine the exact extension complexity, or improve the constants between $\sqrt{n}$ and $24\sqrt{n}$. More generally, which nonnegative ranks occur among cyclic polygons in the all-order jet fiber of a fixed regular polygon?

[source 07] The bounds proved here leave a substantial range below the trivial n -facet and size- n diagonal lifts. The first task is to apply and make explicit the available polygon upper-bound constructions over the chosen real closed field. A lower-bound attack could exploit the structured hierarchy of the exponents rather than only their algebraic independence. No assertion that the elementary upper bounds used here are the best bounds in the literature is intended.

[source 04] The lower bound only uses how many independent parameters are present. It does not exploit their ordered exponents or the exact circle slack identities. A stronger lower-bound certificate might distinguish this family from an arbitrary generic cyclic configuration. Conversely, an explicit small factorization could exploit its highly organized leading regular geometry.

[write] For 04's cyclic family the order is already known to be $\Theta(\sqrt{n})$ ([theorem 70.1](#)); only constants remain. For 07's family, which is not cyclic, the Kwan–Sauermann–Zhao bound does not apply and no upper bound beyond n is proved, so the order of $\text{xc}(P_n)$ is open between $\sqrt{2n}$ and n .

Question 82.2 (Short formulas at finite rational value-group rank). Can comparably strong, uniform separations be obtained from equally short formulas with algebraic real coefficients in a field whose value group has fixed *finite rational rank*?

Our field has valuation rank three but unbounded available rational rank, and the explicit genericity proof uses $2n$ independent exponent directions. An alternative should not merely replace these directions by unspecified algebraically independent real constants. It would need a different lower-bound mechanism, for example a controlled finite combinatorial or elimination certificate. The issue is explicitness and uniform certification, not the bare possibility of scaling a hard finite configuration.

[write] Manuscript 04's [theorem 72.7](#) does give uniform linear separations (order $\sqrt{n}$ against $\log n$) in one field of rational value-group rank two, for instance the real closure of $\mathbb{R}(\omega^{-1}, \omega^{-\omega})$ with $\varepsilon = \omega^{-1}$, but only by the route this question excludes: a single flat element times unspecified real constants algebraically independent over $\mathbb{R}_{\text{alg}}$. The question as posed, with algebraic real coefficients, stays open.

Question 82.3 (Effective precision under restricted input models). For coordinates given by bounded-degree algebraic expressions in a fixed finite list of monomials, can one bound the truncation information needed to decide a specified lift-size inequality?

A useful theorem must state bounds on the expression degrees, coefficient encoding and permitted exponents, and specify whether exact zero tests are available. Quantifier elimination gives finite polynomial tests for a fixed rank question, but a practical theorem would relate their required scale resolution to the input representation. The unrestricted all-orders negative result does not answer this bounded-input problem.

[write] Manuscript 04's [proposition 77.1](#) shows that the exact extension complexity of its explicit family is computable from the finite expression (termination only, with exact zero tests), and its [theorem 74.1](#) shows that no fixed finite truncation order suffices without restrictions, even in rank one. Neither bounds the truncation information; 04's question "Quantitative stability with controlled certificates" below is a companion.

Question 82.4 (Lifting a residue factorization without increasing rank; sources 07 and 04). [source 07] Given S over a convex valuation ring and a factorization of $\bar{S}$, when does that factorization lift to S with the same nonnegative or semidefinite rank? [source 04, "A lifting criterion for positive factorizations"] Find verifiable hypotheses on a factorization of a reduced slack matrix that guarantee lifting to an exact factorization of the same inner dimension, despite a nonzero flat ideal.

[source 07] The descent theorems give only one direction. A first attack is to fix the zero pattern of the factors and study the linearization of the bilinear equations $UV = S$ or $\text{tr}(A_i B_j) = S_{ij}$, together with tangent cones to the positive constraints. The main example warns that agreement in a nonseparated completion cannot itself be a sufficient lifting criterion.

[source 04] [Theorem 71.2](#) rules out lifting based solely on coherent finite-order solutions. A plausible positive theory would need nondegeneracy conditions on the factorization map $(T, U) \mapsto TU$, control of active zero entries, and a genuine Henselian or implicit-function argument in an appropriate local ring. Positivity lies on the boundary for polygon slack matrices, so an ordinary interior implicit-function argument cannot simply be quoted without analysis.

[write] Re-scoped: [theorem 71.2](#) gives, in explicit form, the negative instance that 07's last sentence anticipates. Take for V the coarse valuation ring $\bigcup_{m \geq 0} \varepsilon^{-m} \mathcal{O}$, whose maximal ideal is $\mathfrak{j}_\varepsilon$ (as for V_τ in [proposition 72.10](#)). Then $\overline{S(P_n^\circ)} = \overline{S(R_n)}$, and the reduction of a real factorization $S(R_n) = T_0 U_0$ of inner dimension at most $2m$ is a factorization of the residue matrix that does not lift to $S(P_n^\circ)$ with the same inner dimension (for $m \geq 10$ as stated, and by the same proof for $m \geq 9$, [remark 71.3](#)). So a residue factorization need not lift, even when it is compatible with every finite jet. What remains open is the positive direction: sufficient, checkable hypotheses for lifting.

Question 82.5 (Explicit polynomial obstructions to small lifts). Can the transcendence lower bound for P_n be replaced by a practically checkable polynomial or semialgebraic obstruction for a useful range of n and proposed ranks?

The factorization equations supply a concrete elimination problem. The challenge is to extract a modest certificate rather than a very large quantifier-elimination output. Such a certificate would permit strong lower bounds for chosen algebraic specializations, where full coordinate independence is unavailable.

[write] 04's [corollary 74.6](#) shows that algebraic specializations with the same exact complexity exist near the regular polygon, but by transfer, without such a certificate; see 04's question "Effective real-algebraic approximants" below.

Question 82.6 (Geometry of low-lift loci in a realization space). Inside the realization space of a fixed labeled oriented matroid, what are the valuation strata along which linear or semidefinite lift size can jump?

For each fixed proposed size, the lift-existence locus is semialgebraic by its factorization description. The additional problem is to describe its interaction with valuations and its degenerations at the special reflection locus. One should retain initial information about the *defining relations of the locus*, not only the entries of a slack matrix.

Question 82.7 (Iterated residue chains and complexity profiles). Which finite nonincreasing sequences of lift sizes can occur along a chain of convex residue reductions of one finite polytope?

The current construction gives a large hard stage, a logarithmic-size reference stage, and a fixed simplex shadow. A systematic theory should separate restrictions caused by the common vertex count from restrictions caused by the ordered value group. Constructing prescribed profiles is more informative than producing just one drop.

Question 82.8 (Normal directions and hidden faces). Can one classify the flags of normal cones whose directions disappear under standard part, and relate them to the complexity lost by residue?

[Example 67.5](#) shows that even a vertex of a polytope with a full-dimensional shadow can have no real exposing direction. A refined object should record residue faces together with the next-scale normal cones over them. It must not assume a naive face-poset map, which the same example disproves.

Question 82.9 (Beyond orthants and positive-semidefinite cones). For which ordered-field cones can every factorization of an integral matrix be normalized to integral factors without increasing its size?

The proofs in [Section 73](#) use a bounded normalization cross-section: column sums for the orthant and trace after congruence for the positive-semidefinite cone. Other cones may or may not admit a compatible normalization. A precise criterion would extend the residue-descent theorem to other exact conic lifts without importing unsupported analytic compactness assumptions.

[write] 04's question "Semidefinite and other lift sizes" below asks for the flat-fiber phenomenon for other cones; its normalization clause is this question.

Question 82.10 (A complete certificate path to formal verification; sources 07 and 04). [source 07] What is the smallest reusable real-closed-field library that formally checks the reflection construction, the explicit genericity witness, and the two complexity lower bounds? [source 04, "Formalization milestones"] Can the entire dependency chain be formalized over an abstract real closed ordered field, leaving only the explicit monomial construction to the surreal library?

[source 07] The proof should separate basic ordered-field algebra, finite linear algebra, real-closed-field definability, and surreal monomial semantics. A certificate format for monomial independence is especially promising: the only algebraic-number input here is Eisenstein irreducibility of $X^{2^{n+1}} - 2$, followed by a finite exponent-separation argument.

[source 04] A useful decomposition is: finite convex combinations and exposing functionals; slack matrices and Farkas factorization; the parameter-count lemma; flat ideals and rational congruence; the rational circle identities; and finally the normal-form independence lemma.

The transferred cyclic upper bound is a substantial imported theorem and should remain visibly separate until its proof or a certified transfer route is formalized. Proving the lower-bound and flat-congruence package first would already give a meaningful verified result without pretending that all published polyhedral machinery had been mechanized.

[write] 07's staged design is printed in [section 83.5](#). The repository's `Surreal.Surcomplex.RegularPolygon` (regular polygons at surreal radius) would be a natural source for 04's R_n ; no part of this Part is formalized.

82.2 Separated completion versus flat kernels (source 04)

Question 82.11. In a separated valued field, can a compatible system of positive factorizations in all finite quotient rings fail to lift because the factors do not belong to the field, even though there is no flat kernel? Which completeness or definability assumptions exclude such failure?

Our examples isolate nonseparatedness by using one fixed pair of approximate factors. In a separated but incomplete field, a distinct obstruction could be nonexistence of limits of the compatible factor entries. These mechanisms should be distinguished rather than combined under a general assertion that “formal solutions do not lift.”

82.3 Quantitative stability with controlled certificates (source 04)

Question 82.12. Can finite precision determine extension complexity within a restricted family whose factorization certificates satisfy explicit separation, height, or conditioning bounds?

The counterexamples impose no such uniform bounds. A positive result would have to quantify distance from the relevant semialgebraic rank strata, not merely distance between vertex lists. This connects geometric approximation to exact certificate complexity rather than to coordinate error alone.

[write] Companion to 07's question “Effective precision under restricted input models” above.

82.4 Effective real-algebraic approximants (source 04)

[write] 04 titles this subsection “Effective real-algebraic shadows”; the objects are the approximants of [corollary 74.6](#), not standard parts.

Question 82.13. Replace [corollary 74.6](#) by explicit real algebraic or rational coordinates, with useful degree, height, and running-time bounds, while retaining a provable square-root lower bound near a regular polygon.

The transfer argument certifies existence but hides its complexity. Rational coordinates require additional care: real-algebraic realizability does not automatically imply rational realizability with the same exact extension complexity. A constructive elimination or height argument would be a substantial strengthening.

82.5 Beyond the circle (source 04)

Question 82.14. Identify rational realization charts in higher-dimensional spherical, inscribed, or other constrained polytope families where the chart lower bound of [theorem 76.1](#) can be paired with an equally sharp upper bound.

Preserving a nonsimplicial face lattice may force the perturbation to remain on a proper realization stratum. The dimension relevant to the lower bound is then the dimension of a recoverable chart on that stratum, not the total number of ambient vertex coordinates.

[write] 07's [corollary 75.2](#) gives fixed-dimensional examples (pyramids over its polygons, with simplex shadow), but with a one-sided lower bound only, so it does not answer this question.

82.6 Semidefinite and other lift sizes (source 04)

Question 82.15. What is the corresponding flat-fiber phenomenon for positive semidefinite rank, second-order-cone lifts, or other finite-dimensional conic extension measures?

Finite semialgebraic formulations still support transfer in settings where the cone is first-order definable. However, the parameter budget, the normalization needed before reduction, and the best bounds for regular and cyclic polygons are different. None of the linear extension-complexity estimates in this paper can be relabeled as a semidefinite estimate without a new proof.

[write] Partly answered by manuscript 07: for positive semidefinite rank the phenomenon exists, with unbounded ratio $\text{scx}(P_n)/\text{scx}(Q_n) \geq (2n)^{1/4}/U(n)$ inside one all-order jet fiber ([theorem 64.1](#), parameter budget [theorem 67.4](#)), and the normalization needed before reduction is [theorem 73.4](#). Open: the semidefinite sizes of 04's regular and cyclic polygons on the circle, second-order-cone and other cones (see 07's question "Beyond orthants and positive-semidefinite cones" above).

82.7 Which additional exact data are sufficient? (source 04)

Question 82.16. What compact enrichment of leading Plücker and slack data can certify extension-complexity bounds when all ordinary power jets are identical?

Recording a second scale detects the present examples, but a fixed set of scale tolerances cannot uniformly cover the entire surreal class, by [theorem 74.2](#). A representation-sensitive answer may instead record a finite independent monomial basis and exact algebraic relations. Such data are closer to the effective core of [section 77](#) than to numerical truncations.

[write] Compare 07's remark that a method retaining extra initial forms of polynomial relations or higher coarsenings has strictly more information than the initial slack matrix (end of [section 69](#)), and 07's question "Geometry of low-lift loci in a realization space" above.

Formalization roadmaps, provenance and references

83 Formalization roadmaps (proposals only)

[write] Five of the six manuscripts end with a plan for a Lean development: 05 and 02 (Parts I and II), 06 (Part III), 03 (Part IV) and 07 (Part V); manuscript 04 has no roadmap section, and its formalization proposal is a research question of [Part V](#). The plans are collected here in the order of the Parts, each with a note on where it came from. They are proposals. None of them has been carried out: no declaration of the ProveIt Lean development states a result of this report, and no manuscript ran a Lean build. The collection does contain ingredients that some plans name: the ordered-field coordinate geometry of `Algebra/SurrealNumbers/Surreal/Algebra/Geometry.lean` (`Surreal.Complexify.dot`, `cross`, `cross_def`, `ptolemy`, `heron` and their relatives) and `Surreal.Foundations.SignSequence.omnific_bounded_iff`, the one declaration that any Part uses as an input ([Part IV](#)). Whether the other prerequisites a plan names exist in Mathlib was not checked in the merge.

83.1 Roadmap of manuscript 05 (finite-scale geometry of surreal liftings)

[write] Printed from [section 21](#) of [Part I](#), where it is the source’s subsection “A theorem-by-theorem Lean plan”. It is a proposal: no statement of Parts I and II is formalized.

The most reusable starting point is not the full class of surreals. Work first with a set-sized Hahn-style normal-form structure over $\mathbb{R}$, equipped with coefficient uniqueness, leading signs, and the fact that every nonempty support subset has a largest exponent in the growth convention. The desired modules are proposed components, not existing declarations:

Coefficient compression. Formalize the rank-increasing selection as finite-dimensional linear algebra on coefficient vectors. Prove the key invariant: before the next selected exponent, every coefficient row belongs to the already selected span. Derive kernel equality, leading-term equality, and the exact count of attained leading exponents.

Affine quotient. Use the kernel of the augmented real matrix as the dependence space. Formalize the affine-basis projection isomorphism and coefficientwise gauge. This part is independent of normal forms and should work over any suitable coefficient field.

Certificates. Prove cofactor vectors and barycentric slack vectors belong to the kernel. Prove that their signs imply the listed supporting faces, rather than trusting an external convex-hull routine. The external Python program may produce rational witnesses, but a Lean checker should verify those witnesses by rational equalities and inequalities.

Optimality and shadows. The degree lower bound is a rank-nullity theorem about polynomial coefficient maps. The linear shadow path needs only an order-preserving real-linear standard-part map and a finite real specialization theorem. Neither proof should require a formal theory of tropical fans.

Toric extension. Formalize conformal decomposition and the finite Graver certificate separately. Its ordered-weight proof does not need real closedness or a surreal-specific polynomial ring.

Only after these modules are proved should they be instantiated with the repository’s constructed surreal field. The normal-form, real-embedding, and standard-part interfaces must be checked against that implementation. No claim is made that the whole plan is already supported by existing Mathlib declarations, and no Lean build was run for this manuscript.

83.2 Roadmap of manuscript 02 (surreal polytopes at finite and transfinite scales)

[write] Manuscript 02’s subsection “A proposed Lean development, not an implemented one”, from its section on algorithms and verification (section 27). It is a proposal as well.

The most useful first formalization target is not all of surreal convexity. It is the finite coefficient compression lemma over a real vector space, followed by an abstract leading-coefficient sign interface. One can then separate the theory into three modules.

Finite linear algebra. Define the affine relation space L_A , the affine-height quotient, barycentric slacks, and zero-label cells. Prove the quotient dimension, annihilator description, rank minimality, and finite real sign bound. These statements make sense independently of surreal arithmetic.

An ordered-series interface. Provide a type of well-ordered coefficient families, prefix restriction, and a theorem that the sign of a nonzero real linear evaluation is its first nonzero coefficient sign. The greedy construction needs a least unspanned position and finite-dimensional rank induction. Normal-form integration should occur only after this interface is proved for the repository’s actual surreal construction.

Geometric and counterexample interfaces. Formalize the lower-basis certificate and its marked refinement relation. For the negative examples, the finite identities $1 - (1 + t) \sum_{k=0}^N (-t)^k = (-1)^{N+1} t^{N+1}$ and (26.2) should come first. The inspected `Surreal/Algebra/Geometry.lean` already provides the relevant cross-product algebra, but not the polytope statements of this paper. An ordinary power-series realization of the reciprocal construction can precede its embedding into surreal normal forms. The final transfinite endpoint needs the explicit order comparison $0 < \omega^{-\omega} < \omega^{-m}$ for ordinary m .

No uncompiled Lean skeleton is included as supposed evidence. A future implementation should map each theorem to a checked declaration and record all imported axioms, rather than infer verification from a shared notation or neighboring file.

83.3 Roadmap of manuscript 06 (affine residues and mixed-volume scales)

[write] Printed from manuscript 06, Section “Exact verification and a formalization route” (section 42), subsection “A modular Lean development”; “the article” below is manuscript 06, printed as Part III. This is a proposal only: at the write phase no result of Part III is formalized, and no Lean file in the ProveIt tree mentions mixed volumes. The research question “Formal certificates in the repository” stays with Part III’s questions in section 43.

A modular Lean development

The existing repository’s constructed real closed field provides a potential target for instantiation, but most of the article should be formalized generically first. The mathematical dependencies suggest four layers.

Ordered-field finite geometry. Represent an input polytope by a finite vertex family. Prove the maximum-simplex Cramer bound, the simplex/cube inclusions, and the parallelotope sandwich over a linearly ordered field. These steps require only finite maxima, matrix determinants, convex combinations, and linear independence. They do not require surreal arithmetic.

Convex valuation algebra. Introduce the finite-element subring, its maximal ideal, and residue map, or use an abstract convex valuation ring with an ordered residue field. Prove the positive-sum value rule and the module-basis theorem. Package the mixed determinant ideal independently of polytopes. The probe reconstruction theorem then reduces to matrix integrality and the adjugate formula.

Finite mixed volume. Either develop determinant triangulation volume and its finite geometric properties, or provide a separately verified abstract mixed-volume interface containing the classical properties explicitly used here. The latter is an intermediate theorem package, not a substitute for constructing mixed volume. The determinant squeeze requires only monotonicity, translation invariance, positive homogeneity, multilinearity, and the segment determinant formula.

Genericity and discrete exchange. Formalize the multivariate finite-grid lemma, coefficient-wise residue of finite polynomials, the inner-segment construction, and the determinant exchange identity. The clone proof of M -convexity is then finite combinatorics on labelled sets. This stage does not require the full machinery of Lorentzian polynomials or buildings.

The present package contains no Lean source and reports no new successful Lean build. The repository’s own formalization statements remain subject to its ledger and pinned sources [17]. The outline above identifies concrete formalization tasks rather than treating a prospective interface as an already proved theorem.

83.4 Roadmap of manuscript 03 (real traces and lattice-point counting)

[write] Printed from manuscript 03, Section “Repository integration and the exact proof boundary” (section 56), subsection “A realistic formalization sequence”; “the article” below is manuscript 03, printed as Part IV. This is a proposal only. At the write phase the one existing Lean input is `omnific_bounded_iff` (Algebra/SurrealNumbers/Surreal/Foundations/OmnificUnits.lean, line 65), none of Part IV’s own results is formalized, and no Lean file in the ProveIt tree mentions Ehrhart theory, Pólya–Carlson or curve selection. The research question “A checked geometry pipeline” stays with Part IV’s questions in section 58.

A realistic formalization sequence

The most useful integration order follows the finite proof boundary, not the most sophisticated theorem first.

First: the counting bridge and quadrilateral. Represent the integer candidates in a real bounding box by a finite set. Use `omnific_bounded_iff` to identify all bounded omnific lattice points with integer tuples. Prove the two-level sign lemma

$$a + \varepsilon b \geq 0 \iff (a > 0) \vee (a = 0 \wedge b \geq 0) \quad (a, b \in \mathbb{R}),$$

where ε is positive infinitesimal. This makes the quadrilateral count an elementary finite sum and a ceiling identity. It is the smallest self-contained target with a genuinely nontrivial consequence.

Second: finite flags and cell semantics. Formalize the standard-part functional on a finite-dimensional real subspace of the ordered field, its codimension-one kernel, and induction producing a separated basis. Then prove the row expansion and trace formula. A checker need not discover a separated basis automatically: it can accept a supplied finite expansion and certify its algebraic identities, positivity, and separation conditions.

Third: geometry and asymptotics. Prove the shadow convex-hull lemma, interior lifting by barycentric coordinates, and simple-polytope stability. The surface coefficient also requires ordinary lattice-volume estimates and Ehrhart reciprocity. These should be separate reusable results rather than bundled assumptions inside a surreal-specific theorem.

Fourth: imported global inputs. The one-parameter descent proof uses quantifier elimination and Nash curve selection; the analytic conclusion uses Pólya–Carlson. These are distinct large formalization tasks. A conditional formal theorem may expose them as named assumptions, but that is not an unconditional kernel verification of the whole article. The elementary finite counterexample and its exact count do not depend on either curve selection or Pólya–Carlson.

83.5 Roadmap of manuscript 07 (invisible complexity)

[write] Section 14 of manuscript 07, “Formalization design and the trusted boundary”, printed from Part V (Part V). It is a proposal: no statement of Part V is formalized. The scale written ρ in 07 is this report’s $\tau = \omega^{-\omega}$. Manuscript 04, the other source of Part V, has no roadmap section; its formalization proposal is its research question “Formalization milestones”, merged with 07’s last question in section 82.

The repository’s documented real closed surreal field is a natural target, but this manuscript has not been compiled into Lean and does not extend any existing formalization ledger. A staged implementation should avoid beginning with a proper-class polytope type.

83.5.1 A field-generic core

The first layer should work over an abstract ordered field and formalize finite vertex configurations, affine images, orientation determinants, and the binary reflection relation. The four identities in (65.1)–(65.4) are polynomial identities; after denominators are cleared, ring normalization is enough to check them. Their positivity consequences require only the specified inequalities on u, v, w, τ .

An extension certificate can store a finite matrix of inequalities, affine output maps, and a proof that the image is the required hull. For the reflection step, the two crucial facts are the affine endpoint identity and convexity of the relation (66.2). The procedure should expose these facts rather than hide them inside a numerical optimizer.

83.5.2 The valuation and linear-algebra layers

A convex-subring interface should provide the ideal j_s , its intersection characterization, and closure under bounded multiplication. The power-jet kernel, the coarsening ring and the scalar perturbation lemma can then be proved before any surreal-specific instantiation. The nonnegative normalization theorem is finite sums and inequalities.

The semidefinite layer needs positive-semidefinite matrices over a real closed field, kernel/range decomposition, positive square roots and trace invariance. The normalization proof has a clear endpoint: every factor entry is bounded by an integral scalar. This is a better formal interface than a topological sequential-compactness theorem.

83.5.3 The genuine logical dependency

The semidefinite lower bound depends on the statement that a finite ordered-field definable set has coordinates algebraic over its parameter field. One may prove this from quantifier elimination, or isolate it as a separately verified theorem of real closed fields. Replacing it by an unproved claim that every vertex has a rational normal would be incorrect. [Example 67.5](#) should be included as a regression example.

The linear lower bound is more elementary: it requires dimension bounds, Cramer’s rule or Gaussian elimination, and transcendence-degree counting. The PSD factorization–lift correspondence is an additional imported mathematical theorem unless the formal library proves it. A proof of the rank lower bound is not, by itself, a formal proof of the geometric lift lower bound without this bridge.

83.5.4 Surreal instantiation

Only the final layer needs the actual surreal monomial operation. Its interface consists of multiplication of monomials, domination by larger exponents, and nonvanishing of finite sums with distinct supports. The algebraic-number lemma supplies linear independence of the exponent coefficients. All individual configurations can be instantiated inside a small real closed subfield; the set-sized precision theorem can remain a separate class-level result.

Artifact or dependency	Evidence actually supplied here
Orientation and reflection identities	Symbolic exact checks and written proofs.
Finite reference and perturbed polygons	Exact rational checks; not transcendental witnesses.
All-orders stability and independence	Written mathematical proofs.
Generic linear and PSD lower bounds	Written field-generic proofs, with stated classical imports.
Repository foundations	Documentation consulted; no independent rebuild.
Lean implementation	Proposed architecture only; no Lean source represented as checked.

[write] The one related Lean module, `Surreal.Surcomplex.RegularPolygon` (regular polygons at surreal radius: `vertex`, `vertex_radius`, `side_length`, `area_eq`), is used by neither manuscript.

A Provenance of the report

[write] This appendix was written in the merge.

A.1 Sources

The report was built in the batch-58 write phase from six manuscripts, all dated 30 September 2026 and delivered on that day in six archives (arrival commit b2b626d); they were placed in commit b30441a, which staged manuscript 05's `article.tex` and `README.md` as delivered and every other staged file under its manuscript's prefix. Manuscripts are numbered by arrival within the batch (manuscript 01, on a Keller map, went to another report).

No.	Title (archive; main file)	Delivered PDF, author line	Pin	Printed as
05	<i>Finite-Scale Geometry of Surreal Liftings</i> (<code>surreal_liftings_research.zip</code> ; <code>surreal_liftings/article.tex</code>)	25 pp. A4; research manuscript prepared for Vladimir Reshetnikov, an AI-assisted contribution to the ProveIt research program	5e0f4e0	Part I , base (whole)
02	<i>Surreal Polytopes at Finite and Transfinite Scales</i> (<code>surreal_polytopes_research.zip</code> ; <code>surreal_polytopes/article.tex</code>)	28 pp. Letter; research article prepared for Vladimir Reshetnikov	6996fee	Part I (merged) and Part II
06	<i>Surreal Polytopes: Affine Residues, Mixed-Volume Scales, and Simultaneous Segment Models</i> (<code>Surreal_Polytopes_Research_Package.zip</code> ; <code>Surreal_Polytopes/article.tex</code>)	27 pp. A4; research article prepared for Vladimir Reshetnikov	6996fee	Part III (whole)
03	<i>Infinitesimal Boundary Geometry of Surreal Polytopes: Lexicographic traces, one-parameter descent, and Ehrhart universality</i> (<code>surreal_polytope_research.zip</code> ; <code>surreal_polytopes/surreal_polytopes.tex</code>)	28 pp. Letter; research manuscript prepared for Vladimir Reshetnikov	6996fee	Part IV (whole)
07	<i>Invisible Complexity in Surreal Polytopes: Explicit polygons with identical all-order jets and identical initial slack matrices</i> (<code>surreal_polytopes.zip</code> ; <code>surreal_polytopes/surreal_polytopes.tex</code>)	26 pp. A4; research manuscript prepared for Vladimir Reshetnikov	6996fee	Part V , base (whole)
04	<i>Beyond-All-Orders Extension Complexity in Surreal Polygon Geometry: Flat deformations, classical face lattices, and a sharp-order complexity gap</i> (<code>surreal_polytopes_beyond_all_orders.zip</code> ; <code>surreal_polytopes_beyond_all_orders/article.tex</code>)	26 pp. A4; research manuscript prepared for Vladimir Reshetnikov, developed with ChatGPT; independent review and formalization pending	6996fee	Part V (merged)

The pin 6996fee is commit 6996fee43cc97b6c16351def7507d59a95bf62f0 (30 September 2026, 16:07:39 UTC); 5e0f4e0 is 5e0f4e05046bacda3c6ba2643f19b91aa632436b, its second parent. Archives 02, 03 and 07 all wrap an inner directory named surreal_polytopes/; they are different packages.

A.2 What each manuscript contributed

Manuscript 05 (base; Part I). The frame of Part I: the affine gauge and dependence space, the simultaneous leading-term compression ([theorem 6.1](#) and [corollary 6.3](#)), the flag classification ([theorem 8.2](#)), the exact germ degrees ([theorems 9.1](#) and [9.2](#)) and the failure of a single real parameter ([proposition 9.4](#)); determinant and barycentric certificates, the explicit threshold ([lemma 10.3](#)), the linear realization with the exact shadow and the separation of the two degrees ([theorems 10.4](#) and [10.6](#)); shadows and the canonical refinement chain; the higher-rank tropical bridge; the boundary examples; toric initial ideals through Graver certificates ([theorem 17.2](#)); transfer and shadow arcs ([theorem 18.1](#)); the corrections to the private draft ([section 20](#)); the verifier and the halting-problem obstruction ([proposition 21.1](#)); eight research questions, its dependency ledger and a Lean plan.

Manuscript 02 (Parts I and II). In Part I: its introduction (three operations that change scale), its working field and common initial normal-form cuts, the cut-compatible compression ([theorem 7.1](#)) and the transfinite-delay example, the fixed-base comparison-rank theorem with its cut clause ([theorem 11.2](#)), the stage threshold ([theorem 11.4](#)), positive sums and the classification of support-initial histories ([lemma 12.1](#) and [theorem 12.2](#)), the sharp parabolic family and the hexagon ([theorem 12.3](#)), the finite polar identity, the lost face and the optimality band ([propositions 14.1](#) and [14.2](#)), the finite Puiseux model ([theorem 19.1](#)), the certificate workflow and the query-model bound ([proposition 21.2](#)), and a second, covector-level proof of transfer. In Part II: the four-point test, the rational alternation, the universal reciprocal-series histories, the independent transfinite endpoint, finite determinacy for uniform ordinary series and the tail-truncation caveat ([lemma 26.1](#), [theorems 26.2](#), [26.3](#) and [26.6](#), and [proposition 26.7](#)), its recorded tests and proof-dependency audit, eight research directions, the exact recursion and its reproducibility notes; and a Lean plan.

Manuscript 06 (Part III). Moving coordinates over any set-sized real closed $K \supseteq \mathbb{R}$ with its natural valuation: the scale lattice $L(P)$ and the affine residue $\text{Res}(P)$ with its normalization theorem, determinant certificates (at most d^d determinants), the simultaneous inner-segment models that preserve every pure mixed-volume valuation on a finite rational grid, the Grassmannian and clone results, exchange-convexity of valuation profiles over any ordered group, the classification of probes and finite probe sets, the distortion bound, the full characterization corollary, a one-parameter specialization of finite-support Hahn certificates, its examples, ten research questions, its audit appendix and a Lean plan.

Manuscript 03 (Part IV). Shadows and real traces $\text{Tr}_{\mathbb{R}}(P) = P \cap \mathbb{R}^d$ of a surreal polytope: at most $d + 1$ lexicographic levels per inequality, one-parameter descent to the real closure of $\mathbb{R}(t)$, face implantation, the counting functions $(n + 1)^2 - \lceil \alpha n \rceil$ for every irrational $0 < \alpha < 1$ (with an algebraic instance), the second asymptotic coefficient and its attainable interval, rational flags and the dichotomy for counting functions, twelve research questions, a dependency map, its two appendices and a Lean plan. It uses `omnific_bounded_iff` as an input.

Manuscript 07 (base; Part V). The frame of Part V: for every $n \geq 4$, two strictly convex n -gons with the same shadow, oriented matroid, all-order jets and leading slack data, one of logarithmic linear and semidefinite extension complexity and one of complexity at least of order $\sqrt{n}$; generic linear and semidefinite lower bounds; one fixed field of valuation rank three and the order-unit criterion with the sharpness of rank two; residue descent for every convex valuation ring; precision and conservativity theorems; the bounds table; ten research questions and a Lean plan.

Manuscript 04 (Part V). A second construction on one circle: the regular 2^m -gon R_n and a cyclic n -gon P_n° agreeing below every power of ε , with $m \leq \text{xc}(R_n) \leq 2m$ and $\lceil \sqrt{n} \rceil \leq \text{xc}(P_n^\circ) \leq \min(n, \lceil 24\sqrt{n} \rceil)$ ([theorem 70.1](#)); the field- and scale-general criterion that gaps in all-order jet fibres exist exactly when the flat ideal is nonzero, exactly when $v(\varepsilon)$ is not an order unit ([theorem 72.7](#)); the rank-two fields K_n , a finite-order rank-one analogue ([theorem 74.1](#)), the failure of lifting coherent finite-order factorizations ([theorem 71.2](#)), what asymptotic data cannot recover, algebraic finite-accuracy approximants ([corollary 74.6](#)), a rational-chart generalization ([theorem 76.1](#)), an in-principle decision procedure ([proposition 77.1](#)), nine research questions and its repository and notation appendices; second proofs of the flat-ideal lemma, the order-unit theorem, the scalar-extension corollary, the normalization theorem and algebraic realization (with its point case).

A.3 Where the merge had to choose

- *Bases.* Manuscript 05 is the base of Part I and of the report: on the lifting theorem it shares with 02 it assumes no more (an ordered subfield $\mathbb{R} \subseteq K \subseteq \mathbf{No}$, real closure only where needed; 02 assumes a real closed K) and concludes more (exact leading terms, the flag classification, exact germ degrees). Manuscript 07 is the base of Part V: it covers every $n \geq 4$ (04: $n = 2^m$), linear and semidefinite lifts, residue descent for every convex valuation ring (04: the standard part and nonnegative rank only), every fixed dimension, and a sharper parameter count ($dN \leq r(r-1)$ against 04's $s \leq r^2$).
- *Printed once (Part I).* $\text{st conv} = \text{conv st}$ (05 and 02; one proof, both labels); the slack certificate (05's [proposition 10.2](#), carrying 02's label, with 02's statement and proof in [remark 11.1](#)); the failure of a single real parameter (05's [proposition 9.4](#); 02's remark kept and marked as the same result); real-closed-field transfer (05's [theorem 18.1](#), with 02's statement appended in its covector form and 02's proof as a second proof).
- *Printed beside each other (Part I).* 02's [theorems 7.1](#) and [11.2](#) beside 05's [theorem 6.1](#) and [corollary 6.3](#), because 02's cut clauses are new and proved by the argument that also proves the shared clauses; 02's [theorem 11.4](#) beside 05's [lemma 10.3](#), because its constants differ (both valid; both give $1/34$ and $1/10$ on the examples); 02's [theorem 19.1](#) beside the bounded half of [theorem 18.1](#); 02's query-model [proposition 21.2](#) beside 05's halting-problem [proposition 21.1](#) (different obstructions).
- *Printed once (Part V), with both labels and 04's proof as a second proof:* the flat-ideal lemma, the order-unit theorem ([theorem 72.2](#)), the scalar-extension corollary ([corollary 74.7](#)), the normalization theorem ([theorem 73.2](#), 04's statement in a remark), algebraic realization ([theorem 74.3](#), 04's point case in a remark), and precision with 04's tolerances as part (b) of [theorem 74.2](#). Printed beside each other: 04's transcendence-degree proposition beside [theorem 67.1](#) (07's statement does not cover the circle), 04's proposition on Yannakakis's factorization beside 07's displayed factorization, and the two different reflection lemmas; 04's $\text{st conv} = \text{conv st}$ is printed in its conventions with a pointer to [proposition 13.1](#).

- *Moved.* Manuscript 02’s Section 6, recorded tests, audit table, research directions, conclusion and appendices form Part II; its workflow and query bound sit in 05’s computation section. Every manuscript’s Lean roadmap is in [section 83](#). Source appendices became ordinary sections at the end of their Parts. In Part V, 04’s Sections 1–4 are distributed into 07’s first sections, 04’s construction and its Section 7 are sections of their own after 07’s Section 8, 04’s criterion joins 07’s field section, and 04’s charts, computation and appendices are sections before the merged questions.
- *Questions.* Each manuscript’s research questions stay at the end of its material, with [write] cross-references where two manuscripts ask the same thing (05’s questions 1, 2, 4 and 5 and four of 02’s directions; in Part V three of 07’s questions are merged with three of 04’s, and four more are re-scoped by notes; Parts III and IV point to 02’s directions and to each other); a question that another manuscript answers in part is re-scoped by a note rather than printed as open.
- *Paper and layout.* A4 and the base’s fonts; one theorem counter numbered within sections; source numbers are not kept.

A.4 Renamed symbols

Only letters that would otherwise clash inside a Part, or that denote the fixed surreals ω^{-1} and $\omega^{-\omega}$, were renamed; no normalization changed.

Part	Source letter	Printed	Where and why
I	02: $t = \omega^{-1}$	ε	in 02’s material of Part I (working field, example 7.3 , theorems 12.2 and 12.3 , (14.2) , proposition 21.2); t is 05’s real parameter there. 02’s introduction and Part II keep t , the power-series variable evaluated at $t = \varepsilon$
I	05: $\varepsilon_1, \varepsilon_2$	ε, τ	section 16.5 : $\varepsilon_1 = \omega^{-1}$, $\varepsilon_2 = \omega^{-\omega}$
III	06: macro <code>\Span</code> (plain span)	<code>\spanplain</code>	a macro clash with 05’s <code>span_{$\mathbb{R}$}</code> ; printed output unchanged. 06’s $\eta = \omega^{-\omega}$ is kept
V	07: $\rho = \omega^{-\omega}$	τ	main construction, $V_\rho \rightarrow V_\tau$; 07’s re-chosen scale in corollary 72.4 and theorem 74.2 keeps ρ
V	07: auxiliary τ	σ	proof of theorem 74.2
V	04: $\mathcal{I}_\varepsilon, \nu, \Gamma$	$\mathfrak{I}_\varepsilon, \nu, G$	one flat-ideal symbol, one valuation letter and one value-group letter in Part V
V	04: ν_i	μ'_i	monomials in the proof of theorem 74.1
V	04: $P_n, p_i, P_{n,M}$	$P_n^\circ, p_i^\circ, P_{n,M}^\circ$	07’s P_n is a different polygon
V	04: $\mathcal{T} \ni \tau, c_\tau, b_\tau$	$\mathcal{D} \ni \delta, c_\delta, b_\delta$	tolerance set of theorem 74.2
V	04: “algebraic finite-accuracy shadows”	“approximants”	they are real-algebraic approximants, not standard parts (corollary 74.6)

Labels and citation keys received the prefixes of [section 1](#); a few labels were added on unlabelled items that the merged text cites. The report’s `README.md` lists the label counts, the added labels, and the correspondence between the delivered files and the shipped ones.

References

[write] The six manuscripts' reference lists, grouped by Part and numbered consecutively; an entry cited by two manuscripts of one Part is printed once.

Parts I and II (manuscripts 05 and 02)

- [1] Harry Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986. Chapter 5, “Normal Form,” pp. 52–94 [write] (the page range is manuscript 02's; both manuscripts cite this book). doi:10.1017/CBO9780511629143.
- [2] Jesús A. De Loera, Jörg Rambau, and Francisco Santos, *Triangulations: Structures for Algorithms and Applications*, Algorithms and Computation in Mathematics 25, Springer, 2010. doi:10.1007/978-3-642-12971-1.
- [3] Michael Joswig and Ben Smith, “Convergent Hahn series and tropical geometry of higher rank,” *Journal of the London Mathematical Society* **107** (2023), 1450–1481. doi:10.1112/jlms.12716; arXiv:1809.01457. The rank-two decomposition and the concluding questions are relevant comparisons, not new claims of this manuscript.
- [4] Rocco Chirivì, Martina Costa Cesari, Xin Fang, and Peter Littelmann, “Higher rank Gelfand–Kapranov–Zelevinsky fans,” arXiv:2604.20250v2, 13 September 2026. Sections 2–3 concern higher-rank cones and subdivisions; Sections 4–5 concern quasi-valuations and degenerations.
- [5] Saugata Basu, Richard Pollack, and Marie-Françoise Roy, *Algorithms in Real Algebraic Geometry*, 1st edition, Algorithms and Computation in Mathematics 10, Springer, 2003. doi:10.1007/978-3-662-05355-3.
- [6] Saugata Basu and Marie-Françoise Roy, “Quantitative Curve Selection Lemma,” arXiv:1803.00505v3, 19 July 2021; first version 2018. [write] Manuscript 02 cites the 2018 version and uses only its classical qualitative curve-selection and algebraic-Puiseux framework, not a quantitative complexity claim.
- [7] Sonja Petrović, “On the universal Gröbner bases of varieties of minimal degree,” arXiv:0711.2714, 2007. Cited for the classical Graver/universal-Gröbner background; the needed ordered-weight argument is proved in Lemma 17.1.
- [8] Vladimir Reshetnikov, ProveIt repository, Algebra/SurrealNumbers/Surreal/Algebra/Geometry.lean, inspected at commit 5e0f4e0, 30 September 2026. Fixed source link.
- [9] ProveIt, Algebra/SurrealNumbers/README.md and Algebra/SurrealNumbers/docs/README.md, consulted 30 September 2026. Project and report guide. Descriptions of formalization status are repository reports, not a fresh build certification by this manuscript.
- [10] *Finite Convexity and Exact Lexicographic Optimization over the Surreals: A finite-dimensional theorem package and a one-objective compiler for multiscale linear programs*, research draft prepared for the Surreal repository, 22 September 2026, 8 pages. Research-library copy: article(20260923-000126).pdf. Proposition 7.1 and Theorem 9.1, printed pages 5–6, are examined in Section 20. No public repository location for this copy is assumed. [write] The draft is not in ProveIt; see the note at the head of section 20.
- [11] L. van den Dries, Alfred Tarski's elimination theory for real closed fields, *Journal of Symbolic Logic* **53**(1) (1988), 7–19. doi:10.2307/2274424.
- [12] M. Joswig, G. Loho, B. Lorenz, and B. Schröter, Linear programs and convex hulls over fields of Puiseux fractions, in *Mathematical Aspects of Computer and Information Sciences (MACIS 2015)*, Lecture Notes in Computer Science 9582, Springer, 2016, pp. 429–445. doi:10.1007/978-3-319-32859-1_37. arXiv:1507.08092.

- [13] M. Wortel, Lexicographic cones and the ordered projective tensor product, arXiv preprint, 2018. Proposition 4.4. [arXiv:1812.04830](https://arxiv.org/abs/1812.04830).
- [14] C. W. Lee and F. Santos, Subdivisions and triangulations of polytopes, in J. E. Goodman, J. O'Rourke, and C. D. Tóth (eds.), *Handbook of Discrete and Computational Geometry*, 3rd ed., CRC Press, 2017, Chapter 16, pp. 415–447. In particular, Theorem 16.4.1. [Author-hosted chapter](#).
- [15] J. Bochnak, M. Coste, and M.-F. Roy, *Real Algebraic Geometry*, Ergebnisse der Mathematik und ihrer Grenzgebiete, 3rd series, vol. 36, Springer, 1998. [doi:10.1007/978-3-662-03718-8](https://doi.org/10.1007/978-3-662-03718-8).
- [16] V. Reshetnikov and contributors, *ProveIt*, GitHub repository, inspected 30 September 2026. Pinned commit 6996fee43cc97b6c16351def7507d59a95bf62f0. Within Algebra/SurrealNumbers/, consulted README.md, docs/README.md, and the pinned source Surreal/Algebra/Geometry.lean. [Pinned subproject](#). See the accompanying source notes for the distinction between repository claims, inspected source, and formal builds not performed.

Part III (manuscript 06)

- [17] Vladimir Reshetnikov and contributors, *ProveIt: Algebra/SurrealNumbers*, repository README (which links a formalization ledger), inspected at commit 6996fee43cc97b6c16351def7507d59a95bf62f0, 30 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers/README.md>. Repository assertions are distinguished from the new proofs in this article.
- [18] ProveIt research collection, *Reversion Beyond Archimedean Valuations: Strong Hahn convergence, sharp Newton depth, and minimal finite coefficient certificates*, 29 September 2026, with repository editorial notes; inspected at the same commit as [17]. https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Analysis/Transseries/docs/series-and-transseries/Reversion_Beyond_Archimedean_Valuations/article.tex.
- [19] Philip Ehrlich, *The absolute arithmetic continuum and the unification of all numbers great and small*, Bulletin of Symbolic Logic **18** (2012), no. 1, 1–45. DOI: [10.2178/bsl/1327328438](https://doi.org/10.2178/bsl/1327328438).
- [20] Michel Coste, *An Introduction to Semialgebraic Geometry*, lecture notes, Institut de Recherche Mathématique de Rennes, 2002. Author-institution archive: <https://univ-rennes.hal.science/hal-05407113>.
- [21] Xavier Allamigeon, Ricardo D. Katz, and Pierre-Yves Strub, *Formalizing the Face Lattice of Polyhedra*, in Automated Reasoning—IJCAR 2020, Lecture Notes in Computer Science **12167**, Springer, 2020, 185–203. DOI: [10.1007/978-3-030-51054-1_11](https://doi.org/10.1007/978-3-030-51054-1_11). Open text: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7324146/>.
- [22] Rolf Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, second expanded edition, Encyclopedia of Mathematics and its Applications **151**, Cambridge University Press, 2014. ISBN 9781107601017. Especially the chapters on Minkowski addition, mixed volumes, and inequalities for mixed volumes.
- [23] Gennadiy Averkov, Katherina von Dichter, Simon Richard, and Ivan Soprunov, *Mixed volumes of zonoids and the absolute value of the Grassmannian*, arXiv:2404.02842v2, 1 November 2024. <https://arxiv.org/html/2404.02842v2>. The cited version supplies the motivating connection and proposed tropical direction; no assertion about the complete status of its open questions after that version is required here.
- [24] Petter Brändén and June Huh, *Lorentzian polynomials*, Annals of Mathematics (2) **192** (2020), no. 3, 821–891. Preprint: <https://arxiv.org/abs/1902.03719>.
- [25] Luca Battistella, Kevin Kühn, Arne Kuhrs, Martin Ulirsch, and Alejandro Vargas, *Buildings, valuated matroids, and tropical linear spaces*, Journal of the London Mathematical Society **109** (2024), no. 1, e12850. DOI: [10.1112/jlms.12850](https://doi.org/10.1112/jlms.12850). Preprint: <https://arxiv.org/abs/2304.09146>.

Part IV (manuscript 03)

- [26] V. Reshetnikov, *ProveIt: Surreal numbers project README*, repository documentation, consulted September 30, 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Algebra/SurrealNumbers>. The live README was used for orientation; pinned substantive sources follow.
- [27] V. Reshetnikov, *The surreal and surcomplex reports*, `Algebra/SurrealNumbers/docs/README.md`, ProveIt commit 6996fee43cc97b6c16351def7507d59a95bf62f0. [Commit-pinned report guide on GitHub](#).
- [28] V. Reshetnikov, *Units and finite elements of the actual omnific ring*, `OmniFicUnits.lean`, ProveIt commit 6996fee43cc97b6c16351def7507d59a95bf62f0. [Commit-pinned Lean source on GitHub](#).
- [29] H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986. <https://doi.org/10.1017/CB09780511629143>.
- [30] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, Algorithms and Computation in Mathematics 10, Springer, 2003. <https://doi.org/10.1007/978-3-662-05355-3>.
- [31] S. Basu and M.-F. Roy, *Quantitative Curve Selection Lemma*, arXiv:1803.00505, 2018. <https://arxiv.org/abs/1803.00505>.
- [32] J. F. Fernando, *On a Nash curve selection lemma through finitely many points*, arXiv:2504.03348, 2025. <https://arxiv.org/abs/2504.03348>. *Revista Matemática Iberoamericana*, <https://doi.org/10.4171/rmi/1554>. Cited here for its explicit account of classical Nash curve selection, not for a claim that this manuscript requires its stronger multi-point result.
- [33] M. Conforti, M. De Santis, M. Di Summa, and F. Rinaldi, *Scanning integer points with lex-inequalities: a finite cutting plane algorithm for integer programming with linear objective*, *4OR* 19 (2021), 531–548. <https://doi.org/10.1007/s10288-020-00459-6>.
- [34] S. V. Sam and K. M. Woods, *A finite calculus approach to Ehrhart polynomials*, *Electronic Journal of Combinatorics* 17 (2010), R68. <https://doi.org/10.37236/340>. <https://arxiv.org/abs/0904.0679>.
- [35] D. Cristofaro-Gardiner, T. X. Li, and R. P. Stanley, *New examples of period collapse*, arXiv:1509.01887, 2015. <https://arxiv.org/abs/1509.01887>. See also the authors' *Irrational triangles with polynomial Ehrhart functions*, *Discrete & Computational Geometry* 61 (2019), 227–246, <https://doi.org/10.1007/s00454-018-0036-7>.
- [36] J. P. Bell, K. Gunn, K. D. Nguyen, and J. C. Saunders, *A general criterion for the Pólya–Carlson dichotomy and application*, arXiv:2206.00862, 2022. <https://arxiv.org/abs/2206.00862>. The classical Pólya–Carlson theorem is stated as Theorem 1.1.
- [37] T. Yu, *An analytic proof of multivariate Pólya–Carlson theorem*, *Acta Mathematica Sinica, English Series* 41 (2025), 2707–2712. <https://doi.org/10.1007/s10114-025-4213-3>. Preprint: *A multivariate version of Pólya–Carlson theorem*, arXiv:2312.00428, <https://arxiv.org/abs/2312.00428>.
- [38] R. P. Stanley, *Differentiably finite power series*, *European Journal of Combinatorics* 1 (1980), 175–188. [https://doi.org/10.1016/S0195-6698\(80\)80051-5](https://doi.org/10.1016/S0195-6698(80)80051-5).

Part V (manuscripts 07 and 04)

- [39] Vladimir Reshetnikov, *ProveIt: machine-checked mathematics and research artifacts*, repository documentation, accessed 30 September 2026. <https://github.com/VladimirReshetnikov/ProveIt>. Snapshot metadata: 6996fee43cc97b6c16351def7507d59a95bf62f0. [source 07]

- [40] Vladimir Reshetnikov, *ProveIt, Algebra/SurrealNumbers: README and report catalogue*, accessed 30 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Algebra/SurrealNumbers>. Documentation reports formalization status; no independent build was performed for this manuscript. [source 07]
- [41] *Finite Convexity and Exact Lexicographic Optimization over the Surreals: A finite-dimensional theorem package and a one-objective compiler for multiscale linear programs*, research draft prepared for the Surreal repository, 22 September 2026. Prior research-library artifact titled `article(20260923-000126).pdf`; consulted for scope and overlap. [source 07] [write] A private draft, not part of ProveIt; it could not be checked.
- [42] Harry Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986. <https://doi.org/10.1017/CBO9780511629143>. [source 07]
- [43] Samuel Fiorini, Thomas Rothvoß and Hans Raj Tiwary, *Extended formulations for polygons*, *Discrete & Computational Geometry* **48** (2012), 658–668. <https://arxiv.org/abs/1107.0371>. <https://doi.org/10.1007/s00454-012-9421-9>. [sources 07 and 04]
- [44] Volker Kaibel and Kanstantsin Pashkovich, *Constructing Extended Formulations from Reflection Relations*, arXiv:1011.3597, 2010. <https://arxiv.org/abs/1011.3597>. [sources 07 and 04] [write] Source 04 cites the conference version: in *Integer Programming and Combinatorial Optimization (IPCO 2011)*, Lecture Notes in Computer Science **6655**, Springer, 2011, 287–300, https://doi.org/10.1007/978-3-642-20807-2_23.
- [45] Mihalis Yannakakis, *Expressing combinatorial optimization problems by linear programs*, *Journal of Computer and System Sciences* **43** (1991), 441–466. [https://doi.org/10.1016/0022-0000\(91\)90024-Y](https://doi.org/10.1016/0022-0000(91)90024-Y). [sources 07 and 04]
- [46] João Gouveia, Pablo A. Parrilo and Rekha R. Thomas, *Lifts of convex sets and cone factorizations*, *Mathematics of Operations Research* **38** (2013), 248–264. <https://arxiv.org/abs/1111.3164>. <https://doi.org/10.1287/moor.1120.0575>. [source 07]
- [47] João Gouveia, Richard Z. Robinson and Rekha R. Thomas, *Worst-Case Results for Positive Semidefinite Rank*, arXiv:1305.4600v2, 30 May 2014. <https://arxiv.org/abs/1305.4600>. In particular, Theorem 2.1 and its real-field parameter-count argument. [source 07]
- [48] Hamza Fawzi, João Gouveia, Pablo A. Parrilo, Richard Z. Robinson and Rekha R. Thomas, *Positive semidefinite rank*, *Mathematical Programming* **153** (2015), 133–177. <https://arxiv.org/abs/1407.4095>. <https://doi.org/10.1007/s10107-015-0922-1>. [source 07]
- [49] Assia Mahboubi and Cyril Cohen, *Formal proofs in real algebraic geometry: from ordered fields to quantifier elimination*, arXiv:1201.3731, 2012. <https://arxiv.org/abs/1201.3731>. [source 07]
- [50] A. Tarski, *A Decision Method for Elementary Algebra and Geometry*, prepared with the assistance of J. C. C. McKinsey, University of California Press, 1951; RAND report R-109. <https://www.rand.org/pubs/reports/R109.html>. [source 04]
- [51] G. E. Collins, Quantifier elimination for real closed fields by cylindrical algebraic decomposition, in *Automata Theory and Formal Languages*, Lecture Notes in Computer Science **33**, Springer, 1975, 134–183. https://doi.org/10.1007/3-540-07407-4_17. [source 04]
- [52] L. van den Dries and P. Ehrlich, Fields of surreal numbers and exponentiation, *Fundamenta Mathematicae* **167** (2001), 173–188. <https://doi.org/10.4064/fm167-2-3>. [source 04]
- [53] M. Kwan, L. Saueremann, and Y. Zhao, Extension complexity of low-dimensional polytopes, *Transactions of the American Mathematical Society* **375** (2022), 4209–4250. <https://doi.org/10.1090/tran/8614>. Preprint, version 3: <https://arxiv.org/abs/2006.08836>. [source 04]

- [54] V. Reshetnikov et al., *ProveIt: Surreal-number library and research collection*, repository snapshot 6996fee43cc97b6c16351def7507d59a95bf62f0, inspected September 30, 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers/README.md>. [source 04]
- [55] *The surreal and surcomplex reports*, ProveIt reader's guide, same snapshot and access date. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers/docs/README.md>. [source 04]
- [56] *Surreal and Surcomplex Numbers in Symbolic Computer Algebra*, report README in ProveIt, same snapshot and access date. <https://github.com/VladimirReshetnikov/ProveIt/blob/6996fee43cc97b6c16351def7507d59a95bf62f0/Algebra/SurrealNumbers/docs/foundations-and-computation/computer-algebra/README.md>. [source 04]